Exact Recurrences, Algebraic Independence, and Certified Statistical Recovery

Vertex balance, Bessel modules, and entropy transport

KEIO

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Abstract

We study exact differential equations and polynomial recurrences for ensemble traces, Bessel determinants and moments, together with arithmetic classifications and certified mixture estimation. The analytic method combines vertex visibility, a Wronskian balance identity and Laurent operator division. For circular ensembles it gives minimal differential order $n+1$ for every $\beta > 0$ and nonnegative integer twist, and exact moment-recurrence invariants. Jacobi traces, orthogonal-group moments, even-parameter vertex integrals, decaying Bessel moments and planar-walk moments are treated with explicit parameter restrictions.

At every positive even Dyson parameter, the normalized circular integral generates the same irreducible Bessel symmetric power. An invertible Laurent jet gauge identifies its function and arithmetic-value fields across parameters. Its unipotent block gives minimum recurrence degree $n-1$ for every integer twist, including exceptional parameters missed by exponent divisibility. A distinct three-particle family at $\beta = 2q-1$ has degree one; minimum-order degrees are classified for all three-particle parameters. Jacobi degree bounds are extended through zero indicial parameters and beyond a positivity condition, with a full one- and two-particle classification. Exponential-tail moment uniqueness also proves optimality on every real-shifted parity lattice for decaying Bessel products.

The arithmetic part determines algebraicity under sampling and quasipolynomial weighting, multipoint jet relations and field laws. The statistical part gives known-support binomial and Poisson recovery rates, entropy–transport bounds and finite-grid likelihood certificates. Weighted Franel recurrences, Apéry limits and split-beta orthogonality provide further applications. Construction, minimality, arithmetic specialization and computational approximation are distinguished throughout; noninteger-twist minimality and residual resonant degree problems remain explicitly open.

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1 Scope and organization

This volume has four parts. Part I develops scalar equations and minimal recurrences, beginning with determinants and then extending the method to positive-parameter ensembles, orthogonal groups and Bessel moments. Part II develops differential Galois theory, local monodromy, algebraicity and arithmetic specialization. Part III treats entropy–transport bounds and certified mixture estimation. Part IV treats weighted Franel sequences and arithmetic classifications. The analytical and statistical strands use different invariants: exponential factors and local monodromy in the former, posterior relative entropy and likelihood scores in the latter.

General-parameter recurrences and their refinements

The six ensemble and moment theorems are stated together in Section 11. Their proofs establish visibility of every relevant vertex, absence of nonzero apparent singularities, and the transfer of indicial factors to recurrence coefficients. The following refinements use this structure together with the arithmetic part.

1. **Even-parameter modules and arithmetic fields.** Theorems 20.1–20.3 identify the circular integral at every $\beta = 2m$, $m \geq 1$, and integer $q \geq 0$ with the same Bessel symmetric power. The complete initial jets are related by invertible Laurent matrices, with an explicit determinant. Their function fields and numerical jet fields at nonzero algebraic arguments are independent of m, q at fixed dimension.

2. **Exceptional twist degrees.** Theorem 21.2 proves minimum coefficient degree $n-1$ for every positive even β and every integer twist, among eventual recurrences of arbitrary order. Theorem 21.4 proves an infinite family with minimum degree one at $n = 3$, $\beta = 2q-1$. Theorem 21.5 classifies degrees at minimum order for every three-particle parameter. The latter is not a claim about every higher-order recurrence at the remaining non-even exceptional parameters.

3. **Jacobi resonance strata.** Theorem 22.2 replaces the positivity condition by an explicit finite resonance set, gives a multiplicity-sensitive degree lower bound everywhere, and proves the exact first-resonance drop. Corollary 22.4 classifies the one- and two-particle cases, including an explicit exceptional density equation.

4. **Shifted Bessel moment lattices.** Theorem 19.3 proves minimum width $\lceil n/2 \rceil$ and minimum degree $n+1$ for $\int_0^\infty t^{2N+r} f(t)\, dt$, every real $r > -1$, and every nonzero decaying linear combination of degree-n Bessel products. Its leading-factor conclusion holds for arbitrary eventual recurrences.

5. **A partial result for real twists.** Proposition 23.2 constructs a compact distribution transform for every real $q \geq 0$ and locates its possible singularities. Nonvanishing at all candidate vertices, and hence minimality for general noninteger q, is not concluded.

Arithmetic and statistical results

Theorem 32.1 gives the algebraicity dichotomy for holomorphic fixed-parity homogeneous Laurent Bessel observables, including arithmetic sampling and quasipolynomial weighting. Theorems 34.3–34.6 determine multipoint jet relations, and Theorem 34.8 gives the componentwise field laws. In statistics, Theorems 42.3 and 42.4 and Corollary 42.6 give deterministic finite grids that retain the stated mixing-law recovery rates; Theorem 42.5 gives the corresponding Gaussian likelihood guarantee. Classical antecedents are cited where used. An application of a classical theorem is not counted as an independent new method.

Coverage and parameter restrictions

Family	Conclusions and restrictions
Circular ensembles	Shape for $\beta > 0$, real $q > -1/2$; minimal differential order for integer $q \geq 0$. Untwisted recurrence width and degree for every $\beta > 0$ (Theorems 11.1, 11.3).
Twisted circular moments	Width for every $\beta > 0$ and integer $q \geq 0$; degree outside the stated finite exceptional set, and at every positive even β (Theorem 21.2).
Jacobi traces	Minimal transform order for every $a, b > -1$, $\beta > 0$. Degree and density refinements according to the resonance set (Theorem 22.2).
Orthogonal groups	Minimal moment recurrences for types B, D; the type B path interpretation restricts zero steps at the boundary (Theorem 11.5).
Even-parameter vertex integrals	Exact sorted origin orders and explicit solution basis (Theorem 11.6).
Bessel and planar-walk moments	Minimal widths and degrees, with leading-factor divisibility (Theorem 11.7).
Weighted Franel sequences	Minimal widths, including arithmetic subsequences; no general minimum-degree classification (Theorem 43.2).

Notation

A recurrence is eventual unless its index range is stated. Its width is the difference between extreme nonzero shifts; its degree is the maximum degree of its polynomial coefficients. The two minima are distinct optimization questions. Symbols are local to their sections: in statistics N is sample size, t replication, and M a known Poisson support endpoint. Algebraicity of a generating function means algebraicity of its germ over $\mathbb{C}(z)$; it is distinct from algebraicity of a numerical value.

Part I
Vertex balance, scalar equations, and minimal recurrences

2 The three combinatorial families and principal results

2.1 Three sequences

For $\ell \geq 1$ and $n \geq 0$ let

- $u_\ell(n)$ be the number of permutations of $[n] = \{1, \dots, n\}$ whose longest increasing subsequence has length at most ℓ;
- $y_k(n)$ be the number of standard Young tableaux with n cells and at most k rows; by the Robinson–Schensted correspondence, $y_k(n)$ is also the number of involutions of $[n]$ with no decreasing subsequence of length $k+1$;
- $m_\ell(n)$ be the number of perfect matchings of $[2n]$ with no $(\ell+1)$-crossing (equivalently, with no $(\ell+1)$-nesting) [7].

By the Robinson–Schensted correspondence, $u_\ell(n) = \sum_\lambda (f^\lambda)^2$ over partitions $\lambda \vdash n$ with at most ℓ rows, where f^λ is the number of standard tableaux of shape λ. In random matrix language,

$$u_\ell(n) = \mathbb{E}_{U(\ell)} |\operatorname{tr} U|^{2n}, \qquad m_\ell(n) = \mathbb{E}_{\mathrm{Sp}(2\ell)} (\operatorname{tr} U)^{2n}, \tag{2.1}$$

with Haar measure on the unitary and the compact symplectic group. The first identity is due to Gessel and Rains [12, 19]. The second follows from the classical identification of $(\ell+1)$-noncrossing perfect matchings with oscillating tableaux of height at most ℓ, which count the $\mathrm{Sp}(2\ell)$-invariants in $(\mathbb{C}^{2\ell})^{\otimes 2n}$ [2, 21, 7].

Write $b_m(t) = I_m(2t) = \sum_{j \geq 0} t^{2j+|m|}/(j!\,(j+|m|)!)$ for the modified Bessel functions. The basic generating functions are Bessel determinants:

$$\sum_{n \geq 0} u_\ell(n) \frac{t^{2n}}{(n!)^2} = D_\ell(t) := \det\big[b_{i-j}(t)\big]_{i,j=1}^{\ell} = \mathbb{E}_{U(\ell)} e^{2t \operatorname{Re} \operatorname{tr} U}, \tag{2.2}$$

$$Y_{2r}(t) = \det\big[b_{i-j} + b_{i+j-1}\big]_{i,j=1}^{r}, \qquad Y_{2r+1}(t) = e^t \det\big[b_{i-j} - b_{i+j}\big]_{i,j=1}^{r}, \tag{2.3}$$

$$\sum_{n \geq 0} m_\ell(n) \frac{t^{2n}}{(2n)!} = \mathbb{E}_{\mathrm{Sp}(2\ell)} e^{t \operatorname{tr} U} = \det\big[b_{i-j} - b_{i+j}\big]_{i,j=1}^{\ell} = e^{-t}\, Y_{2\ell+1}(t), \tag{2.4}$$

where $Y_k(t) = \sum_n y_k(n) t^n/n!$. Here (2.2) is Gessel's theorem [12], (2.3) is due to Gordon and Gessel [12], and (2.4) follows from Weyl's integration formula for $\mathrm{Sp}(2\ell)$ and Andreief's identity (Lemma 8.1), together with (2.3). In particular all three sequences are P-recursive.

2.2 Conjectures and claims

Bergeron, Favreau and Krob [3] computed differential equations and recurrences for small heights and conjectured their general shape. For tableaux, the conjecture states that Y_k satisfies an equation of order $\lfloor k/2 \rfloor + 1$ with coefficients of degree at most $\lfloor k/2 \rfloor$, leading coefficient $t^{\lfloor k/2 \rfloor}$, and the coefficient of the m-th derivative divisible by t^{m-1}. It was checked for $k \leq 11$ by Bergeron and Gascon [4], listed as open in Mishna's survey [18, Conj. 9], and proved in the manuscript [15]. Theorem 8.2 below recalls that result, and Appendix B contains a complete proof.

For pairs of tableaux, Bergeron and Gascon [4] wrote down the equations satisfied by

$$G_\ell(x) := D_\ell(\sqrt{x}) = \sum_{n\ge 0} u_\ell(n)\frac{x^n}{(n!)^2}$$

for $\ell \le 7$, of orders $3,\dots,8$, each with leading term $x^\ell\,\partial_x^{\ell+1}$. They state that these "settle the corresponding conjectures for the cases $k \le 7$". We formulate this as follows.

Conjecture 2.1 (Bergeron–Favreau–Krob, Bergeron–Gascon; pairs)**.** *For every $\ell \ge 1$ there are polynomials $q_0,\dots,q_{\ell+1}$ of degree at most ℓ with $q_{\ell+1} = x^\ell$ and $x^{m-1} \mid q_m$ for $m \ge 1$, such that $\sum_{m=0}^{\ell+1} q_m(x) G_\ell^{(m)}(x) = 0$.*

Forrester and Wei [11] recently derived scalar differential equations for unitary matrix integrals $\langle (\det U)^q e^{x^{1/2}\operatorname{tr}(U+U^\dagger)}\rangle_{U(\ell)}$ from a first-order matrix system. For $q = 0$ the integral is $G_\ell(x)$ [11, (4.6)]. Their Proposition 4.1 is stated for a differential-difference system that covers these integrals and their β-analogues. For $q = 0$ it asserts an equation $(x^\ell\partial^{\ell+1} + q_\ell(x)x^{\ell-1}\partial^\ell + \cdots + q_0(x))G_\ell = 0$ in which "the degree in x of the coefficient of each d^j/dx^j is $[(\ell-1+j)/2]$". The elimination procedure behind it is described rather than carried out in full. Appendix B of [11], due to Bornemann, argues that the order $\ell+1$ is optimal, using Siegel's theorem on the algebraic independence of I_0 and I_0'.

For the sequence itself, Forrester and Wei note that the conjectured recurrence form of [3], with $\lfloor \ell/2 \rfloor + 1$ terms beyond the first, is not sharp: for $\ell = 4$ a recurrence with only three terms holds. They state the corrected form as Proposition 4.3, obtained "(by inspection)": the sequence satisfies a recurrence of order $\lfloor (\ell-1)/2 \rfloor + 1$. From the cases $\ell = 2,\dots,5$ they observe "an obvious pattern" that the degree of the polynomial coefficients equals $\ell - 1$, which they call "difficult to anticipate". Here we prove the recurrence structure and the corresponding order and degree lower bounds, also treating tableaux and matchings.

2.3 Main results

Throughout, θ denotes the Euler operator $x\,d/dx$ or $t\,d/dt$, and for an operator written in the form $\sum_k x^k P_k(\theta)$ we call the polynomials P_k its *θ-coefficients*. We write $K = \lceil \ell/2 \rceil$, and for $0 \le b \le \ell$

$$\lambda_b = 2(\ell - 2b), \qquad \gamma_b = \tfrac{1}{2}\big((\ell-b)^2 + b^2\big). \tag{2.5}$$

Theorem 2.2 (The Bessel–Toeplitz operator)**.** *Let $\ell \ge 1$.*

(a) G_ℓ satisfies no linear differential equation of order at most ℓ over $\mathbb{C}(x)$. Its minimal operator, normalized to have coprime polynomial coefficients, is

$$L_\ell = \sum_{m=0}^{\ell+1} q_m(x)\,\partial_x^m, \qquad q_{\ell+1} = x^\ell,$$

$$x^{m-1} \mid q_m \ (1 \le m \le \ell+1), \qquad \deg q_m = \Big\lfloor \frac{\ell - 1 + m}{2} \Big\rfloor.$$

Every operator of order $\ell+1$ with polynomial coefficients annihilating G_ℓ is a polynomial multiple of L_ℓ.

(b) L_ℓ has no singular point in $\mathbb{C}\setminus\{0\}$. One has $xL_\ell = \sum_{k=0}^{K} x^k P_k(\theta)$ with $\deg P_k = \ell + 1 - 2k$, where

$$P_0(\theta) = \prod_{b=0}^{\ell} (\theta + b(\ell - b)), \qquad \sum_{k=0}^{K} \operatorname{lc}(P_k)\,\mu^{\ell+1-2k} = \prod_{b=0}^{\ell} (\mu - \ell + 2b),$$

and, if ℓ is even, $P_K(\theta) = \operatorname{lc}(P_K)\,(\theta + \ell^2/8)$. In particular the local exponents of L_ℓ at $x = 0$ are the $\ell+1$ integers $-b(\ell-b)$, $0 \le b \le \ell$.

(c) At $x = \infty$ the operator L_ℓ has a basis of formal solutions $e^{\lambda_b \sqrt{x}}\, x^{-\gamma_b/2}(1 + O(x^{-1/2}))$, $0 \le b \le \ell$.

The same statements hold in the variable $t = \sqrt{x}$ for $D_\ell(t)$; see Theorem 5.8. There the minimal operator has leading coefficient t^ℓ, coefficients of degree at most ℓ, and exponents $-2b(\ell - b)$ at $t = 0$.

Corollary 2.3. *The shape asserted in Proposition 4.1 of [11] holds for $q = 0$, with the exact degrees of the coefficients; the order $\ell + 1$ is minimal, which is Bornemann's claim; and Conjecture 2.1 holds for every $\ell \ge 1$. The integrality statement is completed constructively in Section 10.*

Theorem 2.4 (Permutations with bounded increasing subsequences)**.** *Let $\ell \ge 1$, $K = \lceil \ell/2 \rceil$, and let $P_0, \dots, P_K$ be as in Theorem 2.2. Put*

$$R_0(n) = \prod_{b=1}^{\ell-1} (n + b(\ell - b)), \qquad R_k(n) = P_k(n-k)\,\big[(n-1)(n-2)\cdots(n-k+1)\big]^2 \quad (1 \le k \le K).$$

(a) $\sum_{k=0}^{K} R_k(n)\, u_\ell(n-k) = 0$ for all $n \ge 1$, with $u_\ell(m) = 0$ for $m < 0$. Each R_k has degree exactly $\ell - 1$, and

$$R_K(n) = (-1)^K \prod_{j=0}^{K-1} (\ell - 2j)^2 \cdot \big[(n-1)\cdots(n-K+1)\big]^2 \cdot \begin{cases} 1, & \ell \text{ odd}, \\ n - K + \ell^2/8, & \ell \text{ even}. \end{cases}$$

(b) The characteristic polynomial is $\sum_{k=0}^{K} \mathrm{lc}(R_k)\lambda^{K-k} = \prod_{j=0}^{K-1} (\lambda - (\ell - 2j)^2)$.

(c) Every linear recurrence with polynomial coefficients satisfied by $u_\ell(n)$ for all large n has order at least K and coefficients of degree at least $\ell - 1$. Its leading coefficient, written as the coefficient of $u_\ell(n)$ in a relation among $u_\ell(n), u_\ell(n-1), \dots$, is divisible by $R_0(n)$. Every such recurrence of order K is a polynomial multiple of the one in (a).

For $\ell = 4$ the recurrence in (a) is

$$(n+3)^2(n+4)\, u_4(n) - 2(10n^3 + 31n^2 + 11n - 12)\, u_4(n-1) + 64n(n-1)^2\, u_4(n-2) = 0,$$

which is [11, (4.7)]. Table 3 lists the cases $\ell \le 7$.

Corollary 2.5. *Proposition 4.3 of [11] holds. The minimal order of a recurrence for $u_\ell(n)$ is exactly $\lfloor (\ell-1)/2 \rfloor + 1 = \lceil \ell/2 \rceil$, the minimal degree is exactly $\ell - 1$, and the primitive minimal recurrence is unique up to a nonzero constant factor. In particular a recurrence of the form conjectured in [3], with $\lfloor \ell/2 \rfloor + 1$ terms beyond the first, exists; its order is minimal exactly when ℓ is odd, and for even ℓ the minimal order is one less.*

Twisted integrals

For $q \in \mathbb{Z}$ put

$$D_{\ell,q}(t) = \det\big[b_{i-j+q}(t)\big]_{i,j=1}^{\ell} = \mathbb{E}_{U(\ell)}(\det U)^q\, e^{2t\,\mathrm{Re\,tr}\,U}.$$

Since $D_{\ell,-q} = D_{\ell,q}$, we assume $q \ge 0$. By Lemma 6.1,

$$G_{\ell,q}(x) := x^{-q\ell/2} D_{\ell,q}(\sqrt{x}) = \sum_{n \ge 0} u_{\ell,q}(n) \frac{x^n}{n!\,(n+q\ell)!}, \qquad u_{\ell,q}(n) = \sum_{\mu} f^\mu f^{\mu + (q^\ell)},$$

where μ runs over the partitions of n with at most ℓ parts and $\mu + (q^\ell)$ adds q to each of the ℓ parts. Thus $G_{\ell,0} = G_\ell$. Up to a constant factor, $G_{\ell,q}$ is the hypergeometric function ${}_0F_1(q + \ell; (x)^\ell)$ with Jack parameter 1 [11, (2.4), (2.5)]. For $\beta = 2$ and $n = \ell$, the function in [11, Prop. 4.1] is a constant multiple of ${}_0F_1(2\ell + a + b; (x)^\ell)$ [11, (3.10) and Rem. 3.4]; the choice $a + b = q - \ell$, that is $B_p = p(\ell + q - p)$ in the system [11, (3.6)], gives $G_{\ell,q}$. The case $q = \ell$ arises in the random matrix theory of joint moments of characteristic polynomials and their derivatives [16], [11, §4.3].

Theorem 2.6 (Twisted integrals). *Let $\ell \geq 1$, $q \geq 0$ and $K = \lceil \ell/2 \rceil$.*

(a) $G_{\ell,q}$ satisfies no linear differential equation of order at most ℓ over $\mathbb{C}(x)$. Its minimal operator $L_{\ell,q} = \sum_m q_m(x)\partial_x^m$, normalized to have coprime polynomial coefficients, has the shape of Theorem 2.2(a): $q_{\ell+1} = x^\ell$, $x^{m-1} \mid q_m$ and $\deg q_m = \lfloor (\ell - 1 + m)/2 \rfloor$. It has no singular point in $\mathbb{C} \setminus \{0\}$, and $xL_{\ell,q} = \sum_{k=0}^{K} x^k P_k(\theta)$ with $\deg P_k = \ell + 1 - 2k$, $\sum_k \operatorname{lc}(P_k)\mu^{\ell+1-2k} = \prod_b (\mu - \ell + 2b)$,

$$P_0(\theta) = \prod_{i=0}^{\ell} (\theta + i(\ell + q - i)),$$

and, for even ℓ, $P_K(\theta) = \operatorname{lc}(P_K)(\theta + \ell^2/8 + q\ell/2)$. At infinity $L_{\ell,q}$ has the formal solutions $e^{\lambda_b \sqrt{x}} x^{-(\gamma_b + q\ell)/2}(1 + O(x^{-1/2}))$.

(b) The minimal operator of $D_{\ell,q}(t)$ has order $\ell + 1$ and no singular point in $\mathbb{C} \setminus \{0\}$. Its exponents at $t = 0$ are $q(\ell - 2i) - 2i(\ell - i)$, $0 \leq i \leq \ell$. Its leading coefficient is t^ℓ if one of these exponents is 0, and $t^{\ell+1}$ otherwise.

(c) $u_{\ell,q}$ satisfies a recurrence of order K whose coefficients have degree $\ell - 1$, with leading coefficient $\prod_{i=1}^{\ell-1}(n + i(\ell + q - i))$, and it satisfies no recurrence of smaller order.

Corollary 2.7. *For $\beta = 2$ and every $q \in \mathbb{Z}_{\geq 0}$, the shape asserted in Proposition 4.1 of [11] holds for ${}_0F_1(q + \ell; (x)^\ell)$, with the exact degrees, and the order $\ell + 1$ is minimal. In particular this holds for $q = \ell$, and Bornemann's optimality claim [11, App. B] also holds for the Hankel determinant w_ℓ treated there. Section 10 also establishes integrality in this exact normalization.*

For tableaux let $r = \lfloor k/2 \rfloor$, $\varepsilon = k - 2r$, $e_b = b^2 - 2br - \varepsilon b$ and $\lambda_j^{(k)} = \varepsilon + 2r - 4j$ for $0 \leq j \leq r$.

Theorem 2.8 (Tableaux of bounded height). *Let $k \geq 2$ and put $R = r + 1 - [4 \mid k]$, where $[4 \mid k]$ is 1 if 4 divides k and 0 otherwise. There are polynomials $c_0, \ldots, c_R$ of degree at most r with*

$$\sum_{i=0}^{R} c_i(N)\, y_k(N - i) = 0 \quad (N \geq 1), \qquad c_0(N) = \prod_{b=1}^{r} (N + b(k - b)),$$

and $c_R(N) = \prod_{j=0}^{r} (-\lambda_j^{(k)}) \cdot (N-1)(N-2)\cdots(N-r)$ if $4 \nmid k$, while $c_R(N) = \prod_{j \neq r/2} (-\lambda_j^{(k)}) \cdot (N-1)\cdots(N-r+1)\,(N-r+r^2/2)$ if $4 \mid k$. The characteristic polynomial is $\prod_{j : \lambda_j^{(k)} \neq 0} (\lambda - \lambda_j^{(k)})$. Every recurrence satisfied by y_k for large N has order at least R and degree at least r, and its leading coefficient is divisible by c_0. Every recurrence of order R is a polynomial multiple of this one.

Theorem 2.9 (Noncrossing matchings). *Let $\ell \geq 1$ and $K = \lceil \ell/2 \rceil$. There are polynomials $d_0, \ldots, d_K$ of degree exactly ℓ with*

$$\sum_{i=0}^{K} d_i(N)\, m_\ell(N - i) = 0 \quad (N \geq 1), \qquad d_0(N) = \prod_{b=1}^{\ell} (2N + b(2\ell + 1 - b)),$$

and $d_K(N) = \prod_{j=0}^{\ell} (-\lambda_j) \cdot (2N-1)(2N-2)\cdots(2N-\ell)$ if ℓ is odd, while $d_K(N) = \kappa_\ell (2N - 1)\cdots(2N - \ell + 1)\big(2N - \ell + \frac{\ell(\ell+1)}{2}\big)$ with $\kappa_\ell = \prod_{j \neq \ell/2} (-\lambda_j)$ if ℓ is even. The characteristic polynomial is $\prod_{j < \ell/2} (\lambda - 4(\ell - 2j)^2)$. Every recurrence satisfied by m_ℓ for large N has order at least K and degree at least ℓ, and its leading coefficient is divisible by d_0. Every recurrence of order K is a polynomial multiple of this one.

In Theorems 2.8 and 2.9 the coefficients are normalized as in Section 8; only their ratios matter. Finally, the recurrences translate into differential equations for the underlying distributions.

Theorem 2.10 (Distributions of traces)**.** *Let $\ell \geq 2$ and $K = \lceil \ell/2 \rceil$. The distribution of $|\operatorname{tr} U|^2$, U Haar distributed in $U(\ell)$, has a density ρ_ℓ on $[0, \ell^2]$ that is annihilated, in the sense of distributions on $\mathbb{R}$, by*

$$\mathcal{L}_\ell = \sum_{j=0}^{K} w^j \, R_{K-j}(K - j - 1 - \theta), \qquad \theta = w\frac{d}{dw}.$$

The operator $\mathcal{L}_\ell$ has order $\ell - 1$ and is Fuchsian on $\mathbb{P}^1$. One has $\mathcal{L}_\ell = w^{K-1}\mathcal{L}_\ell^\circ$, where $\mathcal{L}_\ell^\circ$ has polynomial coefficients and leading coefficient $c\, w^{\lfloor \ell/2 \rfloor} \prod_{j=0}^{K-1}(w - (\ell - 2j)^2)$ with $c \neq 0$. No nonzero operator of order less than $\ell - 1$ annihilates ρ_ℓ in the sense of distributions on $\mathbb{R}$, and among such annihilators of order $\ell - 1$ the operator $\mathcal{L}_\ell$ is unique up to a factor in $\mathbb{C}(w)$. Its exponents at $w = 0$ are $0, 0, 1, 1, \ldots, K - 2, K - 2$, together with $\ell^2/8 - 1$ when ℓ is even. Its exponents at $w = \infty$, in the coordinate $1/w$, are $1 - b(\ell - b)$, $1 \leq b \leq \ell - 1$. Analogous statements hold for $\operatorname{tr} U$ on $\mathrm{Sp}(2\ell)$, with order ℓ, and for tableaux, with order $\lfloor k/2 \rfloor$; see Section 9.

2.4 Method

The proof of Theorem 2.2 has four steps.

Step 1: the solution space. We write $D_\ell(t)$ as a symmetric multilinear Toeplitz form in solutions of Bessel's equation. Since the differential Galois group of Bessel's equation is SL_2, the solution space of the minimal operator is spanned by $\ell + 1$ *vertex integrals* Ψ_b. In Ψ_b, b of the ℓ eigenvalue variables run over the negative real axis instead of the unit circle.

Step 2: infinity. Laplace's method at the critical point with b coordinates equal to -1 and $\ell - b$ equal to 1 shows that $\Psi_b \sim C_b e^{\lambda_b t} t^{-\gamma_b}$, with pairwise distinct rates λ_b.

Step 3: the origin. Here the vertex integrals are genuinely hard to evaluate: pairs of them share the same leading exponent, and the leading terms cancel in complicated ways. We only prove the crude bound $\Psi_b = O(t^{-2\lfloor b/2 \rfloor(\ell - \lfloor b/2 \rfloor)} |\log t|^b)$. It comes from expanding a mixed Toeplitz determinant and from an elementary inequality for the total displacement of a permutation.

Step 4: the squeeze. The crude bound gives a lower bound for the sum of the local exponents at 0. Abel's identity for the Wronskian, combined with the asymptotics at infinity, gives the corresponding upper bound, with a deficit equal to the total contribution of the apparent singularities. The two bounds coincide because of the identity

$$\sum_{b=0}^{\ell} \gamma_b - 2\sum_{b=0}^{\ell} b(\ell - b) = \binom{\ell + 1}{2}.$$

This forces both the absence of apparent singularities and the exact exponents. The shape of the operator then follows as in [15]. For the twisted determinants the crude bound involves the order statistics of $2i(\ell - i) - q(\ell - 2i)$, whose sum does not depend on q, so the same squeeze applies.

The passage to recurrences uses two simple principles, Lemmas 3.6 and 3.2. First, every recurrence of order M yields an annihilating operator whose *span* in the variable is M, and such an operator can carry at most M (for t: $2M$) solutions with distinct nonzero exponential types. Second, the lowest θ-coefficient of that operator is essentially the leading coefficient of the recurrence. It must vanish at every negative exponent of the minimal operator at the origin, with multiplicity. The exponents $-b(\ell - b)$ thus become the factor R_0 of every recurrence for u_ℓ, and the $\ell - 1$ negative exponents force the degree bound $\ell - 1$.

The same scheme applies verbatim to the Hankel (orthogonal and symplectic) Bessel determinants (2.3)–(2.4), whose minimal operators were determined in [15]. Table 1 summarizes the data that drive all the results.

	$D_\ell(t)$ (unitary)	$Y_k(t)$ (tableaux)	$e^{-t}Y_{2\ell+1}(t)$ (symplectic)
order of minimal operator	$\ell+1$	$r+1$	$\ell+1$
rates at ∞	$2(\ell-2b)$	$\varepsilon+2r-4j$	$2(\ell-2j)$
exponents γ at ∞ ($e^{\lambda t}t^{-\gamma}$)	γ_b, see (2.5)	$\gamma_j^{(k)}$, see (8.1)	$\gamma_j^{(2\ell+1)}$
exponents at 0	$-2b(\ell-b)$	$-b(k-b)$	$-b(2\ell+1-b)$
minimal recurrence: order	$\lceil \ell/2 \rceil$	$r+1-[4 \mid k]$	$\lceil \ell/2 \rceil$
minimal recurrence: degree	$\ell-1$	r	ℓ

Table 1: Local data of the three Bessel determinants, with $k = 2r+\varepsilon$. The indices run over $0 \le b \le \ell$ for D_ℓ, over $0 \le b, j \le r$ for Y_k, and over $0 \le b, j \le \ell$ in the symplectic case. For Y_k and the symplectic case the exponents at 0 are distinct; for D_ℓ the values for b and $\ell - b$ coincide.

2.5 Scope and prior work

Table 2 lists the statements from the literature that are settled here.

Statement	Source	Previous status	Here
Shape of the equation for G_ℓ (pairs conjecture)	[3, 4]	equations computed for $\ell \le 7$	Cor. 2.3
Shape and degrees $\lfloor(\ell-1+j)/2\rfloor$, $q=0$	[11, Prop. 4.1]	proof sketched	Thm. 2.2
The same for $\beta=2$ and all $q \in \mathbb{Z}_{\ge 0}$	[11, Prop. 4.1]	proof sketched; $q=\ell$: $\ell \le 5$ listed	Thm. 2.6
Order $\ell+1$ is optimal ($q=0$ and $q=\ell$)	[11, App. B]	argued	Cor. 2.3, 2.7
Recurrence of order $\lfloor(\ell-1)/2\rfloor+1$ for u_ℓ	[11, Prop. 4.3]	"by inspection"	Thm. 2.4
Degree $\ell-1$ of that recurrence	[11]	observed for $\ell \le 5$	Thm. 2.4
Recurrence form with $\lfloor \ell/2 \rfloor + 1$ terms	[3]	conjectured	Cor. 2.5
Minimal recurrences for y_k and m_ℓ		not determined	Thms. 2.8, 2.9

Table 2: Statements established here and their relation to the cited literature. The table records mathematical coverage, not a comprehensive priority claim.

The existence of differential equations of order $\ell+1$ for G_ℓ and of the recurrences of order $\lceil \ell/2 \rceil$ was known in the sense of [4, 11]. The results proved here include:

- a complete proof of the shape of the minimal operator, including the exact coefficient degrees, the exponents, and the absence of apparent singularities;
- minimality of the order through an explicit differential-Galois argument, complementing the Bessel-module construction of [11, Appendix B];
- the recurrence of order $\lceil \ell/2 \rceil$, with explicit extreme coefficients, minimality in order and degree, and uniqueness;
- the same shape and minimality for the twisted integrals with $(\det U)^q$, $q \in \mathbb{Z}$. For $q=\ell$, Keating and Wei derived an equation of order $\ell+1$ [16], and [11, Prop. 4.6] lists the equations for $\ell \le 5$.

For tableaux, the minimal operators are taken from [15]. The equations of the right shape also follow from the Forrester–Kumar system for Jacobi traces [10], as explained in [15]. The recurrence results for tableaux and matchings include minimality and explicit extreme coefficients; recurrences for matchings with small crossing number were computed in [8]. For the ℓ-step planar random walk, whose generating function is $I_0(2t)^\ell$, the recurrence of order $\lceil \ell/2 \rceil$ and degree $\ell - 1$ was given in general by Verrill [22]; see [5]. Remark 45.1 proves order minimality. Theorem 45.3 proves degree minimality by combining a local splitting lemma with the iterated Borel bound; the numerical exponents at zero alone do not supply this conclusion.

For every $\beta > 0$ and real $q \geq 0$, Corollary 10.4 constructs the Jack-family scalar equation. Theorem 11.1 extends the coefficient-degree statement to $q > -1/2$ and proves minimality for nonnegative integer twists. Theorem 11.4 treats general Jacobi parameters. The even-parameter Galois and arithmetic refinements are in Theorems 20.1–20.3; noninteger-twist minimality remains open here.

Organization

Section 3 collects facts on linear differential operators, including the filtration and squeeze lemmas. Section 4 records the dictionaries between recurrences and operators. Section 5 proves Theorem 2.2, Section 6 extends it to the twisted determinants (Theorem 2.6), and Section 7 proves Theorem 2.4. Section 8 recalls the Hankel case and proves Theorems 2.8 and 2.9. Section 9 treats the distributions (Theorem 2.10) and explains the transport principle behind them. Appendix A describes the computer checks.

3 Linear differential operators

Let $\partial = d/dt$ and $\theta = t\partial$, and write $x^{\underline{m}} = x(x-1)\cdots(x-m+1)$ for the falling factorial, so that $t^m \partial^m = \theta^{\underline{m}}$. An operator $L = \sum_{m=0}^{n} p_m(t)\partial^m$ with $p_m \in \mathbb{C}[t]$ and $p_n \neq 0$ is *primitive* if the p_m have no common zero. For a primitive operator, a point $\tau \in \mathbb{C}$ is singular if $p_n(\tau) = 0$; it is *apparent* if all local solutions are holomorphic at τ.

3.1 θ-forms

After left multiplication by a power of t, every operator with polynomial coefficients becomes a finite sum $\sum_k t^k Q_k(\theta)$ with $Q_k \in \mathbb{C}[\theta]$. Since $t^k \theta^i = \sum_{m \leq i} S(i,m)\, t^{k+m}\partial^m$, where $S(i,m)$ are Stirling numbers of the second kind, the ∂-order of such a sum is $\max_k \deg Q_k$.

Definition 3.1. A *θ-form* is an operator $A = \sum_{k=k_0}^{k_1} t^k Q_k(\theta)$ with $Q_k \in \mathbb{C}[\theta]$, $k_0 \leq k_1$ integers and $Q_{k_0} Q_{k_1} \neq 0$. Its *span* is $k_1 - k_0$. Its *lowest coefficient* is Q_{k_0} and its *top coefficient* is Q_{k_1}.

A θ-form is Fuchsian at 0 if and only if $\deg Q_{k_0} = \max_k \deg Q_k$. In that case Q_{k_0} is, up to a constant factor, the indicial polynomial at 0; this is Fuchs' criterion [13]. Similarly, A is Fuchsian at ∞ if and only if $\deg Q_{k_1} = \max_k \deg Q_k$.

3.2 Expansions at the origin and the filtration lemma

Let $\mathcal{N}$ be the space of functions f on a slit punctured disc $\{0 < |t| < \delta\} \setminus (-\delta, 0]$ that can be written as

$$f = \sum_{\nu \geq \nu_0,\ \nu \in \mathbb{Z}} t^\nu P_\nu(\log t), \tag{3.1}$$

with polynomials P_ν of bounded degree and the series converging absolutely on the slit disc. Equivalently, f is a finite sum of products of a power of $\log t$ and a convergent Laurent series

with finitely many negative powers. For $0 \neq f \in \mathcal{N}$ let $\operatorname{ord} f = \min\{\nu : P_\nu \neq 0\}$; the term $t^{\operatorname{ord} f} P_{\operatorname{ord} f}(\log t)$ is the *leading part*. If $|f(t)| \leq Ct^{-a}|\log t|^c$ for small $t > 0$, then $\operatorname{ord} f \geq -a$, because the leading part dominates all other terms as $t \to 0^+$.

Lemma 3.2 (filtration)**.** *Let $A = \sum_{k=k_0}^{k_1} t^k Q_k(\theta)$ be a θ-form, and let $V \subset \mathcal{N}$ be a finite-dimensional space of solutions of $Af = 0$. For $\nu \in \mathbb{Z}$ put $V_{\geq\nu} = \{f \in V : f = 0 \text{ or } \operatorname{ord} f \geq \nu\}$. Then*

$$\dim V_{\geq\nu}/V_{\geq\nu+1} \leq \operatorname{mult}_\nu(Q_{k_0}),$$

the multiplicity of ν as a root of Q_{k_0}, which is 0 if $Q_{k_0}(\nu) \neq 0$. Consequently $\dim V_{\geq\nu}$ is at most the number of roots of Q_{k_0} in $\mathbb{Z}_{\geq\nu}$, counted with multiplicity.

Proof. Write $\Lambda = \log t$ and $D = d/d\Lambda$ acting on polynomials in Λ. Then $\theta(t^\nu P(\Lambda)) = t^\nu(\nu + D)P(\Lambda)$. Let $f \in V_{\geq\nu}$ with expansion (3.1). In Af, the terms $t^k Q_k(\theta)f$ with $k > k_0$ involve only powers t^μ with $\mu \geq \nu + k > \nu + k_0$. Hence the coefficient of $t^{\nu+k_0}$ in $Af = 0$ gives $Q_{k_0}(\nu + D)P_\nu = 0$. Let $m = \operatorname{mult}_\nu(Q_{k_0})$. Taylor expansion gives $Q_{k_0}(\nu + D) = D^m\,(c + DH(D))$ with $c \neq 0$ and a polynomial H. The factor $c + DH(D)$ is invertible on polynomials, because D is nilpotent there. Therefore $D^m P_\nu = 0$, that is, $\deg P_\nu < m$. The linear map $V_{\geq\nu} \to \mathbb{C}[\Lambda]_{<m}$, $f \mapsto P_\nu$, has kernel $V_{\geq\nu+1}$, which proves the first claim. For the second, note that $V_{\geq\mu} = 0$ for large μ, because a finite-dimensional space of nonzero elements of $\mathcal{N}$ has bounded orders. Summing the first claim over $\mu \geq \nu$ gives the bound. □

Lemma 3.3 (squeeze)**.** *Let L be an operator of order n that is Fuchsian at 0, with indicial polynomial I, and suppose that its local solution space V at 0 is contained in $\mathcal{N}$. Then the roots of I are integers, $\nu_1 \geq \cdots \geq \nu_n$. If $f_1, \ldots, f_n$ is any basis of V with $\operatorname{ord} f_1 \geq \cdots \geq \operatorname{ord} f_n$, then $\operatorname{ord} f_i \leq \nu_i$ for all i. In particular*

$$\sum_{i=1}^{n} \operatorname{ord} f_i \leq \sum_{i=1}^{n} \nu_i,$$

with equality if and only if $\operatorname{ord} f_i = \nu_i$ for all i. In the case of equality, $\dim V_{\geq\nu} = \#\{i : \nu_i \geq \nu\}$ for every $\nu \in \mathbb{Z}$.

Proof. Apply Lemma 3.2 to the θ-form of $t^{n-\operatorname{ord}_0 p_n} L$, whose lowest coefficient is I. The span of $\{f_i : \operatorname{ord} f_i \geq \nu\}$ lies in $V_{\geq\nu}$, so

$$\#\{i : \operatorname{ord} f_i \geq \nu\} \leq \dim V_{\geq\nu} \leq \#\{\text{roots of } I \text{ in } \mathbb{Z}_{\geq\nu}\}.$$

Taking $\nu \leq \min_i \operatorname{ord} f_i$ shows that I, which has degree n, has n integer roots. For a given i, choosing $\nu = \operatorname{ord} f_i$ gives $i \leq \#\{j : \nu_j \geq \operatorname{ord} f_i\}$, hence $\nu_i \geq \operatorname{ord} f_i$. The remaining assertions follow, since equality of the sums forces $\operatorname{ord} f_i = \nu_i$, and then the two outer quantities in the display coincide. □

3.3 Abel's identity and the top degree

The next two lemmas are standard; we recall short proofs from [15].

Lemma 3.4 (Abel)**.** *Let $L = \sum_{m=0}^{n} p_m \partial^m$ be primitive, and let W be the Wronskian of a basis of solutions on a simply connected domain. Then $W'/W = -p_{n-1}/p_n$, and:*

(a) if τ is a regular singular point with exponents $\rho_1, \ldots, \rho_n$, the residue of W'/W at τ is $\sum_i \rho_i - \binom{n}{2}$;

(b) if $p_n(\tau) = 0$ and all solutions are holomorphic at τ, the residue of W'/W at τ is a positive integer.

Proof. The identity is Abel's. For (a), write $(t-\tau)^{n-A}L = \sum_m s_m(t)(t-\tau)^m\partial^m$ with $A = \operatorname{ord}_\tau p_n$ and $s_n(\tau) \neq 0$. The indicial polynomial is $\sum_m s_m(\tau)\rho^{\underline{m}} = s_n(\tau)\big(\rho^n - \big(\binom{n}{2} - s_{n-1}(\tau)/s_n(\tau)\big)\rho^{n-1} + \cdots\big)$, and the residue of $-p_{n-1}/p_n$ is $-s_{n-1}(\tau)/s_n(\tau)$. For (b), W is holomorphic at τ, so the residue is $\operatorname{ord}_\tau W \geq 0$. If $W(\tau) \neq 0$, Cramer's rule applied to the linear system that the coefficients p_m/p_n satisfy shows that all p_m/p_n are holomorphic at τ. Then $(t-\tau)$ divides every p_m, contradicting primitivity. □

Lemma 3.5 (top degree)**.** *Let $L = \sum_{m=0}^{n} p_m\partial^m$, $d = \max_m \deg p_m$, and $\chi(\lambda) = \sum_m [t^d]p_m\,\lambda^m$. Suppose there are solutions $y_1, \dots, y_n$, pairwise distinct $\lambda_1, \dots, \lambda_n$, constants $C_j \neq 0$ and γ_j with*

$$y_j^{(m)}(t) = C_j e^{\lambda_j t} t^{-\gamma_j}\big(\lambda_j^m + O(t^{-1/2})\big) \qquad (t \to +\infty,\ 0 \leq m \leq n).$$

Then $\chi(\lambda) = [t^d]p_n \cdot \prod_j(\lambda - \lambda_j)$; in particular $\deg p_n = d$.

Proof. We have $0 = C_j^{-1}e^{-\lambda_j t}t^{\gamma_j}Ly_j = t^d\chi(\lambda_j) + O(t^{d-1/2})$. So χ is a nonzero polynomial of degree at most n with the n distinct roots λ_j. Hence $\deg\chi = n$ and $[t^d]p_n \neq 0$. □

3.4 Exponential types bound the span

The following lemma converts exponential types at infinity into lower bounds for the order of recurrences.

Lemma 3.6 (types)**.** *Let $A = \sum_{k=k_0}^{k_1} t^k Q_k(\theta)$ be a θ-form. Suppose $y_1, \dots, y_N$ are solutions of $Ay = 0$ on an interval (T, ∞) such that, for some $C_j \neq 0$ and $\gamma_j \in \mathbb{R}$,*

$$y_j^{(m)}(t) = C_j e^{\lambda_j t} t^{-\gamma_j}\big(\lambda_j^m + O(t^{-1/2})\big) \qquad (t \to +\infty,\ 0 \leq m \leq \max_k \deg Q_k),$$

where $\lambda_1, \dots, \lambda_N$ are pairwise distinct and nonzero*. Then $N \leq k_1 - k_0$.*

Proof. Write $d_k = \deg Q_k$ for $Q_k \neq 0$. From $\theta^i = \sum_{m\leq i} S(i,m)t^m\partial^m$ and $S(i,i) = 1$ we get $\theta^i y_j = C_j e^{\lambda_j t}t^{-\gamma_j}\big(\lambda_j^i t^i + O(t^{i-1/2})\big)$. Hence

$$t^k Q_k(\theta)y_j = C_j e^{\lambda_j t}t^{-\gamma_j}\big(\operatorname{lc}(Q_k)\lambda_j^{d_k}t^{k+d_k} + O(t^{k+d_k-1/2})\big).$$

Let $w = \max_k(k + d_k)$ and $S = \{k : Q_k \neq 0,\ k + d_k = w\}$. Then $Ay_j = 0$ gives $h(\lambda_j) = 0$ for $h(\lambda) = \sum_{k\in S}\operatorname{lc}(Q_k)\lambda^{w-k}$. Now $h(\lambda) = \lambda^{w-\max S}g(\lambda)$, where $g(\lambda) = \sum_{k\in S}\operatorname{lc}(Q_k)\lambda^{\max S-k}$ has degree $\max S - \min S$ and $g(0) \neq 0$. The λ_j are N distinct nonzero roots of g, so $N \leq \max S - \min S \leq k_1 - k_0$. □

3.5 Galois orbits

Let E be a Picard–Vessiot extension of $\mathbb{C}(t)$ with differential Galois group G.

Lemma 3.7. *For a D-finite element $f \in E$ over $\mathbb{C}(t)$, the $\mathbb{C}$-span U of the orbit $\{\sigma(f) : \sigma \in G\}$ is finite-dimensional, and it is the solution space in E of the minimal monic operator $L_f \in \mathbb{C}(t)[\partial]$ annihilating f. In particular $\operatorname{ord} L_f = \dim U$.*

Proof. Since L_f has coefficients in $\mathbb{C}(t)$, it annihilates every $\sigma(f)$, so $U \subseteq \operatorname{Sol}_E(L_f)$ and $\dim U \leq \operatorname{ord} L_f$. Conversely, U is finite-dimensional and G-stable. The monic operator L_U with solution space U, built from Wronskians of a basis, is therefore fixed by G, so its coefficients lie in $\mathbb{C}(t)$ by the Galois correspondence [23]. Since $L_U f = 0$, minimality gives $\operatorname{ord} L_f \leq \dim U$. □

4 Recurrences and operators

We record the three dictionaries used later. Let $a : \mathbb{Z}_{\ge 0} \to \mathbb{C}$ be a sequence and suppose

$$\sum_{s=0}^{M} c_s(n)\, a(n+s) = 0 \qquad (n \ge n_0), \tag{4.1}$$

with polynomials c_s of degree at most d and $c_0 c_M \neq 0$. We call M the *order*, d the *degree* and c_M the *leading coefficient* of the recurrence. In the backward normalization $\sum_{i=0}^{M} c_{M-i}(N-M)\, a(N-i) = 0$, the leading coefficient is the coefficient of $a(N)$.

Lemma 4.1 (exponential generating function)**.** *Let $Y(t) = \sum_n a(n) t^n/n!$ and assume (4.1). Then Y is annihilated by the θ-form*

$$A = \sum_{s=0}^{M} t^{M-s}\, (\theta - s)^{\underline{n_0}}\, c_s(\theta - s)\, \theta^{\underline{s}},$$

which has span M, θ-degree at most $d + M + n_0$, lowest coefficient $\theta^{\underline{M+n_0}}\, c_M(\theta - M)$ and top coefficient $\theta^{\underline{n_0}}\, c_0(\theta)$.

Proof. Since $\partial^s Y = \sum_n a(n+s) t^n/n!$ and θ acts on $t^n/n!$ by multiplication with n, the operator $B = \sum_s c_s(\theta)\partial^s$ satisfies $BY = \sum_n \big(\sum_s c_s(n) a(n+s)\big) t^n/n!$. This is a polynomial of degree less than n_0, so it is killed by $\theta^{\underline{n_0}}$. Using $\partial^s = t^{-s}\theta^{\underline{s}}$ and $t^M \theta^{\underline{n_0}} = (\theta - M)^{\underline{n_0}} t^M$, we obtain $t^M \theta^{\underline{n_0}} B = A$. Finally, $(\theta - M)^{\underline{n_0}} \theta^{\underline{M}} = \theta^{\underline{M+n_0}}$. □

Lemma 4.2 (double exponential generating function)**.** *Let $G(x) = \sum_n a(n) x^n/(n!)^2$, $\theta = x\, d/dx$, and assume (4.1). Then G is annihilated by*

$$O = \sum_{k=0}^{M} x^k\, (\theta + k)^{\underline{M+n_0}}\, c_{M-k}(\theta + k - M)\, \big(\theta^{\underline{M-k}}\big)^2,$$

a θ-form of span M with lowest coefficient $\theta^{\underline{M+n_0}}\, c_M(\theta - M)\, \big(\theta^{\underline{M}}\big)^2$. Conversely, if $\sum_{k=0}^{M} x^k P_k(\theta)$ annihilates G, then

$$\sum_{k=0}^{M} P_k(N-k)\, \big(N^{\underline{k}}\big)^2\, a(N-k) = 0 \qquad \text{for all } N \ge 0, \tag{4.2}$$

where terms with $N - k < 0$ are omitted.

Proof. For the converse, the coefficient of x^N in $x^k P_k(\theta) G$ is $P_k(N-k) a(N-k)/((N-k)!)^2$, and $(N!/(N-k)!)^2 = (N^{\underline{k}})^2$. For the direct statement, put $P_k(\theta) = c_{M-k}(\theta + k - M)(\theta^{\underline{M-k}})^2$ and $O_0 = \sum_k x^k P_k(\theta)$. For $N \ge M$, $(N-k)^{\underline{M-k}} = (N-k)!/(N-M)!$, so the coefficient of x^N in $O_0 G$ equals

$$\frac{1}{((N-M)!)^2} \sum_{k=0}^{M} c_{M-k}(N-M)\, a(N-k),$$

which vanishes for $N - M \ge n_0$ by (4.1). For $N < M$, every term contains a factor $(N-k)^{\underline{M-k}}$ with $N - k \ge 0 > N - M$, which vanishes. Hence $O_0 G$ is a polynomial of degree less than $M + n_0$ and is killed by $\theta^{\underline{M+n_0}}$. Finally, $\theta^{\underline{M+n_0}} x^k = x^k (\theta + k)^{\underline{M+n_0}}$. □

Lemma 4.3 (moments)**.** *Let ρ be a compactly supported distribution on $\mathbb{R}$ with moments $\mu_n = \langle \rho, w^n \rangle$. For polynomials P_k the following are equivalent:*

(i) $\sum_{k=0}^{M} w^k P_k(\theta)\rho = 0$ in the sense of distributions, where $\theta = w\, d/dw$;

(ii) $\sum_{k=0}^{M} P_k(-n-k-1)\,\mu_{n+k} = 0$ *for all* $n \geq 0$.

Proof. The formal adjoint of θ is $-\theta - 1$. Hence $\langle w^k P_k(\theta)\rho, w^n\rangle = \langle \rho, P_k(-\theta-1)w^{n+k}\rangle = P_k(-n-k-1)\mu_{n+k}$. A compactly supported distribution whose moments all vanish is zero, since its Fourier transform is an entire function whose Taylor coefficients vanish. □

5 The unitary Bessel–Toeplitz operator

Fix $\ell \geq 1$ and recall $K = \lceil \ell/2 \rceil$, $\lambda_b = 2(\ell - 2b)$ and $\gamma_b = \frac{1}{2}((\ell-b)^2 + b^2)$ from (2.5).

5.1 Two contour sequences

For $t > 0$ and $m \in \mathbb{Z}$ put

$$i_m(t) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{2t\cos\varphi} e^{im\varphi}\, d\varphi = I_m(2t),$$
$$k_m(t) = \int_0^{\infty} e^{-t(s+s^{-1})}(-s)^m \frac{ds}{s} = 2(-1)^m K_m(2t).$$

see [9, 10.32.3, 10.32.10]. Both are integrals $\int_\gamma e^{t(z+z^{-1})} z^m \frac{dz}{z}$, up to the factor $(2\pi i)^{-1}$ in the first case: γ is the unit circle for i_m, and the negative real axis traversed from 0 to $-\infty$ for k_m.

Lemma 5.1. *For* $h \in \{i, k\}$ *and* $m \in \mathbb{Z}$ *one has* $h_{-m} = h_m$, $h'_m = h_{m-1} + h_{m+1}$ *and* $t(h_{m-1} - h_{m+1}) = m\,h_m$. *Consequently there are rational functions* $A_m, B_m \in \mathbb{Q}(t)$, *the same for* $h = i$ *and* $h = k$, *such that* $h_m = A_m h_0 + B_m h'_0$. *Moreover* h_0 *satisfies Bessel's equation*

$$T h := t\,h'' + h' - 4t\,h = 0, \tag{5.1}$$

and i_0, k_0 *form a basis of its solutions on* $(0, \infty)$.

Proof. Differentiation under the integral sign multiplies the integrand by $z + z^{-1}$, which gives the second relation. Next, $\int_\gamma d\big(e^{t(z+z^{-1})} z^m\big) = 0$: the circle is closed, and on the negative axis $e^{t(z+z^{-1})}$ vanishes to infinite order at both ends. This gives $t(h_{m+1} - h_{m-1}) + m h_m = 0$. The substitution $z \mapsto z^{-1}$ maps each contour to itself, reversing the orientation and changing the sign of dz/z, so $h_{-m} = h_m$. The two relations give $h_1 = h_{-1} = h'_0/2$, and they express every h_m through h_0 and its derivatives. They also give $h'_1 = h_0 + h_2$ and $h_0 - h_2 = h_1/t$, hence $h''_0 = 4h_0 - h'_0/t$, which is (5.1); then all h_m lie in $\mathbb{Q}(t)h_0 + \mathbb{Q}(t)h'_0$. Finally, $I_0(2t)$ grows and $K_0(2t)$ decays as $t \to \infty$, so i_0, k_0 are independent. □

Equation (5.1) is the modified Bessel equation of order 0 in the variable $2t$. By Kolchin's theorem [17] (see also [23]), its differential Galois group over $\mathbb{C}(t)$ acts on the solution space $\mathrm{Sol}(T)$ as the full group $\mathrm{SL}(\mathrm{Sol}(T)) \cong \mathrm{SL}_2(\mathbb{C})$. In particular it acts transitively on $\mathrm{Sol}(T) \setminus \{0\}$.

5.2 The Toeplitz form

For $g \in \mathrm{Sol}(T)$ let $g_\bullet = (g_m)_{m\in\mathbb{Z}}$, $g_m = A_m g + B_m g'$; thus $(i_0)_\bullet = i$ and $(k_0)_\bullet = k$. For sequences $h^{(1)}, \ldots, h^{(\ell)}$ of this form put

$$F(h^{(1)}, \ldots, h^{(\ell)}) = \frac{1}{\ell!} \sum_{\sigma,\tau \in S_\ell} \mathrm{sgn}(\sigma)\,\mathrm{sgn}(\tau) \prod_{j=1}^{\ell} h^{(j)}_{\sigma(j)-\tau(j)}.$$

This is $\mathbb{C}$-multilinear and symmetric, and $F(h,\dots,h)=\det[h_{p-p'}]_{p,p'=1}^{\ell}$. Indeed, substituting $j=\tau^{-1}(p')$ turns the double sum into $\ell!$ copies of the Leibniz expansion. For $g\in\mathrm{Sol}(T)$ put

$$\Phi(g)=F(g_\bullet,\dots,g_\bullet)=\det\big[A_{p-p'}g+B_{p-p'}g'\big]_{p,p'=1}^{\ell};$$

it is a homogeneous polynomial of degree ℓ in g,g' with coefficients in $\mathbb{Q}(t)$. By (2.2), $\Phi(i_0)=D_\ell$.

For $0\le b\le\ell$ define the *vertex integrals*

$$\Psi_b=F(\underbrace{k,\dots,k}_{b},\underbrace{i,\dots,i}_{\ell-b}).$$

Lemma 5.2. *Let $z_j=-s_j$ for $j\le b$ and $z_j=e^{i\varphi_j}$ for $j>b$, $f(z)=\sum_{j=1}^{\ell}(z_j+z_j^{-1})$ and $\Delta(z)=\prod_{p<p'}(z_{p'}-z_p)$. Then*

$$\Psi_b(t)=\frac{1}{\ell!}\int_{(0,\infty)^b\times(-\pi,\pi)^{\ell-b}}e^{tf(z)}\,\Delta(z)\Delta(z^{-1})\prod_{j\le b}\frac{ds_j}{s_j}\prod_{j>b}\frac{d\varphi_j}{2\pi},$$

and Ψ_b is real on $(0,\infty)$. Moreover, $\Phi(\alpha i_0+\beta k_0)=\sum_{b=0}^{\ell}\binom{\ell}{b}\alpha^{\ell-b}\beta^b\,\Psi_b$ for $\alpha,\beta\in\mathbb{C}$.

Proof. Expand $\Delta(z)\Delta(z^{-1})=\det[z_j^{p-1}]\det[z_j^{1-p}]=\sum_{\sigma,\tau}\operatorname{sgn}(\sigma\tau)\prod_j z_j^{\sigma(j)-\tau(j)}$ and apply Fubini to the integral representations of i_m and k_m. Since the i_m and k_m are real and F has integer coefficients, Ψ_b is real. The last identity follows from multilinearity and symmetry of F, because $(\alpha i_0+\beta k_0)_\bullet=\alpha i+\beta k$. □

Proposition 5.3. *Let L_ℓ^t be the minimal monic operator over $\mathbb{C}(t)$ annihilating $D_\ell(t)$. Then its solution space is*

$$\mathrm{Sol}(L_\ell^t)=V:=\mathrm{span}_{\mathbb{C}}\{\Psi_0,\dots,\Psi_\ell\}.$$

Proof. Let E be the Picard–Vessiot extension of $\mathbb{C}(t)$ for (5.1), realized by functions on $(0,\infty)$, and G its differential Galois group. Since Φ is a differential polynomial with coefficients in $\mathbb{C}(t)$, $D_\ell=\Phi(i_0)\in E$ and $\sigma(D_\ell)=\Phi(\sigma i_0)$ for $\sigma\in G$. By Kolchin's theorem $\{\sigma i_0\}=\mathrm{Sol}(T)\setminus\{0\}$. Hence the span U of the orbit of D_ℓ is the span of $\{\Phi(g):g\in\mathrm{Sol}(T)\}$. By the polarization identity

$$F(h^{(1)},\dots,h^{(\ell)})=\frac{1}{\ell!}\sum_{\emptyset\neq S\subseteq[\ell]}(-1)^{\ell-|S|}F(h_S,\dots,h_S),\qquad h_S=\sum_{j\in S}h^{(j)},$$

U contains every $F(g_\bullet^{(1)},\dots,g_\bullet^{(\ell)})$ with $g^{(j)}\in\mathrm{Sol}(T)$, in particular every Ψ_b. Conversely, expanding $\Phi(\alpha i_0+\beta k_0)$ by Lemma 5.2 shows $U\subseteq V$. Now apply Lemma 3.7. □

5.3 Asymptotics at infinity

Lemma 5.4 (vertex asymptotics)**.** *For $0\le b\le\ell$ and $m\ge0$, as $t\to+\infty$,*

$$\Psi_b^{(m)}(t)=C_b\,e^{\lambda_b t}t^{-\gamma_b}(\lambda_b^m+O(t^{-1})),\qquad C_b=\frac{(-1)^{b(b-1)/2}\,4^{b(\ell-b)}}{\ell!\,(2\pi)^{\ell-b}}\,\mathcal{M}_{\ell-b}\,\mathcal{M}_b\neq0,$$

where $\mathcal{M}_n=\int_{\mathbb{R}^n}e^{-|x|^2}\Delta(x)^2\,dx>0$ is Mehta's integral. More precisely, Ψ_b has an asymptotic expansion $e^{\lambda_b t}t^{-\gamma_b}\sum_{i\ge0}c_{b,i}t^{-i}$ with $c_{b,0}=C_b$, and the expansions of the derivatives are the termwise derivatives.

Proof. On the domain of Lemma 5.2 we have $z + z^{-1} = 2\cos\varphi \le 2$ on the circle and $z + z^{-1} = -(s + s^{-1}) \le -2$ on the negative axis. So $f \le \lambda_b$, with equality only at the vertex v^* where $s_j = 1$ for $j \le b$ and $\varphi_j = 0$ for $j > b$. Introduce the global coordinates

$$x_j = 2\sin(\varphi_j/2) \in (-2, 2) \quad (j > b), \qquad y_j = s_j^{1/2} - s_j^{-1/2} \in \mathbb{R} \quad (j \le b).$$

In these coordinates the phase is *exactly* quadratic: $2\cos\varphi_j = 2 - x_j^2$ and $-(s_j + s_j^{-1}) = -2 - y_j^2$, so $f = \lambda_b - |x|^2 - |y|^2$. Also $d\varphi_j = (1 - x_j^2/4)^{-1/2} dx_j$ and $ds_j/s_j = 2s_j^{1/2}(1 + s_j)^{-1} dy_j$, with Jacobian factors equal to 1 at v^*.

Next consider the factors of $\Delta(z)\Delta(z^{-1}) = \prod_{p<p'}(2 - z_p/z_{p'} - z_{p'}/z_p)$. A circle–circle factor equals $4\sin^2((\varphi_p - \varphi_{p'})/2)$. It is an analytic function of $(x_p, x_{p'})$ near 0 that vanishes to second order on $x_p = x_{p'}$, so it equals $(x_p - x_{p'})^2$ times an analytic function with value 1 at 0. Similarly, a factor with both variables on the axis equals $-(s_p - s_{p'})^2/(s_p s_{p'}) = -(y_p - y_{p'})^2 \cdot$(analytic, 1 at 0). A mixed factor equals $2 + e^{i\varphi}/s + se^{-i\varphi}$, which is 4 at v^*. Hence, near v^*,

$$e^{tf}\Delta(z)\Delta(z^{-1})\, d(\cdots) = e^{\lambda_b t} e^{-t(|x|^2+|y|^2)}\, \Delta(x)^2\Delta(y)^2\, H(x, y)\, dx\, dy,$$

with H analytic and $H(0) = (-1)^{b(b-1)/2} 4^{b(\ell-b)} (2\pi)^{b-\ell}/\ell!$.

Fix $\delta > 0$ so small that H is analytic on the ball $B_\delta = \{|w| \le \delta\}$, $w = (x, y)$. Outside B_δ we have $f \le \lambda_b - \delta^2$. We estimate this part in the original variables (s, φ), where the integrand of Lemma 5.2 is e^{tf} times a function bounded by a polynomial in the $s_j^{\pm 1}$. Writing $e^{tf} = e^{(t-1)f} e^f$ and using $\int e^f \prod_j (s_j + s_j^{-1})^N \prod_j \frac{ds_j}{s_j} < \infty$, we see that the part outside B_δ contributes $O(e^{(\lambda_b - \delta^2)t})$ for $t \ge 1$. On B_δ write H as its Taylor polynomial of degree $2J$ plus a remainder bounded by $C|w|^{2J+1}$; the remainder contributes $O(t^{-\gamma_b - J - 1/2})$. A monomial w^α contributes

$$\int_{\mathbb{R}^\ell} e^{-t|w|^2}\Delta(x)^2\Delta(y)^2 w^\alpha\, dw + O(e^{-\delta^2 t/2}) = t^{-\gamma_b - |\alpha|/2} \int_{\mathbb{R}^\ell} e^{-|w|^2}\Delta(x)^2\Delta(y)^2 w^\alpha\, dw + O(e^{-\delta^2 t/2}).$$

Here we used that $\Delta(x)^2\Delta(y)^2$ is homogeneous of degree $(\ell - b)(\ell - b - 1) + b(b - 1) = 2\gamma_b - \ell$. The integral vanishes for $|\alpha|$ odd, by the symmetry $w \mapsto -w$. This gives the expansion in integral powers of t^{-1}, with leading coefficient $H(0)\mathcal{M}_{\ell-b}\mathcal{M}_b = C_b$.

For $\Psi_b^{(m)}$ the amplitude is multiplied by $f^m = (\lambda_b - |w|^2)^m$. Monomial by monomial, the resulting expansion is the termwise derivative, since $\int |w|^2 e^{-t|w|^2}(\cdots) = -\frac{d}{dt}\int e^{-t|w|^2}(\cdots)$. Its leading term is $\lambda_b^m C_b$. □

Corollary 5.5. *The functions $\Psi_0, \dots, \Psi_\ell$ are linearly independent, so* $\operatorname{ord} L_\ell^t = \dim V = \ell + 1$.

Proof. The rates λ_b are pairwise distinct and $C_b \ne 0$. □

5.4 The origin

Every element of V is a polynomial in i_0, i_0', k_0, k_0' with coefficients in $\mathbb{Q}(t)$. Moreover i_0 is entire and $k_0 = -2(\log t)\, i_0 +$ (entire) [9, 10.31.2]. Hence $V \subset \mathcal{N}$, every element of V has moderate growth at 0, and 0 is a regular singular point of L_ℓ^t (Fuchs).

Lemma 5.6 (displacement)**.** *(a) Let $S = \{s_1 < \cdots < s_j\} \subseteq [\ell]$ and let σ be a permutation of S. Then $\sum_{s\in S} |\sigma(s) - s| \le 2p(\ell - p)$ with $p = \lfloor j/2 \rfloor$.*

(b) Let $\pi \in S_\ell$ and $S \subseteq [\ell]$ with $|S| = j$. Then

$$\sum_{s\in S} |\pi(s) - s| - \sum_{s\notin S} |\pi(s) - s| \le 2p(\ell - p), \qquad p = \lfloor j/2 \rfloor.$$

Proof. (a) We have

$$\sum_s |\sigma(s) - s| = \sum_s \max(s, \sigma(s)) - \sum_s \min(s, \sigma(s)).$$

The $2j$ numbers $\max(s, \sigma(s))$ and $\min(s, \sigma(s))$ form the multiset in which every element of S occurs twice. So the difference is at most the sum of the j largest elements of that multiset minus the sum of the j smallest, which is $2\sum_{i=1}^{p}(s_{j+1-i} - s_i)$. Since $s_{j+1-i} \le \ell + 1 - i$ and $s_i \ge i$, this is at most $2\sum_{i=1}^{p}(\ell + 1 - 2i) = 2p(\ell - p)$.

(b) For $s \in S$, follow the cycle of π through s until it returns to S: $s \to \pi(s) \to \cdots \to \pi^r(s) =: \sigma(s) \in S$, with $\pi^i(s) \notin S$ for $0 < i < r$. This defines a permutation σ of S. By the triangle inequality,

$$|\pi(s) - s| \le |\sigma(s) - s| + \sum_{0<i<r} |\pi^{i+1}(s) - \pi^i(s)|.$$

Every $s' \notin S$ occurs at most once as an intermediate point $\pi^i(s)$. Summing over $s \in S$ therefore gives $\sum_{s\in S} |\pi(s) - s| \le \sum_{s\in S} |\sigma(s) - s| + \sum_{s'\notin S} |\pi(s') - s'|$. Now apply (a). □

Lemma 5.7 (origin bound)**.** *There is a constant C such that for $0 < t \le 1$ and $0 \le b \le \ell$*

$$|\Psi_b(t)| \le C\, t^{-2\lfloor b/2\rfloor(\ell - \lfloor b/2\rfloor)}\,(1 + |\log t|)^b.$$

Consequently $\operatorname{ord}\Psi_b \ge -2\lfloor b/2\rfloor(\ell - \lfloor b/2\rfloor)$.

Proof. On $(0, 1]$ we have $|i_m(t)| \le e\, t^{|m|}$, since $|I_m(2t)| \le t^{|m|}e^{t^2}/|m|!$. We also have $|k_m(t)| \le C\, t^{-|m|}(1 + |\log t|)$. Indeed, $k_{-m} = k_m$, and for $m > 0$ splitting the integral at $s = 1$ and using $e^{-t(s+s^{-1})} \le e^{-ts}$ for $s \ge 1$ gives $|k_m(t)| \le 1/m + \Gamma(m)t^{-m}$. For $m = 0$ the symmetry $s \mapsto s^{-1}$ gives $k_0(t) = 2\int_1^\infty e^{-t(s+s^{-1})}ds/s \le 2\int_1^\infty e^{-ts}ds/s$, which is $O(1 + |\log t|)$. Let $T_I = [i_{p-p'}]$ and $T_K = [k_{p-p'}]$. By Lemma 5.2, $\binom{\ell}{b}\Psi_b$ is the coefficient of $\alpha^{\ell-b}\beta^b$ in $\det(\alpha T_I + \beta T_K)$. Expanding this determinant column by column,

$$\binom{\ell}{b}\Psi_b = \sum_{|S|=b}\sum_{\pi\in S_\ell} \operatorname{sgn}(\pi) \prod_{s\in S} k_{\pi(s)-s} \prod_{s\notin S} i_{\pi(s)-s}.$$

Each term is bounded by $C\, t^{-E(\pi,S)}(1+|\log t|)^b$, where $E(\pi, S) = \sum_{s\in S}|\pi(s)-s| - \sum_{s\notin S}|\pi(s)-s|$. By Lemma 5.6(b), $E(\pi, S) \le 2\lfloor b/2\rfloor(\ell - \lfloor b/2\rfloor)$. □

5.5 The squeeze and the shape of the operator

Theorem 5.8. *Let $\ell \ge 1$ and let $L_\ell^t = \sum_{m=0}^{\ell+1} p_m(t)\partial^m$ be the minimal operator of D_ℓ, normalized to be primitive. Then:*

(i) $\operatorname{ord} L_\ell^t = \ell + 1$*; after scaling, $p_{\ell+1} = t^\ell$, $\deg p_m \le \ell$, and $t^{m-1} \mid p_m$ for $1 \le m \le \ell + 1$. Moreover $p_m(-t) = (-1)^{m+1}p_m(t)$.*

(ii) *L_ℓ^t has no singular point in $\mathbb{C} \setminus \{0\}$.*

(iii) *The exponents at $t = 0$ are the integers $-2b(\ell - b)$, $0 \le b \le \ell$, and $\operatorname{ord}\Psi_b = -2\lfloor b/2\rfloor(\ell - \lfloor b/2\rfloor)$ for each b.*

(iv) *At $t = \infty$, L_ℓ^t has the basis of formal solutions $e^{\lambda_b t}t^{-\gamma_b}\sum_{i\ge 0} c_{b,i}t^{-i}$, $0 \le b \le \ell$. Moreover $\sum_m [t^\ell]p_m\,\lambda^m = \prod_{b=0}^{\ell}(\lambda - \lambda_b)$.*

(v) $tL_\ell^t = \sum_{k=0}^{K} t^{2k} Q_{2k}(\theta)$ *with* $\deg Q_{2k} = \ell + 1 - 2k$ *and* $\mathrm{lc}(Q_{2k}) = [\lambda^{\ell+1-2k}]\chi(\lambda)$*, where* $\chi(\lambda) = \prod_b (\lambda - \lambda_b)$*. The subleading coefficients are given by*

$$\sum_{k=0}^{K} [\theta^{\ell-2k}] Q_{2k} \cdot \lambda^{\ell-2k} = \sum_{b=0}^{\ell} \gamma_b \prod_{b' \neq b} (\lambda - \lambda_{b'}) - \frac{\lambda}{2}\chi''(\lambda). \tag{5.2}$$

Proof. By Proposition 5.3 and Corollary 5.5, $n := \operatorname{ord} L_\ell^t = \ell + 1$ and $\mathrm{Sol}(L_\ell^t) = V$ has basis $\Psi_0, \dots, \Psi_\ell$.

Singular points. Every element of V extends analytically along all paths in $\mathbb{C} \setminus \{0\}$. Hence every zero $\tau \neq 0$ of p_n is an apparent singularity. By Lemma 3.4(b) the residue κ_τ of W'/W at τ is a positive integer, where W is the Wronskian of $\Psi_0, \dots, \Psi_\ell$.

Abel at infinity. The function $R = W'/W = -p_{n-1}/p_n$ is rational. Its poles are at 0 and at the apparent points, and all of them are simple. At 0 this is Fuchs' criterion; at an apparent point it holds because W is holomorphic there. Hence

$$R = Q(t) + \frac{\kappa_0}{t} + \sum_\tau \frac{\kappa_\tau}{t - \tau}, \qquad W = c\, e^{\int Q}\, t^{\kappa_0} \prod_\tau (t - \tau)^{\kappa_\tau},$$

with $Q \in \mathbb{C}[t]$ and $\kappa_0 = \sum \nu_i - \binom{n}{2}$ by Lemma 3.4(a), where $\nu_1, \dots, \nu_n$ are the exponents at 0. On the other hand, Lemma 5.4 gives

$$W(t) = \Big(\prod_b C_b\Big) e^{(\sum_b \lambda_b)t}\, t^{-\sum_b \gamma_b} \big(\textstyle\prod_{b<b'} (\lambda_{b'} - \lambda_b) + O(t^{-1})\big),$$

and $\sum_b \lambda_b = 0$. Comparing, $Q = 0$ and $\kappa_0 + \sum_\tau \kappa_\tau = -\sum_b \gamma_b$, that is,

$$\sum_{\tau \neq 0} \kappa_\tau = \binom{n}{2} - \sum_b \gamma_b - \sum_i \nu_i. \tag{5.3}$$

The squeeze. By Lemma 5.7 and Lemma 3.3,

$$\sum_i \nu_i \;\geq\; \sum_b \operatorname{ord} \Psi_b \;\geq\; -2 \sum_{b=0}^{\ell} \lfloor b/2 \rfloor (\ell - \lfloor b/2 \rfloor) = -2 \sum_{b=0}^{\ell} b(\ell - b) = -\frac{(\ell - 1)\ell(\ell + 1)}{3}.$$

The middle equality holds because the multisets $\{\lfloor b/2 \rfloor (\ell - \lfloor b/2 \rfloor)\}_{b=0}^{\ell}$ and $\{b(\ell - b)\}_{b=0}^{\ell}$ coincide: both consist of the values $p(\ell - p)$ for $0 \leq p < \ell/2$, each twice, together with $\ell^2/4$ once if ℓ is even. Since $\sum_b \gamma_b = \sum_b b^2 = \ell(\ell + 1)(2\ell + 1)/6$, the right-hand side of (5.3) is at most

$$\frac{\ell(\ell+1)}{2} - \frac{\ell(\ell+1)(2\ell+1)}{6} + \frac{(\ell-1)\ell(\ell+1)}{3} = \frac{\ell(\ell+1)}{6}\big(3 - (2\ell+1) + 2(\ell-1)\big) = 0.$$

As each $\kappa_\tau \geq 1$, there are no apparent singularities, which proves (ii). All the inequalities above are therefore equalities. In particular $\operatorname{ord} \Psi_b = -2\lfloor b/2 \rfloor (\ell - \lfloor b/2 \rfloor)$ for every b, because each term was bounded separately. By the equality case of Lemma 3.3, the exponents are the numbers $\operatorname{ord} \Psi_b$, that is, the multiset $\{-2b(\ell - b)\}$. This proves (iii).

Shape. By (ii), $p_n = ct^A$ with $c \neq 0$. By Lemma 5.4 and Lemma 3.5, $\deg p_m \leq A$ for all m and $\sum_m [t^A] p_m \lambda^m = c \prod_b (\lambda - \lambda_b)$. Fuchs' criterion at 0 gives $\operatorname{ord}_0 p_m \geq A - n + m$, and the indicial polynomial is $I(\rho) = \sum_m [t^{A-n+m}] p_m\, \rho^{\underline{m}}$. If $A \leq n - 2$, the terms with $m < n - A$ vanish, so $\rho^{\underline{n-A}}$ divides I and 1 would be an exponent; this contradicts (iii), as all exponents are ≤ 0. Hence $A \geq n - 1$. Since 0 is an exponent, $[t^{A-n}] p_0 = I(0) = 0$. Hence t^{A-n+1} divides every p_m, and primitivity gives $A \leq n - 1$. So $A = \ell$, which proves the first part of (i) and the second part of (iv).

Formal solutions. By Lemma 5.4, applying L_ℓ^t termwise to the asymptotic expansion of Ψ_b gives the asymptotic expansion of $L_\ell^t \Psi_b = 0$. So each expansion is a formal solution, and they are independent because the λ_b are distinct. This proves (iv).

Parity. Since $I_m(-2t) = (-1)^m I_m(2t)$, conjugation by $\mathrm{diag}((-1)^p)$ shows $D_\ell(-t) = D_\ell(t)$. The operator obtained from L_ℓ^t by $t \mapsto -t$, namely $\sum_m p_m(-t)(-1)^m \partial^m$, is primitive of order n and annihilates D_ℓ. By uniqueness of the minimal operator it is a constant multiple of L_ℓ^t, and comparing leading terms the constant is -1. This gives $p_m(-t) = (-1)^{m+1} p_m(t)$.

The θ-coefficients. Write $tL_\ell^t = \sum_m r_m(t)\theta^{\underline{m}}$ with $r_m = p_m/t^{m-1}$ for $m \geq 1$ and $r_0 = tp_0$, so that $\deg r_m \leq \ell + 1 - m$. Thus $tL_\ell^t = \sum_k t^k Q_k(\theta)$ with $\deg Q_k \leq \ell + 1 - k$. By the parity just proved, tL_ℓ^t is invariant under $t \mapsto -t$, and θ is invariant too; hence $Q_k = 0$ for odd k. Comparing the coefficients of $t^{\ell+1}\partial^m$ gives $[t^\ell]p_m = [\theta^m]Q_{\ell+1-m}$. Hence $\chi(\lambda) = \sum_k [\theta^{\ell+1-k}]Q_k \, \lambda^{\ell+1-k}$. The polynomial $\chi(\lambda) = \lambda^{[\ell \text{ even}]} \prod_{b<\ell/2}(\lambda^2 - \lambda_b^2)$ has nonzero coefficients at all powers $\lambda^{\ell+1-2k}$, $0 \leq k \leq K$: up to sign they are elementary symmetric functions of the positive numbers λ_b^2. Hence $\deg Q_{2k} = \ell + 1 - 2k$, with $\mathrm{lc}(Q_{2k}) = [\lambda^{\ell+1-2k}]\chi$.

For (5.2), substitute a formal solution $y = e^{\lambda t} t^{-\gamma} g$, with $\lambda = \lambda_b$, $\gamma = \gamma_b$ and $g = 1 + O(t^{-1})$, into tL_ℓ^t. Since $\theta(e^{\lambda t} t^{-\gamma} g) = e^{\lambda t} t^{-\gamma}(\lambda t - \gamma + \theta)g$, we must expand $Q_{2k}(\lambda t + \theta - \gamma)$ applied to g. Write $d_k = \ell + 1 - 2k$. The terms of order t^{d_k} and $t^{d_k - 1}$ come from the leading monomial and from the subleading one. In the leading monomial $(\lambda t + \theta - \gamma)^{d_k}$, the contribution with one factor $\theta - \gamma$ in position p acting on $t^{d_k - p} g$ gives $\lambda^{d_k - 1} t^{d_k - 1}(d_k - p - \gamma)$. Summing over p gives $\lambda^{d_k-1} t^{d_k-1}(\binom{d_k}{2} - d_k\gamma)$. Collecting the coefficient of t^ℓ and using $\chi(\lambda_b) = 0$, we get, for every b,

$$\sum_k \lambda_b^{d_k - 1} [\theta^{d_k - 1}] Q_{2k} + \frac{\lambda_b}{2} \chi''(\lambda_b) - \gamma_b \chi'(\lambda_b) = 0.$$

The polynomial obtained from the left side of (5.2) by adding $\frac{\lambda}{2}\chi''(\lambda)$ has degree at most ℓ. At the $\ell + 1$ points λ_b it takes the values $\gamma_b \chi'(\lambda_b) = \gamma_b \prod_{b' \neq b}(\lambda_b - \lambda_{b'})$. Lagrange interpolation gives (5.2). □

Remark 5.9. The squeeze shows a little more than (iii): the crude bound of Lemma 5.7 is sharp in the power of t for every single vertex integral (the logarithmic factor need not be). Pairs of vertex integrals share the same order at 0, namely Ψ_{2p} and Ψ_{2p+1}. Computing their leading terms directly would require controlling the cancellations among the $\ell!^2$ terms of the Toeplitz expansion. The squeeze avoids this entirely. High-precision evaluation confirms the orders for $\ell \leq 6$ (Appendix A).

5.6 Proof of Theorem 2.2

Put $x = t^2$, so that $\theta_t = 2\theta_x$ and $G_\ell(x) = D_\ell(t)$. By Theorem 5.8(v),

$$\Theta := tL_\ell^t = \sum_{k=0}^{K} x^k \, Q_{2k}(2\theta_x)$$

is an operator in x annihilating G_ℓ. If G_ℓ satisfied an equation of order at most ℓ in x, then D_ℓ would satisfy one in t, contradicting Theorem 5.8(i). So the minimal operator of G_ℓ has order $\ell + 1$, and $\Theta = xL_\ell$ up to normalization. We normalize so that $P_k(\theta) := 2^{-\ell-1} Q_{2k}(2\theta)$ gives a monic P_0. Then $\deg P_k = \ell + 1 - 2k$ and $\mathrm{lc}(P_k) = 2^{-2k}\,\mathrm{lc}(Q_{2k})$, so

$$\sum_k \mathrm{lc}(P_k)\mu^{\ell+1-2k} = 2^{-\ell-1}\chi(2\mu) = \prod_b (\mu - \ell + 2b).$$

The polynomial P_0 is the indicial polynomial at $x = 0$. By Theorem 5.8(iii) its roots are half the t-exponents, so $P_0(\theta) = \prod_b (\theta + b(\ell - b))$.

If ℓ is even, the rate $\lambda_{\ell/2}$ vanishes, and $\Psi_{\ell/2}(\sqrt{x}) = C\,x^{-\gamma_{\ell/2}/2}(1+O(x^{-1/2}))$ is a solution. Applying Θ, the top term $x^K P_K(\theta)x^{-\gamma_{\ell/2}/2} = P_K(-\gamma_{\ell/2}/2)\,x^{K-\gamma_{\ell/2}/2}$ must vanish, because all other terms are of lower order. So $P_K(\theta) = \mathrm{lc}(P_K)(\theta+\ell^2/8)$, as $\gamma_{\ell/2} = \ell^2/4$. This proves (b), except for the absence of singular points in $\mathbb{C}^*$, which follows from Theorem 5.8(ii) because $\tau \mapsto \tau^2$ maps $\mathbb{C}^*$ onto $\mathbb{C}^*$. Part (c) is Theorem 5.8(iv) rewritten in x.

It remains to prove (a). Write $L_\ell = \sum_m q_m \partial_x^m$. Since $x^k\theta^i = \sum_{m\le i} S(i,m)x^{k+m}\partial_x^m$, the polynomial xq_m is a combination of monomials x^{k+m} coming from $x^k P_k(\theta)$ with $\deg P_k \ge m$. Hence $x^{m-1} \mid q_m$, $q_{\ell+1} = x^\ell$, and $\deg q_m \le k^*+m-1 = \lfloor(\ell-1+m)/2\rfloor$, where $k^* = \lfloor(\ell+1-m)/2\rfloor$ is the largest k with $\ell+1-2k \ge m$. The coefficient of x^{k^*+m} in xq_m comes only from P_{k^*}. It equals $\mathrm{lc}(P_{k^*}) \neq 0$ if $\deg P_{k^*} = m$, and $\binom{m+1}{2}\mathrm{lc}(P_{k^*}) + [\theta^m]P_{k^*}$ if $\deg P_{k^*} = m+1$.

In the second case put $d = m+1 = \ell+1-2k^*$. By (5.2) and $\mathrm{lc}(Q_{2k}) = [\lambda^{\ell+1-2k}]\chi$, this coefficient equals $2^{d-1-\ell-1}$ times the coefficient of λ^{d-1} in $\frac{\lambda}{2}\chi''(\lambda) + \sum_b \gamma_b \prod_{b'\neq b}(\lambda-\lambda_{b'})$. Write $\varepsilon_\ell = [\ell \text{ even}]$, $c_j = \lambda_j^2 > 0$ for $j < K$, and $\chi = \lambda^{\varepsilon_\ell}\prod_{j<K}(\lambda^2 - c_j)$. Pairing b with $\ell-b$ (and keeping $b = \ell/2$ alone if ℓ is even), a direct computation shows that this coefficient is

$$(-1)^{k^*}\Big(\tbinom{d}{2}\sigma_{k^*}(c) + 2\sum_{j<K}\gamma_j\,\sigma_{k^*}(c_{\neq j}) + \varepsilon_\ell\,\gamma_{\ell/2}\,\sigma_{k^*}(c)\Big),$$

where σ_i denotes the elementary symmetric functions and $c_{\neq j}$ the list with c_j removed. All terms are nonnegative. If $k^* < K$, the middle term is positive; if $k^* = K$, then $d = 1$, so $\varepsilon_\ell = 1$ and the last term is positive. Hence $\deg q_m = \lfloor(\ell-1+m)/2\rfloor$ exactly.

Finally, $q_1(0) = P_0(1) = \prod_b(1+b(\ell-b)) \neq 0$ and $q_{\ell+1} = x^\ell$ has no zero in $\mathbb{C}^*$. Hence L_ℓ is primitive, and every operator of order $\ell+1$ with polynomial coefficients annihilating G_ℓ is $r\,L_\ell$ with $r \in \mathbb{C}(x)$, necessarily a polynomial. □

Proof of Corollary 2.3. For $q = 0$ the function in [11, Prop. 4.1] is G_ℓ [11, (4.6)], and Theorem 2.2(a) gives exactly the asserted shape. The optimality of the order is the first statement of Theorem 2.2(a). Conjecture 2.1 follows from (a), since $\lfloor(\ell-1+m)/2\rfloor \le \ell$. □

Remark 5.10. Appendix B of [11], due to Bornemann, discusses optimality using algebraic independence of the Bessel functions. Here minimality follows from Proposition 5.3 and Corollary 5.5; it also follows from irreducibility of the ℓth symmetric power of the standard representation of SL_2.

6 Twisted Toeplitz determinants

Fix $\ell \ge 1$ and $q \ge 0$. The arguments of Section 5 extend to $D_{\ell,q}$. The main new ingredient is an inequality for displacements with a drift, Lemma 6.3. In addition, the leading coefficient in the variable t can now be $t^{\ell+1}$, and the gauge factor $x^{q\ell/2}$ produces an extra term in the exact-degree argument.

6.1 The functions

Lemma 6.1. *For $q \in \mathbb{Z}$ one has $D_{\ell,-q} = D_{\ell,q}$ and $D_{\ell,q}(t) = \mathbb{E}_{U(\ell)}(\det U)^q\, e^{2t\,\mathrm{Re}\,\mathrm{tr}\,U}$. For $q \ge 0$,*

$$D_{\ell,q}(t) = \sum_{n\ge 0} u_{\ell,q}(n)\,\frac{t^{2n+q\ell}}{n!\,(n+q\ell)!}, \qquad u_{\ell,q}(n) = \sum_\mu f^\mu f^{\mu+(q^\ell)},$$

where μ runs over the partitions of n with at most ℓ parts. In particular $G_{\ell,q}$ is a power series with $G_{\ell,q}(0) = f^{(q^\ell)}/(q\ell)! > 0$, and $D_{\ell,q}(-t) = (-1)^{q\ell}D_{\ell,q}(t)$.

Proof. Transposition and $b_{-m} = b_m$ give $D_{\ell,-q} = D_{\ell,q}$. By Heine's identity, that is, Andreief's identity on the unit circle (see [1]), $\mathbb{E}_{U(\ell)} \prod_j \varphi(z_j) = \det[\hat{\varphi}_{i-j}]_{i,j=1}^{\ell}$, where $z_1, \dots, z_\ell$ are the eigenvalues of U and $\hat{\varphi}_m$ is the m-th Fourier coefficient of φ on the unit circle. For $\varphi(z) = z^{-q} e^{t(z+z^{-1})}$ we have $\hat{\varphi}_m = b_{m+q}$. Hence $\mathbb{E}(\det U)^{-q} e^{2t \operatorname{Re}\operatorname{tr} U} = D_{\ell,q}$, and replacing q by $-q$ gives the stated formula.

For the expansion, write $e^{2t \operatorname{Re}\operatorname{tr} U} = \sum_{a,c \ge 0} \frac{t^{a+c}}{a!\,c!} (\operatorname{tr} U)^a \overline{\operatorname{tr} U}^{\,c}$ and use $(\operatorname{tr} U)^a = \sum_{\lambda \vdash a} f^\lambda s_\lambda(U)$, where $s_\lambda(U) = 0$ if λ has more than ℓ parts. Since $(\det U)^q s_\mu(U) = s_{\mu + (q^\ell)}(U)$ and $\overline{\det U} = (\det U)^{-1}$, Schur orthogonality gives $\mathbb{E}\,(\det U)^{-q} s_\lambda(U) \overline{s_\mu(U)} = \delta_{\lambda, \mu+(q^\ell)}$. Collecting terms in $\mathbb{E}(\det U)^{-q} e^{2t \operatorname{Re}\operatorname{tr} U}$ gives the expansion. The parity follows from $b_m(-t) = (-1)^m b_m(t)$ after conjugation by $\operatorname{diag}((-1)^i)$. □

6.2 Vertex integrals with a twist

For sequences $h^{(1)}, \dots, h^{(\ell)}$ as in Section 5 put

$$F_q(h^{(1)}, \dots, h^{(\ell)}) = \frac{1}{\ell!} \sum_{\sigma,\tau \in S_\ell} \operatorname{sgn}(\sigma) \operatorname{sgn}(\tau) \prod_{j=1}^{\ell} h^{(j)}_{\sigma(j)-\tau(j)+q},$$

$\Phi_q(g) = F_q(g_\bullet, \dots, g_\bullet)$ for $g \in \mathrm{Sol}(T)$, and $\Psi_b^{(q)} = F_q(k, \dots, k, i, \dots, i)$ with b copies of k. As before, F_q is multilinear and symmetric, $F_q(h, \dots, h) = \det[h_{p-p'+q}]_{p,p'=1}^{\ell}$, and $\Phi_q(i_0) = D_{\ell,q}$. Let $L^t_{\ell,q}$ be the minimal operator of $D_{\ell,q}$ over $\mathbb{C}(t)$, normalized to be primitive.

Proposition 6.2. *(a) $\Psi_b^{(q)}$ is given by the integral of Lemma 5.2 with the additional factor $\prod_j z_j^q$ in the integrand. It is real on $(0, \infty)$, and $\Phi_q(\alpha i_0 + \beta k_0) = \sum_b \binom{\ell}{b} \alpha^{\ell-b} \beta^b \Psi_b^{(q)}$.*

(b) $\mathrm{Sol}(L^t_{\ell,q}) = V_q := \operatorname{span}_{\mathbb{C}}\{\Psi_0^{(q)}, \dots, \Psi_\ell^{(q)}\}$.

(c) Lemma 5.4 holds for $\Psi_b^{(q)}$, with the same λ_b and γ_b and with C_b replaced by $(-1)^{qb} C_b$. In particular $\operatorname{ord} L^t_{\ell,q} = \dim V_q = \ell + 1$.

(d) $V_q \subset \mathcal{N}$, and 0 is a regular singular point of $L^t_{\ell,q}$.

Proof. Parts (a), (b) and (d) are proved like Lemma 5.2, Proposition 5.3 and the first paragraph of Section 5.4; $\Psi_b^{(q)}$ is real because the i_m and k_m are real. For (c), the additional factor $\prod_{j \le b} (-s_j)^q \prod_{j > b} e^{iq\varphi_j}$ is real-analytic near the vertex v^*, takes the value $(-1)^{qb}$ there, and is bounded by a polynomial in the $s_j^{\pm 1}$. After absorbing it into H, the proof of Lemma 5.4 applies without change; the vanishing of the odd Gaussian moments used there does not depend on H. The rates λ_b are pairwise distinct, which gives the last assertion as in Corollary 5.5. □

6.3 Displacement with drift

Lemma 6.3. *For $0 \le i \le \ell$ put $f(i) = 2i(\ell - i) - q(\ell - 2i)$, and let $f_{(0)} \le f_{(1)} \le \cdots \le f_{(\ell)}$ be these $\ell+1$ numbers in increasing order. Let $\pi \in S_\ell$ and $S \subseteq [\ell]$ with $|S| = b$, put $d_s = \pi(s) - s + q$, and*

$$E_q(\pi, S) = \sum_{s \in S} |d_s| - \sum_{s \notin S} |d_s|.$$

Then $E_q(\pi, S) \le f_{(b)}$. Moreover $\sum_{b=0}^{\ell} f_{(b)} = \frac{(\ell-1)\ell(\ell+1)}{3}$.

For $q = 0$ we have $f_{(b)} = 2\lfloor b/2 \rfloor (\ell - \lfloor b/2 \rfloor)$, so the lemma contains Lemma 5.6(b).

Proof. Let $S_+ = \{s \in S : d_s > 0\}$, $S_- = S \setminus S_+$ and $j = |S_+|$, and put $A = \sum_{s\in S_+} d_s$ and $B = -\sum_{s\in S_-} d_s \ge 0$. Then $\sum_{s\in S} |d_s| = A + B$ and $\sum_{s\in S} d_s = A - B$. Since $\sum_{s=1}^{\ell} d_s = q\ell$, we have $\sum_{s\notin S} |d_s| \ge |q\ell - A + B|$, and hence

$$E_q(\pi, S) \le A + B - |q\ell - A + B| = \min(2A - q\ell,\ 2B + q\ell).$$

For a set $T \subseteq [\ell]$ with m elements, both $\sum_{s\in T}(\pi(s) - s)$ and $\sum_{s\in T}(s - \pi(s))$ are at most $\sum_{i=1}^{m}((\ell + 1 - i) - i) = m(\ell - m)$. Applied to $T = S_+$ and $T = S_-$, this gives $A \le j(\ell - j) + jq$ and $B \le (b - j)(\ell - b + j) - (b - j)q$. Substituting, one finds

$$2A - q\ell \le f(j), \qquad 2B + q\ell \le f(j'), \qquad j' = j + \ell - b.$$

So $E_q(\pi, S) \le c := \min(f(j), f(j'))$, where $0 \le j \le j' \le \ell$. Since f is a concave quadratic function, $f(i) \ge c$ for each of the $\ell - b + 1$ integers $i \in [j, j']$. Hence at most b of the numbers $f(0), \dots, f(\ell)$ are smaller than c, which means $f_{(b)} \ge c$. Finally, $\sum_b f_{(b)} = \sum_i f(i) = 2\sum_i i(\ell - i)$ because $\sum_{i=0}^{\ell}(\ell - 2i) = 0$. □

Lemma 6.4. *There is a constant C such that $|\Psi_b^{(q)}(t)| \le C\, t^{-f_{(b)}}(1 + |\log t|)^b$ for $0 < t \le 1$ and $0 \le b \le \ell$. Consequently* $\operatorname{ord} \Psi_b^{(q)} \ge -f_{(b)}$.

Proof. By Proposition 6.2(a), $\binom{\ell}{b}\Psi_b^{(q)}$ is the coefficient of $\alpha^{\ell-b}\beta^b$ in $\det(\alpha T_I^{(q)} + \beta T_K^{(q)})$, where $T_I^{(q)} = [i_{p-p'+q}]$ and $T_K^{(q)} = [k_{p-p'+q}]$. Expanding as in the proof of Lemma 5.7, every term is bounded by $C\, t^{-E_q(\pi,S)}(1 + |\log t|)^b$, and Lemma 6.3 applies. □

6.4 The shape

Theorem 6.5. *Let $n = \ell + 1$, write $L_{\ell,q}^t = \sum_{m=0}^{n} p_m(t)\partial^m$, and put $e_i = q(\ell - 2i) - 2i(\ell - i)$ for $0 \le i \le \ell$.*

(i) $L_{\ell,q}^t$ has order $\ell + 1$ and no singular point in $\mathbb{C} \setminus \{0\}$.

(ii) The exponents at $t = 0$ are $e_0, \dots, e_\ell$, and $\operatorname{ord} \Psi_b^{(q)} = -f_{(b)}$ for every b.

(iii) After scaling, $p_n = t^A$, where $A = \ell$ if $0 \in \{e_0, \dots, e_\ell\}$ and $A = \ell + 1$ otherwise. Moreover $\deg p_m \le A$ and $\operatorname{ord}_0 p_m \ge m + A - n$ for all m, and $\sum_m [t^A] p_m \lambda^m = \chi(\lambda) = \prod_{b=0}^{\ell}(\lambda - \lambda_b)$.

(iv) At $t = \infty$, $L_{\ell,q}^t$ has a basis of formal solutions $e^{\lambda_b t} t^{-\gamma_b} \sum_{i\ge 0} c_{b,i}^{(q)} t^{-i}$, $0 \le b \le \ell$, with $c_{b,0}^{(q)} = (-1)^{qb} C_b$.

(v) $t^{n-A} L_{\ell,q}^t = \sum_{k=0}^{K} t^{2k} Q_{2k}(\theta)$ with $Q_0(\theta) = \prod_{i=0}^{\ell}(\theta - e_i)$, $\deg Q_{2k} = \ell + 1 - 2k$ and $\operatorname{lc}(Q_{2k}) = [\lambda^{\ell+1-2k}]\chi(\lambda)$, and (5.2) holds for these Q_{2k}.

For $q = 0$ this is Theorem 5.8. For $q \ge 1$ the exponent 0 occurs only when $q(\ell - 2i) = 2i(\ell - i)$ for some $1 \le i < \ell/2$. For $\ell \le 4$ the only such pairs (ℓ, q) are $(3, 4)$ and $(4, 3)$; further examples are $(5, 12)$, $(6, 8)$ and $(8, 6)$.

Proof. By Proposition 6.2, $\operatorname{Sol}(L_{\ell,q}^t) = V_q$ has the basis $\Psi_0^{(q)}, \dots, \Psi_\ell^{(q)}$, with the same rates λ_b and exponents γ_b at infinity as for $q = 0$. The parts *Singular points*, *Abel at infinity* and *The squeeze* of the proof of Theorem 5.8 therefore apply, with Lemma 6.4 in place of Lemma 5.7. They give

$$\sum_{\tau\neq 0} \kappa_\tau = \binom{n}{2} - \sum_b \gamma_b - \sum_i \nu_i \le \binom{n}{2} - \sum_b \gamma_b + \sum_b f_{(b)} = 0,$$

by Lemma 6.3 and the computation in that proof. Hence there are no apparent singularities, $\operatorname{ord}\Psi_b^{(q)} = -f_{(b)}$ for all b, and, by the equality case of Lemma 3.3, the exponents at 0 form the multiset $\{-f_{(b)}\} = \{-f(i)\} = \{e_i\}$. This proves (i) and (ii).

For (iii), $p_n = ct^A$ by (i). Lemma 3.5, applied with the asymptotics of Proposition 6.2(c), gives $\deg p_m \le A$ and the formula for χ. By Fuchs' criterion $\operatorname{ord}_0 p_m \ge A - n + m$, and the indicial polynomial is $I(\rho) = \sum_m [t^{A-n+m}]p_m\, \rho^{\underline{m}}$. All e_i are congruent to $q\ell$ modulo 2. Suppose first that 0 is not an exponent. If $A < n$, the term $m = 0$ of I vanishes and $I(0) = 0$, which is impossible; so $A \ge n$. Then t^{A-n} divides every p_m, and primitivity gives $A = n$. Suppose now that 0 is an exponent. Then $q\ell$ is even, so 1 is not an exponent. If $A \le n-2$, the terms $m \le 1$ of I vanish and $\rho^{\underline{2}}$ divides I, which is impossible; so $A \ge n-1$. Since $I(0) = 0$, the power t^{A-n+1} divides every p_m, and primitivity gives $A \le n-1$.

Part (iv) follows from Proposition 6.2(c) as in the proof of Theorem 5.8.

For (v), put $\Theta_q = t^{n-A}L_{\ell,q}^t$. By (iii), $\Theta_q = \sum_m r_m(t)\theta^{\underline{m}}$ with $r_m = t^{n-A-m}p_m \in \mathbb{C}[t]$ of degree at most $n-m$. So $\Theta_q = \sum_k t^k Q_k(\theta)$ with $\deg Q_k \le n-k$, and $Q_0(\theta) = \sum_m r_m(0)\theta^{\underline{m}} = I(\theta)$. By Lemma 6.1, $D_{\ell,q}(-t) = \pm D_{\ell,q}(t)$. As in the proof of Theorem 5.8, the operator obtained from $L_{\ell,q}^t$ by the substitution $t \mapsto -t$ is a constant multiple of $L_{\ell,q}^t$. Comparing the terms with ∂^n, the constant is $(-1)^{A+n}$, so Θ_q is invariant under $t \mapsto -t$. Hence $Q_k = 0$ for odd k. The remaining assertions follow as in the proof of Theorem 5.8(v). Comparing coefficients of $t^n\partial^m$ gives $[t^A]p_m = [\theta^m]Q_{n-m}$, which determines the degrees and leading coefficients from χ. The computation with the formal solutions of (iv) gives (5.2). □

Remark 6.6. The bound of Lemma 6.3 is attained for every b. Indeed, $E_q(\pi, S)$ is an integer, so if every term in the proof of Lemma 6.4 had $E_q(\pi,S) < f_{(b)}$, we would get $\operatorname{ord}\Psi_b^{(q)} > -f_{(b)}$, contradicting Theorem 6.5(ii). We also confirmed this by enumeration for $\ell \le 6$ and $q \le 8$.

6.5 Proof of Theorem 2.6

Part (b) is Theorem 6.5(i)–(iii).

(a) Put $x = t^2$. By definition $G_{\ell,q}(t^2) = t^{-q\ell}D_{\ell,q}(t)$, and by Lemma 6.1 this is a power series in t^2. Conjugation by $t^{q\ell}$ replaces θ_t by $\theta_t + q\ell$, and $\theta_t = 2\theta_x$. By Theorem 6.5(v) the θ-form

$$\Theta' = \sum_{k=0}^{K} x^k\, Q_{2k}(2\theta_x + q\ell)$$

annihilates $G_{\ell,q}$. As in the proof of Theorem 2.2, $G_{\ell,q}$ satisfies no equation of order at most ℓ, because $D_{\ell,q}(t) = t^{q\ell}G_{\ell,q}(t^2)$ would then satisfy one. Put $P_k(\theta) = 2^{-\ell-1}Q_{2k}(2\theta + q\ell)$. Then $\deg P_k = \ell + 1 - 2k$, the leading coefficients of the P_k are those of Theorem 2.2(b), and

$$P_0(\theta) = \prod_{i=0}^{\ell}\Big(\theta + \frac{q\ell - e_i}{2}\Big) = \prod_{i=0}^{\ell}(\theta + i(\ell + q - i)).$$

For even ℓ, the solution $x^{-q\ell/2}\Psi_{\ell/2}^{(q)}(\sqrt{x}) = C\,x^{-(\gamma_{\ell/2}+q\ell)/2}(1+O(x^{-1/2}))$ gives $P_K(\theta) = \operatorname{lc}(P_K)(\theta + \ell^2/8 + q\ell/2)$, exactly as in the proof of Theorem 2.2. The formal solutions at infinity are those of Theorem 6.5(iv), multiplied by $x^{-q\ell/2}$.

All roots of P_0 are ≤ 0, and $P_0(0) = 0$. Hence the last three paragraphs of the proof of Theorem 2.2 apply to $L_{\ell,q} := x^{-1}\sum_k x^k P_k(\theta)$. They give $q_{\ell+1} = x^\ell$, so that $L_{\ell,q}$ has no singular point in $\mathbb{C}\setminus\{0\}$, as well as $x^{m-1} \mid q_m$, $\deg q_m \le \lfloor(\ell - 1 + m)/2\rfloor$, and primitivity, because $q_1(0) = P_0(1) = \prod_i(1 + i(\ell + q - i)) \ne 0$. For the exact degrees, the only change concerns the case $\deg P_{k^*} = m + 1 =: d$. Write $Q_{2k^*}(u) = a_d u^d + a_{d-1}u^{d-1} + \cdots$. Then

$[\theta^{d-1}]P_{k^*} = 2^{d-\ell-2}(a_{d-1} + d\,q\ell\,a_d)$, so the coefficient $\binom{d}{2}\operatorname{lc}(P_{k^*}) + [\theta^{d-1}]P_{k^*}$ is $2^{d-\ell-2}$ times the coefficient of λ^{d-1} in

$$\frac{\lambda}{2}\chi''(\lambda) + \sum_b \gamma_b \prod_{b'\neq b}(\lambda - \lambda_{b'}) + q\ell\,\chi'(\lambda).$$

Compared with the case $q = 0$, the additional term is $q\ell\,d\,a_d = (-1)^{k^*} d\,q\ell\,\sigma_{k^*}(c)$, in the notation of that proof. It has the same sign $(-1)^{k^*}$ as the other terms there, so the coefficient does not vanish, and $\deg q_m = \lfloor(\ell - 1 + m)/2\rfloor$ in all cases.

(c) Write $g(n) = u_{\ell,q}(n)/(n!\,(n+q\ell)!)$ for the Taylor coefficients of $G_{\ell,q}$. Comparing coefficients of x^N in $xL_{\ell,q}G_{\ell,q} = 0$ gives $\sum_k P_k(N-k)g(N-k) = 0$. Multiply by $N!\,(N+q\ell)!$ and, for $N \geq 1$, divide by $N(N+q\ell)$, which are the factors of $P_0(N)$ with $i = 0$ and $i = \ell$. This gives

$$\sum_{k=0}^{K} R_k^{(q)}(N)\,u_{\ell,q}(N-k) = 0 \quad (N \geq 1), \qquad R_0^{(q)}(N) = \prod_{i=1}^{\ell-1}(N + i(\ell + q - i)),$$

with $R_k^{(q)}(N) = P_k(N-k)\,(N-1)^{\underline{k-1}}\,(N+q\ell-1)^{\underline{k-1}}$ for $k \geq 1$; all $R_k^{(q)}$ have degree $\ell - 1$. Conversely, suppose that $u_{\ell,q}$ satisfies a recurrence of order M. Since $(n+q\ell)!/n!$ is a polynomial in n, the sequence $a(n) = (n!)^2 g(n) = n!\,u_{\ell,q}(n)/(n+q\ell)!$ satisfies a recurrence of order M as well. By Lemma 4.2, $G_{\ell,q} = \sum_n a(n)x^n/(n!)^2$ is annihilated by a θ-form of span M. In the variable t, after conjugation by $t^{q\ell}$, this is a θ-form of span $2M$ annihilating $D_{\ell,q}$. As in the proof of Theorem 2.4(c), Lemma 3.6 and Proposition 6.2(c) give $2K \leq 2M$. □

Proof of Corollary 2.7. By [11, (2.4), (2.5)], the function ${}_0F_1(q+\ell; (x)^\ell)$ with Jack parameter $\alpha = 1$ is a constant multiple of $x^{-q\ell/2}D_{\ell,q}(\sqrt{x}) = G_{\ell,q}(x)$, and by [11, (3.10)] it is, for $\beta = 2$, $n = \ell$ and $a + b = q - \ell$, a constant multiple of the function in [11, Prop. 4.1]. Theorem 2.6(a) gives the shape asserted there, with exact degrees, and shows that the order $\ell + 1$ is minimal. For the Hankel determinant $w_\ell(z) = c_\ell(-1)^{\ell(\ell-1)/2}z^{-\ell^2}\det[b_{j+k+1}(z)]_{j,k=0}^{\ell-1}$ of [11, App. B], reversing the order of the columns gives $w_\ell(z) = c_\ell z^{-\ell^2}D_{\ell,\ell}(z) = c_\ell\,G_{\ell,\ell}(z^2)$. If w_ℓ satisfied an equation of order at most ℓ in z, so would $D_{\ell,\ell}$, contradicting Theorem 6.5(i). □

7 Permutations with bounded increasing subsequences

Proof of Theorem 2.4. (a) By Theorem 2.2(b), the operator $xL_\ell = \sum_{k=0}^{K} x^k P_k(\theta)$ annihilates the double exponential generating function G_ℓ of u_ℓ. The converse part of Lemma 4.2 gives

$$\sum_{k=0}^{K} P_k(N-k)\big(N^{\underline{k}}\big)^2 u_\ell(N-k) = 0 \qquad (N \geq 0).$$

For $N \geq 1$ divide by N^2. Since $P_0(\theta) = \theta^2\prod_{b=1}^{\ell-1}(\theta + b(\ell - b))$, the term $k = 0$ becomes $R_0(N)u_\ell(N)$. For $k \geq 1$ we have $(N^{\underline{k}})^2/N^2 = ((N-1)^{\underline{k-1}})^2$. This is the recurrence in (a). The degree of R_k is $\deg P_k + 2k - 2 = \ell - 1$, by Theorem 2.2(b). The leading coefficient of P_K is the coefficient of $\mu^{\ell+1-2K} = \mu^{[\ell \text{ even}]}$ in

$$\prod_{b=0}^{\ell}(\mu - \ell + 2b) = \mu^{[\ell \text{ even}]}\prod_{j=0}^{K-1}(\mu^2 - (\ell - 2j)^2),$$

namely $(-1)^K\prod_{j<K}(\ell - 2j)^2$. Together with $P_K(\theta) = \operatorname{lc}(P_K)(\theta + \ell^2/8)$ for even ℓ, this gives the formula for R_K.

(b) We have $\operatorname{lc}(R_k) = \operatorname{lc}(P_k)$. The identity just displayed, divided by $\mu^{[\ell \text{ even}]}$, gives

$$\sum_k \operatorname{lc}(P_k)\,\mu^{2(K-k)} = \prod_{j<K}(\mu^2 - (\ell - 2j)^2).$$

Now put $\lambda = \mu^2$.

(c), order. Suppose u_ℓ satisfies a recurrence of order M for $n \geq n_0$. By Lemma 4.2, G_ℓ is annihilated by a θ-form $O = \sum_{k=0}^{M} x^k O_k(\theta_x)$ of span M. Substituting $x = t^2$, $\theta_x = \theta_t/2$, we obtain the θ-form $O_t = \sum_k t^{2k} O_k(\theta_t/2)$ of span $2M$, which annihilates $D_\ell(t)$. By minimality of L_ℓ^t we have $O_t = QL_\ell^t$ for some operator Q over $\mathbb{C}(t)$, so all Ψ_b are solutions of O_t. By Lemma 5.4 the Ψ_b with $\lambda_b \neq 0$ satisfy the hypotheses of Lemma 3.6; there are $2K$ of them. Hence $2K \leq 2M$.

(c), degree and leading coefficient. Write the recurrence as in (4.1), with leading coefficient c_M. By Lemma 4.2, the lowest coefficient of O is

$$O_0(\theta) = \theta^{\underline{M+n_0}}\, c_M(\theta - M)\, (\theta^{\underline{M}})^2.$$

Let V_x be the local solution space of L_ℓ at $x = 0$. It is spanned by the functions $\Psi_b(\sqrt{x})$, and $\operatorname{ord}_x \Psi_b(\sqrt{x}) = \frac{1}{2} \operatorname{ord}_t \Psi_b$. Here $\Psi_b(\sqrt{x})$ lies in the space $\mathcal{N}$ of Section 3, taken in the variable x. Indeed, $tL_\ell^t = \sum_k t^{2k} Q_{2k}(\theta)$ contains only even powers of t, so the part of an element of V formed by the odd powers of t is again a solution. Its leading exponent would be an odd root of Q_0, but all roots of Q_0 are even, so this part vanishes. By Theorem 5.8(iii) and the equality case of Lemma 3.3 (applied in the variable x, where the exponents are the roots of P_0), we get $\dim(V_x)_{\geq\nu}/(V_x)_{\geq\nu+1} = \operatorname{mult}_\nu(P_0)$ for all $\nu \in \mathbb{Z}$. Since $O = Q'L_\ell$, the space V_x consists of solutions of O. Lemma 3.2 then gives $\operatorname{mult}_\nu(P_0) \leq \operatorname{mult}_\nu(O_0)$. For $\nu < 0$ the factors $\theta^{\underline{M+n_0}}$ and $(\theta^{\underline{M}})^2$ do not vanish, so

$$\operatorname{mult}_\nu\big(c_M(\theta - M)\big) \geq \operatorname{mult}_\nu(P_0) \qquad (\nu < 0).$$

The negative roots of P_0, with multiplicity, are $-b(\ell - b)$ for $1 \leq b \leq \ell - 1$. Therefore $\prod_{b=1}^{\ell-1}(\theta + b(\ell - b))$ divides $c_M(\theta - M)$. Equivalently, $R_0(n + M)$ divides $c_M(n)$, that is, $R_0(N)$ divides the coefficient of $u_\ell(N)$ in the backward normalization. In particular $\deg c_M \geq \ell - 1$.

(c), uniqueness. Let $\mathcal{R} = \sum_{s=0}^{K} c_s(n) S^s$ be the recurrence operator of (a), written in forward form with shift S, and let $\mathcal{R}' = \sum_{s=0}^{K} c'_s(n) S^s$ be another recurrence of order K, valid for large n. The operator $c_K\mathcal{R}' - c'_K\mathcal{R}$ has order less than K and annihilates u_ℓ for large n. By the order bound it is zero, so $c_K c'_s = c'_K c_s$ for all s. The coefficients c_s have no common factor: dividing by one would produce a recurrence of degree $\ell - 2$, contradicting the degree bound. Hence c'_K/c_K is a polynomial, and $\mathcal{R}' = (c'_K/c_K)\mathcal{R}$. □

Proof of Corollary 2.5. Theorem 2.4 gives a recurrence of order $K = \lceil \ell/2 \rceil = \lfloor (\ell - 1)/2 \rfloor + 1$ and degree $\ell - 1$, and shows that neither number can be lowered. The minimal recurrence is unique up to a constant by (c). For odd ℓ, $K = \lfloor \ell/2 \rfloor + 1$. For even ℓ, $K = \ell/2 < \lfloor \ell/2 \rfloor + 1$, so a recurrence of order $\lfloor \ell/2 \rfloor + 1$, as in [11, (4.8)], exists but is not minimal. □

Remark 7.1. The leading coefficients in Table 3 illustrate Theorem 2.4(c): the factor $R_0(n) = \prod_{b=1}^{\ell-1}(n + b(\ell - b))$ is a square times $(n + \ell^2/4)$ when ℓ is even, and a square when ℓ is odd. Its roots record the exponents at 0 of the Bessel–Toeplitz operator. The trailing coefficient R_K records the exponent $\ell^2/8$ of the non-exponential solution at infinity when ℓ is even. The characteristic roots $(\ell - 2j)^2$ are the squared critical values of $|\operatorname{tr} U|$ on $U(\ell)$, attained at matrices with j eigenvalues -1 and $\ell - j$ eigenvalues $+1$, up to a common phase.

Remark 7.2. We note without proof the asymptotic structure that one expects from the formal solutions of L_ℓ at infinity. Each characteristic root $(\ell - 2b)^2$ with $b < \ell/2$ carries a formal solution $(\ell - 2b)^{2n} n^{(1-(\ell-b)^2-b^2)/2}(1 + O(n^{-1}))$. This is what the double Borel transform of the formal solution $e^{2(\ell-2b)\sqrt{x}} x^{-\gamma_b/2}$ of L_ℓ predicts. For $b = 0$ it is consistent with Regev's asymptotics $u_\ell(n) \sim c_\ell\, \ell^{2n} n^{-(\ell^2-1)/2}$ [20]. Nothing in this paper depends on it.

ℓ	$R_0(n),\ R_1(n),\dots,R_K(n)$ in $\sum_k R_k(n)u_\ell(n-k)=0$
2	$n+1,\quad -2(2n-1)$
3	$(n+2)^2,\quad -(10n^2+2n-3),\quad 9(n-1)^2$
4	$(n+3)^2(n+4),\quad -2(10n^3+31n^2+11n-12),\quad 64n(n-1)^2$
5	$(n+4)^2(n+6)^2,\quad -(35n^4+322n^3+843n^2+400n-375),$
	$(n-1)^2(259n^2+622n+45),\quad -225(n-2)^2(n-1)^2$
6	$(n+5)^2(n+8)^2(n+9),\quad -4(14n^5+273n^4+1840n^3+4757n^2+2758n-2352),$
	$4(n-1)^2(196n^3+1608n^2+3167n+444),\quad -1152(n-2)^2(n-1)^2(2n+3)$
7	$(n+6)^2(n+10)^2(n+12)^2,$
	$-(84n^6+2940n^5+38958n^4+238440n^3+634756n^2+429858n-343035),$
	$(n-1)^2(1974n^4+36336n^3+213240n^2+407840n+82425),$
	$-2(n-2)^2(n-1)^2(6458n^2+42646n+49875),\quad 11025(n-3)^2(n-2)^2(n-1)^2$

Table 3: The minimal recurrences of Theorem 2.4 for $2\le\ell\le 7$, valid for $n\ge 1$ with $u_\ell(m)=0$ for $m<0$. They were checked exactly against the values $u_\ell(n)=\sum_\lambda (f^\lambda)^2$ computed from the hook length formula (Appendix A). For $\ell=2$ one recovers the Catalan recurrence; the case $\ell=4$ is [11, (4.7)].

8 The Hankel case: tableaux and matchings

8.1 Jacobi integrals

Let $\Delta(u)=\prod_{p<q}(u_q-u_p)$.

Lemma 8.1. *Let $k=2r+\varepsilon$ with $r\ge 1$, $\varepsilon\in\{0,1\}$, and $w(u)=(2-u)^{\varepsilon-\frac12}(2+u)^{\frac12}$ on $(-2,2)$. Then*

$$Y_k(t)=\frac{e^{\varepsilon t}}{r!\,(2\pi)^r}\int_{[-2,2]^r} e^{t(u_1+\cdots+u_r)}\,\Delta(u)^2\prod_{p=1}^{r} w(u_p)\,du.$$

Moreover, for $\ell\ge 1$, $\mathbb{E}_{\mathrm{Sp}(2\ell)}e^{t\,\mathrm{tr}\,U}=\det[b_{i-j}-b_{i+j}]_{i,j=1}^{\ell}$, which by (2.3) equals $e^{-t}Y_{2\ell+1}(t)$.

Proof. The first formula is proved in [15, Part I]. It follows from (2.3) and the integral $b_m=\frac{1}{\pi}\int_0^\pi e^{2t\cos\varphi}\cos(m\varphi)\,d\varphi$. One then uses the product formulas for $\cos((i-j)\varphi)\pm\cos((i+j-1+\varepsilon)\varphi)$, Andreief's identity and the substitution $u=2\cos\varphi$. For the second, Weyl's integration formula expresses the eigenvalue angles $\varphi_1,\dots,\varphi_\ell\in[0,\pi]$ of $U\in\mathrm{Sp}(2\ell)$ through the density proportional to $\prod_{p<q}(\cos\varphi_p-\cos\varphi_q)^2\prod_p\sin^2\varphi_p$, and $\mathrm{tr}\,U=\sum_p 2\cos\varphi_p$. The identities $\det[\sin(i\varphi_p)]=c\prod_p\sin\varphi_p\prod_{p<q}(\cos\varphi_q-\cos\varphi_p)$ and $b_{i-j}-b_{i+j}=\frac{2}{\pi}\int_0^\pi e^{2t\cos\varphi}\sin(i\varphi)\sin(j\varphi)\,d\varphi$, together with Andreief's identity, give the determinant [1, 14]. The normalization follows from the value 1 at $t=0$. □

In particular $\mathbb{E}_{\mathrm{Sp}(2\ell)}(\mathrm{tr}\,U)^N$ is the N-th moment of the law of $\sum_p 2\cos\varphi_p$, and these moments vanish for odd N since $-I\in\mathrm{Sp}(2\ell)$.

8.2 The minimal operator of Y_k

We recall the theorem of [15], in the form needed here. Let $k=2r+\varepsilon\ge 2$ and put

$$e_b=b^2-2br-\varepsilon b=-b(k-b),\qquad \lambda_j^{(k)}=\varepsilon+2r-4j,\qquad \gamma_j^{(k)}=(r-j)\bigl(r-j+\varepsilon-\tfrac12\bigr)+j\bigl(j+\tfrac12\bigr),\tag{8.1}$$

for $0\le b,j\le r$.

Theorem 8.2 ([15]). *Let $k \geq 2$. The minimal operator $L_k = \sum_{m=0}^{r+1} p_m(t)\partial^m$ of Y_k, normalized to be primitive, has order $r+1$, with $p_{r+1} = t^r$, $\deg p_m \leq r$ and $t^{m-1} \mid p_m$. It has no singular point in $\mathbb{C} \setminus \{0\}$. Its solution space has a basis $y_0, \dots, y_r$, with y_0 a nonzero multiple of Y_k, such that:*

(i) y_b has an expansion in $\mathcal{N}$ with leading part $C_b' t^{e_b}$, $C_b' \neq 0$; thus the exponents at 0 are the distinct integers $e_0 = 0 > e_1 > \cdots > e_r$;

(ii) $y_j^{(m)}(t) = C_j e^{\lambda_j^{(k)} t} t^{-\gamma_j^{(k)}} \big((\lambda_j^{(k)})^m + O(t^{-1})\big)$ as $t \to +\infty$ for all $m \geq 0$, with $C_j \neq 0$; and

$$\sum_m [t^r] p_m \, \lambda^m = \chi_k(\lambda) := \prod_{j=0}^{r} (\lambda - \lambda_j^{(k)}).$$

Outline. A complete proof, following [15], is given in Appendix B. It follows the scheme of Section 5. The one-variable functions are the Laplace transforms of w over $[-2,2]$ and of $(2-u)^{\varepsilon-\frac{1}{2}}|2+u|^{\frac{1}{2}}$ over $(-\infty,-2]$. They solve a second-order equation gauge equivalent to (5.1), whose Galois group is SL_2. The Toeplitz form is replaced by the Hankel form $F(g_1,\dots,g_r) = \sum_{\sigma,\tau} \operatorname{sgn}(\sigma\tau) \prod_p g_p^{(\sigma(p)+\tau(p)-2)}$, and the vertex integrals are taken over $(-\infty,-2]^b \times [-2,2]^{r-b}$. Laplace's method at the vertices gives (ii). The substitution $u_p = -s_p/t$ on the rays gives (i) directly, with $C_b' \neq 0$; here no squeeze is needed, because the exponents are distinct. Abel's identity together with

$$\sum_{j=0}^{r} \gamma_j^{(k)} + \sum_{b=0}^{r} e_b = \binom{r+1}{2}$$

excludes apparent singularities, and the shape follows as in the proof of Theorem 5.8. □

8.3 Proof of Theorem 2.8

Write $p_{m,i} = [t^i]p_m$. Comparing the coefficients of $t^M/M!$ in $L_k Y_k = 0$, and using $[t^M/M!]\, t^i \partial^m Y_k = M^{\underline{i}}\, y_k(M+m-i)$, we get

$$\sum_s C_s(M)\, y_k(M+s) = 0 \quad (M \geq 0), \qquad C_s(M) = \sum_{m-i=s} p_{m,i}\, M^{\underline{i}}. \tag{8.2}$$

Since $m-1 \leq i \leq r$, the shifts $s = m-i$ lie in $[-r,1]$, and $\deg C_s \leq r$.

The top shift $s=1$ collects the terms with $i = m-1$. The indicial polynomial at 0 is $I(\rho) = \sum_m p_{m,m-1} \rho^{\underline{m}} = \prod_{b=0}^{r} (\rho - e_b)$; it is monic because $p_{r+1} = t^r$, and its roots are the e_b by Theorem 8.2(i). Since $(M+1)^{\underline{m}} = (M+1) M^{\underline{m-1}}$,

$$C_1(M) = \frac{I(M+1)}{M+1} = \prod_{b=1}^{r} (M+1+b(k-b)).$$

The bottom shift $s=-r$ has $C_{-r}(M) = p_{0,r} M^{\underline{r}} = \chi_k(0) M^{\underline{r}}$, and $\chi_k(0) = \prod_j (-\lambda_j^{(k)})$ vanishes exactly when some $\lambda_j^{(k)} = 0$, that is, when $\varepsilon = 0$ and r is even, i.e. $4 \mid k$.

In that case the bottom shift is $s = 1-r$, with

$$C_{1-r}(M) = p_{0,r-1} M^{\underline{r-1}} + p_{1,r} M^{\underline{r}} = M^{\underline{r-1}} (p_{0,r-1} + p_{1,r}(M-r+1)).$$

To identify the linear factor, write $tL_k = \sum_l t^l Q_l(\theta)$ as in the proof of Theorem 5.8. Then $Q_{r+1} = p_{0,r} = 0$ and $Q_r(\theta) = p_{0,r-1} + p_{1,r}\theta$. The solution $y_{r/2}$ has rate 0 and behaves like $C\, t^{-\gamma}(1 + O(t^{-1}))$ with $\gamma = \gamma_{r/2}^{(k)} = r^2/2$. In $tL_k y_{r/2}$ the top power $t^{r-\gamma}$ has coefficient $Q_r(-\gamma)$,

so $Q_r(-r^2/2) = 0$. Moreover $p_{1,r} = [\lambda]\chi_k = \prod_{j\neq r/2}(-\lambda_j^{(k)}) \neq 0$. Setting $N = M+1$ and $c_i(N) = C_{1-i}(N-1)$ gives the recurrence of Theorem 2.8, with c_0 and c_R as stated.

The leading coefficient of C_s in M is $p_{s+r,r} = [\lambda^{s+r}]\chi_k$. Hence $\sum_s \mathrm{lc}(C_s)\lambda^{s+r} = \chi_k(\lambda)$, and dividing by λ when $4 \mid k$ gives the characteristic polynomial.

Order. Let a recurrence of order M' hold for large N. Lemma 4.1 gives a θ-form of span M' annihilating Y_k. All y_j are solutions of it, by minimality of L_k. The number of j with $\lambda_j^{(k)} \neq 0$ is $r+1-[4 \mid k] = R$, so Lemma 3.6 gives $M' \geq R$.

Degree and leading coefficient. By Lemma 4.1, the lowest coefficient of that θ-form is

$$\theta^{\underline{M'+n_0}}\, c_{M'}(\theta - M'),$$

where $c_{M'}$ is the leading coefficient in forward normalization. By Lemma 3.2 applied to y_b, which has leading exponent e_b, every e_b is a root of the lowest coefficient. For $b \geq 1$, $e_b \leq 1-2r-\varepsilon < 0$, so e_b is a root of $c_{M'}(\theta - M')$. The e_b are distinct, so $\prod_{b=1}^{r}(\theta - e_b)$ divides $c_{M'}(\theta - M')$. In the backward normalization this says that $c_0(N) = \prod_b (N - e_b)$ divides the coefficient of $y_k(N)$. Uniqueness follows as in the proof of Theorem 2.4(c). □

Example 8.3. For $k = 8$ one has $r = 4$, $\lambda^{(8)} = (8, 4, 0, -4, -8)$ and $e = (0, -7, -12, -15, -16)$. The minimal recurrence has order 4 and degree 4, with

$$c_0(N) = (N+7)(N+12)(N+15)(N+16), \qquad c_4(N) = 1024\,(N-1)(N-2)(N-3)(N+4),$$

and characteristic polynomial $(\lambda-8)(\lambda-4)(\lambda+4)(\lambda+8)$. The factor $N+4 = N - r + r^2/2$ comes from the rate-zero solution, whose exponent at infinity is $\gamma_2^{(8)} = 8$.

8.4 Proof of Theorem 2.9

Let $Z_\ell = \mathbb{E}_{\mathrm{Sp}(2\ell)} e^{t\,\mathrm{tr}\,U} = e^{-t} Y_{2\ell+1}$, and let $\mu_N = \mathbb{E}(\mathrm{tr}\,U)^N$, so that $\mu_{2n} = m_\ell(n)$ by (2.1) and $\mu_{2n+1} = 0$. If $L_{2\ell+1}(\partial)$ annihilates $Y_{2\ell+1}$, then $L' = L_{2\ell+1}(\partial+1) = \sum_m p_m(t)(\partial+1)^m$ annihilates Z_ℓ.

The operator L' has the same shape as $L_{2\ell+1}$. It also has the same indicial polynomial at 0, because $[t^{m-1}]$ of the coefficient of ∂^m is unchanged: $p_{m'}$ is divisible by $t^{m'-1}$. Its solutions are $e^{-t}y_j$, so the rates are $\lambda_j^{(2\ell+1)} - 1 = 2(\ell - 2j) = \lambda_j$, with the same $\gamma_j^{(2\ell+1)}$. The exponents at 0 are $e_b = -b(2\ell+1-b)$, and $\sum_m [t^\ell] p'_m \lambda^m = \chi'(\lambda) := \prod_{j=0}^{\ell}(\lambda - \lambda_j)$.

As in (8.2), $\sum_{s=s_{\min}}^{1} C'_s(M)\mu_{M+s} = 0$ for $M \geq 0$. Here $C'_1(M) = \prod_{b=1}^{\ell}(M+1+b(2\ell+1-b))$, and $s_{\min} = -\ell$ if ℓ is odd, while $s_{\min} = 1-\ell$ if ℓ is even (then $\chi'(0) = 0$). In both cases $s_{\min}$ is odd.

Take $M = 2N-1$ odd. Then the terms with even s involve odd moments and vanish, and we obtain

$$\sum_{i=0}^{K} d_i(N)\, m_\ell(N-i) = 0, \qquad d_i(N) = C'_{1-2i}(2N-1), \qquad K = \tfrac{1-s_{\min}}{2} = \lceil \ell/2 \rceil.$$

The extreme coefficients are computed exactly as in the proof of Theorem 2.8. The rate-zero solution for even ℓ has exponent $\gamma_{\ell/2}^{(2\ell+1)} = \ell(\ell+1)/2$ at infinity, which produces the factor $2N - \ell + \ell(\ell+1)/2$.

Since $\chi'(-\lambda) = (-1)^{\ell+1}\chi'(\lambda)$, the nonzero coefficients of χ' sit at the powers $\lambda^{s+\ell}$ with s odd. Hence $\mathrm{lc}(C'_{1-2i}) = [\lambda^{1-2i+\ell}]\chi' \neq 0$; up to sign these are elementary symmetric functions of the numbers $4(\ell-2j)^2$. So each d_i has degree exactly ℓ, and the characteristic polynomial is $\prod_{j<\ell/2}(\lambda - 4(\ell-2j)^2)$.

For the lower bounds, a recurrence of order M' and degree d for m_ℓ, valid for large N, yields a recurrence of order $2M'$ and degree d for μ: replace N by $M''/2$, and note that for odd M'' all terms vanish. Lemma 4.1 and Lemma 3.6, applied to the $\ell+1-[\ell \text{ even}]$ solutions $e^{-t}y_j$ with $\lambda_j \neq 0$, give $2M' \geq \ell+1-[\ell \text{ even}]$, i.e. $M' \geq \lceil \ell/2 \rceil$. Lemma 3.2 applied to the ℓ distinct negative exponents e_b shows that $\prod_b (M''-e_b)$ divides the leading coefficient of the μ-recurrence. In terms of $N = M''/2$, this is $\prod_b (2N-e_b) = d_0(N)$. Uniqueness is proved as before. □

Example 8.4. For $\ell = 4$ (perfect matchings without 5-crossings) the minimal recurrence is

$$\prod_{b=1}^{4}(2N+b(9-b))\, m_4(N)+d_1(N)\, m_4(N-1)+4096\,(N-1)(N+3)(2N-1)(2N-3)\, m_4(N-2)=0,$$

with d_1 of degree 4, $d_0(N) = (2N+8)(2N+14)(2N+18)(2N+20)$, and characteristic polynomial $(\lambda-64)(\lambda-16)$.

9 Distributions of traces and the transport principle

9.1 The transport principle

The recurrences of Sections 7 and 8 are statements about moments, and by Lemma 4.3 they are equivalent to differential equations for the underlying measures. The following classical observation explains why the exponential rates of the Laplace side become the singular points of the measure side.

Proposition 9.1. *Let ρ be a finite measure on $\mathbb{R}$ with compact support and $Z(t) = \int e^{tw}\, d\rho(w)$. Let ϕ be the ring isomorphism of Weyl algebras with $\phi(t) = -\partial_w$ and $\phi(\partial_t) = w$. Then for every $A \in \mathbb{C}[t]\langle\partial_t\rangle$,*

$$AZ = 0 \iff \phi(A)\rho = 0 \quad \textit{(in the sense of distributions).}$$

If $A = \sum_i a_i(t)\partial_t^i$ with $\max_i \deg a_i = d$, then $\phi(A)$ has order at most d in ∂_w, and its coefficient of ∂_w^d is $(-1)^d \sum_i [t^d]a_i\, w^i$.

Proof. For monomials, $t^a\partial_t^i Z = \int w^i\, \partial_w^a(e^{tw})\, d\rho = (-1)^a \int e^{tw}\, d(\partial_w^a(w^i\rho)) = \int e^{tw}\, d(\phi(t^a\partial_t^i)\rho)$. The Fourier–Laplace transform is injective on compactly supported distributions. The last assertion follows from $\phi(t^a\partial^i) = (-1)^a\partial_w^a w^i$. □

Thus, when the minimal operator of Z has leading coefficient a power of t and coefficients of degree at most d, the measure is annihilated by an operator of order d. The leading coefficient of that operator is the polynomial $\sum_i [t^d]a_i\, w^i$, whose roots are the exponential rates λ of Z at infinity. For example, D_ℓ is the Laplace transform of the law of $2\operatorname{Re}\operatorname{tr} U$. By Theorem 5.8 and Proposition 9.1, this law is annihilated by an operator of order ℓ with leading coefficient $\pm\prod_{b=0}^{\ell}(w-2(\ell-2b))$, whose zeros are exactly the critical values of $2\operatorname{Re}\operatorname{tr} U$. For the law of $|\operatorname{tr} U|^2$ we argue directly from Theorem 2.4.

9.2 Proof of Theorem 2.10

Write the recurrence of Theorem 2.4 in forward form: for $n \geq 0$, $\sum_{j=0}^{K} c_j(n)u_\ell(n+j) = 0$ with $c_j(n) = R_{K-j}(n+K)$. By (2.1), $u_\ell(n)$ is the n-th moment of the law of $|\operatorname{tr} U|^2$, a probability measure on $[0,\ell^2]$. For $\ell \geq 2$ it is absolutely continuous, since $|\operatorname{tr} U|^2$ is a nonconstant real-analytic function of the eigenvalue angles; its density is ρ_ℓ. Lemma 4.3 with $P_j(s) = c_j(-s-j-1) = R_{K-j}(K-j-1-s)$ shows that $\mathcal{L}_\ell\rho_\ell = 0$.

All P_j have degree $\ell-1$, so $\mathcal{L}_\ell$ has order $\ell-1$. Since $\theta^{\ell-1} = w^{\ell-1}\partial_w^{\ell-1} + \cdots$, its leading coefficient is

$$w^{\ell-1}\sum_j \mathrm{lc}(P_j)\,w^j = (-1)^{\ell-1}w^{\ell-1}\sum_j \mathrm{lc}(R_{K-j})\,w^j = (-1)^{\ell-1}w^{\ell-1}\prod_{i=0}^{K-1}(w-(\ell-2i)^2),$$

by Theorem 2.4(b). The lowest and top θ-coefficients P_0 and P_K both have the maximal degree $\ell-1$, so $\mathcal{L}_\ell$ is Fuchsian at 0 and ∞. The remaining singular points are simple zeros of the leading coefficient, hence regular singular. For $j<K$ the polynomial $R_{K-j}(n)$ contains the factor $[(n-1)\cdots(n-K+j+1)]^2$, so $P_j(\theta) = R_{K-j}(K-j-1-\theta)$ is divisible by $\theta^{\underline{K-j-1}} = w^{K-j-1}\partial_w^{K-j-1}$. Hence every term $w^jP_j(\theta)$ is left divisible by w^{K-1}, which gives the factorization $\mathcal{L}_\ell = w^{K-1}\mathcal{L}_\ell^\circ$ and the leading coefficient of $\mathcal{L}_\ell^\circ$.

Let A be a nonzero operator of order d with polynomial coefficients such that $A\rho_\ell = 0$ in the sense of distributions on $\mathbb{R}$. Then w^dA is a θ-form of θ-degree d annihilating ρ_ℓ, and Lemma 4.3 yields a recurrence of degree at most d for u_ℓ. Hence $d \ge \ell-1$ by Theorem 2.4(c). If A and B are two such annihilators of order $\ell-1$, then $\mathrm{lc}(B)A - \mathrm{lc}(A)B$ annihilates ρ_ℓ and has smaller order, so it vanishes.

The exponents at 0 are the roots of $P_0(\theta) = R_K(K-1-\theta)$. By the formula for R_K, these are $0,1,\dots,K-2$, each twice, and $\ell^2/8-1$ when ℓ is even. The roots of the top coefficient $P_K(\theta) = R_0(-1-\theta) = \prod_b(b(\ell-b)-1-\theta)$ are $b(\ell-b)-1$. In the local coordinate $1/w$ at infinity, this means that the exponents are $1-b(\ell-b)$, $1\le b\le \ell-1$.

For the symplectic case, the recurrence $\sum_{s=s_{\min}}^{1} C'_s(M)\mu_{M+s} = 0$ of Section 8 has order $1-s_{\min} = \ell+1-[\ell \text{ even}]$ and degree ℓ for the full moment sequence of $\operatorname{tr}U$. By Lemma 4.3, the law of $\operatorname{tr}U$ on $\mathrm{Sp}(2\ell)$ is annihilated, in the sense of distributions, by $A_\ell = \sum_k w^kP_k(\theta)$ with $P_k(\theta) = C'_{k+s_{\min}}(-\theta-k-1-s_{\min})$. As above, A_ℓ has order ℓ, it is Fuchsian, and its leading coefficient is $\pm w^{-s_{\min}}\chi'(w)$. For $s\le 0$ every term of $C'_s(M) = \sum_{m-i=s} p'_{m,i}M^{\underline{i}}$ contains the factor $M^{\underline{-s}}$, so $P_k(\theta)$ is divisible by $\theta^{\underline{-s_{\min}-k}}$ for $k\le -s_{\min}$. Hence $A_\ell = w^{-s_{\min}}A_\ell^\circ$, where A_ℓ° has polynomial coefficients and leading coefficient $\pm\chi'(w) = \pm\prod_{j=0}^{\ell}(w-2(\ell-2j))$. Thus the finite singular points of A_ℓ° lie among the critical values $2(\ell-2j)$ of $\operatorname{tr}U$; in particular $w=0$ is an ordinary point when ℓ is odd. No operator of smaller order annihilates the law, because every recurrence for (μ_N) has degree at least ℓ, by the proof of Theorem 2.9. For tableaux, the same argument applied to (8.2) shows the following. The law of $\varepsilon+\sum_p u_p$ under the probability measure proportional to $\Delta(u)^2\prod_p w(u_p)\,du$ on $[-2,2]^r$ is annihilated by a Fuchsian operator of order r with polynomial coefficients and leading coefficient $\pm\chi_k(w) = \pm\prod_{j=0}^{r}(w-\lambda_j^{(k)})$, and by no operator of smaller order. In particular $w=0$ is an ordinary point unless $4\mid k$. □

Remark 9.2. For $\ell=2$ the theorem says that ρ_2 satisfies a first-order equation with exponents $-\frac12$ at 0 and $\frac12$ at 4. Indeed $\rho_2(w) = \frac{1}{2\pi}\sqrt{(4-w)/w}$ on $[0,4]$, whose moments are the Catalan numbers $u_2(n)$. For $\ell=3$, ρ_3 satisfies a second-order Fuchsian equation with singular points $0,1,9,\infty$ and exponents $0,0$ at the origin.

10 A denominator-free construction and arithmetic refinements

This section completes the coefficient-integrality question left aside in Corollaries 2.3 and 2.7. Integrality itself was already asserted in [11, Proposition 4.1]; the point here is an explicit finite formula, with no elimination denominators, and its consequences for the operators and recurrences of this paper. The construction also gives the coefficient shape for the positive-parameter Jack family outside $\beta=2$. Minimality for nonnegative integer twists in that larger family is proved by visibility in Theorem 11.1.

10.1 An Ore continuant and a matching formula

Fix an integer $\ell \geq 1$, write $\theta = x\partial_x$, and take parameters $B_0 = 0, B_1, \ldots, B_\ell$. Put

$$w_j = (j+1)(\ell - j) \qquad (0 \leq j < \ell).$$

In the Ore algebra defined by $\theta x = x(\theta + 1)$ define

$$\mathcal{C}_{\ell+1} = 1, \qquad \mathcal{C}_\ell = \theta + B_\ell, \qquad \mathcal{C}_j = (\theta + B_j)\mathcal{C}_{j+1} - w_j x \mathcal{C}_{j+2} \quad (0 \leq j < \ell). \tag{10.1}$$

A matching M of the path with vertices $0, 1, \ldots, \ell$ is a set of pairwise disjoint edges $\{j, j+1\}$. Let $V(M)$ be its incident vertices and, for an unmatched vertex i, put

$$r_M(i) = \#\{\{j, j+1\} \in M : j > i\}, \qquad w(M) = \prod_{\{j,j+1\} \in M} (j+1)(\ell - j).$$

Empty products equal 1.

Theorem 10.1 (Integral matching formula)**.** *With $K = \lceil \ell/2 \rceil$, one has*

$$\mathcal{C}_0 = \sum_{k=0}^{K} x^k \mathcal{P}_k(\theta), \qquad \mathcal{P}_k(z) = (-1)^k \sum_{\substack{M \text{ a path matching} \\ |M| = k}} w(M) \prod_{i \notin V(M)} (z + B_i + r_M(i)). \tag{10.2}$$

In particular:

1. *$\mathcal{P}_k \in \mathbb{Z}[B_1, \ldots, B_\ell, z]$, $\deg_z \mathcal{P}_k = \ell + 1 - 2k$, and*

$$\mathcal{P}_0(z) = \prod_{i=0}^{\ell} (z + B_i), \qquad \sum_{k=0}^{K} \operatorname{lc}(\mathcal{P}_k) \mu^{\ell+1-2k} = \prod_{i=0}^{\ell} (\mu - \ell + 2i).$$

2. *The operator $\mathcal{L} = x^{-1}\mathcal{C}_0$ belongs to $\mathbb{Z}[B_1, \ldots, B_\ell, x]\langle \partial_x \rangle$. Its order is $\ell + 1$, its leading coefficient is x^ℓ, and, if $\mathcal{L} = \sum_m a_m(x)\partial_x^m$, then*

$$x^{m-1} \mid a_m \quad (m \geq 1), \qquad \deg_x a_m \leq \left\lfloor \frac{\ell - 1 + m}{2} \right\rfloor.$$

3. *If every $B_i \geq 0$, all the degree bounds in (2) are equalities. Moreover $(-1)^k \mathcal{P}_k(z)$ has nonnegative coefficients as a polynomial in $z, B_1, \ldots, B_\ell$.*

Proof. Expand (10.1) recursively. At the leftmost remaining vertex j, either retain $\theta + B_j$, or select the edge $\{j, j+1\}$, contributing $-w_j x$ and deleting both endpoints. Thus the resulting terms are indexed precisely by path matchings. Moving every x to the left increases an unmatched factor $\theta + B_i$ by one for each selected edge strictly to its right. This proves (10.2) and integrality. Every summand for a k-matching is monic of degree $\ell + 1 - 2k$, with sign $(-1)^k$ and positive weight. Consequently that degree is exact.

The polynomial $\sum_k \operatorname{lc}(\mathcal{P}_k)\mu^{\ell+1-2k}$ is the ordinary tridiagonal determinant with zero diagonal and products w_j of opposite off-diagonal entries. It is the characteristic polynomial of $v\partial_u + u\partial_v$ on homogeneous polynomials of degree ℓ in u, v: in the monomial basis the opposite entries have product $(j+1)(\ell - j)$. The basis $(u+v)^{\ell-i}(u-v)^i$ diagonalizes this operator, with eigenvalues $\ell - 2i$. This proves the claimed characteristic polynomial.

Since $B_0 = 0$, $\mathcal{P}_0(0) = 0$. The identity $\theta^d = \sum_{m=0}^{d} S(d, m) x^m \partial_x^m$, with integral Stirling numbers $S(d, m)$, now proves that $x^{-1}\mathcal{C}_0$ has polynomial coefficients. Indeed its only potentially negative power comes from $x^{-1}\mathcal{P}_0(\theta)$, whose constant term is zero. The bounds in (2) follow directly from $\deg \mathcal{P}_k = \ell + 1 - 2k$.

For exactness let $k_* = \lfloor (\ell+1-m)/2 \rfloor$. The coefficient of the largest possible power x^{k_*+m-1} in a_m comes solely from $\mathcal{P}_{k_*}$. If its degree is m, that coefficient is its nonzero leading coefficient. If its degree is $m+1$ and $m \geq 1$, its leading term contributes $S(m+1,m) = \binom{m+1}{2} > 0$ after multiplication by $(-1)^{k_*}$, and all other contributions have the same nonnegative sign. The remaining case is $m = 0$, ℓ even. A maximal matching leaving vertex 0 unmatched contributes a positive multiple of $z + K$ to $(-1)^K \mathcal{P}_K(z)$, so its constant coefficient is positive. This proves (3). □

Proposition 10.2 (Scalarization of the published matrix system)**.** *Suppose $I_0, \dots, I_\ell$ satisfy*

$$(\ell - p)xI_{p+1} = (\theta + B_p)I_p - pI_{p-1} \qquad (0 \leq p \leq \ell), \tag{10.3}$$

where the terms with nonexistent indices are omitted. Then $\mathcal{L}I_\ell = 0$ for the operator of Theorem 10.1. For $B_p = p(\ell + q - p)$, $q \in \mathbb{Z}_{\geq 0}$, this operator equals $L_{\ell,q}$ of Theorem 2.6.

Proof. Set $J_p = (\ell!/p!)I_p$. Descending induction in (10.3) gives

$$J_p = \mathcal{C}_{p+1} I_\ell \qquad (0 \leq p \leq \ell).$$

The equation with $p = 0$ is $\theta J_0 - \ell x J_1 = 0$, hence $\mathcal{C}_0 I_\ell = 0$. System (10.3) is the Forrester–Wei system [11, (3.6)]. For the stated specialization, its last component is a nonzero constant multiple of $G_{\ell,q}$; for $q = 0$ use the modified initial vector in [11, Remark 3.4(1)]. Theorem 2.6 gives minimal order $\ell + 1$. Both $\mathcal{L}$ and $L_{\ell,q}$ have that order and leading coefficient x^ℓ, so they coincide. □

Corollary 10.3 (Polynomial dependence and recurrence integrality)**.** *For every $\ell \geq 1$ the coefficients of $L_{\ell,q}$ belong to $\mathbb{Z}[q,x]$. More precisely, $P_k(\theta)$ of Theorem 2.6 belongs to $\mathbb{Z}[q,\theta]$ and $(-1)^k P_k$ has nonnegative coefficients in (q,θ). The recurrence coefficients*

$$R_0^{(q)}(N) = \prod_{i=1}^{\ell-1} (N + i(\ell + q - i)), \qquad R_k^{(q)}(N) = P_k(N-k)(N-1)^{\underline{k-1}}(N + q\ell - 1)^{\underline{k-1}} \quad (k \geq 1)$$

belong to $\mathbb{Z}[q,N]$ in the normalization of Theorem 2.6(c). These statements hold without an unspecified common denominator or an additional rescaling of the leading coefficient.

Proof. Substitute $B_i = i(\ell - i) + iq$ into Theorem 10.1 and use Proposition 10.2. The displayed recurrence formula is the coefficient extraction already established in the proof of Theorem 2.6(c). □

Corollary 10.4 (Positive-parameter Jack family)**.** *Let $\beta > 0$, $q \geq 0$, and*

$$F_{\ell,\beta,q}(x) = {}_0F_1^{(2/\beta)}\left(q + 1 + \frac{\beta}{2}(\ell - 1); (x)^\ell\right).$$

With $B_p = p(q + \frac{\beta}{2}(\ell - p))$, the operator $\mathcal{L}$ of Theorem 10.1 annihilates $F_{\ell,\beta,q}$. It has order $\ell + 1$, leading coefficient x^ℓ, and the exact coefficient degrees in Theorem 2.2(a). Its coefficients are polynomials over $\mathbb{Z}$ in $q, \beta/2, x$; hence they are integral whenever $q, \beta/2$ are nonnegative integers. This corollary concerns existence and shape; Theorem 11.1 supplies order minimality for nonnegative integer twists.

Proof. The identification of the last component of (10.3) with the displayed Jack hypergeometric function is [11, Proposition 3.3], with the explicit substitution $n = \ell$ and $a + b = q - 1 - \frac{\beta}{2}(\ell - 1)$. Its zero-$q$ normalization is as in Remark 3.4(1). All B_p are nonnegative. Apply Theorem 10.1 and Proposition 10.2. □

10.2 Exactly which twists lower the leading power in the Bessel variable?

Theorem 10.5 (Pythagorean classification of the exceptional twists)**.** *Let $\ell \geq 1$ and $q \in \mathbb{Z}_{\geq 0}$. In the normalization of Theorem 6.5, the leading coefficient of the minimal operator of $D_{\ell,q}(t)$ is t^{ℓ} precisely when*

$$q = 0 \qquad \text{or} \qquad q > 0 \text{ and } \ell^2 + q^2 \text{ is a square.}$$

In the latter case the unique index carrying exponent zero is

$$i_0 = \frac{\ell + q - \sqrt{\ell^2 + q^2}}{2}, \qquad 0 < i_0 < \frac{\ell}{2}.$$

Consequently all positive exceptional pairs are exactly

$$(\ell, q) = d(m^2 - n^2, 2mn) \quad \text{or} \quad (\ell, q) = d(2mn, m^2 - n^2),$$

where $d \geq 1$, $m > n \geq 1$, $\gcd(m, n) = 1$, and m, n have opposite parity. For every other positive twist the leading coefficient is $t^{\ell+1}$.

Proof. Theorem 6.5 says that the lower power occurs exactly when

$$q(\ell - 2i) - 2i(\ell - i) = 0$$

for some integer $i \in [0, \ell]$. For $q = 0$, the indices $0, \ell$ do this. For $q > 0$, the roots of this quadratic in i are

$$i_{\pm} = \frac{\ell + q \pm \sqrt{\ell^2 + q^2}}{2}.$$

The plus root is larger than ℓ, and the minus root belongs to $(0, \ell/2)$. Therefore an allowed integer index exists exactly when the square root is an integer: necessity follows directly from $\sqrt{\ell^2 + q^2} = \ell + q - 2i$; for sufficiency, if $s^2 = \ell^2 + q^2$ then $s \equiv \ell + q \pmod 2$, so $i_- \in \mathbb{Z}$. This also proves uniqueness. The displayed parametrization is the elementary classification of integer right triangles. For completeness, divide a triple by its common divisor; one leg is even and the other odd. Writing the even leg as $2ab$ in the factorization $(s - u)(s + u) = v^2$, the two coprime factors $(s + u)/2$ and $(s - u)/2$ are squares m^2, n^2. Thus $u = m^2 - n^2$, $v = 2mn$, with coprime opposite-parity m, n; restoring the common divisor gives exactly the stated pairs. □

Example 10.6. For $\ell = 3$, the only positive exceptional twist is $q = 4$; for $\ell = 4$, it is $q = 3$. For $\ell = 5$, it is $q = 12$, while for $\ell = 8$ the positive exceptional twists are $q = 6$ and $q = 15$. The classification concerns the primitive scalar operator in t; it does not change the uniform leading coefficient x^{ℓ} of the normalized operator $L_{\ell,q}$ in $x = t^2$.

11 Vertex balance and general-parameter ensembles

11.1 The integrals

Let $\beta > 0$ and let U be distributed according to the circular β-ensemble CβE$_n$, that is, its eigenvalues $e^{i\varphi_1}, \dots, e^{i\varphi_n}$ have density proportional to $\prod_{j<k} |e^{i\varphi_j} - e^{i\varphi_k}|^{\beta}$ on $\mathbb{T}^n$. For $q \in \mathbb{Z}_{\geq 0}$ Forrester and Wei [11, (2.7)] write

$$\mathbb{E}\big[(\det U)^q \, e^{t \operatorname{tr}(U + U^{-1})}\big] = c\, t^{qn}\, G^{\beta}_{n,q}(t^2), \qquad G^{\beta}_{n,q}(x) = {}_0F_1^{(2/\beta)}\Big(q + 1 + \frac{(n-1)\beta}{2}; (x)^n\Big), \quad (11.1)$$

where ${}_0F_1^{(\alpha)}(c; (x)^n)$ is the hypergeometric function of n equal arguments with Jack parameter α [59], and $c \neq 0$. For $\beta = 2$ this is Gessel's Bessel–Toeplitz determinant, whose minimal equations

were determined in Sections 5–6; for $q=0$ it is the generating function $\sum_N \mu_N x^N/(N!)^2$ of the moments $\mu_N=\mathbb{E}|\operatorname{tr}U|^{2N}$. As $\beta\to 0$ the eigenvalues become independent and $G^0_{n,0}(t^2)=I_0(2t)^n$.

For the Jacobi ensemble with parameters $a,b>-1$, whose eigenvalues $s_1,\dots,s_n\in[0,1]$ have density proportional to $\prod_j s_j^a(1-s_j)^b\prod_{j<k}|s_j-s_k|^\beta$, the Laplace transform of the trace is [59], [10, (2.12)]

$$H(x)=\mathbb{E}\,e^{-x\sum_j s_j}={}_1F_1^{(2/\beta)}\Big(\frac{\beta}{2}(n-1)+a+1;\ \beta(n-1)+a+b+2;\ (-x)^n\Big). \tag{11.2}$$

For $\beta=1$ the trace is Pillai's V statistic of multivariate analysis; for $\beta=2$ and $a,b\in\{\pm\frac12\}$ it is, up to an affine change, the trace of a Haar-distributed orthogonal or symplectic matrix (Section 18).

11.2 What was known

Forrester and Kumar [10, Cor. 2.5] showed that H is, up to normalization, a component of a solution of an $(n+1)$-dimensional first-order system, and Forrester and Wei [11, Prop. 3.3] obtained the analogous system for ${}_0F_1^{(2/\beta)}$ by a confluent limit. Eliminating, [11, Prop. 4.1] asserts, for all β and all parameters, a scalar equation

$$\Big(x^n\frac{d^{n+1}}{dx^{n+1}}+q_n(x)x^{n-1}\frac{d^n}{dx^n}+\cdots+q_1(x)\frac{d}{dx}+q_0(x)\Big)I_n(x)=0$$

in which "the degree in x of the coefficient of each d^j/dx^j is $[(n-1+j)/2]$", with integer coefficients when the parameters B_j of the system are integers. The proof displays the first three elimination steps and then states that the procedure can be repeated. Bornemann [11, App. B] argued that the order $n+1$ is optimal for $\beta=2$ and $q\in\{0,n\}$; no statement about the order is made for $\beta\neq 2$. For the moments, [11, Prop. 4.4] shows that the elimination produces, for general parameters, a recurrence with $\lfloor(n-1)/2\rfloor+2$ terms, and for $\beta=2$ they give the corresponding form "by inspection" [11, Prop. 4.3]; from the cases they computed they observe that the degree of the coefficients is $\ell-1$, which they call difficult to anticipate. Minimality is not addressed. Forrester and Kumar [10, Prop. 1.2] give a vector recurrence for the moments of the Jacobi trace, and in [10, Remark 3.8] they note that eliminating their system for the density of the trace gives a linear differential equation of order N, written out for $N=2$.

The determinant method in Sections 5–6 extends to arbitrary positive Dyson parameters through visibility and an explicit Kac recursion. The results below also cover Jacobi trace laws, orthogonal-group moments, and decaying Bessel products.

11.3 Main results

Throughout, θ is the Euler operator of the variable at hand, $K=\lceil n/2\rceil$, and for an operator $\sum_k x^k P_k(\theta)$ we call the polynomials P_k its θ-coefficients. Put, for $0\le p,b\le n$,

$$\varepsilon_p=p\Big(q+\frac{\beta}{2}(n-p)\Big),\qquad \lambda_b=2(n-2b),\qquad \gamma_b=\frac{n}{2}+\frac{\beta}{2}\Big(\binom{n-b}{2}+\binom{b}{2}\Big). \tag{11.3}$$

Theorem 11.1 (circular ensembles)**.** *Let $\beta>0$, $n\ge 1$ and $q>-\frac12$ real, and $G=G^\beta_{n,q}$.*

(a) G is annihilated by an operator $L=\sum_{m=0}^{n+1}q_m(x)\partial_x^m$ with

$$q_{n+1}=x^n,\qquad x^{m-1}\mid q_m\ (1\le m\le n+1),\qquad \deg q_m=\Big\lfloor\frac{n-1+m}{2}\Big\rfloor\ (0\le m\le n+1).$$

L has no singular point in $\mathbb{C}\setminus\{0\}$, and it is primitive if $q\ge 0$. The coefficients of the q_m lie in $\mathbb{Z}[B_1,\dots,B_n]$, where $B_p=p(q+1+\frac{\beta}{2}(n-p))$; in particular they are integers when all B_p are.

(b) $xL = \sum_{k=0}^{K} x^k P_k(\theta)$ *with* $\deg P_k = n+1-2k$,

$$P_0(\theta) = \prod_{p=0}^{n} (\theta + \varepsilon_p), \qquad \sum_{k=0}^{K} \operatorname{lc}(P_k)\, \mu^{n+1-2k} = \prod_{b=0}^{n} (\mu - n + 2b),$$

and, for even n, $P_K(\theta) = \operatorname{lc}(P_K)\big(\theta + \tfrac{1}{2}(\gamma_{n/2} + qn)\big)$. *The exponents of* L *at* $x = 0$ *are* $-\varepsilon_0, \dots, -\varepsilon_n$.

(c) *At* $x = \infty$, L *has the formal solutions* $e^{\lambda_b \sqrt{x}}\, x^{-(\gamma_b + qn)/2}\big(1 + O(x^{-1/2})\big)$, $0 \le b \le n$.

(d) *If* $q \in \mathbb{Z}_{\ge 0}$, *then* G *satisfies no linear differential equation of order at most* n *over* $\mathbb{C}(x)$; L *is its minimal operator, and every operator of order* $n+1$ *with polynomial coefficients annihilating* G *is a polynomial multiple of* L. *The minimal operator of* $\mathbb{E}[(\det U)^q e^{t \operatorname{tr}(U + U^{-1})}]$ *in* t *has order* $n+1$, *no singular point in* $\mathbb{C} \setminus \{0\}$, *and exponents* $q(n-2p) - \beta p(n-p)$, $0 \le p \le n$, *at* $t = 0$.

Corollary 11.2. *For every* $\beta > 0$ *and all parameters* a, b *with* $q' := \frac{\beta}{2}(n-1) + a + b + 1 > -\frac{1}{2}$, *the function* I_n *of [11, Prop. 3.3], which is a constant multiple of* $G^{\beta}_{n,q'}$ *[11, (3.10)], satisfies the equation of [11, Prop. 4.1] with the coefficient of* d^j/dx^j *of degree exactly* $\lfloor (n-1+j)/2 \rfloor$, *and with integer coefficients when the* B_j *of [11, (3.3)] are integers; the operator obtained there by elimination is the operator* L *of Theorem 11.1 for* $q = q'$. *If* $q' \in \mathbb{Z}_{\ge 0}$, *which is the case relevant to the circular ensembles [11, (3.13), Remark 3.4], the order* $n+1$ *is minimal. Outside the range* $q' > -\frac{1}{2}$ *the degree assertion of [11, Prop. 4.1] can fail (Remark 15.3).*

Theorem 11.3 (moments)**.** *Let* $\beta > 0$, $n \ge 2$, $q = 0$, *and let* $P_0, \dots, P_K$ *be as in Theorem 11.1. Put* $R_0(N) = \prod_{i=1}^{n-1} (N + \frac{\beta}{2} i(n-i))$ *and* $R_k(N) = P_k(N-k)\big[(N-1)(N-2)\cdots(N-k+1)\big]^2$ *for* $1 \le k \le K$.

(a) $\sum_{k=0}^{K} R_k(N)\, \mu_{N-k} = 0$ *for all* $N \ge 1$, *with* $\mu_m = 0$ *for* $m < 0$. *Each* R_k *has degree exactly* $n-1$.

(b) *The characteristic polynomial* $\sum_k \operatorname{lc}(R_k) \lambda^{K-k}$ *is* $\prod_{j < n/2} (\lambda - (n-2j)^2)$, *independent of* β, *and* $R_K(N) = (-1)^K \prod_{j<n/2} (n-2j)^2 \cdot [(N-1)\cdots(N-K+1)]^2$ *for odd* n, *while for even* n *the last factor is multiplied by* $N - K + \frac{1}{2}\gamma_{n/2}$.

(c) *Every recurrence satisfied by* (μ_N) *for all large* N *has order at least* K *and degree at least* $n-1$, *and its leading coefficient, written as the coefficient of* μ_N, *is divisible by* $R_0(N)$. *Every recurrence of order* K *is a polynomial multiple of the one in (a).*

(d) *Let* $q \in \mathbb{Z}_{\ge 1}$ *and* $a_q(N) = \mathbb{E}\big[(\det U)^q (\operatorname{tr} U)^N \overline{\operatorname{tr} U}^{\,N+qn}\big]$. *Then* a_q *satisfies a recurrence of order* K *and degree* $n-1$ *with leading coefficient* $\prod_{i=1}^{n-1}(N + \varepsilon_i)$, *and no recurrence of smaller order. If no* ε_i, $1 \le i \le n-1$, *lies in* $\{1, 2, \dots, qn\}$ *(this excludes only finitely many* β *for given* n, q*), then every recurrence has degree at least* $n-1$ *and leading coefficient divisible by* $\prod_{i=1}^{n-1}(N + \varepsilon_i)$. *If no* ε_i *lies in* $\{qn - K + 1, \dots, qn\}$, *then every recurrence of order* K *has degree at least* $n-1$, *and the recurrence of order* K *is unique up to a polynomial factor.*

For $\beta = 2$, parts (a)–(c) recover Theorem 2.4. Part (d) gives the generic-parameter twisted degree bound. Theorem 21.2 below also covers every positive even β, including all exceptional integer twists, while Theorem 21.4 gives an infinite family where the degree drops. Corollary 15.5 below gives the minimal equation, of order $n-1$, for the density of $|\operatorname{tr} U|^2$.

Let $\varepsilon_p^J = p(a+b+1+\frac{\beta}{2}(2n-p-1))$ and $\gamma_k^J = (n-k)(a+1) + k(b+1) + \beta\big(\binom{n-k}{2} + \binom{k}{2}\big)$, and let $(\star)$ denote the condition $a + b + 1 + \frac{\beta}{2}(n-1) > 0$.

Theorem 11.4 (Jacobi ensembles). *Let $\beta > 0$, $a, b > -1$, $n \geq 1$, and let $m_N = \mathbb{E}(\sum_j s_j)^N$ and ρ^J be the moments and the density of the trace.*

(a) H satisfies no linear differential equation of order at most n. It is annihilated by $L^J = \sum_{m=0}^{n+1} q_m \partial_x^m$ with $q_{n+1} = x^n$, $x^{m-1} \mid q_m$ and $\deg q_m = n$ for $1 \leq m \leq n+1$, and $\deg q_0 = n-1$; L^J has no singular point in $\mathbb{C} \setminus \{0\}$, and $xL^J = \sum_{k=0}^{n} x^k P_k^J(\theta)$ with $P_0^J = \prod_{p=0}^{n}(\theta + \varepsilon_p^J)$, $\sum_k \operatorname{lc}(P_k^J)\mu^{n+1-k} = \mu \prod_{k=1}^{n}(\mu + k)$ and $P_n^J = n!(\theta + \gamma_0^J)$. Its exponents at 0 are $-\varepsilon_p^J$, and at infinity it has the formal solutions $e^{-kx} x^{-\gamma_k^J}(1 + O(x^{-1}))$, $0 \leq k \leq n$. Under $(\star)$, L^J is primitive; it is then the minimal operator of H, and every operator of order $n+1$ with polynomial coefficients annihilating H is a polynomial multiple of it. Without $(\star)$, L^J may be divisible by x.

(b) With $c_0(N) = \prod_{p=1}^{n}(N + \varepsilon_p^J)$ and $c_k(N) = (-1)^k P_k^J(N-k)\,(N-1)\cdots(N-k+1)$,

$$\sum_{k=0}^{n} c_k(N)\, m_{N-k} = 0 \qquad (N \geq 1),$$

all c_k have degree n, and the characteristic polynomial is $\prod_{k=1}^{n}(\lambda - k)$. Every recurrence for (m_N) has order at least n. Under $(\star)$, every recurrence has degree at least n, its leading coefficient is divisible by $c_0(N)$, and the recurrence of order n is unique up to a polynomial factor.

(c) ρ^J is annihilated by an explicit operator of order n with leading coefficient $\pm w^n \prod_{k=1}^{n}(w-k)$. Under $(\star)$, no nonzero operator of order less than n annihilates ρ^J.

Under $(\star)$, part (c) shows that the equation of order N obtained by elimination in [10, Remark 3.8] has minimal order (Remark 17.2).

For $\ell \geq 1$ let $w_B(N) = \mathbb{E}_{\mathrm{SO}(2\ell+1)}(\operatorname{tr} U)^N$ and $w_D(N) = \mathbb{E}_{\mathrm{SO}(2\ell)}(\operatorname{tr} U)^{2N}$. By the Pieri rules, $w_B(N)$ is the number of sequences of dominant weights $0 = \lambda^0, \lambda^1, \ldots, \lambda^N = 0$ of $\mathfrak{so}_{2\ell+1}$ with steps in $\{\pm e_1, \ldots, \pm e_\ell\}$ and, when $\lambda_\ell > 0$, also the step 0; and $w_D(N)$ counts such sequences of length $2N$ for $\mathfrak{so}_{2\ell}$ with steps $\pm e_i$. The representation-theoretic connection with chamber paths is discussed by Grabiner and Magyar [56]. Here the type B count always uses the stated restriction on the zero step. Recurrences for individual small ℓ can be found by guessing, for instance the case SO(5) in [60, A095922].

Theorem 11.5 (orthogonal groups and Weyl chamber walks). *(a) For $\ell \geq 1$, w_B satisfies a recurrence of order $\ell + 1$ and degree ℓ whose leading coefficient is $\prod_{p=1}^{\ell}(N + p(2\ell - p))$, whose last coefficient is a constant times $(N-1)\cdots(N-\ell)$, and whose characteristic polynomial is $\prod_{j=0}^{\ell}(\lambda - 2\ell - 1 + 4j)$. Every recurrence for w_B has order at least $\ell + 1$ and degree at least ℓ, with leading coefficient divisible by $\prod_p(N + p(2\ell - p))$, and every recurrence of order $\ell + 1$ is a polynomial multiple of this one.*

(b) For $\ell \geq 2$, w_D satisfies a recurrence of order $\lceil \ell/2 \rceil$ and degree ℓ whose leading coefficient is $\prod_{p=1}^{\ell}(2N + p(2\ell - 1 - p))$ and whose characteristic polynomial is $\prod_{j<\ell/2}(\lambda - 4(\ell - 2j)^2)$. Every recurrence for w_D has order at least $\lceil \ell/2 \rceil$ and degree at least ℓ, with leading coefficient divisible by $\prod_p(2N + p(2\ell - 1 - p))$, and every recurrence of order $\lceil \ell/2 \rceil$ is a polynomial multiple of this one.

For $\ell = 1$, (a) is the Riordan recurrence $(N+1)w(N) = (N-1)(2w(N-1) + 3w(N-2))$; for $\ell = 2$, (b) is the recurrence of the squared Catalan numbers. Table 4 lists the cases $\ell \leq 4$. The symplectic case is Theorem 2.9, and it is recovered by the same argument.

Theorem 11.6 (even β: the vertex integrals)**.** *Let $\beta = 2m$ with $m \in \mathbb{Z}_{\geq 1}$ and $q \in \mathbb{Z}_{\geq 0}$. The minimal operator of $\mathbb{E}[(\det U)^q e^{t\,\mathrm{tr}(U+U^{-1})}]$ has a basis of solutions $\Psi_0^{(m,q)}, \dots, \Psi_n^{(m,q)}$ (the vertex integrals of Section 16), with $\Psi_b^{(m,q)} \sim C_b e^{\lambda_b t} t^{-\gamma_b}$ as $t \to +\infty$, and*

$$\mathrm{ord}_{t=0}\, \Psi_b^{(m,q)} = -f_{m,(b)} \qquad (0 \leq b \leq n),$$

where $f_{m,(0)} \leq \cdots \leq f_{m,(n)}$ are the numbers $2mi(n-i) - q(n-2i)$, $0 \leq i \leq n$, in increasing order. This is proved without the first-order systems of [10, 11]; the argument also reproves, independently of those systems, the minimality and the statements in the variable t in Theorem 11.1(d) for these β and q.

For the Bessel moments $c_{a,b}(k) = \int_0^\infty t^k I_0(t)^a K_0(t)^b\, dt$ with integers $b > a \geq 0$, put $n = a + b$. Bailey, Borwein, Borwein and Crandall conjectured [51, Conj. 1] (see also [50, 52]), and Borwein and Salvy [54] proved, that these moments satisfy a recursion with $\lceil n/2 \rceil + 1$ terms in steps of two, obtained from the n-th symmetric power of Bessel's equation. The same symmetric power governs the even moments $W_n(2N) = \sum_{k_1+\cdots+k_n=N} \binom{N}{k_1,\dots,k_n}^2$ of the planar random walk with n unit steps, for which Verrill [22] found a recurrence of order $\lceil n/2 \rceil$ and degree $n-1$.

Theorem 11.7 (Bessel moments)**.** *Let $\sum_k t^{2k} P_k^{(n)}(\theta)$ be the symmetric power operator of Section 19, with $P_0^{(n)} = \theta^{n+1}/n!$.*

(a) $\sum_{i=0}^{\lceil n/2 \rceil} P_i^{(n)}(-k-1-2i)\, c_{a,b}(k+2i) = 0$ *for all $k \geq 0$; the coefficient of $c_{a,b}(k)$ is $(-k-1)^{n+1}/n!$, and the coefficient of $c_{a,b}(k+2i)$ has degree $n+1-2i$.*

(b) If $\sum_{s=0}^{M} p_s(k)\, c_{a,b}(k+s) = 0$ for all large k, with polynomials p_s and $p_0 p_M \neq 0$, then $M \geq 2\lceil n/2 \rceil$ and $(k+1)^{n+1}$ divides $p_0(k)$. Every such recursion with $M = 2\lceil n/2 \rceil$ is a polynomial multiple of (a). In particular (a) has minimal order and minimal degree $n+1$.

(c) Every recurrence satisfied by $(W_n(2N))_N$ has order at least $\lceil n/2 \rceil$ and degree at least $n-1$, and its leading coefficient, written as the coefficient of $W_n(2N)$, is divisible by N^{n-1}. Verrill's recurrence is, up to a constant factor, the unique recurrence of order $\lceil n/2 \rceil$ and degree $n-1$.

11.4 The vertex balance method

All the integrals above, except the Bessel moments, are Laplace transforms $F(t) = \int e^{tw} \rho(w)\, dw$ of the law ρ of a trace $w = f(z)$ over a compact domain: the torus $\mathbb{T}^n$, or the cube $[0,1]^n$. The trace has finitely many critical configurations on the domain and its faces, the *vertices*; for $\mathbb{T}^n$ these are the points with all coordinates in $\{\pm 1\}$, and for the cube its corners. The method consists of four moves.

1. Vertex data. Each vertex v produces an exponential solution $e^{f(v)t} t^{-\gamma_v}(1 + \cdots)$ of the minimal operator at infinity, with γ_v computed from the local homogeneity of the weight at v (a Mehta–Selberg exponent). In the presence of a first-order system the same exponents appear as diagonal entries of the residue of a Kac matrix in its eigenbasis, which we compute with $\mathfrak{sl}_2$ (Proposition 14.4).

2. Balance (Theorem 12.5). If a basis of solutions has these exponents at infinity and satisfies crude bounds $\mathrm{ord}_0\, \Psi_v \geq -\beta_v$ at the regular singular point, then the *balance defect*

$$\Delta = \binom{N}{2} - \sum_v \gamma_v + \sum_v \beta_v$$

bounds the total weight of the apparent singularities. When $\Delta = 0$ there are none, and every crude bound is attained. For the circular ensembles the vanishing of Δ is the identity

$\sum_b \gamma_b - \sum_i (\beta i(n-i) - q(n-2i)) = \binom{n+1}{2}$, which is the trace of the residue of the Kac system (Remark 14.5).

3. Visibility (Section 13). Each vertex leaves a point at which ρ is not smooth: localizing the Fourier transform at the critical value, all non-vertex points contribute $O(\tau^{-\infty})$ by integration by parts along translations of clusters of coinciding coordinates, which leave the singular weight invariant, and the vertex contributes $c\,\tau^{-\gamma_v} e^{i\tau f(v)}$ with $c \neq 0$, by a product of two one-sided Erdélyi expansions. Dually, if L annihilates F, then ρ is analytic off the roots of the top-degree polynomial χ_L, whose degree is at most $\operatorname{ord} L$ (Lemma 13.1). Hence the minimal order is at least the number of vertex values, and the span of every annihilating θ-form, which is the order of the corresponding recurrence, is at least the number of nonzero vertex values.

4. Transfer (Lemma 12.2). Once the minimal operator is known to have no singular point in $\mathbb{C} \setminus \{0\}$, every annihilating θ-form is a left multiple of it by a θ-form. So its lowest θ-coefficient, which is essentially the leading coefficient of the recurrence, is divisible by the indicial polynomial at 0. Each negative exponent therefore contributes one to the degree of every recurrence.

For even β the integrals are polynomial in Bessel functions, the solution space is spanned by explicit vertex integrals, and moves 1, 2 and 4 suffice. For general β the first-order systems of [10, 11] give an explicit operator through a three-term *Kac recursion* (Section 14), which has no apparent singularities by construction, and move 3 proves that it is minimal. For Bessel moments, move 4 is applied to the Mellin transform.

Organization

Section 12 collects the facts on operators, including the transfer lemma and the balance theorem. Section 13 proves visibility. Section 14 derives the Kac recursions and the formal solutions at infinity. Sections 15–19 prove Theorems 11.1–11.7, and Section 23 discusses what remains open. Appendix C describes the computations.

12 Operators, θ-forms and the balance theorem

Throughout, θ denotes the Euler operator of the variable at hand ($\theta = x\,d/dx$ or $\theta = t\,d/dt$), $z^{\underline{m}} = z(z-1)\cdots(z-m+1)$, and an operator $L = \sum_{m=0}^{r} p_m \partial^m$ with polynomial coefficients and $p_r \neq 0$ is *primitive* if the p_m have no common zero. The *minimal operator* of a function (or formal series) y is the monic operator of least order in $\mathbb{C}(x)[\partial]$ annihilating y; after clearing denominators it is a primitive operator, unique up to a constant factor. Every operator over $\mathbb{C}(x)$ annihilating y is a left multiple of it.

12.1 θ-forms

A *θ-form* is an operator $A = \sum_{k=k_0}^{k_1} x^k Q_k(\theta)$ with $Q_k \in \mathbb{C}[\theta]$, $k_0 \leq k_1$ in $\mathbb{Z}$, and $Q_{k_0} Q_{k_1} \neq 0$. Its *span* is $k_1 - k_0$, its *lowest coefficient* is Q_{k_0} and its *top coefficient* is Q_{k_1}. Since $\theta = x\partial$, $\partial = x^{-1}\theta$ and $x^k \theta^i = \sum_{m \leq i} S(i,m) x^{k+m} \partial^m$ (Stirling numbers of the second kind), the nonzero elements of $\mathbb{C}[x, x^{-1}][\partial]$ are exactly the θ-forms, and the ∂-order of A is $\max_k \deg Q_k$. The commutation rule

$$x^j M(\theta) \cdot x^k Q(\theta) = x^{j+k} M(\theta + k) Q(\theta) \tag{12.1}$$

shows that the product of two θ-forms $\sum_{j=j_0}^{j_1} x^j M_j(\theta)$ and $\sum_{k=k_0}^{k_1} x^k Q_k(\theta)$ is a θ-form with lowest coefficient $M_{j_0}(\theta + k_0) Q_{k_0}(\theta)$ and top coefficient $M_{j_1}(\theta + k_1) Q_{k_1}(\theta)$. In particular spans add.

Lemma 12.1 (Laurent division). *Let $L \in \mathbb{C}[x][\partial]$ have order r and leading coefficient cx^A with $c \neq 0$. If $O \in \mathbb{C}[x, x^{-1}][\partial]$ and $O = QL$ with $Q \in \mathbb{C}(x)[\partial]$, then $Q \in \mathbb{C}[x, x^{-1}][\partial]$.*

Proof. The leading coefficient of L is a unit of $\mathbb{C}[x, x^{-1}]$, so right division by L can be carried out in $\mathbb{C}[x, x^{-1}][\partial]$: $O = Q'L + R$ with $Q', R \in \mathbb{C}[x, x^{-1}][\partial]$ and $\operatorname{ord} R < r$. By uniqueness of right division in $\mathbb{C}(x)[\partial]$, $Q' = Q$ and $R = 0$. □

Lemma 12.2 (transfer)**.** *Let y have minimal operator L, normalized to have polynomial coefficients, and suppose that its leading coefficient is a monomial cx^A, so that L has no singular point in $\mathbb{C} \setminus \{0\}$. Write $x^e L = \sum_{k=k_0}^{k_1} x^k P_k(\theta)$ for some $e \in \mathbb{Z}$. If a θ-form O annihilates y, then $O = Q \cdot x^e L$ for a θ-form $Q = \sum_{j=j_0}^{j_1} x^j M_j(\theta)$. Consequently:*

(a) $\operatorname{span} O \geq k_1 - k_0$*;*

(b) the lowest coefficient of O is $M_{j_0}(\theta + k_0)P_{k_0}(\theta)$, and the top coefficient is $M_{j_1}(\theta + k_1)P_{k_1}(\theta)$;

(c) if $\operatorname{span} O = k_1 - k_0$*, then $O = x^{j_0} M(\theta)\, x^e L = \sum_k x^{j_0+k} M(\theta + k)P_k(\theta)$ for one polynomial M.*

Proof. By minimality, $O = \tilde{Q}L$ with $\tilde{Q} \in \mathbb{C}(x)[\partial]$, and $\tilde{Q} \in \mathbb{C}[x, x^{-1}][\partial]$ by Lemma 12.1. Put $Q = \tilde{Q}x^{-e}$ and apply (12.1). □

12.2 Recurrences

A sequence $a : \mathbb{Z}_{\geq 0} \to \mathbb{C}$ satisfies a recurrence of *order* M and *degree* d if

$$\sum_{s=0}^{M} c_s(N)\, a(N + s) = 0 \qquad (N \geq n_0) \tag{12.2}$$

with polynomials c_s of degree at most d and $c_0 c_M \neq 0$. The *leading coefficient* is c_M; in the backward form $\sum_{i=0}^{M} c_{M-i}(N - M)\, a(N - i) = 0$ it is the coefficient of $a(N)$. Part (a) of the following lemma is Lemma 4.1 ; part (b) sharpens Lemma 4.2, whose lowest coefficient contains $(\theta^{\underline{M}})^2$ instead of $\theta^{\underline{M}}$, and allows a shift r.

Lemma 12.3 (dictionaries)**.** *(a) Let $Y(t) = \sum_N a(N)t^N/N!$. If (12.2) holds, then Y is annihilated by a θ-form of span M with lowest coefficient $\theta^{\underline{M+n_0}}\, c_M(\theta - M)$. Conversely, if $\sum_{k=0}^{M} t^k P_k(\theta)$ annihilates Y, then $\sum_k P_k(N - k)N^{\underline{k}}\, a(N - k) = 0$ for all $N \geq 0$, terms with $N - k < 0$ being omitted.*

(b) Let $r \in \mathbb{Z}_{\geq 0}$ and $G(x) = \sum_N a(N)x^N/(N!\,(N + r)!)$. If (12.2) holds, then G is annihilated by a θ-form of span M with lowest coefficient $\theta^{\underline{M+n_0}}\, (\theta + r)^{\underline{M}}\, c_M(\theta - M)$. Conversely, if $\sum_{k=0}^{M} x^k P_k(\theta)$ annihilates G, then $\sum_k P_k(N - k)N^{\underline{k}}(N + r)^{\underline{k}}\, a(N - k) = 0$ for all $N \geq 0$.

Proof. For (a) see Lemma 4.1. For (b) put $F(\theta) = \theta^{\underline{M+n_0}}(\theta + r)^{\underline{M}}$ and

$$Q_i(\theta) = \frac{F(\theta + i)\, c_{M-i}(\theta + i - M)}{(\theta + i)^{\underline{i}}\, (\theta + i + r)^{\underline{i}}} \qquad (0 \leq i \leq M).$$

Since $i \leq M$, $(\theta + i)^{\underline{i}}$ divides $(\theta + i)^{\underline{M+n_0}}$ and $(\theta + i + r)^{\underline{i}}$ divides $(\theta + i + r)^{\underline{M}}$, so the Q_i are polynomials, with $Q_0 = F(\theta)c_M(\theta - M)$ and $Q_M \neq 0$. Because $N^{\underline{i}}\,(N - i)! = N!$ and $(N + r)^{\underline{i}}\,(N + r - i)! = (N + r)!$, the coefficient of x^N in $\sum_i x^i Q_i(\theta)G$ is

$$\sum_{i=0}^{M} Q_i(N - i)\frac{a(N - i)}{(N - i)!\,(N - i + r)!} = \frac{F(N)}{N!\,(N + r)!}\sum_{s=0}^{M} c_s(N - M)\, a(N - M + s).$$

For $N < M$, terms with negative sequence indices are omitted; each retained term vanishes because its factorial denominators are nonzero and $F(N) = 0$. For $N \geq M$, the displayed expression vanishes for $N - M \geq n_0$ by (12.2), and for $N < M + n_0$ because $F(N) = 0$. The converse statements follow by comparing coefficients. □

The transfer lemma turns the lowest coefficient of the minimal operator into a lower bound for the degree of every recurrence. We record the form in which it will be used.

Corollary 12.4. *In the situation of Lemma 12.3(a), or of Lemma 12.3(b) with $r = 0$, let the minimal operator L of Y (resp. G) be as in Lemma 12.2, with $k_1 - k_0 = K$. Then every recurrence (12.2) satisfied by a has order $M \geq K$, and the polynomial $\prod_\nu (\theta - \nu)^{m_\nu}$, taken over the roots $\nu \notin \mathbb{Z}_{\geq 0}$ of P_{k_0} with their multiplicities m_ν, divides $c_M(\theta - M)$. In particular $\deg c_M$ is at least the number of roots of P_{k_0} outside $\mathbb{Z}_{\geq 0}$, counted with multiplicity.*

Proof. The operator O of Lemma 12.3 has span M, so $M \geq K$ by Lemma 12.2(a). By Lemma 12.2(b), $P_{k_0}(\theta)$ divides the lowest coefficient of O. The factors $\theta^{\underline{M+n_0}}$ and $\theta^{\underline{M}}$ vanish only on $\mathbb{Z}_{\geq 0}$. □

12.3 The balance theorem

Let $\mathcal{N}$ be the space of functions on a slit punctured disc $\{0 < |t| < \delta\} \setminus (-\delta, 0]$ of the form $\sum_{\nu \geq \nu_0} t^\nu P_\nu(\log t)$ with $\nu \in \mathbb{Z}$, polynomials P_ν of bounded degree, and absolute convergence; for $f \neq 0$ in $\mathcal{N}$, $\operatorname{ord} f$ is the least ν with $P_\nu \neq 0$. We use three facts from Section 3, proved in Lemmas 3.3, 3.4, and 3.5. (i) *Squeeze*: if L of order N is Fuchsian at 0 with local solution space in $\mathcal{N}$, its exponents $\nu_1 \geq \cdots \geq \nu_N$ are integers, and for every basis $f_1, \dots, f_N$ with $\operatorname{ord} f_1 \geq \cdots \geq \operatorname{ord} f_N$ one has $\operatorname{ord} f_i \leq \nu_i$; hence $\sum_i \operatorname{ord} f_i \leq \sum_i \nu_i$ with equality only if $\operatorname{ord} f_i = \nu_i$ for all i. (ii) *Abel*: if W is a Wronskian of $L = \sum p_m \partial^m$ (primitive), then $W'/W = -p_{N-1}/p_N$; at a regular singular point the residue is $\sum_i \nu_i - \binom{N}{2}$; at a zero τ of p_N where all solutions are holomorphic, the residue is an integer $\kappa_\tau \geq 1$. (iii) *Top degree*: if $d = \max \deg p_m$ and L has solutions y_j with $y_j^{(m)} = C_j e^{\lambda_j t} t^{-\gamma_j}(\lambda_j^m + O(t^{-1/2}))$, $C_j \neq 0$, for N distinct λ_j, then $[t^d]p_N \neq 0$ and $\sum_m [t^d] p_m \lambda^m = [t^d] p_N \prod_j (\lambda - \lambda_j)$.

Theorem 12.5 (balance)**.** *Let $L = \sum_{m=0}^{N} p_m(t) \partial^m$ be primitive, and assume:*

(B1) every solution of L continues analytically along every path in $\mathbb{C} \setminus \{0\}$;

(B2) L is Fuchsian at 0, with local solution space contained in $\mathcal{N}$;

(B3) L has a basis of solutions $\Psi_1, \dots, \Psi_N$ on some interval (T, ∞) with

$$\Psi_j^{(m)}(t) = C_j e^{\lambda_j t} t^{-\gamma_j} (\lambda_j^m + O(t^{-1/2})) \qquad (0 \leq m \leq N),$$

where $C_j \neq 0$, $\gamma_j \in \mathbb{R}$, and the λ_j are pairwise distinct;

(B4) $\operatorname{ord} \Psi_j \geq -\beta_j$ for integers β_j, where Ψ_j is continued along $(0, \infty)$ to a slit disc at 0.

Define the balance defect

$$\Delta = \binom{N}{2} - \sum_{j=1}^{N} \gamma_j + \sum_{j=1}^{N} \beta_j.$$

Then $\sum_\tau \kappa_\tau \leq \Delta$, the sum running over the zeros $\tau \neq 0$ of p_N. In particular $\Delta \geq 0$, and p_N has at most Δ distinct zeros in $\mathbb{C} \setminus \{0\}$. If $\Delta = 0$, then $p_N = c t^A$, $\operatorname{ord} \Psi_j = -\beta_j$ for every j, and the exponents of L at 0 are $-\beta_1, \dots, -\beta_N$.

Proof. By (B1), every zero $\tau \neq 0$ of p_N is an apparent singularity, since a disc around τ avoiding 0 is simply connected. Let W be the Wronskian of $\Psi_1, \dots, \Psi_N$. The rational function $W'/W = -p_{N-1}/p_N$ has simple poles only, at 0 by Fuchs' criterion and at the apparent points by Abel's lemma, so $W'/W = Q(t) + \kappa_0/t + \sum_\tau \kappa_\tau/(t - \tau)$ with $Q \in \mathbb{C}[t]$ and $\kappa_0 = \sum_i \nu_i - \binom{N}{2}$,

where ν_i are the exponents at 0. Hence $W = c\,e^{\int Q}\,t^{\kappa_0}\prod_\tau (t-\tau)^{\kappa_\tau}$. By (B3), expanding the determinant,

$$W(t) = \Big(\prod_j C_j\Big) e^{(\sum_j \lambda_j)t}\, t^{-\sum_j \gamma_j}\Big(\prod_{i<j}(\lambda_j-\lambda_i) + O(t^{-1/2})\Big),$$

and the Vandermonde product is nonzero. Comparing the two expressions as $t \to +\infty$ gives $\int Q = (\sum_j \lambda_j)t + \text{const}$ and $\kappa_0 + \sum_\tau \kappa_\tau = -\sum_j \gamma_j$. Therefore

$$\sum_\tau \kappa_\tau = \binom{N}{2} - \sum_j \gamma_j - \sum_i \nu_i \le \binom{N}{2} - \sum_j \gamma_j - \sum_j \operatorname{ord}\Psi_j \le \Delta,$$

by the squeeze (i) and (B4). If $\Delta = 0$, all inequalities are equalities: there are no apparent singularities, so p_N has no zero in $\mathbb{C}\setminus\{0\}$; $\operatorname{ord}\Psi_j = -\beta_j$ for every j, because each term was bounded separately; and by the equality case of the squeeze the exponents are the numbers $\operatorname{ord}\Psi_j$. □

We call a family of solutions satisfying (B3)–(B4) with $\Delta = 0$ *balanced*. The balance theorem is the abstract form of the squeeze of Sections 5–6: crude bounds at the regular singular point and exact data at the irregular point determine the local exponents and exclude apparent singularities as soon as the defect vanishes. The following two lemmas convert this information into the shape of the operator.

Lemma 12.6 (shape at the origin)**.** *Let $L = \sum_{m=0}^{N} p_m \partial^m$ be primitive with $p_N = ct^A$, Fuchsian at 0. If 0 is not an exponent at 0, then $A = N$. If 0 is an exponent and 1 is not, then $A = N-1$. In both cases $\operatorname{ord}_0 p_m \ge m + A - N$ for all m.*

Proof. By Fuchs' criterion $\operatorname{ord}_0 p_m \ge m + A - N$, and the indicial polynomial is

$$I(\rho) = \sum_m [t^{m+A-N}]p_m\, \rho^{\underline{m}}.$$

Primitivity gives some $p_m(0) \neq 0$, hence $A \le N$. If $A \le N-1$, the term $m = 0$ of I vanishes, so $I(0) = 0$; if $A \le N-2$, also the term $m = 1$ vanishes and $I(1) = 0$. This proves the first assertion, and that $A \ge N-1$ in the second case. If 0 is an exponent and $A = N$, then $\operatorname{ord}_0 p_m \ge m$ for all m and $p_0(0) = I(0) = 0$, so all p_m vanish at 0, contradicting primitivity. □

Lemma 12.7 (the η-identity)**.** *Let $\Theta = \sum_{k=0}^{K} t^{2k} Q_{2k}(\theta)$ with $\deg Q_{2k} \le N - 2k$, and suppose*

$$\chi(\lambda) := \sum_k [\theta^{N-2k}]Q_{2k}\,\lambda^{N-2k} = c\prod_{b=1}^{N}(\lambda-\lambda_b), \qquad c \neq 0,$$

with pairwise distinct λ_b. If for every b the operator Θ has a formal solution $e^{\lambda_b t}t^{-g_b}(1 + \sum_{i\ge1} c_{b,i}t^{-i})$, then

$$\sum_k [\theta^{N-2k-1}]Q_{2k}\,\lambda^{N-2k-1} = c\sum_{b=1}^{N} g_b \prod_{b'\neq b}(\lambda-\lambda_{b'}) - \frac{\lambda}{2}\chi''(\lambda). \tag{12.3}$$

Proof. Since $\theta(e^{\lambda t}t^{-g}h) = e^{\lambda t}t^{-g}(\lambda t - g + \theta)h$, we must expand $Q_{2k}(\lambda t + \theta - g)$, applied to $h = 1 + c_1 t^{-1} + \cdots$. Write $d_k = N - 2k$ and $Q_{2k} = a_k\theta^{d_k} + s_k\theta^{d_k-1} + \cdots$. In $(\lambda t + \theta - g)^{d_k}$ the terms with exactly one factor $\theta - g$, in position p from the left, contribute $\lambda^{d_k-1}t^{d_k-1}(d_k - p - g)$; their sum is $\lambda^{d_k-1}t^{d_k-1}(\binom{d_k}{2} - d_k g)$. Collecting the coefficient of t^{N-1} in $e^{-\lambda t}t^{g}\,\Theta(e^{\lambda t}t^{-g}h)$, and noting that the contribution of c_1 is $c_1\chi(\lambda)$, we obtain for $\lambda = \lambda_b$, $g = g_b$:

$$\sum_k s_k\lambda_b^{d_k-1} + \frac{\lambda_b}{2}\chi''(\lambda_b) - g_b\chi'(\lambda_b) = 0.$$

The polynomial $\sum_k s_k\lambda^{d_k-1} + \frac{\lambda}{2}\chi''(\lambda)$ has degree at most $N-1$ and takes the value $g_b\chi'(\lambda_b) = c\,g_b\prod_{b'\neq b}(\lambda_b - \lambda_{b'})$ at the N points λ_b. Lagrange interpolation gives (12.3). □

13 Visibility: vertices leave singularities in the density

All generating functions in Sections 14–18 are Laplace transforms $F(t) = \langle \rho, e^{tw} \rangle$ of compactly supported distributions ρ (the law of a trace, possibly twisted by a character). This section shows that every critical value of the trace that comes from a vertex of the integration domain is a point where ρ fails to be smooth, and that such points bound the order of every differential equation for F from below.

13.1 Dual singular support

Lemma 13.1 (dual singular support)**.** *Let ρ be a compactly supported distribution on $\mathbb{R}$ and $F(t) = \langle \rho_w, e^{tw} \rangle$, an entire function. Let $L = \sum_{m=0}^{r} p_m(t)\partial_t^m \neq 0$ have polynomial coefficients, $d = \max_m \deg p_m$, and*

$$\chi_L(\lambda) = \sum_{m=0}^{r} [t^d] p_m \, \lambda^m,$$

a nonzero polynomial of degree at most r. If $LF = 0$, then ρ is real-analytic on $\mathbb{R} \setminus \chi_L^{-1}(0)$.

Proof. For a polynomial p, $p(t)\partial_t^m F(t) = \langle \rho, w^m p(\partial_w) e^{tw} \rangle = \langle p(-\partial_w)(w^m \rho), e^{tw} \rangle$. Hence $LF = \langle \hat{L}\rho, e^{tw} \rangle$ with $\hat{L} = \sum_m p_m(-\partial_w) \circ w^m$. The distribution $\hat{L}\rho$ has compact support and vanishing Fourier–Laplace transform, so $\hat{L}\rho = 0$, since the Fourier transform is injective on distributions with compact support [57, §7.1]. Writing $p_m = \sum_j c_{m,j} t^j$ we have $\hat{L} = \sum_{j \le d} (-1)^j \partial_w^j \circ a_j(w)$ with $a_j(w) = \sum_m c_{m,j} w^m$, so the coefficient of ∂_w^d in $\hat{L}$ is $(-1)^d a_d = (-1)^d \chi_L$. If $d = 0$, then $\hat{L}$ is multiplication by χ_L and ρ vanishes off the zeros of χ_L. If $d \ge 1$, then on an open interval I on which χ_L has no zero, $\hat{L}\rho = 0$ is equivalent to a system $U' = A(w)U$ for $U = (\rho, \rho', \dots, \rho^{(d-1)}) \in \mathcal{D}'(I)^d$, with A real-analytic on I. If Φ is a fundamental matrix of $Y' = AY$, then $(\Phi^{-1}U)' = 0$ in $\mathcal{D}'(I)$, so $\Phi^{-1}U$ is constant and $U = \Phi \cdot \mathrm{const}$ is real-analytic. □

Corollary 13.2. *Suppose ρ is not C^∞ at the distinct points $w_1, \dots, w_s$.*

(a) Every nonzero operator with polynomial coefficients annihilating F has order at least s.

(b) Every θ-form in t annihilating F has span at least $\#\{i : w_i \neq 0\}$.

Proof. (a) χ_L is a nonzero polynomial of degree at most $\operatorname{ord} L$ vanishing at every w_i. (b) Multiplying by a power of t we may assume $A = \sum_{k=k_0}^{k_1} t^k Q_k(\theta)$ with $k_0 \ge 0$. Let $d_* = \max_k (k + \deg Q_k)$ and $S = \{k : k + \deg Q_k = d_*\}$. The coefficient of $t^{d_*}\partial^m$ in A is $\operatorname{lc}(Q_{d_*-m})$ if $d_* - m \in S$ and 0 otherwise, so $\chi_A(\lambda) = \sum_{k \in S} \operatorname{lc}(Q_k) \lambda^{d_*-k} = \lambda^{d_* - \max S} g(\lambda)$ with $g(0) \neq 0$ and $\deg g = \max S - \min S \le k_1 - k_0$. The nonzero w_i are roots of g. □

Part (b) is the dual form of the types lemma of Lemma 3.6: there, distinct exponential types of solutions at infinity were counted; here we count points of the singular support of the density.

13.2 Oscillatory integrals at a vertex

Lemma 13.3 (Erdélyi)**.** *Let $\mu \in \{1, 2\}$, $\lambda > 0$, $\sigma \in \{1, -1\}$ and $J \ge 1$. For $g \in C^\infty([0, \infty))$ supported in $[0, R]$ and $\tau \ge 1$,*

$$\int_0^\infty e^{i\sigma\tau r^\mu} r^{\lambda-1} g(r)\, dr = \sum_{j<J} \frac{g^{(j)}(0)}{j!} \frac{1}{\mu} \Gamma\Big(\frac{\lambda + j}{\mu}\Big) e^{i\sigma\pi(\lambda+j)/(2\mu)} \tau^{-(\lambda+j)/\mu} + E_J(\tau),$$

where $|E_J(\tau)| \le C\, \tau^{-(\lambda+J)/\mu}$ and C depends only on μ, λ, J, R and on $\max_{i \le M} \|g^{(i)}\|_\infty$ for some $M = M(\mu, \lambda, J)$.

This is Erdélyi's lemma [55]; see also [65]. That the constant depends on finitely many derivatives of g (how many depends on λ) can be read off from the proof, which writes g as its Taylor polynomial plus a remainder vanishing to high order at 0 and integrates by parts in the remainder term.

Let $N_1, N_2 \geq 0$. For $N \geq 1$ we use polar coordinates $u = r\omega$ on $\mathbb{R}^N$, with $du = r^{N-1} dr\, d\omega$, and simplex coordinates $u = r\omega$ on $[0,\infty)^N$, with $r = \sum_j u_j$, ω in the simplex $\Sigma_N = \{\omega \in [0,1]^N : \sum \omega_j = 1\}$ and $du = r^{N-1} dr\, d\sigma(\omega)$, where $d\sigma = d\omega_1 \cdots d\omega_{N-1}$ is the projected Lebesgue measure.

Lemma 13.4 (vertex models)**.** *Let $A_i \geq 0$ be measurable on $\mathbb{R}^{N_i}$ (case Q) or on $[0,\infty)^{N_i}$ (case L), positively homogeneous of degree d_i, integrable and not a.e. zero on the unit sphere (resp. on Σ_{N_i}). Let h be C^∞ with compact support on $\mathbb{R}^{N_1} \times \mathbb{R}^{N_2}$. Put*

$$\gamma_i = \tfrac{1}{2}(N_i + d_i) \ \text{(case Q)}, \qquad \gamma_i = N_i + d_i \ \text{(case L)},$$

and assume $\gamma_1, \gamma_2 > 0$ when $N_1, N_2 \geq 1$. Then, as $\tau \to \infty$,

$$\int e^{i\tau\phi(u,v)} A_1(u) A_2(v) h(u,v)\, du\, dv = c\, h(0)\, \tau^{-\gamma_1-\gamma_2}\big(1 + O(\tau^{-1/2})\big) + O(\tau^{-\gamma_1-\gamma_2-1/2}),$$

where in case Q the integral is over $\mathbb{R}^{N_1} \times \mathbb{R}^{N_2}$, $\phi = -|u|^2 + |v|^2$ and

$$c = \tfrac{1}{4}\Gamma(\gamma_1)\Gamma(\gamma_2) e^{i\pi(\gamma_2-\gamma_1)/2} \int_{S^{N_1-1}} A_1 \int_{S^{N_2-1}} A_2,$$

and in case L the integral is over $[0,\infty)^{N_1} \times [0,\infty)^{N_2}$, $\phi = \sum_j u_j - \sum_j v_j$ and

$$c = \Gamma(\gamma_1)\Gamma(\gamma_2) e^{i\pi(\gamma_1-\gamma_2)/2} \int_{\Sigma_{N_1}} A_1 \int_{\Sigma_{N_2}} A_2.$$

If $N_i = 0$, the corresponding variable, factor and Γ-factor (together with one factor $\frac{1}{2}$ in case Q) are omitted. In particular $c \neq 0$.

Proof. Take case Q with $N_1, N_2 \geq 1$; the other cases are identical. In polar coordinates in u and in v separately, using homogeneity, the integral equals

$$\int_0^\infty \int_0^\infty e^{-i\tau r^2 + i\tau s^2} r^{2\gamma_1-1} s^{2\gamma_2-1} K(r,s)\, dr\, ds, \qquad K(r,s) = \iint A_1(\omega) A_2(\omega') h(r\omega, s\omega')\, d\omega\, d\omega'.$$

Since h is smooth with compact support and the A_i are integrable on the spheres, K is the restriction to $[0,\infty)^2$ of a C^∞ function on $\mathbb{R}^2$ with compact support, and $K(0,0) = h(0) \int A_1 \int A_2$. Apply Lemma 13.3 in s with $\mu = 2$, $\sigma = 1$ and $J > 2\gamma_1 + 1$; the coefficients $\partial_s^j K(r,0)$ are smooth in r, and the remainder is $O(\tau^{-\gamma_2 - J/2})$ uniformly in r. Then apply Lemma 13.3 in r with $\sigma = -1$ to each coefficient; the remainder contributes $O(\tau^{-\gamma_2-J/2})$, which is $O(\tau^{-\gamma_1-\gamma_2-1/2})$. The leading term is $\frac{1}{2}\Gamma(\gamma_1) e^{-i\pi\gamma_1/2} \cdot \frac{1}{2}\Gamma(\gamma_2) e^{i\pi\gamma_2/2} K(0,0)\, \tau^{-\gamma_1-\gamma_2}$. In case L one uses simplex coordinates and $\mu = 1$. Finally $\int A_i > 0$. □

13.3 Non-stationary points

The weights we meet are not smooth on the diagonals, so the usual non-stationary phase argument needs a direction in which the weight is smooth. Translations of a cluster of coinciding coordinates provide one.

Lemma 13.5 (cluster translation)**.** *Let $V \neq 0$ be a constant vector field on $\mathbb{R}^n$, $D \subseteq \mathbb{R}^n$ a closed set with $D + sV = D$ for all $s \in \mathbb{R}$, $U \subset \mathbb{R}^n$ open, and ϕ real-analytic on U with $|V\phi| \geq c_0 > 0$ on U. Let $a = \Phi\, g$, where $\Phi \geq 0$ is locally integrable and invariant under the translations $z \mapsto z + sV$, and $g \in C^\infty$ has compact support in U. Then $\int_D e^{i\tau\phi} a\, dz = O(\tau^{-M})$ for every M.*

Proof. Write $z = z' + sV$ with $z' \perp V$; then $D = D' \times \mathbb{R}$ in these coordinates and Φ does not depend on s. Integrating by parts in s on almost every line gives, without boundary terms,

$$\int_D e^{i\tau\phi}\Phi g\,dz = \frac{-1}{i\tau}\int_D e^{i\tau\phi}\,\Phi\,V\Big(\frac{g}{V\phi}\Big)dz,$$

and $V(g/V\phi)$ is again smooth with compact support in U. Iterate M times. □

In our applications $V = \sum_{j\in C}\partial_{z_j}$ translates a set C of coordinates, and Φ is a product of factors $|\varpi(z_i - z_j)|^\beta$, with $\varpi(s) = s$ or $2\sin(s/2)$, over pairs i, j both in C or both outside C, and of factors depending only on coordinates outside C.

13.4 The circular β-ensemble

For $\beta > 0$, $q \in \mathbb{Z}$ and $n \geq 1$, let $\rho^\beta_{n,q}$ be the image of the complex measure

$$e^{iq\sum_j \varphi_j}\prod_{i<j}|e^{i\varphi_i} - e^{i\varphi_j}|^\beta\,d\varphi_1\cdots d\varphi_n \qquad \text{on } \mathbb{T}^n = (\mathbb{R}/2\pi\mathbb{Z})^n$$

under $f(\varphi) = \sum_j 2\cos\varphi_j$. It is a compactly supported distribution, and $\langle\rho^\beta_{n,q}, e^{tw}\rangle$ is a constant multiple of $\mathbb{E}_{\mathrm{C}\beta\mathrm{E}_n}(\det U)^q e^{t\,\mathrm{tr}(U+U^{-1})}$.

Proposition 13.6. *For every $0 \leq b \leq n$, the distribution $\rho^\beta_{n,q}$ is not C^∞ at $\lambda_b = 2(n-2b)$. More precisely, if $\psi \in C^\infty_c(\mathbb{R})$ equals 1 near λ_b and vanishes near the other $\lambda_{b'}$, then*

$$\langle\rho^\beta_{n,q}, \psi(w)e^{i\tau w}\rangle = \binom{n}{b}c_b\,e^{i\tau\lambda_b}\tau^{-\gamma_b}(1 + O(\tau^{-1/2})), \qquad \gamma_b = \frac{n}{2} + \frac{\beta}{2}\Big(\binom{n-b}{2} + \binom{b}{2}\Big),$$

with $c_b \neq 0$.

Proof. The left side equals $\int_{\mathbb{T}^n} e^{i\tau f}\psi(f)\,a\,d\varphi$, where a is the density above. The critical points of f are the vertices φ^S, $S \subseteq [n]$, with $\varphi^S_j = \pi$ for $j \in S$ and $\varphi^S_j = 0$ otherwise, and $f(\varphi^S) = \lambda_{|S|}$. Let $\varphi^* \in \operatorname{supp}\psi(f)$. If φ^* is not a vertex, some *cluster* $C = \{j : \varphi^*_j = \vartheta\}$ of coinciding coordinates has $\vartheta \notin \{0, \pi\}$. On a small neighbourhood U of φ^*, the factors $|e^{i\varphi_i} - e^{i\varphi_j}|^\beta = |2\sin((\varphi_i - \varphi_j)/2)|^\beta$ with i, j in different clusters are smooth, and $V = \sum_{j\in C}\partial_{\varphi_j}$ satisfies $Vf = -2\sum_{j\in C}\sin\varphi_j \neq 0$. By Lemma 13.5 (on $D = \mathbb{R}^n$, lifting to a chart, with Φ the product of the factors over pairs inside clusters), a smooth cutoff of $\psi(f)a$ to U contributes $O(\tau^{-M})$. By compactness and a partition of unity, the integral equals the sum over $|S| = b$ of the contributions of small neighbourhoods W_S of φ^S, up to $O(\tau^{-M})$; the vertices with $|S| \neq b$ lie outside $\operatorname{supp}\psi(f)$.

Near φ^S put $u_j = 2\sin(\varphi_j/2)$ for $j \notin S$ and $v_j = 2\sin((\varphi_j - \pi)/2)$ for $j \in S$. Then $f = \lambda_b - |u|^2 + |v|^2$ exactly. For $i, j \notin S$, $2\sin((\varphi_i - \varphi_j)/2)$ is a real-analytic function of (u_i, u_j) near 0 that vanishes on $u_i = u_j$ and has partial derivative 1 in u_i at 0; hence it equals $(u_i - u_j)\,g_{ij}(u)$ with g_{ij} real-analytic and $g_{ij}(0) = 1$, and $|2\sin((\varphi_i - \varphi_j)/2)|^\beta = |u_i - u_j|^\beta\,|g_{ij}|^\beta$ with $|g_{ij}|^\beta$ smooth near 0. The same holds for $i, j \in S$ in the variables v. For $i \notin S$, $j \in S$ the factor is smooth and equals 2^β at φ^S. The twist equals $(-1)^{qb}$ at φ^S, and $d\varphi_j = (1 - u_j^2/4)^{-1/2}du_j$. Hence the contribution of W_S is

$$e^{i\tau\lambda_b}\int e^{i\tau(-|u|^2+|v|^2)}|\Delta(u)|^\beta|\Delta(v)|^\beta h_S(u,v)\,du\,dv, \qquad h_S(0) = (-1)^{qb}2^{\beta b(n-b)},$$

with h_S smooth and compactly supported. Lemma 13.4 (case Q, $N_1 = n-b$, $N_2 = b$, $d_1 = \beta\binom{n-b}{2}$, $d_2 = \beta\binom{b}{2}$) gives $c_b e^{i\tau\lambda_b}\tau^{-\gamma_b}(1 + O(\tau^{-1/2}))$ with $c_b \neq 0$ independent of S. Summing over the $\binom{n}{b}$ sets S proves the formula. If $\rho^\beta_{n,q}$ were smooth on a neighbourhood of λ_b, we could choose ψ supported in it; then $\psi\rho^\beta_{n,q}$ would be in C^∞_c and the left side would be $O(\tau^{-M})$ for all M. □

13.5 The β-Jacobi ensemble

For $\beta>0$ and $a,b>-1$ let ρ^{J} be the image of the probability measure

$$\frac{1}{Z}\prod_{j=1}^{n} s_j^{a}(1-s_j)^{b}\prod_{i<j}|s_i-s_j|^{\beta}\,ds \qquad \text{on } [0,1]^n$$

under $(s_j)\mapsto\sum_j s_j$.

Proposition 13.7. *For every $0\le k\le n$, ρ^{J} is not C^{∞} at k. If $\psi\in C_c^{\infty}(\mathbb{R})$ equals 1 near k and vanishes near the other integers in $[0,n]$, then $\langle\rho^{J},\psi e^{i\tau w}\rangle=\binom{n}{k}c_k'e^{i\tau k}\tau^{-\gamma_k^{J}}(1+O(\tau^{-1/2}))$ with $c_k'\neq0$ and*

$$\gamma_k^{J}=(n-k)(a+1)+k(b+1)+\beta\Big(\binom{n-k}{2}+\binom{k}{2}\Big).$$

Proof. The vertices of the cube are the indicator vectors $\mathbf{1}_S$, $S\subseteq[n]$, and $\sum_j s_j=|S|$ at $\mathbf{1}_S$. Let $s^{*}\in[0,1]^n$ with $\sum s_j^{*}\in\operatorname{supp}\psi$. If s^{*} is not a vertex, some cluster C of coinciding coordinates has its common value in $(0,1)$. Choose a small open neighbourhood U of s^{*} in $\mathbb{R}^n$ on which the coordinates in C stay in $(0,1)$ and the factors $s_j^{a}(1-s_j)^{b}$, $j\in C$, and $|s_i-s_j|^{\beta}$, i,j in different clusters, are smooth. Apply Lemma 13.5 with $V=\sum_{j\in C}\partial_{s_j}$, $V\sum_j s_j=|C|\neq0$, on the V-invariant set $D=\{s:0\le s_j\le1 \text{ for } j\notin C\}$; on $U\cap D$ the weight agrees with that of the ensemble, and the factors $s_j^{a}(1-s_j)^{b}$, $j\notin C$, and $|s_i-s_j|^{\beta}$, i,j in a common cluster, go into Φ. So these regions contribute $O(\tau^{-M})$. Near $\mathbf{1}_S$ with $|S|=k$, put $u_j=s_j$ for $j\notin S$ and $v_j=1-s_j$ for $j\in S$; then $\sum_j s_j=k+\sum u_j-\sum v_j$, and the weight is $\prod u_j^{a}\,|\Delta(u)|^{\beta}\cdot\prod v_j^{b}\,|\Delta(v)|^{\beta}\cdot h(u,v)$ with h smooth near 0 and $h(0)=1/Z$. Lemma 13.4 (case L) gives the contribution $c_k'e^{i\tau k}\tau^{-\gamma_k^{J}}(1+O(\tau^{-1/2}))$ with $c_k'\neq0$, where we used $\int_{\Sigma_N}\prod\omega_j^{a}|\Delta(\omega)|^{\beta}d\sigma<\infty$ for $a>-1$. Conclude as before, choosing ψ supported in a neighbourhood of k on which ρ^{J} would be smooth. □

14 Kac recursions: scalar operators from the first-order systems

14.1 The Sylvester–Kac matrix

Let T_n be the $(n+1)\times(n+1)$ matrix, indexed by $0\le p,p'\le n$, with $T_n[p,p-1]=p$, $T_n[p,p+1]=n-p$ and all other entries 0.

Lemma 14.1 (Sylvester–Kac)**.** *(a) T_n has the simple eigenvalues $n-2b$, $0\le b\le n$. Every eigenvector v has $v_0\neq0$ and $v_n\neq0$.*

(b) Define polynomials $r_p(\mu)$ by $r_{-1}=0$, $r_0=1$ and $r_{p+1}=(\mu r_p-p\,r_{p-1})/(n-p)$ for $0\le p<n$. Then

$$\mu\,r_n(\mu)-n\,r_{n-1}(\mu)=\frac{1}{n!}\prod_{b=0}^{n}(\mu-n+2b).$$

(c) Let $S=\operatorname{diag}\binom{n}{p}^{-1/2}$. Then $S^{-1}T_nS$ is the matrix of $J_++J_-=2J_x$ in the standard basis $|j,m\rangle$, $m=j-p$, of the irreducible representation of $\mathfrak{sl}_2$ of spin $j=n/2$.

Proof. (a) On homogeneous polynomials of degree n in X,Y with basis $e_p=X^{n-p}Y^{p}$, the operator $E=Y\partial_X+X\partial_Y$ acts by $Ee_p=(n-p)e_{p+1}+pe_{p-1}$, so its matrix is $T_n^{\mathsf T}$. In the coordinates $X\pm Y$ one has $E=(X+Y)\partial_{X+Y}-(X-Y)\partial_{X-Y}$, with eigenvectors $(X+Y)^{n-b}(X-Y)^{b}$ and eigenvalues $n-2b$. If $T_nv=\mu v$ and $v_0=0$, the rows $0,1,\dots,n-1$ give successively $v_1=\cdots=v_n=0$, since the entries $n-p$ are nonzero; similarly for v_n using the entries p.

(b) Rows $0,\dots,n-1$ of $(T_n-\mu)v=0$ are the recursion for $v_p=r_p(\mu)$ with $v_0=1$, and row n reads $n\,r_{n-1}-\mu r_n=0$. By (a), μr_n-nr_{n-1} vanishes at the $n+1$ eigenvalues; it has degree $n+1$ and leading coefficient $1/n!$.

(c) $S^{-1}T_nS$ has entries $\sqrt{p(n-p+1)}$ in positions $(p,p-1)$ and $(p-1,p)$, and $\langle j,m+1|J_+|j,m\rangle=\sqrt{(j-m)(j+m+1)}=\sqrt{p(n-p+1)}$ for $m=j-p$. □

Part (a) is Sylvester's theorem [61], rediscovered by Kac [58]; see [62].

14.2 The circular ensembles

Let $\beta>0$, $n\ge 1$ and $q>-1$, and put

$$G(x)=G^{\beta}_{n,q}(x)={}_0F_1^{(2/\beta)}\Big(q+1+\frac{(n-1)\beta}{2};\,(x)^n\Big),$$

the hypergeometric function of n equal arguments with Jack parameter $2/\beta$ [59, 11]. For $q\in\mathbb{Z}_{\ge 0}$,

$$\mathbb{E}_{C\beta E_n}\big[(\det U)^q\,e^{t\,\mathrm{tr}(U+U^{-1})}\big]=c\,t^{qn}\,G(t^2),\qquad c\neq 0,\tag{14.1}$$

by [11, (2.7)]; for $q=0$, $c=1$ and $G(x)=\sum_N \mu_N x^N/(N!)^2$ with $\mu_N=\mathbb{E}|\,\mathrm{tr}\,U|^{2N}$. Define

$$B_p=p\Big(q+1+\frac{\beta}{2}(n-p)\Big),\qquad \varepsilon_p=B_p-p=p\Big(q+\frac{\beta}{2}(n-p)\Big)\quad(0\le p\le n),\tag{14.2}$$

operators $R_p\in\mathbb{C}[x][\theta]$ by

$$R_{-1}=0,\quad R_0=1,\quad R_{p+1}=\frac{(\theta-p+B_p)R_p-p\,x\,R_{p-1}}{n-p}\quad(0\le p<n),\tag{14.3}$$

and

$$\mathcal{L}=(\theta-n+B_n)R_n-n\,x\,R_{n-1}.\tag{14.4}$$

Proposition 14.2. $\mathcal{L}G=0$.

Proof. Forrester and Wei [11, Prop. 3.3] consider the system

$$(n-p)\,x\,I_{p+1}=B_pI_p+\theta I_p-p\,I_{p-1}\qquad(0\le p\le n)\tag{14.5}$$

with $B_p=p(a+b+1+\frac{\beta}{2}(2n-p-1))$, and show that its power series solution normalized by $I_0(0)=1$ has first component $I_0={}_0F_1^{(2/\beta)}(\beta(n-1)+a+b+1;(x)^n)$ [11, (3.9)]. The choice $a+b=q-\frac{\beta}{2}(n-1)$ gives $B_p=p(q+1+\frac{\beta}{2}(n-p))$, which is [11, (3.13)], and $I_0=G$. Put $K_p=x^pI_p$. Then $\theta K_p=(p-B_p)K_p+p\,x\,K_{p-1}+(n-p)K_{p+1}$, which for $p<n$ says $K_{p+1}=R_{p+1}G$ by induction, and for $p=n$ says $\mathcal{L}G=0$. □

Forrester and Wei derive (14.5) from the Forrester–Kumar system of Proposition 14.6 below by a confluent limit ($a\to\infty$ with $a+b$ fixed, coefficientwise in the power series). Their Proposition 4.1 eliminates downwards, from I_n; we eliminate upwards, from I_0, which produces the three-term recursion (14.3).

Conversely, if $\mathcal{L}y=0$ for some (formal or actual) function y, then $K_p:=R_py$ satisfy $\theta K_p=(p-B_p)K_p+pxK_{p-1}+(n-p)K_{p+1}$ for all p. So the scalar equation and the system are equivalent, with $y\mapsto(R_py)_p$ and its inverse given by the first component. We verified Proposition 14.2 independently, coefficient by coefficient, on the Jack expansion of G (Appendix C).

Proposition 14.3 (structure)**.** *Write* $\mathcal{L}=\sum_{k\ge 0}x^kP_k(\theta)$ *and* $K=\lfloor(n+1)/2\rfloor=\lceil n/2\rceil$. *Then* $P_k=0$ *for* $k>K$ *and:*

(a) $P_0(\theta) = \frac{1}{n!}\prod_{p=0}^{n}(\theta+\varepsilon_p)$;

(b) $\deg P_k \le n+1-2k$, *and* $\sum_{k=0}^{K}[\theta^{n+1-2k}]P_k\cdot\mu^{n+1-2k} = \frac{1}{n!}\prod_{b=0}^{n}(\mu-n+2b)$; *in particular* $\deg P_k = n+1-2k$ *for* $0\le k\le K$;

(c) $n!\,P_k \in \mathbb{Z}[B_1,\dots,B_n][\theta]$ *for every* k.

Proof. Write $R_p=\sum_k x^k R_{p,k}(\theta)$. By (12.1), (14.3) becomes

$$R_{p+1,k}(\theta) = \frac{(\theta+k-p+B_p)R_{p,k}(\theta) - p\,R_{p-1,k-1}(\theta)}{n-p}.$$

By induction $\deg R_{p,k}\le p-2k$, and the coefficients $\rho_{p,k}=[\theta^{p-2k}]R_{p,k}$ satisfy $\rho_{p+1,k}=(\rho_{p,k}-p\rho_{p-1,k-1})/(n-p)$, so that $\sum_k\rho_{p,k}\mu^{p-2k}=r_p(\mu)$ in the notation of Lemma 14.1(b). Likewise $P_k=(\theta+k-n+B_n)R_{n,k}-nR_{n-1,k-1}$, so $\deg P_k\le n+1-2k$ and $\sum_k[\theta^{n+1-2k}]P_k\mu^{n+1-2k}=\mu r_n-nr_{n-1}$, which is (b) by Lemma 14.1(b). The polynomial $\mu^{[n\text{ even}]}\prod_{b<n/2}(\mu^2-(n-2b)^2)$ has nonzero coefficients at all μ^{n+1-2k}, $0\le k\le K$, and none at higher k. For (a), $R_{p+1,0}=(\theta-p+B_p)R_{p,0}/(n-p)$ gives $P_0=\frac{1}{n!}\prod_{p=0}^{n}(\theta-p+B_p)$, and $B_p-p=\varepsilon_p$. For (c), put $\tilde R_p=\frac{n!}{(n-p)!}R_p$. Then $\tilde R_0=1$ and $\tilde R_{p+1}=(\theta-p+B_p)\tilde R_p-p(n-p+1)\,x\,\tilde R_{p-1}$, so all $\tilde R_p$ have coefficients in $\mathbb{Z}[B_1,\dots,B_n]$, and $n!\,\mathcal{L}=(\theta-n+B_n)\tilde R_n-n\,x\,\tilde R_{n-1}$. □

14.3 Formal solutions at infinity

Put $x=t^2$ and $J_p(t)=t^{-p}K_p$, where $K_p=R_py$ for a solution y of $\mathcal{L}$. The system for K_p becomes

$$\frac{dJ}{dt} = \Big(2T_n+\frac{1}{t}\,\mathcal{D}\Big)J, \qquad \mathcal{D}=\operatorname{diag}(p-2B_p)_{0\le p\le n}, \tag{14.6}$$

with $J_0(t)=y(t^2)$. Let $\lambda_b=2(n-2b)$ and

$$\gamma_b = \frac{n}{2}+\frac{\beta}{4}\Big((n-b)(n-b-1)+b(b-1)\Big) = \frac{n}{2}+\frac{\beta}{2}\Big(\binom{n-b}{2}+\binom{b}{2}\Big), \qquad 0\le b\le n. \tag{14.7}$$

Proposition 14.4. *For every* $0\le b\le n$, *the operator* $\Theta_t:=\sum_k t^{2k}P_k(\theta_t/2)$, *which annihilates* $G(t^2)$, *has a formal solution*

$$e^{\lambda_b t}\,t^{-\gamma_b-qn}\Big(1+\sum_{i\ge1}c_{b,i}t^{-i}\Big).$$

Proof. The system (14.6) has an irregular singular point of Poincaré rank 1 at $t=\infty$ whose leading matrix $2T_n$ has the distinct eigenvalues λ_b. By the formal reduction theory [64, Ch. III], [53, Ch. 3], it has a formal fundamental matrix $\hat F(t)\,t^{\Lambda_1}e^{t\Lambda_0}$ with $\hat F=F_0+F_1t^{-1}+\cdots$, $\Lambda_0=\operatorname{diag}(\lambda_b)$, F_0 a matrix of eigenvectors of $2T_n$, and Λ_1 the diagonal part of $F_0^{-1}\mathcal{D}F_0$. The first component of the b-th column is a formal solution of the scalar equation, of the form $e^{\lambda_b t}t^{\rho_b}((F_0)_{0b}+O(t^{-1}))$ with $(F_0)_{0b}\neq0$ by Lemma 14.1(a).

To compute $\rho_b=(F_0^{-1}\mathcal{D}F_0)_{bb}$, take $F_0=S\Psi$ with Ψ an orthogonal matrix of eigenvectors ψ_b of the symmetric matrix $S^{-1}T_nS$. Since $\mathcal{D}$ is diagonal, $\rho_b=\psi_b^{\mathsf T}\mathcal{D}\psi_b=\langle p\rangle_b-2\langle B_p\rangle_b$, where $\langle f(p)\rangle_b=\sum_p f(p)\psi_b(p)^2$. By Lemma 14.1(c), ψ_b is an eigenvector of J_x with eigenvalue $m_x=n/2-b$, and $p=j-J_z$ with $j=n/2$. For an eigenvector of J_x one has $\langle J_z\rangle=-i\langle[J_x,J_y]\rangle=0$ and, using the rotation $e^{-i\pi J_x/2}$, which commutes with J_x and conjugates J_y to $\pm J_z$, $\langle J_z^2\rangle=\langle J_y^2\rangle=\frac12\big(j(j+1)-m_x^2\big)$. Hence $\langle p\rangle_b=n/2$ and

$$\langle p(n-p)\rangle_b = j^2-\langle J_z^2\rangle = \frac{(n-b)(n-b-1)+b(b-1)}{4}.$$

With $B_p=(q+1)p+\frac{\beta}{2}p(n-p)$ this gives $\rho_b=-(2q+1)\frac{n}{2}-\frac{\beta}{4}\big((n-b)(n-b-1)+b(b-1)\big)=-\gamma_b-qn$. □

For $\beta=2$, $\gamma_b=\frac12((n-b)^2+b^2)$, as in Sections 5–6. The values γ_b coincide with the exponents found at the vertices in Proposition 13.6; the formal theory and the stationary phase computation see the same vertex.

Remark 14.5 (balance as a trace identity). Since Λ_1 is the diagonal part of $F_0^{-1}\mathcal{D}F_0$, $\sum_b \rho_b = \operatorname{tr}\mathcal{D}=\sum_p(p-2B_p)=-\binom{n+1}{2}-2\sum_p\varepsilon_p$. In the variable t the exponents of $G(t^2)$ at 0 are $-2\varepsilon_p$, so

$$\sum_{b=0}^{n}(\gamma_b+qn)-\sum_{p=0}^{n}2\varepsilon_p=\binom{n+1}{2}.$$

This is the vanishing of the balance defect of Theorem 12.5, written for $D^{\beta}_{n,q}=c\,t^{qn}G(t^2)$: with $f(p)=2\varepsilon_p-qn=\beta p(n-p)-q(n-2p)$ it reads $\sum_b\gamma_b-\sum_p f(p)=\binom{n+1}{2}$. In Section 16 this identity is verified by summation and used as an input; here it is the trace of the residue of the Kac system.

14.4 The Jacobi ensembles

Let $\beta>0$, $a,b>-1$, and let

$$H(x)=\mathbb{E}_{\beta\mathrm{J}}\,e^{-x\sum_j s_j}={}_1F_1^{(2/\beta)}\Big(\frac{\beta}{2}(n-1)+a+1;\ \beta(n-1)+a+b+2;\ (-x)^n\Big)$$

be the Laplace transform of the trace of the β-Jacobi ensemble of Section 13 [59, 10]. Define

$$B_p^J=p\Big(a+b+2+\frac{\beta}{2}(2n-p-1)\Big),\qquad D_p=p\Big(b+1+\frac{\beta}{2}(n-p)\Big),\qquad \varepsilon_p^J=B_p^J-p,\tag{14.8}$$

operators $R^J_{-1}=0$, $R^J_0=1$,

$$R^J_{p+1}=\frac{(\theta-p+B_p^J+(n-p)x)R_p^J-D_p\,x\,R^J_{p-1}}{n-p},$$

and $\mathcal{L}^J=(\theta-n+B_n^J)R_n^J-D_n\,x\,R^J_{n-1}$.

Proposition 14.6. *$\mathcal{L}^JH=0$. Writing $\mathcal{L}^J=\sum_k x^kP_k^J(\theta)$, one has $P_k^J=0$ for $k>n$,*

$$P_0^J(\theta)=\frac{1}{n!}\prod_{p=0}^{n}(\theta+\varepsilon_p^J),\qquad \sum_{k=0}^{n}[\theta^{n+1-k}]P_k^J\cdot\mu^{n+1-k}=\frac{1}{n!}\,\mu\prod_{k=1}^{n}(\mu+k),\qquad P_n^J(\theta)=\theta+\gamma_0^J,$$

with $\gamma_0^J=n(a+1)+\beta\binom{n}{2}$. In particular $\deg P_k^J=n+1-k$ for $0\le k\le n$. The operator $\mathcal{L}^J$ has, for every $0\le k\le n$, a formal solution $e^{-kx}x^{-\gamma_k^J}(1+O(x^{-1}))$ at $x=\infty$, with γ_k^J as in Proposition 13.7.

Proof. For $p=0,\dots,n$ let

$$\hat H_p(x)=\binom{n}{p}^{-1}\int_{[0,1]^n}\prod_l s_l^a(1-s_l)^{b'-1}e^{-x\sum_l s_l}\prod_{j<k}|s_k-s_j|^{\beta}\,e_p(1-s_1,\dots,1-s_n)\,ds,$$

with e_p the elementary symmetric polynomial. Forrester and Kumar [10, Cor. 2.5] show that

$$(n-p)\,x\,\hat H_{p+1}=\big((n-p)x+\tilde B_p\big)\hat H_p+\theta\hat H_p-\tilde D_p\hat H_{p-1},$$

where $\tilde B_p=p(a+b'+1+\frac{\beta}{2}(2n-p-1))$ and $\tilde D_p=p(\frac{\beta}{2}(n-p)+b')$. Take $b'=b+1$, so that $\tilde B_p=B_p^J$, $\tilde D_p=D_p$, and $\hat H_0$ is the Selberg integral $S_n(a,b,\beta)$ times H. With $H_p=\hat H_p/S_n(a,b,\beta)$ the system reads

$$\theta H_p=-B_p^JH_p+D_pH_{p-1}+(n-p)\,x\,(H_{p+1}-H_p),\qquad H_0=H.\tag{14.9}$$

With $K_p = x^p H_p$ this becomes $\theta K_p = (p - B_p^J - (n-p)x)K_p + D_p x K_{p-1} + (n-p)K_{p+1}$, and $\mathcal{L}^J H = 0$ follows as in Proposition 14.2. The degree bounds follow from the recursion as in Proposition 14.3, now with $\deg R_{p,k}^J \le p-k$. The top coefficients satisfy $\rho_{p+1}(\mu) = (\mu + (n-p))\rho_p(\mu)/(n-p)$ for the generating polynomials $\rho_p(\mu) = \sum_k [\theta^{p-k}] R_{p,k}^J \mu^{p-k}$ (taken at $x=1$), because the term $D_p x R_{p-1}^J$ has lower total degree; hence the top symbol of $\mathcal{L}^J$ is $\frac{1}{n!}\mu\prod_{k=1}^n(\mu+k)$, all of whose coefficients are positive. The coefficient of x^p in R_p^J is 1 by induction, which gives $P_n^J = \theta + B_n^J - D_n$, and $B_n^J - D_n = \gamma_0^J$.

For the formal solutions, H-system (14.9) reads $H' = (X + x^{-1}Y)H$ with $X = -\mathrm{diag}(n-p) + \mathrm{diag}^+(n-p)$ and $Y = -\mathrm{diag}(B_p^J) + \mathrm{diag}^-(D_p)$. The eigenvalue $-k$ of X has right eigenvector $v_p = \binom{n-k}{p}/\binom{n}{p}$ (so $v_0 = 1$, $v_p = 0$ for $p > n-k$) and a left eigenvector w supported on $p \ge n-k$ with $w_{n-k+1}/w_{n-k} = -k$. By the formal reduction theory the exponent is $\rho_k = wYv/wv$, and only $p \in \{n-k, n-k+1\}$ contribute:

$$\rho_k = -B_{n-k}^J + D_{n-k}\frac{v_{n-k-1}}{v_{n-k}} + D_{n-k+1}\frac{w_{n-k+1}}{w_{n-k}} = -B_{n-k}^J + (k+1)D_{n-k} - k\,D_{n-k+1} = -\gamma_k^J,$$

by a direct computation. □

15 Circular β-ensembles: proofs of Theorems 11.1 and 11.3

Throughout this section $\beta > 0$, $n \ge 1$, $q > -\frac{1}{2}$ is real, $K = \lceil n/2 \rceil$, and $\mathcal{L} = \sum_{k=0}^K x^k P_k(\theta)$ is the operator of Proposition 14.3. Put $\hat{P}_k = n!\,P_k$ and $g_b = \gamma_b + qn$; since $\gamma_b \ge n/2$, we have $g_b > 0$.

15.1 Exact degrees

Set $Q_{2k}(u) = 2^{n+1}\hat{P}_k(u/2)$. Since $\theta_x = \theta_t/2$ for $x = t^2$, the θ-form $\Theta_t = \sum_k t^{2k} Q_{2k}(\theta_t)$ is a constant multiple of the operator of Proposition 14.4. By Proposition 14.3(b),

$$\chi(\lambda) := \sum_k [u^{n+1-2k}] Q_{2k}\,\lambda^{n+1-2k} = 2^{n+1}\prod_{b=0}^n\Big(\frac{\lambda}{2} - n + 2b\Big) = \prod_{b=0}^n(\lambda - \lambda_b).$$

By Proposition 14.4, Θ_t has the formal solutions $e^{\lambda_b t} t^{-g_b}(1 + O(t^{-1}))$. Lemma 12.7 therefore applies with $N = n+1$ and $c = 1$.

Lemma 15.1. *Let $0 \le m \le n$, let $k^* = \lfloor (n+1-m)/2 \rfloor$ and $d = n+1-2k^* \in \{m, m+1\}$. If $d = m+1$, then $\binom{d}{2}\,\mathrm{lc}(\hat{P}_{k^*}) + [\theta^{d-1}]\hat{P}_{k^*} \ne 0$.*

Proof. Since $\hat{P}_k(\theta) = 2^{-n-1}Q_{2k}(2\theta)$, the quantity in question equals $2^{d-n-2}(d(d-1)[u^d]Q_{2k^*} + [u^{d-1}]Q_{2k^*})$. By (12.3) and $[u^d]Q_{2k^*} = [\lambda^d]\chi$, this is 2^{d-n-2} times the coefficient of λ^{d-1} in

$$\frac{\lambda}{2}\chi''(\lambda) + \sum_{b=0}^n g_b \prod_{b' \ne b}(\lambda - \lambda_{b'}).$$

Write $\varepsilon = [n \text{ even}]$ and $c_j = \lambda_j^2 > 0$ for $0 \le j < n/2$, so that $\chi = \lambda^\varepsilon \prod_{j<n/2}(\lambda^2 - c_j)$, and note that there are exactly K indices $j < n/2$. Pairing b with $n-b$ (since $\lambda_{n-b} = -\lambda_b$ and $g_{n-b} = g_b$) and keeping $b = n/2$ alone when n is even, one finds that the coefficient of λ^{d-1} equals

$$(-1)^{k^*}\Big(\binom{d}{2}\sigma_{k^*}(c) + 2\sum_{j<n/2} g_j\,\sigma_{k^*}(c_{\ne j}) + \varepsilon\, g_{n/2}\,\sigma_{k^*}(c)\Big),$$

where σ_i are the elementary symmetric functions and $c_{\ne j}$ is the list c with c_j removed. Indeed, $g_j(\prod_{b' \ne j} + \prod_{b' \ne n-j})(\lambda - \lambda_{b'}) = 2g_j\lambda^{1+\varepsilon}\prod_{i \ne j}(\lambda^2 - c_i)$ and $\frac{\lambda}{2}\chi''$ contributes $\binom{d}{2}[\lambda^d]\chi = (-1)^{k^*}\binom{d}{2}\sigma_{k^*}(c)$. All terms are ≥ 0 because $g_b > 0$. If $k^* < K$, the middle sum is positive, as $c_{\ne j}$ has $K - 1 \ge k^*$ positive entries. If $k^* = K$, then $d = n+1-2K \le 1$, so $d = 1$, n is even, and the last term is positive. □

15.2 Proof of Theorem 11.1

(a) and (b). By Propositions 14.2 and 14.3, $n!\,\mathcal{L} = \sum_{k=0}^{K} x^k \hat{P}_k(\theta)$ annihilates G, with $\hat{P}_0 = \prod_p(\theta+\varepsilon_p)$, $\deg \hat{P}_k = n+1-2k$ and $\sum_k \operatorname{lc}(\hat{P}_k)\mu^{n+1-2k} = \prod_b(\mu-n+2b)$. Since $\deg \hat{P}_0 > \deg \hat{P}_k$ for $k \geq 1$, the operator is Fuchsian at 0 with indicial polynomial $\hat{P}_0$, so the exponents are $-\varepsilon_p$. Using $x^k\theta^i = \sum_{m\leq i} S(i,m)x^{k+m}\partial^m$, write $n!\,\mathcal{L} = x\sum_m q_m\partial^m$. For $m \geq 1$, xq_m is a combination of the monomials x^{k+m} coming from $x^k\hat{P}_k$ with $\deg \hat{P}_k \geq m$; for $m = 0$, $xq_0 = \sum_k x^k\hat{P}_k(0)$ and $\hat{P}_0(0) = 0$. Hence the q_m are polynomials, $x^{m-1} \mid q_m$, $q_{n+1} = x^n$, and $\deg q_m \leq k^*+m-1 = \lfloor (n-1+m)/2 \rfloor$ with k^* as in Lemma 15.1. The coefficient of x^{k^*+m} in xq_m comes from $\hat{P}_{k^*}$ only; it is $\operatorname{lc}(\hat{P}_{k^*}) \neq 0$ if $\deg \hat{P}_{k^*} = m$, and $\binom{m+1}{2}\operatorname{lc}(\hat{P}_{k^*}) + [\theta^m]\hat{P}_{k^*} \neq 0$ if $\deg \hat{P}_{k^*} = m+1$, by Lemma 15.1. So $\deg q_m = \lfloor (n-1+m)/2 \rfloor$. The operator $L_x = \sum_m q_m\partial^m$ has no singular point in $\mathbb{C}\setminus\{0\}$ because $q_{n+1} = x^n$, and for $q \geq 0$ it is primitive because $q_1(0) = \hat{P}_0(1) = \prod_p(1+\varepsilon_p) > 0$. The integrality statement follows from Proposition 14.3(c), since the Stirling numbers $S(i,m)$ are integers.

If n is even, Proposition 14.4 with $b = n/2$ gives a formal solution $x^{-g_{n/2}/2}(1+O(x^{-1/2}))$ of $n!\,\mathcal{L}$. Applying $\sum_k x^k\hat{P}_k(\theta)$, the term $x^K\hat{P}_K(\theta)$ produces $\hat{P}_K(-g_{n/2}/2)\,x^{K-g_{n/2}/2}$ and all other contributions are of lower order. Hence $\hat{P}_K(-g_{n/2}/2) = 0$, and since $\deg \hat{P}_K = 1$, $\hat{P}_K(\theta) = \operatorname{lc}(\hat{P}_K)(\theta + g_{n/2}/2)$.

(c) is Proposition 14.4 in the variable $x = t^2$.

(d) Let $q \in \mathbb{Z}_{\geq 0}$. By (14.1), $D(t) := c\,t^{qn}G(t^2)$ is a constant multiple of $\langle \rho^{\beta}_{n,q}, e^{tw}\rangle$. By Proposition 13.6, $\rho^{\beta}_{n,q}$ is not smooth at the $n+1$ points λ_b, so by Corollary 13.2(a) every nonzero operator annihilating D has order at least $n+1$. If G satisfied an equation of order $s \leq n$ in x, then $G(t^2)$, and hence D, would satisfy one of order s in t. Therefore L_x is the minimal operator of G. If $\tilde{L}$ of order $n+1$ with polynomial coefficients annihilates G, then $\tilde{L} = hL_x$ with $h = u/v \in \mathbb{C}(x)$ in lowest terms; v divides every q_m, hence is constant by primitivity. In the variable t, conjugation by t^{qn} shows that $\sum_k t^{2k}\hat{P}_k((\theta_t - qn)/2)$ annihilates D. This θ-form has order $n+1$, so it is a multiple of the minimal operator of D by an element of $\mathbb{C}(t)$; its coefficient of ∂_t^{n+1} is $2^{-n-1}t^{n+1}$, so the minimal operator has no singular point in $\mathbb{C}\setminus\{0\}$, and its exponents at 0 are the roots $q(n-2p) - \beta p(n-p) = qn - 2\varepsilon_p$ of $\hat{P}_0((\theta - qn)/2)$. □

Remark 15.2. At $\beta = 0$ the recursion (14.3) with $q = 0$ becomes $R_{p+1} = (\theta R_p - pxR_{p-1})/(n-p)$, which is the classical recursion for the n-th symmetric power of the Bessel operator $\theta^2 - x$ (Section 19), in accordance with $\mathbb{E}_{\mathrm{C0E}_n} e^{t\,\mathrm{tr}(U+U^{-1})} = I_0(2t)^n$. Thus $\mathcal{L}$ interpolates between the symmetric power of Bessel's equation ($\beta = 0$, independent eigenvalues), the Bessel–Toeplitz operator of Sections 5–6 ($\beta = 2$), and the even-β operators of Section 16. As $\beta \to 0$ all exponents $-\varepsilon_p$ with $q = 0$ tend to 0, and the degree bound of Theorem 11.3 must then come from the exponent 0 itself; this is done in Theorem 11.7(c) with the sharper dictionary of Lemma 12.3(b).

Proof of Corollary 11.2. By [11, (3.10)], I_n is a constant multiple of ${}_0F_1^{(2/\beta)}(\beta(n-1)+a+b+2;(x)^n) = G^{\beta}_{n,q'}$. Theorem 11.1(a) with $q = q'$ gives an operator of the form [11, (4.1)] with the asserted degrees. In terms of the parameters $B_p^{\mathrm{FW}} = p(a+b+1+\frac{\beta}{2}(2n-p-1))$ of [11, (3.3)] one has $B_p = p(q'+1+\frac{\beta}{2}(n-p)) = B_p^{\mathrm{FW}} + p$, so the integrality statement of Theorem 11.1(a) applies. The operator obtained in [11, §4.1] by eliminating (14.5) downwards from I_n has the leading term $x^n\partial^{n+1}$, annihilates I_n, and, for fixed β, has coefficients that are polynomials in q'. The same holds for L. For $q' \in \mathbb{Z}_{\geq 0}$ both are equal by the uniqueness in Theorem 11.1(d); hence they are equal for all q'. Minimality for $q' \in \mathbb{Z}_{\geq 0}$ is Theorem 11.1(d). □

Remark 15.3. The range $q' > -\frac{1}{2}$ cannot simply be dropped in [11, Prop. 4.1]. For $n = 2$, $\beta = \frac{1}{2}$ and $q' = -\frac{1}{2}$ (for instance $a+b = -\frac{7}{4}$), the elimination gives $L = x^2\partial^3 + \frac{7}{4}x\partial^2 - 4x\partial$, whose coefficient of d^0/dx^0 vanishes identically instead of having degree 0; also L is divisible by x. Here $g_1 = \gamma_1 + 2q' = 0$, which is exactly the case excluded in Lemma 15.1. The minimal order is still

3, as can be proved directly. Put $h(x) = {}_0F_1(\frac{3}{4}; 4x)$ and $R = x\partial_x^2 + \frac{3}{4}\partial_x - 4$. Then $Rh = 0$ and

$$\frac{L}{x} = \partial_x \circ R, \qquad G^{1/2}_{2,-1/2}(x) = \frac{1 + h(x)}{2}.$$

For the second identity, the Jack series gives $G(0) = 1$ and $G'(0) = 2/(3/4) = 8/3$, which are also the initial coefficients of $(1 + h)/2$. The equation $(L/x)G = 0$ determines every later analytic coefficient from these two: if $G = \sum_{j\ge 0} a_j x^j$, then

$$a_{j+2} = \frac{4a_{j+1}}{(j + 2)(j + 7/4)} \qquad (j \ge 0).$$

To prove minimality, termwise integration of the exponential series and the beta integral give, for positive x,

$$h(x) = \frac{\Gamma(3/4)}{\sqrt{\pi}\,\Gamma(1/4)} \int_{-1}^{1} e^{4\sqrt{x}\,s}(1 - s^2)^{-3/4}\, ds.$$

The integral grows faster than every polynomial as $x \to +\infty$. The probability measures obtained by normalizing its integrand concentrate at $s = 1$: for every fixed $\delta > 0$, comparison with a fixed subinterval of $(1 - \delta/2, 1)$ shows that the mass of $[-1, 1 - \delta]$ tends to zero exponentially in $\sqrt{x}$. Differentiation under the integral consequently yields

$$\frac{h'(x)}{h(x)} \sim \frac{2}{\sqrt{x}}.$$

If $A + Bh + Ch' = 0$ with $A, B, C \in \mathbb{C}(x)$ and $C \neq 0$, then $h'/h = -B/C - A/(Ch)$. The second term decays faster than every power of x, whereas a nonzero rational function has an integral leading power of x at infinity. This contradicts the displayed half-integral leading power. If $C = 0$, the growth of h forces $A = B = 0$. Thus $1, h, h'$ are linearly independent over $\mathbb{C}(x)$. Finally

$$h' = 2G', \qquad h = \frac{x}{2}G'' + \frac{3}{8}G', \qquad 1 = 2G - h,$$

so G, G', G'' are linearly independent over $\mathbb{C}(x)$. The displayed third-order annihilator is therefore minimal in order.

15.3 Proof of Theorem 11.3

Let $q = 0$, so that $G(x) = \sum_N \mu_N x^N/(N!)^2$ and $\hat{P}_0 = \theta^2 \prod_{p=1}^{n-1}(\theta + \varepsilon_p)$ with $\varepsilon_p = \frac{\beta}{2}p(n - p) > 0$.

(a) By Lemma 12.3(b), $\sum_k \hat{P}_k(N - k)(N^{\underline{k}})^2\mu_{N-k} = 0$ for all $N \ge 0$. For $N \ge 1$ divide by N^2: the term $k = 0$ becomes $R_0(N)\mu_N$, and $(N^{\underline{k}})^2/N^2 = ((N - 1)^{\underline{k-1}})^2$ for $k \ge 1$. Each R_k has degree $\deg \hat{P}_k + 2k - 2 = n - 1$.

(b) We have $\mathrm{lc}(R_k) = \mathrm{lc}(\hat{P}_k)$, and dividing $\sum_k \mathrm{lc}(\hat{P}_k)\mu^{n+1-2k} = \mu^{\varepsilon} \prod_{j<n/2}(\mu^2 - (n - 2j)^2)$ by μ^{ε} and putting $\lambda = \mu^2$ gives the characteristic polynomial. The formula for R_K follows from $\mathrm{lc}(\hat{P}_K) = [\mu^{\varepsilon}] \prod_b(\mu - n + 2b) = (-1)^K \prod_{j<n/2}(n - 2j)^2$ and Theorem 11.1(b).

(c) By Theorem 11.1(d), L_x is the minimal operator of G; it has leading coefficient x^n and $xL_x = \sum_{k=0}^{K} x^k \hat{P}_k(\theta)$ has span K. Corollary 12.4 shows that every recurrence has order $M \ge K$ and that $\prod_{p=1}^{n-1}(\theta + \varepsilon_p)$ divides $c_M(\theta - M)$; that is, $R_0(N + M)$ divides $c_M(N)$, which is the assertion about the leading coefficient in the backward form, and $\deg c_M \ge n - 1$. If $\mathcal{R} = \sum_{s=0}^{K} c_s(N)S^s$ is the forward form of the recurrence in (a) and $\mathcal{R}' = \sum_{s=0}^{K} c'_s S^s$ another one of order K, then $c_K\mathcal{R}' - c'_K\mathcal{R}$ has order $< K$ and annihilates μ for large N, so it vanishes. The c_s have no common factor, since dividing by one would give a recurrence of degree $< n - 1$. Hence c'_K/c_K is a polynomial and $\mathcal{R}' = (c'_K/c_K)\mathcal{R}$.

(d) Let $q \in \mathbb{Z}_{\geq 1}$. Expanding $e^{t(\operatorname{tr} U + \overline{\operatorname{tr} U})}$ and using the invariance of CβE$_n$ under $U \mapsto e^{i\phi}U$, only the terms $(\operatorname{tr} U)^N \overline{\operatorname{tr} U}^{\,N+qn}$ survive in (11.1), so that $G(x) = c^{-1} \sum_N a_q(N)\, x^N/(N!\,(N+qn)!)$. Comparing coefficients of x^N in $\sum_k x^k \hat{P}_k(\theta) G = 0$ and multiplying by $c\,N!\,(N+qn)!$ gives $\sum_k \hat{P}_k(N-k) N^{\underline{k}} (N+qn)^{\underline{k}}\, a_q(N-k) = 0$. Here $\hat{P}_0(\theta) = \theta(\theta+qn)\prod_{i=1}^{n-1}(\theta+\varepsilon_i)$, since $\varepsilon_0 = 0$ and $\varepsilon_n = qn$. Dividing by $N(N+qn)$ for $N \geq 1$ gives a recurrence of order K with coefficients $\prod_{i=1}^{n-1}(N+\varepsilon_i)$ and $\hat{P}_k(N-k)(N-1)^{\underline{k-1}}(N+qn-1)^{\underline{k-1}}$, all of degree $n-1$.

Conversely, let $\sum_{s=0}^{M} c_s(N) a_q(N+s) = 0$ for $N \geq n_0$. By Lemma 12.3(b) with $r = qn$, G is annihilated by a θ-form of span M with lowest coefficient $\theta^{\underline{M+n_0}}(\theta+qn)^{\underline{M}}\, c_M(\theta - M)$. By Lemma 12.2, $M \geq K$ and $\hat{P}_0$ divides this coefficient. The roots of $\theta^{\underline{M+n_0}}$ are nonnegative, and the roots of $(\theta+qn)^{\underline{M}}$ are $-qn, -qn+1, \ldots, -qn+M-1$. Hence, if I is the multiset of those ε_i, $1 \leq i \leq n-1$, that do not lie in $\{qn-M+1, \ldots, qn\}$, then $\prod_{i\in I}(\theta+\varepsilon_i)$ divides $c_M(\theta - M)$. If no ε_i lies in $\{1, \ldots, qn\}$, then I contains all $n-1$ of them for every M, which gives the degree bound and the divisibility of the leading coefficient. If no ε_i lies in $\{qn-K+1, \ldots, qn\}$, the same holds for $M = K$, and uniqueness of the recurrence of order K follows as in (c). Finally, $\varepsilon_i \in \{1, \ldots, qn\}$ means $\beta = 2(k/i - q)/(n-i)$ for some $k \in \{1, \ldots, qn\}$, which leaves finitely many values. □

Remark 15.4. The hypothesis in Theorem 11.3(d) cannot be dropped. For $n = 3$, $q = 1$, $\beta = 1$ one has $\varepsilon_1 = 2$ and $\varepsilon_2 = 3$, both in $\{1, 2, 3\}$, and the twisted moments satisfy

$$(N+3)\, a_1(N) = (10N+6)\, a_1(N-1) - 9(N-1)\, a_1(N-2),$$

a recurrence of order $K = 2$ and degree $1 < n-1$. Theorem 21.4 proves this identity and its minimality as part of an infinite family. Here $K = 2$, $qn = 3$, and the roots $-\varepsilon_1 = -2$ and $-\varepsilon_2 = -3$ of $\hat{P}_0$ are also roots of the factor $(\theta+3)^{\underline{2}} = (\theta+3)(\theta+2)$ in the proof, so they need not divide the leading coefficient. For $\beta = 2$, the exponent-only test is exceptional, but Theorem 21.2 proves minimum degree $n-1$ for every integer twist.

15.4 The law of $|\operatorname{tr} U|^2$

Corollary 15.5. *Let $\beta > 0$ and $n \geq 2$. The law of $|\operatorname{tr} U|^2$ under CβE$_n$ has a density ρ on $[0, n^2]$, which is annihilated, in the sense of distributions on $\mathbb{R}$, by*

$$\mathcal{M} = \sum_{j=0}^{K} w^j\, R_{K-j}(K-j-1-\theta_w)$$

with R_k as in Theorem 11.3. The operator $\mathcal{M}$ has order $n-1$, and its leading coefficient is

$$(-1)^{n-1} w^{n-1} \prod_{j<n/2} (w - (n-2j)^2).$$

Every nonzero operator with polynomial coefficients annihilating ρ has order at least $n-1$.

Proof. The map $U \mapsto |\operatorname{tr} U|^2$ is a nonconstant real-analytic function of the eigenangles, so the law is absolutely continuous. The recurrence of Theorem 11.3(a) holds for $N \geq 1$ with $\mu_{<0} = 0$, so $\sum_{j=0}^{K} R_{K-j}(N+K)\mu_{N+j} = 0$ for all $N \geq 0$. By Lemma 4.3, a θ-form $\sum_j w^j P_j(\theta)$ annihilates ρ if and only if $\sum_j P_j(-N-j-1)\mu_{N+j} = 0$ for all $N \geq 0$; with $P_j(\theta) = R_{K-j}(K-j-1-\theta)$ this is the recurrence. The order of $\mathcal{M}$ is $\max_j \deg R_{K-j} = n-1$, and the coefficient of ∂_w^{n-1} is $(-1)^{n-1} w^{n-1} \sum_j \operatorname{lc}(R_{K-j}) w^j$, which is given by Theorem 11.3(b). Conversely, an annihilator of order r with polynomial coefficients, multiplied by w^r, is a θ-form $\sum_j w^j P_j(\theta)$ with $\max \deg P_j = r$; the resulting recurrence $\sum_j P_j(-N-j-1)\mu_{N+j} = 0$ is nontrivial, because $\mu_N > 0$ for all N, and has degree at most r. By Theorem 11.3(c), $r \geq n-1$. □

For $\beta = 2$ this is the minimality statement of Theorem 2.10.

16 Even β: the balance theorem with a multiple drift

In this section $\beta = 2m$ with $m \in \mathbb{Z}_{\geq 1}$, and $q \in \mathbb{Z}_{\geq 0}$. We give a second proof of Theorem 11.1(d) and of the t-statements in it for these parameters. It does not use the first-order system of Proposition 14.2, and it replaces visibility by the balance theorem. It also gives information that the first proof does not see: the exact orders at $t = 0$ of the vertex integrals (Theorem 11.6). The only new ingredient compared with Sections 5–6 is the multiple drift inequality, Lemma 16.2.

16.1 The integral as a multilinear form

Let

$$i_k(t) = I_k(2t), \qquad k_k(t) = \int_0^\infty e^{-t(s+s^{-1})}(-s)^k \, \frac{ds}{s} = 2(-1)^k K_k(2t)$$

be the two Bessel sequences of Section 5; both satisfy $h_k' = h_{k-1} + h_{k+1}$ and $t(h_{k-1} - h_{k+1}) = k h_k$, and every h_k is a $\mathbb{Q}(t)$-linear combination of h_0, h_0', where h_0 solves Bessel's equation $Th := th'' + h' - 4th = 0$. On the torus, $|z_p - z_{p'}|^2 = 2 - z_p/z_{p'} - z_{p'}/z_p$, and $\prod_{p<p'}(2 - z_p/z_{p'} - z_{p'}/z_p) = \Delta(z)\Delta(z^{-1})$ with $\Delta(z) = \prod_{p<p'}(z_{p'} - z_p)$. Hence

$$D^{\beta}_{n,q}(t) := \mathbb{E}_{\mathrm{C}\beta\mathrm{E}_n}(\det U)^q e^{t\,\mathrm{tr}(U+U^{-1})} = \frac{1}{Z}\int_{\mathbb{T}^n} \prod_j z_j^q \, e^{t\sum_j (z_j + z_j^{-1})} \big(\Delta(z)\Delta(z^{-1})\big)^m \prod_j \frac{d\varphi_j}{2\pi},$$

with $z_j = e^{i\varphi_j}$ and $Z = \int_{\mathbb{T}^n} (\Delta(z)\Delta(z^{-1}))^m \prod_j d\varphi_j/2\pi$. Since

$$\Delta(z)\Delta(z^{-1}) = \sum_{\sigma,\tau \in S_n} \mathrm{sgn}(\sigma\tau) \prod_j z_j^{\sigma(j)-\tau(j)},$$

$$\big(\Delta(z)\Delta(z^{-1})\big)^m = \sum_{\boldsymbol{\sigma},\boldsymbol{\tau}} \epsilon(\boldsymbol{\sigma},\boldsymbol{\tau}) \prod_j z_j^{\delta_j(\boldsymbol{\sigma},\boldsymbol{\tau})}, \qquad \delta_j = \sum_{r=1}^m (\sigma_r(j) - \tau_r(j)),$$

where $\boldsymbol{\sigma} = (\sigma_1, \dots, \sigma_m)$ and $\boldsymbol{\tau}$ run over S_n^m and $\epsilon = \prod_r \mathrm{sgn}(\sigma_r\tau_r)$. For sequences $h^{(1)}, \dots, h^{(n)}$ put

$$F_q^{(m)}(h^{(1)}, \dots, h^{(n)}) = \frac{1}{n!} \sum_{\boldsymbol{\sigma},\boldsymbol{\tau}} \epsilon(\boldsymbol{\sigma},\boldsymbol{\tau}) \prod_{j=1}^n h^{(j)}_{q+\delta_j(\boldsymbol{\sigma},\boldsymbol{\tau})}.$$

Replacing every σ_r by $\sigma_r \circ \pi$ and every τ_r by $\tau_r \circ \pi$ shows that $F_q^{(m)}$ is symmetric; it is multilinear. Integrating term by term with $\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{2t\cos\varphi} e^{ik\varphi} d\varphi = i_k(t)$ gives $D^{\beta}_{n,q} = \frac{n!}{Z} F_q^{(m)}(i, \dots, i)$. For $0 \leq b \leq n$ define the *vertex integrals*

$$\begin{aligned} \Psi_b^{(m,q)} &= F_q^{(m)}(\underbrace{k, \dots, k}_{b}, \underbrace{i, \dots, i}_{n-b}) \\ &= \frac{1}{n!}\int_{(0,\infty)^b \times (-\pi,\pi)^{n-b}} e^{tf(z)} \prod_j z_j^q \big(\Delta(z)\Delta(z^{-1})\big)^m \prod_{j \leq b} \frac{ds_j}{s_j} \prod_{j>b} \frac{d\varphi_j}{2\pi}. \end{aligned}$$

with $z_j = -s_j$ for $j \leq b$, $z_j = e^{i\varphi_j}$ for $j > b$ and $f(z) = \sum_j (z_j + z_j^{-1})$; the integral formula follows from the integral representations of i_k and k_k by Fubini, as in Lemma 5.2. All $\Psi_b^{(m,q)}$ are real on $(0, \infty)$.

Proposition 16.1. *Let L be the minimal operator of $D^{\beta}_{n,q}$ over $\mathbb{C}(t)$, normalized to be primitive.*

(a) $\mathrm{Sol}(L) = V := \mathrm{span}\{\Psi_0^{(m,q)}, \dots, \Psi_n^{(m,q)}\}$. *Every element of V continues analytically along all paths in $\mathbb{C} \setminus \{0\}$, $V \subset \mathcal{N}$, and L is Fuchsian at 0.*

(b) *For $m' \ge 0$ and $t \to +\infty$, $\partial^{m'}\Psi_b^{(m,q)} = C_b e^{\lambda_b t} t^{-\gamma_b}(\lambda_b^{m'} + O(t^{-1}))$, with an asymptotic expansion in integral powers of t^{-1} that may be differentiated termwise; here γ_b is given by (14.7) with $\beta = 2m$, and*

$$C_b = \frac{(-1)^{qb+m\binom{b}{2}}\, 4^{mb(n-b)}}{n!\,(2\pi)^{n-b}}\, \mathcal{M}_{n-b}^{(m)}\, \mathcal{M}_b^{(m)} \neq 0, \qquad \mathcal{M}_N^{(m)} = \int_{\mathbb{R}^N} e^{-|x|^2} \Delta(x)^{2m}\, dx > 0.$$

In particular $\operatorname{ord} L = n+1$.

(c) $D_{n,q}^{\beta}(-t) = (-1)^{qn} D_{n,q}^{\beta}(t)$.

Proof. (a) $D_{n,q}^{\beta}$ is a polynomial in i_0, i_0' with coefficients in $\mathbb{Q}(t)$. The differential Galois group of Bessel's equation is SL_2 (Kolchin), acting transitively on nonzero solutions, so the span of the Galois orbit of $D_{n,q}^{\beta}$ is spanned by the values $F_q^{(m)}(g_\bullet, \dots, g_\bullet)$, $g \in \mathrm{Sol}(T)$; by polarization and multilinearity this span is V. By Lemma 3.7 it is $\mathrm{Sol}(L)$. Every element of V is a polynomial in i_0, i_0', k_0, k_0' with coefficients in $\mathbb{Q}[t, t^{-1}]$, because the recurrences for h_k only divide by t; i_0 is entire and $k_0 = -2(\log t) i_0 + (\text{entire})$, which gives the remaining assertions.

(b) This is the proof of Lemma 5.4 with the following changes. In the global coordinates $x_j = 2\sin(\varphi_j/2)$ and $y_j = s_j^{1/2} - s_j^{-1/2}$ the phase is $f = \lambda_b - |x|^2 - |y|^2$. A circle–circle factor $2 - z_p/z_{p'} - z_{p'}/z_p$ is $(x_p - x_{p'})^2$ times an analytic function equal to 1 at the vertex, an axis–axis factor is $-(y_p - y_{p'})^2$ times such a function, and a mixed factor equals 4 at the vertex; the twist $\prod_j z_j^q$ equals $(-1)^{qb}$ there. After raising to the m-th power, the amplitude is $\Delta(x)^{2m}\Delta(y)^{2m} H(x,y)$ with H analytic and $H(0) = C_b(\mathcal{M}_{n-b}^{(m)}\mathcal{M}_b^{(m)})^{-1}$. The weight $\Delta(x)^{2m}\Delta(y)^{2m}$ is homogeneous of degree $2\gamma_b - n$, and Laplace's method gives the expansion as in Sections 5–6. The rates are distinct and $C_b \neq 0$, so $\dim V = n+1$.

(c) The substitution $z \mapsto -z$ preserves $\Delta(z)\Delta(z^{-1})$ and the Haar measure, and multiplies $\prod_j z_j^q$ by $(-1)^{qn}$. □

16.2 The multiple drift inequality

Lemma 16.2. *Let $m \ge 1$, $q \ge 0$, and for $0 \le i \le n$ put $f_m(i) = 2mi(n-i) - q(n-2i)$; let $f_{m,(0)} \le \cdots \le f_{m,(n)}$ be these numbers in increasing order. Let $\sigma_1, \tau_1, \dots, \sigma_m, \tau_m \in S_n$, put $d_s = q + \sum_{r=1}^{m} (\sigma_r(s) - \tau_r(s))$, and let $S \subseteq [n]$ with $|S| = b$. Then*

$$E := \sum_{s \in S} |d_s| - \sum_{s \notin S} |d_s| \le f_{m,(b)}, \qquad \text{and} \qquad \sum_{b=0}^{n} f_{m,(b)} = \frac{m(n-1)n(n+1)}{3}.$$

Proof. Let $S_+ = \{s \in S : d_s > 0\}$, $j = |S_+|$, $A = \sum_{s \in S_+} d_s$ and $B = -\sum_{s \in S \setminus S_+} d_s \ge 0$. Since $\sum_{s=1}^{n} d_s = nq$, we have $\sum_{s \notin S} |d_s| \ge |nq - A + B|$, hence $E \le A + B - |nq - A + B| = \min(2A - nq,\ 2B + nq)$. For $T \subseteq [n]$ with $|T| = k$ and $\sigma, \tau \in S_n$, both $\sum_{s \in T}(\sigma(s) - \tau(s))$ and its negative are at most the sum of the k largest elements of $[n]$ minus the sum of the k smallest, which is $k(n-k)$. Summing over r gives $A \le j\big(q + m(n-j)\big)$ and $B \le (b-j)\big(m(n-b+j) - q\big)$. Hence $2A - nq \le f_m(j)$ and $2B + nq \le f_m(j')$ with $j' = j + n - b$, so $E \le \min(f_m(j), f_m(j'))$. Since f_m is concave, $f_m(i) \ge \min(f_m(j), f_m(j'))$ for the $n - b + 1$ integers $i \in [j, j']$; so at most b values of f_m are smaller, that is, $f_{m,(b)} \ge \min(f_m(j), f_m(j')) \ge E$. Finally $\sum_i f_m(i) = 2m \sum_i i(n-i)$, because $\sum_i (n - 2i) = 0$. □

Lemma 16.3. *For $0 < t \le 1$, $|\Psi_b^{(m,q)}(t)| \le C\, t^{-f_{m,(b)}}(1 + |\log t|)^b$. Hence $\operatorname{ord} \Psi_b^{(m,q)} \ge -f_{m,(b)}$.*

Proof. By definition, $\Psi_b^{(m,q)} = \frac{1}{n!} \sum \epsilon(\boldsymbol{\sigma}, \boldsymbol{\tau}) \prod_{j \le b} k_{d_j} \prod_{j > b} i_{d_j}$ with $d_j = q + \delta_j = q + \sum_r (\sigma_r(j) - \tau_r(j))$. Use $|i_k(t)| \le e\, t^{|k|}$ and $|k_k(t)| \le C\, t^{-|k|}(1 + |\log t|)$ on $(0,1]$ (proof of Lemma 5.7), and Lemma 16.2 with $S = \{1, \dots, b\}$. □

16.3 The balance

Theorem 16.4. *Let $\beta = 2m$, $q \in \mathbb{Z}_{\geq 0}$, $N = n+1$, write the minimal operator of $D^{\beta}_{n,q}$ as $L = \sum_{k=0}^{N} p_k \partial^k$ (primitive), and put $e_i = q(n-2i) - \beta i(n-i)$, $0 \leq i \leq n$.*

(i) L has order $n+1$ and no singular point in $\mathbb{C} \setminus \{0\}$.

(ii) The exponents of L at $t = 0$ are $e_0, \dots, e_n$, and $\operatorname{ord} \Psi_b^{(m,q)} = -f_{m,(b)}$ for every b.

(iii) After scaling, $p_N = t^A$ with $A = n$ if $0 \in \{e_i\}$ and $A = n+1$ otherwise; $\deg p_k \leq A$, $\operatorname{ord}_0 p_k \geq k + A - N$, and $\sum_k [t^A] p_k \lambda^k = \chi(\lambda) = \prod_{b=0}^{n} (\lambda - \lambda_b)$.

(iv) $t^{N-A} L = \sum_{k=0}^{K} t^{2k} Q_{2k}(\theta)$, $K = \lceil n/2 \rceil$, with $Q_0(\theta) = \prod_i (\theta - e_i)$, $\deg Q_{2k} = n + 1 - 2k$, $\operatorname{lc}(Q_{2k}) = [\lambda^{n+1-2k}]\chi$, and (12.3) holds with $g_b = \gamma_b$.

Proof. By Proposition 16.1 and Lemma 16.3, the basis $\Psi_b^{(m,q)}$ satisfies (B1)–(B4) of Theorem 12.5 with $\beta_b = f_{m,(b)}$. Its balance defect is

$$\begin{aligned} \Delta &= \binom{n+1}{2} - \sum_b \gamma_b + \sum_b f_{m,(b)} \\ &= \frac{n(n+1)}{2} - \Big(\frac{n(n+1)}{2} + \frac{m(n+1)n(n-1)}{3} \Big) + \frac{m(n-1)n(n+1)}{3} = 0, \end{aligned}$$

where we used $\sum_b \big((n-b)(n-b-1) + b(b-1)\big) = 2\sum_b b(b-1) = \frac{2}{3}(n+1)n(n-1)$. Theorem 12.5 gives (i) and (ii), since the multiset $\{-f_{m,(b)}\}$ equals $\{-f_m(i)\} = \{e_i\}$. All e_i are congruent to qn modulo 2, so 0 and 1 are not both exponents, and Lemma 12.6 gives A and $\operatorname{ord}_0 p_k \geq k + A - N$. The top degree fact (iii) of Section 12, applied to the asymptotics of Proposition 16.1(b), gives $\deg p_k \leq A$ and the formula for χ.

For (iv), $t^{N-A} L = \sum_k r_k(t) \theta^{\underline{k}}$ with $r_k = t^{N-A-k} p_k \in \mathbb{C}[t]$ of degree at most $N-k$, so $t^{N-A} L = \sum_j t^j Q_j(\theta)$ with $\deg Q_j \leq N - j$ and $Q_0 = \sum_k r_k(0) \theta^{\underline{k}}$, the indicial polynomial. By Proposition 16.1(c), the operator obtained from L by $t \mapsto -t$ is primitive, annihilates $D^{\beta}_{n,q}$ and has order N, so it is a constant multiple of L; comparing the terms with ∂^N, $t^{N-A} L$ is invariant under $t \mapsto -t$, and $Q_j = 0$ for odd j. Comparing coefficients of $t^N \partial^k$ gives $[t^A] p_k = [\theta^k] Q_{N-k}$; as $\chi = \lambda^{[n \text{ even}]} \prod_{b < n/2} (\lambda^2 - \lambda_b^2)$ has nonzero coefficients at all λ^{N-2k}, $\deg Q_{2k} = N - 2k$. Applying L termwise to the asymptotic expansion of $\Psi_b^{(m,q)}$ gives the expansion of $L\Psi_b^{(m,q)} = 0$, so $e^{\lambda_b t} t^{-\gamma_b}(1 + O(t^{-1}))$ is a formal solution for every b, and Lemma 12.7 gives (12.3). □

Remark 16.5. Theorem 16.4(ii) says that the crude Toeplitz bound of Lemma 16.3 is attained, in the power of t, by every single vertex integral, and consequently that the bound of Lemma 16.2 is attained for every b by some $(\boldsymbol{\sigma}, \boldsymbol{\tau}, S)$. A direct proof would require controlling the cancellations among the $(n!)^{2m}$ terms of the expansion; the balance avoids this. For $m = 1$ this is Theorem 6.5.

17 Jacobi ensembles: proof of Theorem 11.4

Let $\beta > 0$, $a, b > -1$, $n \geq 1$, and let $\mathcal{L}^J = \sum_{k=0}^{n} x^k P_k^J(\theta)$ be as in Proposition 14.6; put $\hat{P}_k^J = n!\, P_k^J$. Recall

$$\varepsilon_p^J = p\Big(a + b + 1 + \frac{\beta}{2}(2n - p - 1)\Big), \qquad \gamma_k^J = (n-k)(a+1) + k(b+1) + \beta\Big(\binom{n-k}{2} + \binom{k}{2}\Big).$$

We write $(\star)$ for the condition $a + b + 1 + \frac{\beta}{2}(n-1) > 0$. Since $\varepsilon_p^J / p \geq a + b + 1 + \frac{\beta}{2}(n-1)$ for $1 \leq p \leq n$, under $(\star)$ all ε_p^J with $p \geq 1$ are positive. Condition $(\star)$ holds whenever $a + b \geq -1$ and $n \geq 2$, and for $n = 1$ whenever $a + b > -1$.

Proof of Theorem 11.4. (a) By Proposition 14.6, $n!\,\mathcal{L}^J = x\sum_m q_m\partial^m$ annihilates H, where, exactly as in Section 15, $q_m = x^{m-1}\sum_k x^k \sum_i [\theta^i]\hat{P}_k^J S(i,m)$ for $m \ge 1$ and $q_0 = \sum_{k\ge1} x^{k-1}\hat{P}_k^J(0)$, using $\hat{P}_0^J(0) = 0$. Since $\deg \hat{P}_k^J = n+1-k$, the coefficient of ∂^m $(m \ge 1)$ is x^{m-1} times a polynomial of degree at most $n+1-m$, whose top coefficient $\mathrm{lc}(\hat{P}_{n+1-m}^J)$ is a positive elementary symmetric function of $1,\dots,n$. Hence $x^{m-1} \mid q_m$, $\deg q_m = n$ for $1 \le m \le n+1$, and $q_{n+1} = x^n$. The top coefficient of q_0 is $\hat{P}_n^J(0) = n!\,\gamma_0^J > 0$, so $\deg q_0 = n-1$. Under $(\star)$, $q_1(0) = \hat{P}_0^J(1) = \prod_p(1+\varepsilon_p^J) > 0$, so $L^J = \sum_m q_m\partial^m$ is primitive. It is Fuchsian at 0 with indicial polynomial $\hat{P}_0^J$, so its exponents are $-\varepsilon_p^J$, and its formal solutions at infinity are those of Proposition 14.6.

For minimality, $F(t) = H(-t) = \langle \rho^J, e^{tw}\rangle$. By Proposition 13.7, ρ^J is not smooth at the $n+1$ points $0,1,\dots,n$, and Corollary 13.2(a) shows that no nonzero operator of order at most n annihilates F, hence none annihilates H. So the minimal operator of H has order $n+1$; it equals $x^{-c}L^J$ for some $c \ge 0$, and $c = 0$ under $(\star)$, since L^J is then primitive. In that case, as in the proof of Theorem 11.1(d), every operator of order $n+1$ with polynomial coefficients annihilating H is a polynomial multiple of L^J. Without $(\star)$, L^J can be divisible by x: for $n = 2$ one has $q_1(0) = (a+b+\beta+2)(2a+2b+\beta+3)$, $q_0(0) = (2a+\beta+2)(2a+2b+\beta+3)$ and $q_m(0) = 0$ for $m \ge 2$, and $2a+2b+\beta+3 = 0$ is possible with $a, b > -1$.

(b) Since $H(-s) = \sum_N m_N s^N/N!$, the θ-form $\sum_k (-1)^k s^k \hat{P}_k^J(\theta_s)$ annihilates $\sum_N m_N s^N/N!$. By Lemma 12.3(a), $\sum_k (-1)^k \hat{P}_k^J(N-k)N^{\underline{k}}\, m_{N-k} = 0$ for all $N \ge 0$. For $N \ge 1$ divide by N; since $\hat{P}_0^J(N) = N\prod_{p\ge1}(N+\varepsilon_p^J)$ and $N^{\underline{k}}/N = (N-1)^{\underline{k-1}}$, this is the recurrence of (b), and $\deg c_k = \deg \hat{P}_k^J + k - 1 = n$ for all k. The formula for c_n follows from $\hat{P}_n^J = n!(\theta + \gamma_0^J)$. By Proposition 14.6, $\mathrm{lc}(c_k) = (-1)^k \mathrm{lc}(\hat{P}_k^J) = (-1)^k e_k(1,\dots,n)$, where e_k is the elementary symmetric function, so the characteristic polynomial $\sum_k \mathrm{lc}(c_k)\lambda^{n-k}$ is $\prod_{k=1}^n(\lambda - k)$.

Let a recurrence of order M hold for large N. By Lemma 12.3(a) the exponential generating function is annihilated by a θ-form of span M in s; hence H is annihilated by a θ-form of span M in x. The primitive minimal operator is $x^{-c}L^J$ for some $c \ge 0$, and its leading coefficient is a nonzero monomial. Multiplication by a Laurent monomial gives the equivalent form $xL^J = \sum_{k=0}^n x^k \hat{P}_k^J(\theta)$, of span n, to which the transfer lemma applies. By Corollary 12.4, $M \ge n$, and the factors $\theta + \varepsilon_p^J$ with $\varepsilon_p^J \notin \mathbb{Z}_{\le 0}$ divide $c_M(\theta - M)$. Under $(\star)$ all n factors with $p \ge 1$ qualify, so $\deg c_M \ge n$ and $\prod_{p\ge1}(N+\varepsilon_p^J)$ divides the leading coefficient in the backward form. Uniqueness follows as in the proof of Theorem 11.3(c).

(c) As in Corollary 15.5, Lemma 4.3 translates the recurrence of (b), shifted to hold for all $N \ge 0$, into the annihilator $\mathcal{M}^J = \sum_{j=0}^n w^j c_{n-j}(n-j-1-\theta)$ of ρ^J. Its order is $\max_j \deg c_{n-j} = n$, and the coefficient of ∂_w^n is $(-1)^n w^n \sum_k \mathrm{lc}(c_k) w^{n-k} = (-1)^n w^n \prod_{k=1}^n (w-k)$. An annihilator of order r yields a nontrivial recurrence of degree at most r, since $m_N > 0$; under $(\star)$, (b) gives $r \ge n$. □

Remark 17.1. For $\beta = 1$ the trace $\sum_j s_j$ is Pillai's V statistic of multivariate analysis, and for this case the system (14.9) goes back to work of Davis; see [10, Remark 2.6]. Forrester and Kumar [10, Remark 3.8] note that eliminating their system for the density leads to a linear differential equation of order N (after using an integrated form of the equation with $p = 0$), and they write it out for $N = 2$.

Remark 17.2. Assume $(\star)$. Let $\mathcal{A}$ be an operator of order r with polynomial coefficients that annihilates ρ^J on $\mathbb{R}\setminus\{0,1,\dots,n\}$, for instance the equation of [10, Remark 3.8]. Then $\mathcal{A}\rho^J$ is a distribution supported in $\{0,\dots,n\}$, hence a finite combination of derivatives of Dirac masses, and $\prod_{k=0}^n (w-k)^J \mathcal{A}$ annihilates ρ^J in the sense of distributions for large J. This operator has order r, so $r \ge n$ by Theorem 11.4(c). If $\mathcal{A}$ has order n, the same argument shows that $\mathcal{A}$ and $\mathcal{M}^J$ differ by a left factor in $\mathbb{C}(w)$, since otherwise a suitable combination would have order less than n. (They are not equal: for $N = 2$ the leading coefficient in [10, (3.35)] is $t(t-1)(t-2)$, while that of $\mathcal{M}^J$ is $w^2(w-1)(w-2)$.)

18 Classical groups and Weyl chamber walks: proof of Theorem 11.5

18.1 Shifted Jacobi traces

Let $\beta > 0$, $a, b > -1$ satisfy $(\star)$, let $n \geq 1$, $\kappa \neq 0$ and $c \in \mathbb{R}$, and consider the moments $M_N = \mathbb{E}(\kappa \sum_j s_j + c)^N$ in the β-Jacobi ensemble. Their exponential generating function is $F(t) = e^{ct} H(-\kappa t)$, the Laplace transform of the law of $\kappa \sum s_j + c$, whose non-smooth points are $c + \kappa k$, $0 \leq k \leq n$ (Proposition 13.7).

Lemma 18.1. *Let L_F be the minimal operator of F. Then L_F has order $n+1$, leading coefficient a constant times t^n, and*

$$tL_F = \sum_{k=0}^{n+1} t^k \tilde{P}_k(\theta), \qquad \deg \tilde{P}_k \leq n+1-k, \qquad \tilde{P}_0 = \text{const} \cdot \prod_{p=0}^{n} (\theta + \varepsilon_p^J),$$

with $\sum_k [\theta^{n+1-k}] \tilde{P}_k \, \lambda^{n+1-k} = \text{const} \cdot \prod_{k=0}^{n} (\lambda - c - \kappa k)$ and $\tilde{P}_{n+1} = \text{const} \cdot \prod_{k=0}^{n} (c + \kappa k)$.

Proof. Write $L^J = \sum_m q_m(x) \partial_x^m$ as in Theorem 11.4(a): $q_{n+1} = x^n$, $\deg q_m \leq n$, and $\operatorname{ord}_0 q_m \geq m - 1$ for $m \geq 1$. Substituting $x = -\kappa t$ and conjugating by e^{ct}, which replaces ∂_t by $\partial_t - c$, gives the operator $L_F = \sum_j \tilde{q}_j(t) \partial_t^j$ with $\tilde{q}_j = \sum_{m \geq j} \binom{m}{j} (-c)^{m-j} (-\kappa)^{-m} q_m(-\kappa t)$. It annihilates F and is minimal because L^J is. Its coefficients have degree at most n, $\tilde{q}_{n+1}$ is a constant times t^n, and $\operatorname{ord}_0 \tilde{q}_j \geq j - 1$ for $j \geq 1$. Hence $t \tilde{q}_j \partial^j = t^{1-j} \tilde{q}_j \, \theta^{\underline{j}}$ with $t^{1-j} \tilde{q}_j$ a polynomial of degree at most $n + 1 - j$, which gives the θ-form with $\deg \tilde{P}_k \leq n + 1 - k$. Since $\deg \tilde{P}_0 = n + 1$ is the largest degree, $\tilde{P}_0$ is the indicial polynomial at 0, which is unchanged by the substitution and the conjugation. The top coefficients are $[\theta^{n+1-k}] \tilde{P}_k = [t^n] \tilde{q}_{n+1-k}$, so their generating polynomial is $\chi_F(\lambda) = \sum_j [t^n] \tilde{q}_j \lambda^j$. From the formula for $\tilde{q}_j$, $\chi_F(\lambda) = (-\kappa)^n \chi^J((\lambda - c)/(-\kappa))$ with $\chi^J(\mu) = \sum_m [x^n] q_m \, \mu^m = \sum_k \operatorname{lc}(\hat{P}_k^J) \mu^{n+1-k} = \mu \prod_{k=1}^{n} (\mu + k)$ by Proposition 14.6 (note $[x^n] q_0 = 0$). Hence χ_F is a constant times $\prod_{k=0}^{n} (\lambda - c - \kappa k)$. Finally $\tilde{P}_{n+1} = [t^n] \tilde{q}_0 = \chi_F(0)$. □

Proposition 18.2. *Assume $(\star)$ and $c + \kappa k \neq 0$ for $0 \leq k \leq n$. Then (M_N) satisfies*

$$\sum_{k=0}^{n+1} \tilde{c}_k(N) M_{N-k} = 0 \quad (N \geq 1),$$
$$\tilde{c}_k(N) = \tilde{P}_k(N-k) \, (N-1)^{\underline{k-1}} \quad (k \geq 1), \qquad \tilde{c}_0(N) = \tilde{P}_0(N)/N.$$

a recurrence of order $n+1$ and degree n, with leading coefficient a constant times $\prod_{p=1}^{n} (N + \varepsilon_p^J)$, last coefficient a constant times $(N-1) \cdots (N-n)$, and characteristic polynomial $\prod_{k=0}^{n} (\lambda - c - \kappa k)$. Every recurrence satisfied by (M_N) has order at least $n+1$ and degree at least n, and its leading coefficient in the backward form is divisible by $\prod_{p=1}^{n} (N + \varepsilon_p^J)$. Every recurrence of order $n+1$ is a polynomial multiple of this one.

Proof. By Lemma 18.1, tL_F has span exactly $n+1$, because $\tilde{P}_{n+1} \neq 0$. Lemma 12.3(a) gives the recurrence, and dividing by N uses $\tilde{P}_0(0) = 0$; the degree is at most $\max_k (n + 1 - k + k - 1) = n$, with equality for $k = 0$. The last coefficient is $\tilde{P}_{n+1}(N-1)^{\underline{n}}$. Lemma 12.2 and Corollary 12.4, with the n negative roots $-\varepsilon_p^J$ of $\tilde{P}_0$, give the lower bounds, and uniqueness follows as before. □

Proposition 18.3. *Assume $(\star)$, $a = b$, $\kappa \neq 0$ and $c = -\kappa n/2$, so that the law of $\kappa \sum s_j + c$ is symmetric and $M_{2N+1} = 0$. Put $M'_N = M_{2N}$ and $K = \lceil n/2 \rceil$. Then $\tilde{P}_k = 0$ for odd k, tL_F has span $2K$, and*

$$\sum_{k=0}^{K} \hat{c}_k(N) M'_{N-k} = 0 \quad (N \geq 1), \qquad \hat{c}_0(N) = \frac{\tilde{P}_0(2N)}{2N},$$

with $\hat{c}_k(N) = \tilde{P}_{2k}(2N-2k)\,(2N-1)^{\underline{2k-1}}$ *for* $k \ge 1$*, a recurrence of order* K *and degree* n *with leading coefficient a constant times* $\prod_{p=1}^{n}(2N+\varepsilon_p^J)$ *and characteristic polynomial* $\prod_{j<n/2}(\lambda - \kappa^2(n/2-j)^2)$*. Every recurrence satisfied by* (M'_N) *has order at least* K *and degree at least* n*, with leading coefficient divisible by* $\prod_{p=1}^{n}(2N+\varepsilon_p^J)$*; every recurrence of order* K *is a polynomial multiple of this one.*

Proof. Since F is even, the operator obtained from L_F by $t \mapsto -t$ is a constant multiple of L_F, and comparing the terms with ∂^{n+1} shows that tL_F is invariant; hence $\tilde{P}_k = 0$ for odd k. The rates $c+\kappa k = \kappa(k-n/2)$ are symmetric, and $\prod_k(\lambda - \kappa(k-n/2)) = \lambda^{[n \text{ even}]}\prod_{j<n/2}(\lambda^2 - \kappa^2(n/2-j)^2)$ has nonzero coefficients at all λ^{n+1-2k}, $0 \le k \le K$; so the top coefficients of $\tilde{P}_{2k}$ are nonzero for $k \le K$ and the span is $2K$. Comparing coefficients of $t^{2N}/(2N)!$ in $tL_F F = 0$ (Lemma 12.3(a)) and dividing by $2N$ gives the recurrence; its degree is at most $\max_k(n+1-2k+2k-1) = n$.

Conversely, suppose that $\sum_{s=0}^{M} c_s(N) M'_{N+s} = 0$ for $N \ge n_0$. Then the full sequence satisfies

$$\sum_s c_s(j/2)\, M_{j+2s} = 0 \qquad (j \ge 2n_0),$$

because odd moments vanish; it has order $2M$ and leading coefficient $c_M(j/2)$. By Lemma 12.3(a), F is annihilated by a θ-form of span $2M$ whose lowest coefficient is $\theta^{\underline{2M+2n_0}} c_M((\theta-2M)/2)$. Lemma 12.2 gives $2M \ge 2K$ and shows that $\prod_{p\ge1}(\theta+\varepsilon_p^J)$ divides $c_M((\theta-2M)/2)$, that is, $\prod_p(2N+2M+\varepsilon_p^J)$ divides $c_M(N)$. Uniqueness follows as before. □

For $a = b = \frac12$, $\beta = 2$, $\kappa = 4$, $c = -2n$ this is the symplectic case $\mathbb{E}_{\mathrm{Sp}(2n)}(\operatorname{tr} U)^{2N}$, and Proposition 18.3 recovers the minimal recurrence for noncrossing matchings in Theorem 2.9, with leading coefficient $\prod_p(2N + p(2n+1-p))$.

18.2 Orthogonal groups

Let $u_j = 2\cos\varphi_j$ and $s_j = (2+u_j)/4 = \cos^2(\varphi_j/2) \in [0,1]$. By Weyl's integration formula, the eigenangles $\pm\varphi_1, \dots, \pm\varphi_\ell$ of a Haar-distributed element of $\mathrm{SO}(2\ell+1)$ have density proportional to $\prod_{i<j}(\cos\varphi_i - \cos\varphi_j)^2 \prod_j \sin^2(\varphi_j/2)$, and those of $\mathrm{SO}(2\ell)$ have density proportional to $\prod_{i<j}(\cos\varphi_i - \cos\varphi_j)^2$. Since $d\varphi = -ds/\sqrt{s(1-s)}$ and $\sin^2(\varphi/2) = 1-s$, the variables s_j form a $\beta = 2$ Jacobi ensemble with

$$(a,b) = (-\tfrac12, \tfrac12) \text{ for } \mathrm{SO}(2\ell+1), \qquad (a,b) = (-\tfrac12, -\tfrac12) \text{ for } \mathrm{SO}(2\ell),$$

and $\operatorname{tr} U = 1 + \sum_j u_j = 4\sum_j s_j + 1 - 2\ell$, respectively $\operatorname{tr} U = 4\sum_j s_j - 2\ell$. In both cases $n = \ell$, and $\varepsilon_p^J = p(2\ell - p)$, respectively $p(2\ell-1-p)$. Condition $(\star)$ holds ($a+b+1 = 1$, respectively $a+b+1+(\ell-1) = \ell - 1 > 0$ for $\ell \ge 2$).

Proof of Theorem 11.5. For $\mathrm{SO}(2\ell+1)$ apply Proposition 18.2 with $\kappa = 4$, $c = 1-2\ell$; the rates $c + 4k = 1 - 2\ell + 4k$ are odd, hence nonzero, and equal $2\ell + 1 - 4j$ with $j = \ell - k$. For $\mathrm{SO}(2\ell)$, $\ell \ge 2$, apply Proposition 18.3 with $\kappa = 4$, $c = -2\ell$, so that $\kappa^2(n/2-j)^2 = 4(\ell - 2j)^2$. The identification of $w_B(N) = \mathbb{E}_{\mathrm{SO}(2\ell+1)}(\operatorname{tr} U)^N$ and $w_D(N) = \mathbb{E}_{\mathrm{SO}(2\ell)}(\operatorname{tr} U)^{2N}$ with numbers of sequences of dominant weights follows from $\mathbb{E}_G \chi_V(U)^N = \dim(V^{\otimes N})^G$ and the Pieri rules for tensoring with the defining representation V [56]: $V_\lambda \otimes V = \bigoplus V_{\lambda \pm e_i}$ over dominant $\lambda \pm e_i$, plus V_λ itself for $\mathrm{SO}(2\ell+1)$ when $\lambda_\ell > 0$. The type B path model here retains the restriction on zero steps; in rank one this gives Riordan, rather than Motzkin, numbers. □

group	minimal recurrence, coefficients of $w(N), w(N-1), \dots$
SO(3)	$N+1, \quad -2(N-1), \quad -3(N-1)$
SO(5)	$(N+3)(N+4), \quad -3(N-1)(N+2), \quad -(N-1)(13N+4), \quad 15(N-2)(N-1)$
SO(7)	$(N+5)(N+8)(N+9), \quad -4(N-1)(N+4)(N+7), \quad -2(N-1)(17N^2+109N+99),$ $4(N-2)(N-1)(19N+13), \quad 105(N-3)(N-2)(N-1)$
SO(4)	$(N+1)^2, \quad -4(2N-1)^2$
SO(6)	$(N+2)(N+3)^2, \quad -4N(2N-1)(5N+7), \quad 36(N-1)(2N-3)(2N-1)$
SO(8)	$(N+3)(N+5)(N+6)^2, \quad -4(2N-1)(10N^3+77N^2+157N+50),$ $256(N-1)(N+1)(2N-3)(2N-1)$

Table 4: Minimal recurrences of Theorem 11.5, valid for $N \geq 1$ with $w(m) = 0$ for $m < 0$. For $\mathrm{SO}(2\ell+1)$ the sequence is $w_B(N)$; for $\mathrm{SO}(2\ell)$ it is $w_D(N) = \mathbb{E}(\operatorname{tr} U)^{2N}$. They were derived from the θ-forms and checked against the walk counts obtained by dynamic programming over dominant weights (Appendix C). SO(3) gives the Riordan numbers, SO(4) the squared Catalan numbers; the SO(5) recurrence is also recorded in [60, A095922].

19 Bessel moments: proof of Theorem 11.7

Let $n \geq 1$. The modified Bessel equation of order 0 reads $\theta^2 y = t^2 y$, with solutions $I_0(t)$, $K_0(t)$. Define $S_0 = 1$, $S_{j+1} = (\theta S_j - j\, t^2 S_{j-1})/(n-j)$ for $0 \leq j < n$ (with $S_{-1} = 0$), and

$$\mathcal{B}_n = \theta S_n - n\, t^2 S_{n-1} = \sum_{k=0}^{K} t^{2k} P_k^{(n)}(\theta), \qquad K = \lceil n/2 \rceil .$$

Lemma 19.1. *$\mathcal{B}_n$ annihilates y^n for every solution y of $\theta^2 y = t^2 y$, and its solution space is* $\operatorname{span}\{I_0^i K_0^{n-i} : 0 \leq i \leq n\}$. *Moreover $P_0^{(n)} = \theta^{n+1}/n!$, $\deg P_k^{(n)} = n+1-2k$, and*

$$\sum_k [\theta^{n+1-2k}] P_k^{(n)} \mu^{n+1-2k} = \frac{1}{n!} \prod_{b=0}^{n} (\mu - n + 2b).$$

For even n, $P_K^{(n)}(\theta) = \mathrm{lc}(P_K^{(n)})(\theta + n/2)$. The minimal operator of $I_0^a K_0^b$, for any $a, b \geq 0$ with $a+b = n$, is $\mathcal{B}_n$.

Proof. For a solution y put $Y_j = y^{n-j}(\theta y)^j$. Then $\theta Y_j = (n-j) Y_{j+1} + j\, t^2 Y_{j-1}$, so $Y_j = S_j y^n$ by induction and $\mathcal{B}_n y^n = 0$. The products $I_0^i K_0^{n-i}$ have the distinct exponential rates $2i - n$ at $+\infty$, so they are independent and span the $(n+1)$-dimensional solution space. The statements on $P_k^{(n)}$ follow from the recursion exactly as in Proposition 14.3, with $x = t^2$ of weight 2 and θ of weight 1 (Lemma 14.1(b)); $P_0^{(n)} = \theta^{n+1}/n!$ because $S_{j+1,0} = \theta S_{j,0}/(n-j)$. For even n the solution $(I_0 K_0)^{n/2} = (2t)^{-n/2}(1 + O(t^{-1}))$ has rate 0, which forces $P_K^{(n)}(-n/2) = 0$ as in the proof of Theorem 11.1. Finally, by Kolchin's theorem the differential Galois group of the Bessel equation acts on its solution space as SL_2 [17, 23]; the span of the orbit of $I_0^a K_0^b$ is a nonzero SL_2-stable subspace of the irreducible representation Sym^n, hence all of it, and Lemma 3.7 gives the last assertion. □

Lemma 19.2 (Moment uniqueness with an exponential tail)**.** *Let $h \in L^1(0, \infty)$ satisfy $\int_0^\infty e^{\eta t} |h(t)|\, dt < \infty$ for some $\eta > 0$. If all its nonnegative integer moments vanish, or if just all its nonnegative even moments vanish, then $h = 0$ almost everywhere.*

Proof. For all integer moments, the function $F(z) = \int_0^\infty e^{zt} h(t)\,dt$ is holomorphic on $\operatorname{Re} z < \eta$ and has all derivatives zero at zero. The identity theorem gives $F = 0$, and uniqueness of the Fourier transform on $L^1(\mathbb{R})$ gives $h = 0$ after extending h by zero. For even moments, extend h evenly to $H(t) = h(|t|)$ on $\mathbb{R}$. All moments of H vanish, since its odd moments vanish by symmetry. Its bilateral transform $\int_{\mathbb{R}} e^{zt} H(t)\,dt$ is holomorphic on $|\operatorname{Re} z| < \eta$ and has all derivatives zero at zero. The same identity and Fourier uniqueness argument applies. □

Proof of Theorem 11.7. Let $f = I_0^a K_0^b$ with $b > a \geq 0$, $n = a + b$, and $\mathcal{M}(z) = \int_0^\infty t^{z-1} f(t)\,dt$ for $\operatorname{Re} z > 0$. Thus $c_{a,b}(k) = \mathcal{M}(k+1)$. For every fixed j, near zero $\theta^j f = O((1 + |\log t|)^b)$, while at infinity it is bounded by a polynomial times $e^{-(b-a)t}$. All boundary terms therefore vanish in repeated Euler integration by parts, giving

$$\int_0^\infty t^{z-1} t^{2i} P(\theta) f(t)\,dt = P(-z-2i)\mathcal{M}(z+2i).$$

Applying this to $\mathcal{B}_n f = 0$ and setting $z = k+1$ proves (a).

For (b), suppose $\sum_{s=0}^M p_s(k) c_{a,b}(k+s) = 0$ for all integers $k \geq k_0$, with $k_0 \geq 0$, and define

$$O = \sum_{s=0}^M t^s p_s(-\theta - 1 - s), \qquad g = Of.$$

Integration by parts gives

$$\int_0^\infty t^k g(t)\,dt = \sum_{s=0}^M p_s(k) c_{a,b}(k+s) = 0 \qquad (k \geq k_0).$$

The function $t^{k_0} g(t)$ is integrable at zero and has an exponential tail. Lemma 19.2 implies $g = 0$. Thus the operator O, with lowest t-power zero and span M, annihilates f. By Lemma 19.1, the minimal operator $\mathcal{B}_n$ has leading differential coefficient $t^{n+1}/n!$, lowest coefficient $\theta^{n+1}/n!$, and span $2K$, where $K = \lceil n/2 \rceil$. Lemma 12.2 now gives $M \geq 2K$ and

$$\theta^{n+1} \mid p_0(-\theta - 1),$$

equivalently $(k+1)^{n+1} \mid p_0(k)$. When $M = 2K$, the equality case gives $O = m(\theta)\mathcal{B}_n$ for a polynomial m. Since $m(\theta) t^{2i} = t^{2i} m(\theta + 2i)$, the corresponding recurrence is the common polynomial $m(-k-1)$ times (a). Its lowest coefficient has degree $n + 1 + \deg m$. This proves the order, degree, and uniqueness statements in (b).

For (c), write $A_N = W_n(2N)$. The multinomial identity gives

$$G(x) = I_0(2\sqrt{x})^n = \sum_{N \geq 0} \frac{A_N}{(N!)^2} x^N.$$

To distinguish the two variable conventions, define

$$Q_i^{(n)}(u) = 2^{\,2i-n-1} P_i^{(n)}(2u), \qquad \mathcal{B}_n^x = \sum_{i=0}^K x^i Q_i^{(n)}(\theta_x).$$

Indeed, $x = t^2/4$ gives $\theta_t = 2\theta_x$, so $\mathcal{B}_n^x$ is $2^{-(n+1)}\mathcal{B}_n$ after that change of variable. Consequently $\mathcal{B}_n^x G = 0$, its span is K, and

$$Q_0^{(n)}(u) = \frac{u^{n+1}}{n!}, \qquad \deg Q_i^{(n)} = n + 1 - 2i.$$

Its leading differential coefficient is $x^{n+1}/n!$. It is minimal: a differential equation of order at most n for G would pull back to one of order at most n for $I_0(t)^n$, contradicting Lemma 19.1.

Suppose $\sum_{s=0}^{M} c_s(N)A_{N+s} = 0$ for integers $N \geq n_0$, with $n_0 \geq 0$ and $c_0 c_M \neq 0$. Lemma 12.3(b), with its factorial shift equal to zero, gives an annihilator of G with span M and lowest coefficient

$$\theta^{\underline{M+n_0}}\theta^{\underline{M}}c_M(\theta - M).$$

Here $u^{\underline{j}} = u(u-1)\cdots(u-j+1)$. By Lemma 12.2, $M \geq K \geq 1$ and the displayed coefficient is divisible by θ^{n+1}. Each falling factorial contains θ exactly once, and therefore

$$\theta^{n-1} \mid c_M(\theta - M).$$

Equivalently $(N+M)^{n-1} \mid c_M(N)$; after writing the recurrence backward, its coefficient of A_N is divisible by N^{n-1}. In particular its coefficient degree is at least $n-1$.

Conversely coefficient extraction in $\mathcal{B}_n^x G = 0$ yields

$$\sum_{i=0}^{K} Q_i^{(n)}(N-i)\big(N^{\underline{i}}\big)^2 A_{N-i} = 0.$$

Every coefficient is divisible by N^2: for $i = 0$ this follows from $n+1 \geq 2$, and for $i \geq 1$ it follows from the square of the falling factorial. Dividing by N^2 gives a recurrence of order K and degree $n-1$, with leading coefficient $N^{n-1}/n!$. The coefficients of this recurrence have no nonconstant polynomial common divisor, since dividing by such a divisor would contradict the degree lower bound just established. Ore division over $\mathbb{C}(N)$ shows that any two recurrences of minimum order K are rational-function multiples: a nonzero remainder would be an eventual recurrence of lower order. Primitivity of the displayed recurrence then makes the multiplier polynomial whenever both recurrences have polynomial coefficients. If both have degree $n-1$, the multiplier is constant. This applies in particular to Verrill's recurrence. □

Theorem 19.3 (Minimal recurrences on shifted parity lattices)**.** *Let $n \geq 1$, $K = \lceil n/2 \rceil$, and let*

$$f(t) = \sum_{\substack{a+b=n \\ b>a\geq 0}} \lambda_{a,b} I_0(t)^a K_0(t)^b$$

be a nonzero linear combination with complex coefficients. For any real $r > -1$, put

$$u_r(N) = \int_0^\infty t^{2N+r} f(t)\, dt \qquad (N \geq 0).$$

Then

$$\sum_{i=0}^{K} P_i^{(n)}(-2N - r - 1 - 2i)u_r(N+i) = 0. \tag{19.1}$$

Every eventual polynomial-coefficient recurrence $\sum_{s=0}^{M} q_s(N)u_r(N+s) = 0$ with $q_0 q_M \neq 0$ satisfies

$$M \geq K, \qquad (2N + r + 1)^{n+1} \mid q_0(N).$$

Thus the minimum order is K and the minimum coefficient degree, among recurrences of any order, is $n+1$. Every recurrence of order K is a polynomial multiple of (19.1); at order K and degree $n+1$ it is unique up to a nonzero constant.

Proof. Every nonzero vector in the solution space of $\mathcal{B}_n$ has minimal operator $\mathcal{B}_n$: its differential-Galois orbit spans a nonzero invariant subspace of the irreducible representation $\mathrm{Sym}^n(\mathbb{C}^2)$ of SL_2. The argument of Lemma 19.1 therefore applies to the indicated linear combination. All its summands have positive exponential decay rates at infinity, and its fixed Euler derivatives have at most logarithmic growth at zero. Integration by parts in $\mathcal{B}_n f = 0$ gives (19.1).

For a recurrence valid at integers $N \ge N_0$, with $N_0 \ge 0$, form

$$O = \sum_{s=0}^{M} t^{2s} q_s\Big(\frac{-\theta - r - 1 - 2s}{2}\Big).$$

The endpoint estimates justify integration by parts and give

$$\int_0^\infty t^{2N+r}(Of)(t)\,dt = 0 \qquad (N \ge N_0).$$

The function $h(t) = t^{2N_0+r}(Of)(t)$ is integrable and has an exponential tail. All its even moments vanish, so Lemma 19.2 gives $Of = 0$. The operator O has lowest t-power zero and span $2M$. Lemma 12.2 gives $2M \ge 2K$ and

$$\theta^{n+1} \mid q_0\Big(\frac{-\theta - r - 1}{2}\Big),$$

which is the required divisibility. The displayed recurrence attains both the order bound and the coefficient-degree bound. When $M = K$, the equality case of that lemma gives $O = m(\theta)\mathcal{B}_n$ for a polynomial m. Commuting $m(\theta)$ across t^{2i} shows that its recurrence is the common polynomial $m(-2N-r-1)$ times (19.1). The lowest coefficient has degree $n+1+\deg m$, proving uniqueness. □

Remark 19.4. The identity (19.1) is the classical symmetric-power Mellin recurrence evaluated at a shifted argument. The additional assertion is minimality when a relation is known only on one shifted parity lattice, and for any nonzero vector in the decaying solution space. Restricting the original recurrence to that lattice does not alone prove this assertion; even-moment uniqueness supplies the missing implication. No assertion for strides greater than two is made here.

Remark 19.5. For $n = 5$ the recursion (a), multiplied by 120, is

$$\begin{aligned}(k+1)^6 c(k) - (35k^4 + 280k^3 + 882k^2 + 1288k + 731)\,c(k+2)\\ + (259k^2 + 1554k + 2435)\,c(k+4) - 225\,c(k+6) = 0,\end{aligned}$$

the recursion for $\int_0^\infty t^k K_0(t)^5 dt$ found by Bailey, Borwein, Broadhurst and Glasser [52] and proved in general by Borwein and Salvy [54]. Theorem 11.7 shows that it cannot be shortened and that no recursion of lower degree exists. The same holds for the moments $\int_0^\infty t^k I_0^a K_0^b dt$ with $b > a$, which satisfy the same recursion.

20 Even Dyson parameters: a common Bessel module and common jet fields

The multiple-drift argument gives more than the scalar operator at even Dyson parameters. It supplies an everywhere invertible comparison with the same Bessel symmetric power, independently of the Dyson parameter and of the integer twist. Throughout this section let $n, m \ge 1$ and $q \ge 0$ be integers, let $\beta = 2m$, and put

$$y(x) = I_0(2\sqrt{x}), \qquad \widehat{G}^{2m}_{n,q}(x) = t^{-nq} D^{2m}_{n,q}(t), \quad x = t^2.$$

The normalized function is entire: in the exponential expansion, invariance under simultaneous rotation of all z_j forces the powers of $\sum z_j$ and $\sum z_j^{-1}$ to differ by nq. In particular, if $a^{2m}_{n,q}(N) = \mathbb{E}[(\det U)^q (\operatorname{tr} U)^N \overline{\operatorname{tr} U}^{\,N+nq}]$, then

$$\widehat{G}^{2m}_{n,q}(x) = \sum_{N \ge 0} \frac{a^{2m}_{n,q}(N)}{N!(N+nq)!}\, x^N. \tag{20.1}$$

Thus the displayed definition involves no choice of square root. It differs from the Jack-normalized function of Section 15 by a nonzero rational constant: $\widehat{G}^{2m}_{n,q} = c\,G^{2m}_{n,q}$. Indeed the ensemble identity gives $c \neq 0$, and the constant coefficient of the finite Laurent-integral expansion makes c rational. All module and field assertions below therefore hold equally for $G^{2m}_{n,q}$.

Let $R = \mathbb{Q}[x, x^{-1}]$ and let the linear forms $C_k(x; X, Y) \in R[X, Y]$ be those of (24.1); they satisfy $t^{-k} I_k(2t) = C_k(t^2; y(t^2), y'(t^2))$ for every integer k. For the permutations and signs in Section 16, define

$$H^{(m)}_{n,q}(x; X, Y) = \frac{1}{Z} \sum_{\boldsymbol{\sigma},\boldsymbol{\tau}} \epsilon(\boldsymbol{\sigma}, \boldsymbol{\tau}) \prod_{j=1}^{n} C_{q+\delta_j}(x; X, Y). \tag{20.2}$$

Here Z is the positive normalization integral of that section. It is a nonzero integer, since it is the constant coefficient of the Laurent polynomial $(\Delta(z)\Delta(z^{-1}))^m$.

Theorem 20.1 (The common even-parameter module)**.** *For all the parameters above,*

$$\widehat{G}^{2m}_{n,q}(x) = H^{(m)}_{n,q}(x; y(x), y'(x)),$$

where $H^{(m)}_{n,q}$ is a nonzero homogeneous form of degree n over R. The minimal differential operator is irreducible of order $n+1$ over $\mathbb{C}(x)$, with differential Galois group $\mathrm{Sym}^n(\mathrm{SL}_2)$ in its solution representation. Its local monodromy at zero is a single block $J_{n+1}(1)$. Order and irreducibility persist over every finite algebraic extension of $\mathbb{C}(x)$, and ramified pullback at zero preserves this block size.

Proof. For each tuple of permutations one has $\sum_j \delta_j = 0$. Substituting $I_{q+\delta_j}(2t) = t^{q+\delta_j} C_{q+\delta_j}(t^2; y, y')$ in the finite expansion of the integral therefore produces the common factor t^{nq} and proves the identity. Every summand of (20.2) is homogeneous of degree n. The form is nonzero: Proposition 16.1(b), with $b = 0$, gives a nonzero leading asymptotic coefficient for $D^{2m}_{n,q}$.

Theorem 30.1, applied to this form with $d = n$ and $\varepsilon = 0$, now proves the assertions. Explicitly, the rank-two equation $xy'' + y' - y = 0$ has group SL_2 and local block $J_2(1)$; polarization of the nonzero form embeds its irreducible nth symmetric power into the Picard–Vessiot field. The image is the solution space of the minimal scalar equation. The symmetric power of $J_2(1)$ has one block of size $n+1$. Connectedness of the group gives stability under finite algebraic base extension, and raising a unipotent block to a positive integral power preserves its size in characteristic zero. □

Put

$$\mathbf{J}^{(m)}_{n,q} = (\widehat{G}^{2m}_{n,q}, (\widehat{G}^{2m}_{n,q})', \dots, (\widehat{G}^{2m}_{n,q})^{(n)})^{\mathsf{T}}, \qquad \mathbf{m}_n = (y^n, y^{n-1}y', \dots, (y')^n)^{\mathsf{T}}.$$

Define $P^{(m)}_{n,q}$ by

$$\mathbf{J}^{(m)}_{n,q} = P^{(m)}_{n,q}(x)\mathbf{m}_n. \tag{20.3}$$

Equivalently, its row with index j, $0 \le j \le n$, is the coefficient vector of $\mathscr{D}^j H^{(m)}_{n,q}$, where $\mathscr{D} = \partial_x + Y\partial_X + x^{-1}(X - Y)\partial_Y$.

Theorem 20.2 (An everywhere invertible comparison)**.** *The matrix $P^{(m)}_{n,q}$ belongs to $\mathrm{GL}_{n+1}(R)$, and*

$$\det P^{(m)}_{n,q}(x) = c^{(m)}_{n,q}\, x^{-n(n+1)(m(n-1)+3q)/6}, \qquad c^{(m)}_{n,q} = \det P^{(m)}_{n,q}(1) \in \mathbb{Q}^*. \tag{20.4}$$

Consequently, for fixed n and any $m, m' \ge 1$, $q, q' \ge 0$, the two complete jets are related by a matrix $P^{(m')}_{n,q'}(P^{(m)}_{n,q})^{-1} \in \mathrm{GL}_{n+1}(R)$.

Proof. All entries belong to R. The determinant is nonzero, since a dependence among its rows would give an equation of order at most n for $\widehat{G}^{2m}_{n,q}$, contrary to Theorem 20.1. By Theorem 16.4, the t-equation has no singularity away from zero. The unramified change $x = t^2$ and multiplication by t^{-nq} preserve ordinariness locally at every nonzero point. The x-equation consequently has no finite singularity except possibly zero.

The matrix $P^{(m)}_{n,q}$ intertwines the symmetric-power connection and the scalar companion connection. Both connections are holomorphic at each nonzero point. A fundamental matrix for the former is carried to a fundamental matrix for the latter; Abel's identity therefore excludes a zero of $\det P^{(m)}_{n,q}$ there. A nonzero Laurent polynomial with no zero in $\mathbb{C}^*$ is a monomial, so its determinant is cx^s for an integer s.

The exponents $q(n-2i) - 2mi(n-i)$ of the t-equation, given by Theorem 16.4, become

$$\rho_i = -i(q + m(n-i)), \qquad 0 \le i \le n,$$

after multiplication by t^{-nq} and passage to $x = t^2$. To justify this passage also in resonant cases, the explicit Laurent map (20.2) identifies the x-module with the Bessel symmetric power; its local solutions have integral-power Laurent–log expansions. Their pullbacks, multiplied by t^{nq}, span the t-solution space, so the indicial exponents transform by $\nu \mapsto (\nu - nq)/2$, with multiplicity. The trace of the connection on $\mathbf{m}_n$ is $-n(n+1)/(2x)$. The residue of the trace of the companion connection is $\sum_i \rho_i - n(n+1)/2$, by Abel's indicial formula. Taking the trace of the gauge equation at zero thus gives

$$s = \sum_{i=0}^{n} \rho_i = -\frac{n(n+1)(m(n-1)+3q)}{6}.$$

Evaluation at $x = 1$ proves that c is nonzero and rational. The adjugate formula then shows that the inverse matrix also has entries in R. □

Corollary 20.3 (Parameter-independent jet fields and arithmetic relations)**.** *Let $K = \mathbb{C}(x)$. For every positive even Dyson parameter and every nonnegative integer twist,*

$$K(\mathbf{J}^{(m)}_{n,q}) = K(y^n, y^{n-1}y', \dots, (y')^n) = K(y, y')^{\mu_n}. \tag{20.5}$$

Under the change of coordinates $\mathbf{Z} = (P^{(m)}_{n,q})^{-1}\mathbf{J}^{(m)}_{n,q}$, the ideal of all algebraic relations is generated by

$$Z_i Z_{j+1} - Z_{i+1} Z_j, \qquad 0 \le i < j \le n-1.$$

Its transcendence degree is two, and any two distinct entries of $\mathbf{J}^{(m)}_{n,q}$ are algebraically independent over K.

At every $\alpha \in \overline{\mathbb{Q}}^*$ these statements hold over $\overline{\mathbb{Q}}$ for the corresponding numerical jets, with the invertible matrix $P^{(m)}_{n,q}(\alpha)$. In particular all the derivative values of orders $0, \dots, n$ are nonzero and transcendental. For a fixed n their complete numerical jet field is independent of m and q. The function fields and numerical fields in dimensions d, e obey the gcd–lcm laws of Corollary 24.5.

Proof. The invertible Laurent gauge reduces all assertions to the degree-n monomials in y, y'. These two functions are algebraically independent. Their monomials have the displayed rational-normal-cone ideal, and generate $K(y'/y, y^n) = K(y, y')^{\mu_n}$, as in Theorem 24.4. Any two rows of an invertible matrix give nonproportional homogeneous forms of the same positive degree in two independent variables; such forms are algebraically independent.

For a nonzero algebraic α, the two values $y(\alpha), y'(\alpha)$ are algebraically independent by the Siegel–Shidlovskii argument in Theorem 24.6. This argument applies the E-function theorem to $I_0(2t)$ and its t-derivative at a nonzero algebraic square root of α. Equation (20.4) makes the

specialized coordinate change invertible for every such α, so the same polynomial-kernel and homogeneous-form proofs apply to the values. The field laws follow from the fixed fields of the scalar groups μ_d and μ_e exactly as in Corollary 24.5. □

Remark 20.4. These conclusions use the integer Laurent expansion at $\beta = 2m$. They do not assert the same Bessel module or arithmetic independence for non-even Dyson parameters. For $m = 1$ they recover the determinant results of Section 24; for $m > 1$ they identify all complete jets with those same fields, even though their scalar operators and exponents at zero differ.

21 Exact degrees for twisted moments

The shifted factorials in the twisted generating function can absorb negative integral exponents in the recurrence dictionary. Local monodromy supplies a degree obstruction that is unaffected by this cancellation. For every positive even value of the ensemble parameter, it settles the exceptional integer twists as well.

Lemma 21.1 (The double Borel transform of a rational germ)**.** *Let $A(z) = \sum_{N\ge 0} a_N z^N$ be nonzero, rational and holomorphic at zero. The minimal differential equation of*

$$A_2(x) = \sum_{N\ge 0} \frac{a_N x^N}{(N!)^2}$$

is regular singular or ordinary at zero, and every local monodromy Jordan block there has size at most two.

Proof. Partial fractions give, apart from finitely many initial coefficients,

$$a_N = \sum_{j=1}^{s} p_j(N)\lambda_j^N, \qquad p_j \in \mathbb{C}[N], \quad \lambda_j \neq 0.$$

Consequently A_2 is a polynomial plus a finite sum of $p_j(\theta)I_0(2\sqrt{\lambda_j x})$, with $\theta = x\partial_x$. Each Bessel function belongs to a regular singular rank-two module whose local monodromy at zero satisfies $(T - I)^2 = 0$. A rational differential image, a direct sum, and a differential subquotient preserve this annihilating-polynomial identity. The polynomial term belongs to a trivial rank-one module. These observations prove both assertions. □

Theorem 21.2 (Minimal degree for all even-parameter integer twists)**.** *Let $m, n \ge 1$ and $q \ge 0$ be integers, set $\beta = 2m$, and put*

$$a_N = \mathbb{E}_{\mathrm{C}\beta\mathrm{E}_n}\big[(\det U)^q (\operatorname{tr} U)^N \overline{\operatorname{tr} U}^{\,N+qn}\big].$$

Every nonzero polynomial-coefficient recurrence for (a_N) that holds for all sufficiently large indices has degree at least $n - 1$. The smallest recurrence order and the smallest coefficient degree are respectively

$$K = \lceil n/2 \rceil, \qquad n - 1.$$

The recurrence obtained from the Kac system attains both bounds and is primitive. Every polynomial-coefficient recurrence of order K is a polynomial multiple of it; in particular, the recurrence attaining both bounds is unique up to a nonzero constant factor. These conclusions include every exceptional integer twist at $\beta = 2$.

Proof. The case $n = 1$ gives $a_N = 1$ and is immediate. Assume $n \geq 2$, set $r = qn$, and write

$$A(z) = \sum_{N\geq 0} a_N z^N, \qquad G(x) = \sum_{N\geq 0} \frac{a_N x^N}{N!(N+r)!}.$$

The normalization of G differs from the normalized Jack function by a nonzero constant, which has no effect on its differential module. The estimate $|a_N| \leq n^{2N+r}$ proves that A is convergent near zero. The Kac recurrence in Theorem 11.3 has the backward form

$$\sum_{k=0}^{K} R_k(N) a_{N-k} = 0 \quad (N \geq 1), \qquad R_0(N) = \prod_{i=1}^{n-1} (N + i(q + m(n-i))), \qquad \deg R_k \leq n-1,$$

with negative-index terms omitted. Thus

$$RA = R_0(0)a_0, \qquad R = \sum_{k=0}^{K} z^k R_k(\theta + k).$$

The operator R has order $n-1$, is Fuchsian at zero, and has lowest θ-coefficient $R_0(\theta)$. The operator θR is a homogeneous annihilator of A. Its lowest θ-coefficient is $\theta R_0(\theta)$, of degree n, and all its other θ-coefficients have degree at most n; hence it is Fuchsian at zero as well. Its minimal right factor, the minimal equation of A, is therefore regular singular or ordinary there.

By Theorem 20.1, G generates the irreducible nth symmetric power of the rank-two Bessel module. Its local monodromy at zero is the single block $J_{n+1}(1)$. An exact coefficient identity gives

$$A_2(x) := \sum_{N\geq 0} \frac{a_N x^N}{(N!)^2} = \prod_{j=1}^{r} (\theta + j) G(x), \tag{21.1}$$

where the empty product is one. This differential image is nonzero, since $a_0 \neq 0$ and the displayed coefficient multipliers are nonzero for $N \geq 0$. A nonzero differential image of an irreducible module is isomorphic to that module. The minimal equation of A_2 thus also has the block $J_{n+1}(1)$ at zero.

Suppose a recurrence of degree $d \geq 1$ exists. Lemma 45.2 bounds every local monodromy block of A by d. Theorem 27.3, applied twice, bounds the unit-eigenvalue blocks of A_2 by $d+2$. It follows that $n+1 \leq d+2$, and hence $d \geq n-1$. If $d = 0$, the same recurrence lemma makes A rational, whereas Lemma 21.1 bounds the blocks of A_2 by two. This contradicts $n+1 \geq 3$. The lower bound is proved for recurrences of arbitrary order and with any finite initial segment omitted.

The Kac recurrence has order K and degree $n-1$, and its order is minimal by Theorem 11.3. Its coefficients cannot have a nonconstant common factor, since cancelling that factor would give an eventual recurrence of degree below $n-1$. Two order-K annihilators over $\mathbb{C}(N)$ are proportional: cancel their highest-shift coefficients, and minimality excludes a nonzero annihilator of smaller order. If one is primitive with polynomial coefficients, a rational proportionality factor giving polynomial coefficients must itself be polynomial, by Bezout's identity for their greatest common divisor. This proves the remaining assertions. □

Remark 21.3. The proof gives the least coefficient degree among recurrences of every order, as well as uniqueness at the least order. It does not assert that $R_0(N)$ divides the leading coefficient of every higher-order recurrence in an exceptional case. That stronger divisibility statement does not follow from the Jordan-block bound.

Theorem 21.4 (An infinite family with a degree drop)**.** *Let $q \geq 1$ be an integer, take $n = 3$ and $\beta = 2q - 1$, and define a_N as in Theorem 21.2, using this value of β. Then, for every $N \geq 1$,*

$$(N + 4q - 1)a_N = 2(5N + 6q - 3)a_{N-1} - 9(N-1)a_{N-2}, \tag{21.2}$$

with negative-index terms omitted. The least recurrence order is two and the least coefficient degree is one, even among eventual recurrences of arbitrary order. The displayed recurrence is primitive and unique up to a nonzero constant among recurrences attaining both bounds.

Proof. For $n = 3$, direct elimination in the Kac system gives, for arbitrary $\beta > 0$ and integer $q \geq 0$, the coefficients

$$\begin{aligned} R_0(N) &= (N + \beta + q)(N + \beta + 2q), \\ R_1(N) &= -10N^2 - 6\beta N - 30qN + 10N - 9\beta q + 3\beta - 18q^2 + 15q - 3, \\ R_2(N) &= 9(N - 1)(N + 3q - 1). \end{aligned}$$

At $\beta = 2q - 1$ they have the common factor $N + 3q - 1$ and become

$$(N + 3q - 1)(N + 4q - 1,\ -2(5N + 6q - 3),\ 9(N - 1)).$$

Since $N + 3q - 1 \neq 0$ for $N \geq 1$, this proves (21.2) for all such indices.

To exclude degree zero at any order, let $A(z) = \sum_{N \geq 0} a_N z^N$. It is analytic at zero and satisfies

$$\left[z(1 - z)(1 - 9z)\partial_z + (4q - 1) - (12q + 4)z + 9z^2\right] A = (4q - 1)a_0. \tag{21.3}$$

A rational solution cannot have a pole at an ordinary point of this equation. At its two other finite singularities $z = 1$ and $z = 1/9$, the exponents of its homogeneous equation are respectively $q - \frac{1}{2}$ and $3q - \frac{3}{2}$. Neither is a negative integer, so a rational solution of the inhomogeneous equation cannot have a pole there either: the leading polar term would have to solve the indicial equation of the homogeneous operator. Analyticity excludes a pole at zero. A rational A would therefore be a polynomial. But if A had degree $D \geq 0$ and nonzero leading coefficient c, the left side of (21.3) would have leading term $9(D + 1)cz^{D+2}$, which cannot equal the nonzero constant on the right. Thus A is not rational, and degree zero is impossible.

The order bound in Theorem 11.3 gives order at least two. The three linear coefficients in (21.2) have no common root, so the recurrence is primitive. The same minimal-order proportionality argument as in Theorem 21.2 proves its uniqueness. □

The specialization $q = 1$ gives the exceptional $(n, q, \beta) = (3, 1, 1)$ recurrence of Remark 15.4.

Corollary 21.5 (The three-particle minimum-order classification)**.** *For $n = 3$, every $\beta > 0$ and every integer $q \geq 0$, the least degree among recurrences of the minimum order two is*

$$\begin{cases} 1, & q \geq 1 \text{ and } \beta = 2q - 1, \\ 2, & \text{otherwise.} \end{cases}$$

The primitive order-two recurrence is unique up to scale in every case.

Proof. Use the three quadratic coefficients R_0, R_1, R_2 in the preceding proof. Since the roots of R_0 are negative, $N - 1$ cannot be a common factor. For $q = 0$, this already excludes any common factor. For $q \geq 1$, the only remaining possibility is $N + 3q - 1$. The condition $R_0(1 - 3q) = 0$ forces $\beta = 2q - 1$ or $\beta = q - 1$. The first is the displayed degree drop. For the second,

$$R_1(1 - 3q) = -3q(3q - 1) \neq 0,$$

so no cancellation occurs. The primitive quadratic recurrence therefore has degree two except in the stated family, where it has degree one. Minimal order is two; hence all recurrences of that order are rational multiples of the primitive one. Bezout's identity makes every polynomial-coefficient multiple a polynomial multiple, which establishes the degree assertion and uniqueness. □

22 Jacobi resonances beyond the positivity condition

The condition ($\star$) is sufficient, but not necessary, for the degree statements. The following refinement includes zero indicial parameters and isolates the finite collection of resonance hyperplanes on which the previous argument can lose a degree.

Lemma 22.1 (Multiplicity in the exponential-generating-function transfer)**.** *In the exponential-generating-function case of Lemma 12.3, suppose that the lowest coefficient of a minimal θ-form is $P(\theta)$ and that the transfer lemma applies. If ν is a root of P of multiplicity m_ν, every recurrence of order M has leading coefficient c_M satisfying*

$$\prod_{\nu\notin\mathbb{Z}_{\geq 0}} (\theta-\nu)^{m_\nu} \prod_{\nu\in\mathbb{Z}_{\geq 0}} (\theta-\nu)^{\max(m_\nu-1,0)} \;\mid\; c_M(\theta-M).$$

Proof. The lowest coefficient supplied by the dictionary is $\theta^{\underline{M+n_0}} c_M(\theta-M)$ and is divisible by $P(\theta)$. The falling factorial has only simple roots, all nonnegative integers. Comparison of the multiplicity at each root proves the assertion. □

For the Jacobi ensemble, define the set of distinct positive-integer resonances and a polynomial by

$$J=\{j\in\mathbb{Z}_{>0} : \varepsilon_p^J=-j \text{ for some } 1\leq p\leq n\}, \qquad D_n(N)=\frac{\prod_{p=1}^n (N+\varepsilon_p^J)}{\prod_{j\in J}(N-j)}.$$

Each denominator factor cancels one, and only one, copy of that factor in the numerator, so D_n is a polynomial. Moreover $\varepsilon_p^J > -p$, because $a+b+1>-1$ and $\beta>0$; consequently $J\subseteq\{1,\dots,n-1\}$.

Theorem 22.2 (Jacobi degrees away from, and at, the first resonance)**.** *Let $n\geq 1$, $\beta>0$, and $a,b>-1$, without assuming ($\star$).*

1. *Every polynomial recurrence for the Jacobi trace moments has order at least n. Its leading coefficient in backward form is divisible by $D_n(N)$, and its degree is at least $n-|J|$.*
2. *If J is empty, all the conclusions of Theorem 11.4 hold: L^J is primitive, the moment recurrence has minimal order n and minimal degree n, and the density has minimal differential order n. In particular zero values of ε_p^J are permitted. Every recurrence of order n is a polynomial multiple of the displayed recurrence of Theorem 11.4.*
3. *If $J=\{1\}$, each coefficient $c_k(N)$ of that displayed moment recurrence is divisible by $N-1$. Dividing all coefficients by $N-1$ gives a recurrence valid for $N\geq 2$, of minimal order n and minimal degree $n-1$, with leading coefficient $D_n(N)$. Every recurrence of order n is a polynomial multiple of this reduced recurrence. The Jacobi trace density has minimal differential order $n-1$.*

Proof. Visibility gives differential order $n+1$ for H throughout the stated parameter range. The scalar operator L^J has leading coefficient x^n; its primitive normalization therefore still has monomial leading coefficient. The θ-form xL^J has span n and lowest coefficient

$$P_0(\theta)=\theta\prod_{p=1}^n(\theta+\varepsilon_p^J).$$

Multiplication by a power of x in passing to the primitive normalization does not change this lowest coefficient. The transfer lemma proves the order bound. At $\theta=0$ the first factor of P_0 absorbs the one zero of the falling factorial in Lemma 22.1; any additional zero factors remain. At each distinct $j\in J$ the falling factorial can absorb at most one factor $\theta-j$. Every other factor

remains. After passing from forward to backward form, this is precisely the asserted divisibility by $D_n(N)$, proving (1).

If J is empty, no ε_p^J equals -1, and $q_1(0) = \prod_{p=0}^{n}(1+\varepsilon_p^J) \neq 0$. Thus L^J is primitive. The already constructed recurrence has degree n, and (1) proves optimality. For uniqueness, the leading coefficient of any order-n recurrence is a polynomial multiple of that of the constructed recurrence. Subtracting the corresponding polynomial multiple gives a recurrence of smaller order, which is zero by the order bound. A density annihilator of order r gives a moment recurrence of degree at most r; the constructed density equation has order n. This proves (2), including the density assertion.

Suppose now that $J = \{1\}$. Then $q_1(0) = 0$. For $m \geq 2$, $x^{m-1} \mid q_m$, and hence $q_m(0) = 0$. Since $L^J H = 0$ and $H(0) = 1$, evaluation at zero gives $q_0(0) = 0$. Equivalently, $\hat{P}_1^J(0) = 0$. The leading recurrence coefficient satisfies $c_0(1) = 0$, and $c_1(1) = -\hat{P}_1^J(0) = 0$. For $k \geq 2$, the formula $c_k(N) = (-1)^k \hat{P}_k^J(N-k)\underline{(N-1)^{k-1}}$ directly gives $c_k(1) = 0$. Division by $N-1$ therefore produces polynomial coefficients of degree $n-1$ and does not change recurrence order. The recurrence is asserted only for $N \geq 2$: its value at $N = 1$ cannot be recovered by division. Its leading coefficient is D_n, so (1) proves optimality and the preceding subtraction argument gives uniqueness at order n. The vanishing of every $q_m(0)$ also shows that x divides L^J. Thus L^J/x annihilates H and all of its coefficient polynomials have degree at most $n-1$. Its Fourier–Laplace dual annihilates ρ^J and has order at most $n-1$. Conversely every density annihilator gives a recurrence of no greater degree, so (1) forces order at least $n-1$. □

Remark 22.3. For fixed n, the excluded set in part (2) is the finite union

$$p\left(a+b+1+\frac{\beta}{2}(2n-p-1)\right) = -j, \qquad 2 \leq p \leq n, \quad 1 \leq j \leq p-1.$$

Thus part (2) applies on both sides of ($\star$), except on these hyperplanes. Part (1) is a lower bound on every remaining hyperplane; we do not assert that it is always attained. The shifted-trace and symmetric even-moment conclusions of Propositions 18.2 and 18.3 also hold with ($\star$) replaced by $J = \varnothing$: the primitive operator and its indicial polynomial are unchanged by analytic exponential conjugation and nonzero scaling, and the preceding multiplicity argument replaces the use of strictly negative roots.

Corollary 22.4 (The one- and two-particle cases)**.** *For $n = 1$ the minimal recurrence order, recurrence degree, and density differential order are all 1 for every $a, b > -1$. The recurrence is*

$$(N+a+b+1)m_N - (N+a)m_{N-1} = 0 \qquad (N \geq 1).$$

For $n = 2$ the minimal recurrence order is always 2. The minimum recurrence degree and density differential order both equal

$$\begin{cases} 1, & 2a+2b+\beta+3 = 0, \\ 2, & 2a+2b+\beta+3 \neq 0. \end{cases}$$

On the exceptional hyperplane, necessarily $0 < \beta < 1$, and the reduced recurrence is

$$\left(N+\frac{\beta-1}{2}\right)m_N - (3N+4a+2\beta)m_{N-1} + 2(N+2a+\beta)m_{N-2} = 0 \qquad (N \geq 2).$$

The density satisfies, in the sense of distributions,

$$w(w-1)(w-2)(\rho^J)'(w) = \left(\frac{\beta-3}{2}w^2 - (4a+2\beta)w + 4a + 2\beta + 2\right)\rho^J(w).$$

On each of $(0,1)$ and $(1,2)$ it is therefore a constant multiple of

$$w^{2a+\beta+1}|w-1|^{-(\beta+1)/2}(2-w)^{-2a-2},$$

where the two constants are not asserted to be equal.

Proof. For $n = 1$, $\varepsilon_1^J = a + b + 1 > -1$, so J is empty. For $n = 2$, $\varepsilon_1^J = a + b + 1 + \beta > -1$ and $\varepsilon_2^J = 2a + 2b + 2 + \beta > -2$. The only possible positive-integer resonance is therefore $\varepsilon_2^J = -1$, equivalent to the stated hyperplane. The preceding theorem proves the orders and degrees. On this hyperplane the scalarization gives

$$\begin{aligned}\hat{P}_0^J(\theta) &= \theta(\theta-1)\left(\theta + \frac{\beta-1}{2}\right),\\ \hat{P}_1^J(\theta) &= \theta(3\theta + 4a + 2\beta + 3),\\ \hat{P}_2^J(\theta) &= 2(\theta + 2a + \beta + 2).\end{aligned}$$

These formulas give the reduced recurrence by coefficient extraction. They also give the primitive operator

$$x\partial_x^3 + \left(3x + \frac{\beta+3}{2}\right)\partial_x^2 + (2x + 4a + 2\beta + 6)\partial_x + 4a + 2\beta + 4.$$

For $H(x) = \langle \rho^J, e^{-xw}\rangle$, replace ∂_x by multiplication by $-w$ and x by ∂_w, retaining operator order. This gives the displayed first-order density equation. Partial fractions and integration on each interval yield the power expression. Finally $a + b > -2$ and $a + b = -(\beta+3)/2$ imply $\beta < 1$. □

Remark 22.5 (The rank-one even orthogonal case). The preceding multiplicity refinement also includes SO(2), which was outside ($\star$). Its trace is $2\cos\varphi$ with uniform φ, so its even moments are $w_D(N) = \binom{2N}{N}$ and obey

$$N w_D(N) - 2(2N-1)w_D(N-1) = 0 \qquad (N \geq 1).$$

This recurrence has minimal order and degree 1. It is also the $n = 1$, $a = b = -1/2$, $\kappa = 4$ case of the extended symmetric-trace statement, for which $\varepsilon_1^J = 0$.

23 Parameter boundaries and remaining analytic questions

23.1 Noninteger twists

The scalar construction and formal solutions in Theorem 11.1 hold for real $q > -1/2$. Minimality is established here for nonnegative integer twists. For real $q \geq 0$ the following remains a question.

Conjecture 23.1. *For every $\beta > 0$, $n \geq 1$ and real $q \geq 0$, the function $G_{n,q}^{\beta}$ has differential order $n + 1$ over $\mathbb{C}(x)$.*

The integer-twist proof does not extend merely by replacing the character $e^{iq\sum\varphi_j}$ by a branch for non-integral q. Such a branch is not a smooth function on the torus. Moreover, when $qn \notin \mathbb{Z}$, the function $t^{qn}G_{n,q}^{\beta}(t^2)$ is not entire at $t = 0$, because $G_{n,q}^{\beta}(0) = 1$; consequently it cannot be the Fourier–Laplace transform of a compactly supported distribution on the real line. There is, however, a compact distribution representation for the ungauged function, for every real $q \geq 0$.

Proposition 23.2 (A compact transform for real twists)**.** *Let $\beta > 0$, $n \geq 1$, $q \geq 0$, and put $F(t) = G_{n,q}^{\beta}(t^2)$. Then*

$$|F(t)| \leq \exp(2n|\operatorname{Re} t|) \qquad (t \in \mathbb{C}).$$

There is a unique distribution $\sigma_{n,q}^{\beta}$ supported in $[-2n, 2n]$ such that

$$F(t) = \langle \sigma_{n,q}^{\beta}, e^{tw}\rangle.$$

This distribution is real-analytic away from the points $2(n - 2b)$, $0 \leq b \leq n$.

Proof. Let $A = S^{-1}T_nS$ be the real symmetric matrix of Lemma 14.1(c), and let

$$D = \operatorname{diag}(-p(2q+1+\beta(n-p)))_{p=0}^{n}.$$

Thus D is negative semidefinite and $\|A\| = n$. Consider the power-series solution $Y(t) = \sum_{k\ge 0} y_k t^k$ of

$$Y' = \left(2A + \frac{D}{t}\right) Y, \qquad y_0 = e_0.$$

Its coefficients are uniquely determined by

$$(kI - D)y_k = 2Ay_{k-1} \qquad (k \ge 1).$$

Since $\|(kI-D)^{-1}\| \le 1/k$, $\|y_k\| \le (2n)^k/k!$, and Y is entire. The tridiagonal form of A implies $(y_k)_p = 0$ unless $p \le k$ and $p \equiv k \pmod 2$. Consequently $J = SY$ satisfies the system (14.6); its first component is an even entire function. The elimination in Section 14 shows that $J_0(t) = g(t^2)$, where $\mathcal{L}g = 0$ and $g(0) = 1$. For every integer $N \ge 1$ the constant-block coefficient

$$P_0(N) = \frac{1}{n!}\prod_{p=0}^{n}(N+\varepsilon_p)$$

is nonzero, because $\varepsilon_p \ge 0$. The coefficient recurrence therefore gives a unique power-series solution normalized at zero, so $g = G^{\beta}_{n,q}$.

For a fixed angle ϑ put $Z(r) = Y(re^{i\vartheta})$. For $r > 0$,

$$\frac{d}{dr}\|Z(r)\|^2 = 4\cos\vartheta\, Z(r)^* AZ(r) + \frac{2}{r} Z(r)^* DZ(r) \le 4n|\cos\vartheta|\,\|Z(r)\|^2.$$

Since $\|Z(0)\| = 1$, integration gives $\|Y(t)\| \le e^{2n|\operatorname{Re} t|}$. The first diagonal entry of S is 1, hence the same bound holds for $F = J_0$. The Paley–Wiener–Schwartz theorem applied to $F(iz)$ gives the unique compactly supported distribution with support in $[-2n, 2n]$; see [57, §7.3].

Finally the operator $\Theta_t = \sum_k t^{2k} P_k(\theta_t/2)$ annihilates F. By Proposition 14.3, its top polynomial-degree coefficient is

$$\chi_{\Theta_t}(w) = \sum_k \operatorname{lc}(P_k)(w/2)^{n+1-2k} = \frac{1}{n!}\prod_{b=0}^{n}\left(\frac{w}{2} - n + 2b\right).$$

Lemma 13.1 now proves the asserted analyticity. □

Proposition 23.2 proves existence and localizes the possible singularities. It does not show that all those points are singular. To obtain minimal order $n+1$ for arbitrary real $q \ge 0$ by visibility, one still needs a nonvanishing singular term at every point $2(n-2b)$. In particular, the formal existence of all exponential types for the operator does not establish their presence in this particular solution. The non-integer-twist minimality conjecture remains open here. The Hankel-contour representation mentioned after [11, (2.3)] is another possible route to these connection coefficients, but the compact real-density vertex argument above does not apply to those contours without further work.

23.2 Residual exceptional degree questions

Theorem 21.2 determines the least coefficient degree for all positive even β and integer twists, at every recurrence order. Theorem 21.4 shows that a degree drop can occur at other parameters. A classification of all exceptional non-even parameters is not claimed. The Jacobi refinements above likewise distinguish exact conclusions from the residual resonant cases. Explicit vertex connection bases for arbitrary non-even β, and controlled local expansions on Hankel contours, remain outside the present proofs.

Part II
Monodromy, algebraicity, and arithmetic specialization

24 Galois groups, jet fields, and values of twisted determinants

The absence of nonzero finite singularities in Theorem 2.6(a) has consequences beyond the order of an operator. It makes the change from Bessel monomials to scalar jets invertible at *every* nonzero argument. We determine that change of coordinates over the Laurent polynomial ring, its determinant, the complete algebraic relation ideal, and the arithmetic specialization. These statements cover every integer twist $q \geq 0$.

Put $R = \mathbb{Q}[x, x^{-1}]$, $K = \mathbb{C}(x)$ and

$$y(x) = I_0(2\sqrt{x}) = \sum_{n\geq 0} \frac{x^n}{(n!)^2}, \qquad xy'' + y' - y = 0.$$

The power series, rather than a choice of square root, defines y. For integers m define linear forms $C_m \in R[X, Y]$ by

$$C_0 = X, \qquad C_1 = Y, \qquad C_{m+1} = x^{-1}(C_{m-1} - mC_m) \quad (m \geq 1), \qquad C_{-m} = x^m C_m \quad (m \geq 1). \tag{24.1}$$

For $\ell \geq 1$ and $q \geq 0$ put

$$H_{\ell,q}(x; X, Y) = \det[C_{q+i-j}(x; X, Y)]_{i,j=1}^{\ell}. \tag{24.2}$$

The Bessel recurrence gives $t^{-m} I_m(2t) = C_m(t^2; y(t^2), y'(t^2))$ for all $m \in \mathbb{Z}$. Factoring t^{q+i-j} from the determinant therefore proves

$$G_{\ell,q}(x) = H_{\ell,q}(x; y(x), y'(x)). \tag{24.3}$$

Thus $H_{\ell,q}$ is a nonzero homogeneous form of degree ℓ. Its nonvanishing also follows from the positive constant term of $G_{\ell,q}$ in Lemma 6.1.

24.1 The full group and algebraic base changes

Theorem 24.1 (Galois group and local monodromy)**.** *For every $\ell \geq 1$ and integer $q \geq 0$, the minimal operator of $G_{\ell,q}$ is irreducible over K and has differential Galois group*

$$\mathrm{Sym}^{\ell}(\mathrm{SL}_2) \cong \begin{cases} \mathrm{SL}_2, & \ell \text{ odd}, \\ \mathrm{PSL}_2, & \ell \text{ even}. \end{cases}$$

Here the group is specified in its $(\ell+1)$-dimensional solution representation. Its local monodromy at $x = 0$ is a single unipotent Jordan block $J_{\ell+1}(1)$, independently of q. The order $\ell + 1$ and irreducibility remain unchanged over every finite algebraic extension of K. A ramified algebraic pullback at zero preserves the size of this local block.

Proof. The Bessel connection for y, y' has Galois group SL_2. Indeed its Wronskian is a nonzero constant times x^{-1}, so its Galois group is contained in SL_2; the pullback $x = t^2$ is equation (5.1), whose group is SL_2 by the Bessel result used in Section 5. In particular y, y' are algebraically independent over K. To see the latter assertion directly, choose a second solution v. The orbit of y consists of $ay + bv$, $(a, b) \neq (0, 0)$. A polynomial relation for y, y' would consequently vanish at

$(ay + bv, ay' + bv')$ for all such constants a, b. The nonzero Wronskian makes this an invertible linear substitution in a, b, forcing the polynomial to be zero.

Let V be the two-dimensional Bessel solution space. Polarization of $u \mapsto H_{\ell,q}(x; u, u')$ defines an equivariant linear map from $\mathrm{Sym}^{\ell} V$ into the Picard–Vessiot field. This map is nonzero by (24.3). Since $\mathrm{Sym}^{\ell} V$ is irreducible, it is injective. Its image is finite-dimensional and the orbit of $G_{\ell,q}$ spans it. The minimal scalar equation has precisely this solution space, so its group is the displayed image of SL_2. The kernel of the symmetric-power representation is trivial for odd ℓ and $\{I, -I\}$ for even ℓ.

At zero the rank-two Bessel equation has one analytic solution and a second solution containing a nonzero logarithmic term. Its local monodromy is a nontrivial unipotent block of size two. Its ℓth symmetric power is a single block of size $\ell + 1$: the logarithm acts on homogeneous binary forms by a nonzero multiple of $X\partial_Y$, with one chain of that length. The rational polynomial solution map above does not change local monodromy.

The group $\mathrm{Sym}^{\ell}(\mathrm{SL}_2)$ is connected. A finite algebraic extension of the base preserves its identity component, hence its full group and irreducible representation. Ramification of index $e \geq 1$ raises the local monodromy to its eth power; in characteristic zero a unipotent block retains its size under this operation. □

24.2 An everywhere invertible Laurent gauge

Let

$$\mathbf{J}_{\ell,q} = (G_{\ell,q}, G'_{\ell,q}, \dots, G^{(\ell)}_{\ell,q})^{\mathsf{T}}, \qquad \mathbf{m}_{\ell} = (y^{\ell}, y^{\ell-1}y', \dots, (y')^{\ell})^{\mathsf{T}}.$$

Define $P_{\ell,q}(x)$ by the coefficient identities

$$\mathbf{J}_{\ell,q} = P_{\ell,q}(x)\mathbf{m}_{\ell}. \tag{24.4}$$

It is computed without differential elimination: the jth row is the coefficient vector of $\mathscr{D}^j H_{\ell,q}$ in the monomials $X^{\ell-i}Y^i$, where

$$\mathscr{D} = \partial_x + Y\partial_X + x^{-1}(X - Y)\partial_Y.$$

Thus every entry of $P_{\ell,q}$ lies in R.

Theorem 24.2 (Exact determinant of the jet gauge)**.** *One has $P_{\ell,q} \in \mathrm{GL}_{\ell+1}(R)$ and*

$$\det P_{\ell,q}(x) = c_{\ell,q} x^{-\ell(\ell+1)(\ell+3q-1)/6}, \qquad c_{\ell,q} = \det P_{\ell,q}(1) \in \mathbb{Q} \setminus \{0\}. \tag{24.5}$$

In particular both $P_{\ell,q}$ and its inverse are regular and invertible at every $x \in \mathbb{C}^*$.

Proof. If the rows of $P_{\ell,q}$ were dependent over K, (24.4) would give an equation of order at most ℓ for $G_{\ell,q}$, contradicting Theorem 2.6(a). Hence its determinant is nonzero. Write $\mathbf{m}'_{\ell} = A_{\ell}\mathbf{m}_{\ell}$. The diagonal entry of A_{ℓ} belonging to $y^{\ell-i}(y')^i$ is $-i/x$, whence

$$\operatorname{tr} A_{\ell} = -\frac{\ell(\ell+1)}{2x}.$$

Let $B_{\ell,q}$ be the companion matrix of the monic minimal scalar equation. Theorem 2.6(a) says that the ratio of its two leading polynomial coefficients is a/x for a constant a. Put

$$s_{\ell,q} = \sum_{i=0}^{\ell} -i(\ell + q - i) = -\frac{\ell(\ell+1)(\ell+3q-1)}{6}.$$

Comparison of the coefficient of ρ^{ℓ} in its indicial polynomial gives $a = \ell(\ell+1)/2 - s_{\ell,q}$. Therefore $\operatorname{tr} B_{\ell,q} = -a/x$. Differentiating (24.4) and using the independence of its monomials yields $P'_{\ell,q} = B_{\ell,q}P_{\ell,q} - P_{\ell,q}A_{\ell}$. Taking traces after multiplying by $P_{\ell,q}^{-1}$ gives

$$\frac{(\det P_{\ell,q})'}{\det P_{\ell,q}} = \operatorname{tr} B_{\ell,q} - \operatorname{tr} A_{\ell} = \frac{s_{\ell,q}}{x}.$$

Its nonzero rational solution is exactly (24.5), with $c_{\ell,q} \in \mathbb{Q}^{*}$. Since $x^{s_{\ell,q}}$ is a unit of R, the adjugate formula proves that $P_{\ell,q}^{-1}$ also has entries in R. □

Proposition 24.3 (The rational gauge constant)**.** *The constant in* (24.5) *has the closed form*

$$c_{\ell,q} = (-1)^{\binom{\ell+1}{3}+q\binom{\ell+1}{2}} \prod_{j=1}^{\ell} (j!)^{2\ell-2j+1}.$$

Proof. Put $N = \binom{\ell+1}{2}$, $t = \sqrt{x}$, $u(x) = I_0(2t)$ and $v(x) = 2K_0(2t)$. Their x-Wronskian is $-x^{-1}$, so the determinant of the symmetric-power fundamental matrix is $(-1)^N x^{-N}$. By Proposition 6.2(a), its image under $P_{\ell,q}$ has columns

$$\binom{\ell}{b}(f_b, f_b', \dots, f_b^{(\ell)})^{\mathsf{T}}, \qquad f_b(x) = t^{-q\ell}\Psi_b^{(q)}(t), \quad 0 \le b \le \ell.$$

The vertex asymptotics in that proposition give

$$f_b(x) \sim (-1)^{qb} C_b e^{2(\ell-2b)t} t^{-\gamma_b - q\ell}.$$

Their differentiated expansions show that the leading Wronskian factor is the Vandermonde determinant in $\ell - 2b$, multiplied by $t^{-N}\prod_b f_b$. This remains valid when one of these numbers is zero: the determinant of the leading coefficient matrix is still the nonzero Vandermonde determinant. Its value is $(-2)^N \prod_{j=1}^{\ell} j!$. Comparing with (24.5) consequently gives

$$c_{\ell,q} = (-1)^{qN} 2^N \left(\prod_{j=1}^{\ell} j!\right) \left(\prod_{b=0}^{\ell} \binom{\ell}{b} C_b\right).$$

For clarity, the normalization of the Mehta integral in Lemma 5.4 is

$$\mathcal{M}_n = \pi^{n/2} 2^{-n(n-1)/2} \prod_{j=1}^{n} j!.$$

This follows by applying Andreief to monic Hermite polynomials, whose squared norms for the weight e^{-x^2} are $\sqrt{\pi}\, 2^{-j} j!$. Put $A_b = \prod_{j=1}^{b} j!$, $A_0 = 1$. Substitution into the displayed formula for C_b yields

$$\prod_{b=0}^{\ell} C_b = (-1)^{\binom{\ell+1}{3}} 2^{-N} \frac{\prod_{b=0}^{\ell} A_b^2}{(\ell!)^{\ell+1}}, \qquad \prod_{b=0}^{\ell} \binom{\ell}{b} = \frac{(\ell!)^{\ell+1}}{A_{\ell}^2}.$$

The powers of π cancel. Combining these identities gives $c_{\ell,q} = (-1)^{\binom{\ell+1}{3}+qN} A_{\ell} \prod_{b=0}^{\ell-1} A_b^2$, which is the asserted product. □

24.3 Algebraic jet relations and twist fields

Theorem 24.4 (The jet cone)**.** *Set* $\mathbf{Z} = P_{\ell,q}^{-1}\mathbf{J}_{\ell,q}$*. The ideal of all polynomial relations over* K *among the coordinates* $Z_0, \dots, Z_\ell$ *is generated by*

$$Z_i Z_{j+1} - Z_{i+1} Z_j, \qquad 0 \le i < j \le \ell - 1. \tag{24.6}$$

For $\ell = 1$ *this is the zero ideal. In particular*

$$\operatorname{trdeg}_K K(\mathbf{J}_{\ell,q}) = 2,$$

and every two distinct entries of $\mathbf{J}_{\ell,q}$ *are algebraically independent over* K*. These assertions persist after finite algebraic extension of* K*. Moreover*

$$K(\mathbf{J}_{\ell,q}) = K(y^\ell, y^{\ell-1}y', \dots, (y')^\ell) = K(y, y')^{\mu_\ell}, \tag{24.7}$$

where μ_ℓ *acts by simultaneous scalar multiplication. The index of this field in* $K(y, y')$ *is exactly* ℓ*.*

Proof. The coordinates $Z_i = y^{\ell-i}(y')^i$ are the degree-ℓ monomials in two algebraically independent elements. Their relation ideal is the ideal of the affine cone over the rational normal curve. An elementary proof of its generators is as follows. The minors (24.6) allow a pair of indices whose gap is at least two to be replaced by a closer pair with the same sum. Repeating this operation makes all indices differ by at most one. Thus every product of r coordinates is congruent to the unique such balanced product having the same sum of indices. These are exactly the equivalence classes of monomials with the same image in $K[y, y']$. Grouping terms according to their images proves that the minors generate the kernel.

Two distinct rows of $P_{\ell,q}$ are linearly independent. The corresponding homogeneous forms A, B of degree ℓ are therefore nonproportional, and are algebraically independent. Indeed A/B is a nonconstant rational function of Y/X, while common scaling of X, Y changes the degree-ℓ magnitude. More formally, splitting a putative relation into homogeneous pieces would make A/B algebraic over K, a contradiction. This proves pairwise independence of the scalar jets. An algebraic extension of the coefficient field cannot decrease their transcendence degree.

Finally, writing $r = y'/y$ shows that the middle field in (24.7) is $K(r, y^\ell)$. Since r, y are algebraically independent, $K(r, y)/K(r, y^\ell)$ is a cyclic extension of degree ℓ, with precisely the stated scalar action. □

Corollary 24.5 (Twist invariance and the gcd–lcm law)**.** *For fixed* ℓ*, every integer twist* $q \ge 0$ *gives the same complete jet field. Denote it by* $\mathcal{F}_\ell$*. For positive integers* d, e*,*

$$\mathcal{F}_d \mathcal{F}_e = \mathcal{F}_{\gcd(d,e)}, \qquad \mathcal{F}_d \cap \mathcal{F}_e = \mathcal{F}_{\operatorname{lcm}(d,e)}.$$

Thus complete jets in any two coprime positive dimensions recover $K(y, y')$*. For a fixed dimension, the jets at twists* q *and* r *are related by a matrix* $P_{\ell,r}P_{\ell,q}^{-1} \in \mathrm{GL}_{\ell+1}(R)$*.*

Proof. The last assertion and twist invariance follow from (24.4). Put $L = \operatorname{lcm}(d, e)$. In the cyclic Galois extension $K(y, y')/K(y, y')^{\mu_L}$, the two fields are fixed by μ_d and μ_e. Their compositum is fixed by the intersection $\mu_{\gcd(d,e)}$, and their intersection is fixed by the group they generate, μ_L. □

24.4 Every nonzero algebraic argument

Theorem 24.6 (Exact arithmetic jet relations for every twist)**.** *Let $\ell \geq 1$, $q \geq 0$ be integers and let $\alpha \in \overline{\mathbb{Q}} \setminus \{0\}$. Then*

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(G_{\ell,q}(\alpha), G'_{\ell,q}(\alpha), \dots, G^{(\ell)}_{\ell,q}(\alpha)) = 2.$$

Every two distinct numbers in this tuple are algebraically independent. Under the invertible algebraic coordinate change

$$\mathbf{Z} = P_{\ell,q}(\alpha)^{-1} \mathbf{J}_{\ell,q}(\alpha),$$

all polynomial relations are generated by (24.6). In particular each displayed derivative value is nonzero and transcendental, and $G'_{\ell,q}(\alpha)/G_{\ell,q}(\alpha)$ is transcendental.

Proof. It is essential here to apply the E-function theorem in the variable t, not to assume that $I_0(2\sqrt{x})$ is an E-function in x. Put $b(t) = I_0(2t)$. Its exponential-series coefficients of index $2n$ are $\binom{2n}{n}$, and its odd coefficients are zero. They are integers of exponential growth. The same is true of the shifted coefficient sequence of b'. Thus b, b' are E-functions, satisfying

$$\frac{d}{dt}\begin{pmatrix} b \\ b' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 4 & -1/t \end{pmatrix}\begin{pmatrix} b \\ b' \end{pmatrix}.$$

They are algebraically independent over $\mathbb{C}(t)$ by the Bessel Galois argument in the proof of Theorem 24.1. The Siegel–Shidlovskii theorem [24, Theorem 1.1], whose exceptional points are contained in the zeros of t times the common denominator of the system, proves algebraic independence of $b(s), b'(s)$ for every nonzero algebraic s.

Choose s with $s^2 = \alpha$. Since

$$y(\alpha) = b(s), \qquad y'(\alpha) = \frac{b'(s)}{2s},$$

the two base values are algebraically independent over $\overline{\mathbb{Q}}$. Theorem 24.2 proves that $P_{\ell,q}(\alpha)$ is an invertible algebraic matrix. We can therefore repeat the polynomial-kernel and homogeneous-form arguments of Theorem 24.4, over $\overline{\mathbb{Q}}$ and with these two independent numbers in place of y, y'. This proves the exact ideal, transcendence degree and pairwise independence. Each individual coordinate is a nonzero positive-degree homogeneous form in two independent numbers and is consequently transcendental. An algebraic ratio of two distinct jet coordinates would give a linear algebraic relation between them, so their ratio is transcendental as well. □

Corollary 24.7 (Zeros and arithmetic field recovery)**.** *Every complex zero of $G^{(j)}_{\ell,q}$, for $0 \leq j \leq \ell$, is transcendental. At each fixed $\alpha \in \overline{\mathbb{Q}}^*$, the numerical jet fields*

$$\mathcal{F}_\ell(\alpha) = \overline{\mathbb{Q}}(G^{(j)}_{\ell,q}(\alpha) : 0 \leq j \leq \ell)$$

are independent of q and satisfy the same gcd–lcm laws as in Corollary 24.5. In particular two coprime dimensions recover $\overline{\mathbb{Q}}(y(\alpha), y'(\alpha))$ exactly.

Proof. Theorem 24.6 excludes every nonzero algebraic zero. At zero all the indicated derivatives are strictly positive rational numbers, by the coefficient expansion in Lemma 6.1; hence zero is excluded as well. The two base values are algebraically independent, so their field is a rational function field in two generators over $\overline{\mathbb{Q}}$. The proof of Corollary 24.5 applies verbatim to it. □

Remark 24.8 (What the arithmetic conclusion concerns). The result concerns the normalized Bessel determinant and its ordinary derivatives at nonzero algebraic arguments. It does not assert that arbitrary periods, ordinary generating functions, or unrelated numerical constants are transcendental. The no-exception assertion uses the Laurent-unit determinant (24.5); abstract symmetric-power irreducibility alone would not supply it.

25 Exact exponents at every nonzero trace threshold

The indicial exponent described heuristically in the original final remarks can be determined from the first two coefficient layers of the minimal recurrence. This is a statement about the scalar equation; it does not by itself assert a nonzero singular amplitude for a chosen physical branch of the density.

Theorem 25.1 (The radial unitary thresholds)**.** *Let $\ell \geq 2$, put $d = \ell - 1$ and $K = \lceil \ell/2 \rceil$, and let $\mathcal{L}_\ell$ be the density operator in Theorem 2.10. For every $0 \leq j < K$, the nonzero point $c_j = (\ell - 2j)^2$ is a regular singular point. Its full indicial multiset is*

$$0, 1, \dots, \ell - 3, \quad \frac{(\ell - j)^2 + j^2 - 3}{2}.$$

The list $0, \dots, \ell - 3$ is empty when $\ell = 2$. The same statement holds for the polynomial operator $\mathcal{L}_\ell^\circ$ displayed in that theorem, since the removed power of w is invertible at c_j. Repeated exponents in this list are counted with multiplicity.

Proof. Write $h_k = d + 2 - 2k$ and denote the top two coefficients by

$$P_k(\theta) = r_k \theta^{h_k} + p_k \theta^{h_k - 1} + \cdots, \qquad R_k(n) = r_k n^d + s_k n^{d-1} + \cdots.$$

When $h_k = 0$, set $p_k = 0$. The normalization of Theorem 2.4 gives, for every $0 \leq k \leq K$,

$$s_k = p_k - k(d + 1 - k) r_k. \tag{25.1}$$

For $k = 0$ this follows from $R_0 = P_0/n^2$. For $k \geq 1$, expand $P_k(n - k) \prod_{a=1}^{k-1} (n - a)^2$ to its top two powers.

Set

$$H(w) = \sum_{k=0}^{K} r_k w^{K-k} = \prod_{j=0}^{K-1} (w - c_j), \qquad F(w) = \sum_{k=0}^{K} \{p_k + (k^2 - k - d) r_k\} w^{K-k}.$$

The density operator is $\sum_{k=0}^{K} w^{K-k} R_k(k - 1 - \theta)$, where $\theta = w\partial_w$. Since $\theta^d = w^d \partial_w^d + \binom{d}{2} w^{d-1} \partial_w^{d-1} + \cdots$, its top two ordinary derivative coefficients are

$$a_d(w) = (-1)^d w^d H(w),$$
$$a_{d-1}(w) = (-1)^d w^{d-1} \{\tbinom{d}{2} H(w) - F(w)\}.$$

At a nonzero simple root c of H, the indicial polynomial is proportional to

$$\rho^{\underline{d-1}} \left(\rho - d + 1 + \frac{a_{d-1}(c)}{a_d'(c)} \right).$$

Thus the extra exponent is $d - 1 + F(c)/(cH'(c))$.

It remains to evaluate this ratio without computing all the P_k. Fix $j < K$, put $\mu = \ell - 2j$, $c = \mu^2$, $\gamma = ((\ell - j)^2 + j^2)/2$, and $\epsilon = \ell + 1 - 2K \in \{0, 1\}$. Define

$$C(v) = \sum_{k=0}^{K} r_k v^{h_k} = v^\epsilon H(v^2).$$

The formal solution of Theorem 2.2(c), in $x = t^2$, is $e^{2\mu t} t^{-\gamma} (1 + O(t^{-1}))$. Substitution into $\sum_k t^{2k} P_k(\frac{1}{2} t \partial_t)$ gives at the next power after the leading characteristic equation

$$\sum_{k=0}^{K} p_k \mu^{h_k - 1} = \frac{\gamma}{2} C'(\mu) - \frac{\mu}{4} C''(\mu). \tag{25.2}$$

Indeed the subleading term of $(\frac{1}{2}t\partial_t)^{h_k}$ on the displayed formal solution is $\{h_k(h_k-1)/4 - h_k\gamma/2\}\mu^{h_k-1}t^{h_k-1}$. The next coefficient of $1+O(t^{-1})$ multiplies $C(\mu)=0$.

With $E=v\partial_v$ and $k=(d+2-h_k)/2$, direct expansion gives

$$\sum_k (k^2-k-d)r_k v^{h_k} = \frac{1}{4}\{v^2C''(v) - (2d+1)vC'(v) + d(d-2)C(v)\}.$$

Multiply (25.2) by μ and add this identity at $v=\mu$. The second derivative terms cancel, and $C(\mu)=0$, so

$$\mu^\epsilon F(c) = \frac{2\gamma-2d-1}{4}\mu C'(\mu) = \left(\gamma - d - \frac{1}{2}\right)\mu^\epsilon cH'(c).$$

The extra exponent is therefore $\gamma - 3/2$, as asserted. □

Lemma 25.2 (A simple exponential rate gives a density exponent)**.** *Suppose a polynomial differential operator has the form*

$$L = t^d\chi(\partial_t) + t^{d-1}\psi(\partial_t) + \text{terms of smaller degree in } t,$$

where $d\geq 1$. If λ is a simple root of χ and $e^{\lambda t}t^{-\gamma}(1+O(t^{-1}))$ is a formal solution, then the Fourier operator under $t\mapsto -\partial_w$, $\partial_t\mapsto w$ has at $w=\lambda$ the indicial multiset

$$0, 1, \ldots, d-2, \gamma-1.$$

Proof. The first subleading formal coefficient gives $\psi(\lambda)=\gamma\chi'(\lambda)$. The two highest derivative coefficients of the Fourier operator are $(-1)^d\chi(w)$ and $(-1)^d(d\chi'(w)-\psi(w))$. Its leading coefficient has a simple zero at λ, hence the point is regular singular. The simple-zero indicial formula used above gives the remaining exponent $d-1-(d-\gamma)=\gamma-1$. □

Corollary 25.3 (Tableau and symplectic trace thresholds)**.** *For the tableau trace law with $k=2r+\varepsilon\geq 2$, the Fourier transform of L_k has at each $\lambda_j^{(k)}=\varepsilon+2r-4j$ the exponents*

$$0, 1, \ldots, r-2, \qquad (r-j)(r-j+\varepsilon-\tfrac{1}{2}) + j(j+\tfrac{1}{2}) - 1.$$

For the trace law on $\mathrm{Sp}(2\ell)$, the Fourier transform of the operator of $e^{-t}Y_{2\ell+1}$ has at $2(\ell-2j)$ the exponents

$$0, 1, \ldots, \ell-2, \qquad (\ell-j)(\ell-j+\tfrac{1}{2}) + j(j+\tfrac{1}{2}) - 1.$$

These statements include a zero threshold whenever it occurs.

Proof. Apply Lemma 25.2 to the degree-r operator in Theorem 8.2 and its degree-ℓ symplectic gauge transform. Their characteristic roots are distinct and their formal powers are the $\gamma_j^{(k)}$ in (8.1). □

Remark 25.4. Integral differences among these exponents permit logarithmic solutions, but do not force a logarithm in every solution. In particular the indicial data do not replace a connection formula for the actual probability density. All Fourier transforms above are formed before any cancellation by a function vanishing at a threshold.

26 Exact recurrence widths after arithmetic sampling

For a nonzero sequence a_n, its *eventual recurrence width* is the least L for which

$$\sum_{j=0}^{L} p_j(n) a_{n-j} = 0$$

holds for every sufficiently large integer n, with $p_j \in \mathbb{C}(n)$ and $p_0 p_L \neq 0$. Clearing denominators gives the same minimum with polynomial coefficients. The following result concerns every sampling interval, without a genericity assumption.

Theorem 26.1 (Recurrence widths under arithmetic sampling)**.** *Let $d \geq 1$, $0 \leq b < d$, $\ell \geq 1$, and $q \geq 0$ be integers. Then*

$$\operatorname{width}\bigl(u_{\ell,q}(dn+b)\bigr) = \operatorname{width}\bigl(m_\ell(dn+b)\bigr) = \left\lceil \frac{\ell}{2} \right\rceil .$$

For tableaux, put $r = \lfloor k/2 \rfloor$. For every $k \geq 2$,

$$\operatorname{width}\bigl(y_k(dn+b)\bigr) = \begin{cases} r+1, & k \text{ odd}, \\ 2\lceil r/2 \rceil, & k \text{ even}, \ d \text{ odd}, \\ \lceil r/2 \rceil, & k \text{ even}, \ d \text{ even}. \end{cases}$$

All these conclusions remain valid after multiplying the sampled sequence by $c^n h(n)$, where $c \in \mathbb{C}^*$ and $0 \neq h \in \mathbb{C}(n)$, with the finitely many poles omitted.

The untwisted assertion $q = 0$ applies to permutations with bounded increasing subsequences. For even tableau height, sampling on an even progression halves the original exact width. This reduction is caused by antipodal Bessel components; it is independent of the residue b.

Lemma 26.2 (Sampling upper bound)**.** *If a sequence satisfies an eventual recurrence of width R, then every arithmetic subsequence satisfies one of width at most R.*

Proof. Outside finitely many indices, a recurrence of width R gives a first-order system $v_{n+1} = A(n)v_n$ for a vector of R consecutive terms, with $A(n)$ over $\mathbb{C}(n)$. On indices $n = dm + b$ this becomes

$$v_{d(m+1)+b} = A(dm+b+d-1) \cdots A(dm+b) v_{dm+b}.$$

Each of $R+1$ consecutive sampled outputs is a rational row vector applied to v_{dm+b}. These rows are linearly dependent over $\mathbb{C}(m)$, yielding a recurrence of width at most R after deleting zero extreme coefficients. Since the sampled sequences in the theorem have positive terms, the relation cannot have width zero. □

Lemma 26.3 (Distinct simple components cannot cancel)**.** *Suppose $V_1, \ldots, V_s$ are pairwise nonisomorphic simple differential modules over $\mathbb{C}(t)$, realized as differential submodules of a common field of meromorphic functions, and $0 \neq f_j \in V_j$. The differential module generated by $\sum_j f_j$ is their direct sum.*

Proof. The sum of pairwise nonisomorphic simple submodules is direct. It is semisimple, and any submodule is the direct sum of a subset of these simple summands: this follows inductively from simplicity and the fact that a nonzero homomorphism between simple modules is an isomorphism. The submodule generated by $\sum_j f_j$ projects nontrivially to each summand, so it contains every one of them. □

Lemma 26.4 (A factorial budget for a sampled recurrence)**.** *Let $p, h \geq 1$, $c \geq 0$, and $\alpha_1, \dots, \alpha_p \geq 0$ be integers, and put*

$$H(t) = \sum_{n \geq 0} \frac{a_n t^{p(hn+c)}}{\prod_{i=1}^{p} (hn + c + \alpha_i)!}.$$

Suppose the minimal scalar equation of H has formal solutions at infinity of the form $e^{\beta_j t} t^{\mu_j}(1 + O(t^{-1}))$, with ν distinct nonzero numbers β_j. Then every eventual recurrence of (a_n) has width at least $\nu/(ph)$.

Proof. Clear the denominators of a recurrence $\sum_{j=0}^{L} p_j(n) a_{n-j} = 0$ and let $D = \max_j \deg p_j$. With $\theta = t\partial_t$, form

$$P = \sum_{j=0}^{L} t^{phj} p_j\left(\frac{\theta}{ph} + j - \frac{c}{h}\right) \prod_{i=1}^{p} \left(\frac{\theta}{p} + \alpha_i\right)^{\overline{h(L-j)}}.$$

For sufficiently large n, the coefficient of $t^{p(hn+c)}$ in PH equals the proposed recurrence divided by $\prod_i (hn + c - hL + \alpha_i)!$. Consequently PH is a polynomial. A polynomial in θ annihilates this finite remainder.

For $P = \sum_m b_m(t) \partial_t^m$, the largest degree of the polynomials b_m is $D + phL$. Its leading polynomial in the derivative variable is

$$\sigma_P(\xi) = \sum_{\deg p_j = D} \operatorname{lc}(p_j) \left(\frac{\xi}{ph}\right)^{D} \left(\frac{\xi}{p}\right)^{ph(L-j)}.$$

The powers for distinct j differ, so this polynomial is nonzero. After a power of ξ is removed, its degree is at most phL. The additional polynomial in θ used to remove the finite remainder changes only this power of ξ.

Every homogeneous annihilator is a left multiple of the minimal operator of H. It therefore annihilates each of its formal solutions. Substituting $e^{\beta_j t} t^{\mu_j}(1 + O(t^{-1}))$ shows that $\sigma_P(\beta_j) = 0$. Since the β_j are distinct and nonzero, $\nu \leq phL$. □

Lemma 26.5 (A Laurent Bessel form has a short parity recurrence)**.** *Let $r \geq 1$, $y(t) = I_0(2t)$, and let $\Phi \in \mathbb{C}[t, t^{-1}][X_0, X_1]$ be homogeneous of degree r. Suppose $F(t) = \Phi(t, y(t), y'(t))$ is nonzero, holomorphic at zero and $F(-t) = (-1)^p F(t)$, where $p \in \{0, 1\}$. Write*

$$F(t) = \sum_{n \geq 0} a_n \frac{t^{2n+p}}{(2n+p)!}.$$

Then (a_n) and all its arithmetic subsequences have eventual recurrence width at most $\lceil r/2 \rceil$.

Proof. For $0 \leq j \leq r$, differentiation of y^r using $y'' = 4y - t^{-1}y'$ gives

$$\partial_t^j(y^r) = r^{\underline{j}} y^{r-j} (y')^j + \sum_{i<j} a_{j,i}(t) y^{r-i} (y')^i, \qquad a_{j,i} \in \mathbb{C}[t, t^{-1}].$$

The change-of-basis matrix is triangular with nonzero constant diagonal entries. Its inverse therefore also has Laurent-polynomial entries. It follows that $F = R(y^r)$ for a differential operator $R \in \mathbb{C}[t, t^{-1}]\langle \partial_t \rangle$ of order at most r. Replacing R by its parity-p part under $(t, \partial_t) \mapsto (-t, -\partial_t)$, we may take every monomial $t^a \partial_t^j$ occurring in R to satisfy $a + j \equiv p \pmod 2$.

Put

$$W_r(n) = \sum_{n_1 + \cdots + n_r = n} \binom{n}{n_1, \dots, n_r}^2, \qquad y(t)^r = \sum_{n \geq 0} W_r(n) \frac{t^{2n}}{(n!)^2}.$$

Verrill's recurrence has width $K = \lceil r/2 \rceil$ [22, Theorem 1 and equation (15)]; its Bessel derivation is recalled in Remark 45.1. For $r = 1$ this is simply $W_1(n) = 1$. The contribution of $t^a \partial_t^j(y^r)$ to a_n is

$$(2n+p)! \, \frac{(2n+2c)^{\underline{j}}}{((n+c)!)^2} \, W_r(n+c), \qquad c = \frac{p+j-a}{2} \in \mathbb{Z},$$

for every sufficiently large n. Dividing by $\binom{2n}{n}$ makes its coefficient of $W_r(n+c)$ a rational function of n. Thus $a_n/\binom{2n}{n}$ is a finite rational-function linear combination of shifts of W_r.

A recurrence of width K expresses every such shift through K consecutive values, using forward or backward shifts outside finitely many poles. Hence $K+1$ consecutive values of any rational linear combination are linearly dependent over $\mathbb{C}(n)$, and its width is at most K. Finally $\binom{2n+2}{n+1}/\binom{2n}{n} = 2(2n+1)/(n+1)$ is rational, so restoring this hypergeometric factor preserves the width. Lemma 26.2 handles further sampling. □

Proof of Theorem 26.1. Let $\zeta = e^{2\pi i/d}$ and choose $\omega_j = e^{\pi i j/d}$, $0 \le j < d$, so $\omega_j^2 = \zeta^j$. Root-of-unity filtering gives

$$H(t) = \frac{1}{d} \sum_{j=0}^{d-1} \zeta^{-bj} G_{\ell,q}(\zeta^j t^2) = \sum_{n \ge 0} \frac{u_{\ell,q}(dn+b) t^{2(dn+b)}}{(dn+b)!(dn+b+q\ell)!}.$$

Each term on the left generates a simple differential module: it is a rational gauge of the ℓth symmetric power of the Bessel module at scale ω_j. Simplicity follows from the SL_2 Galois group and irreducibility of its symmetric powers, as in the proof of the original minimal-order theorem. Its formal exponential rates are

$$2\omega_j(\ell - 2i), \qquad 0 \le i \le \ell.$$

Two such sets could coincide only if their scales differed by a sign: compare the two extreme points of largest modulus. The chosen ω_j never differ by a sign. Thus the modules are pairwise nonisomorphic, and Lemma 26.3 shows that the filtered function contains all of them. Furthermore all their nonzero rates are distinct: a ratio of two nonzero real rates can have modulus one only when it is 1 or -1. There are exactly $2d\lceil \ell/2 \rceil$ such rates. Lemma 26.4, with $p = 2$, $h = d$, $c = b$, and $(\alpha_1, \alpha_2) = (0, q\ell)$, gives the required lower bound. The recurrence of Theorem 2.6 and Lemma 26.2 give the matching upper bound.

For matchings put $M_\ell(t) = \sum_{N \ge 0} m_\ell(N) t^{2N}/(2N)!$. It is an even Bessel determinant generating the ℓth symmetric-power module, with rates $2(\ell - 2i)$. The same filter $d^{-1} \sum_{j<d} \zeta^{-bj} M_\ell(\omega_j t)$ selects $N \equiv b \pmod d$. The simple-module argument gives the same $2d\lceil \ell/2 \rceil$ distinct nonzero rates. Apply the budget with $p = 1$, $h = 2d$, $c = 2b$, and $\alpha_1 = 0$; Theorem 2.9 supplies the upper bound.

For tableaux write $k = 2r + \varepsilon$, put $R = r + 1 - [4 \mid k]$, and use the filter

$$\frac{1}{d} \sum_{j=0}^{d-1} \zeta^{-bj} Y_k(\zeta^j t) = \sum_{n \ge 0} y_k(dn+b) \frac{t^{dn+b}}{(dn+b)!}.$$

Its components are simple, being exponential twists of Sym^r of a scaled Bessel module. Their rate sets are

$$\zeta^j \{\varepsilon + 2r - 4i : 0 \le i \le r\}.$$

If $\varepsilon = 1$, the mean of the unscaled set is 1, so different scales give different sets. No two unscaled rates are negatives of one another, since all are congruent to the same odd number modulo 4. All $d(r+1)$ rates in the filter are therefore nonzero and distinct. If $\varepsilon = 0$ and d is odd, no two dth roots of unity are antipodal; the same argument gives dR distinct nonzero rates. The budget

with $p = 1$, $h = d$, $c = b$ yields the lower bound R, and Theorem 2.8 gives the upper bound in these cases.

It remains to consider even $k = 2r$ and even d. Write $p \equiv b \pmod 2$ and put $F_p(t) = (Y_{2r}(t) + (-1)^p Y_{2r}(-t))/2$. This is a homogeneous Laurent-polynomial Bessel form of degree r; it is nonzero, since its coefficients count tableaux of sizes $2n + p$ and are positive. By Lemma 26.5, $y_{2r}(2n + p)$ has width at most $K = \lceil r/2 \rceil$. Our sampled sequence is a further arithmetic subsequence of it, so it also has width at most K.

In the d-term filter above, pair indices j and $j + d/2$. Their sum is a nonzero scalar multiple of $F_p(\zeta^j t)$. For $0 \le j < d/2$ these functions generate pairwise nonisomorphic simple modules, with rate sets $\zeta^j\{2r - 4i : 0 \le i \le r\}$. The scales are distinct modulo sign, and each module has $2K$ distinct nonzero rates. Consequently the entire filter has dK such rates. Lemma 26.4, still with $p = 1$, $h = d$, $c = b$, gives the matching lower bound K. This proves the last line of the classification.

Finally multiplication by $c^n h(n)$ is an invertible change of unknown in recurrences over $\mathbb{C}(n)$; it changes their coefficients but not their extreme shifts. □

Corollary 26.6 (Exact width for every homogeneous parity observable)**.** *Let $r \ge 1$, $p \in \{0, 1\}$, and $\Phi \in \mathbb{C}[t, t^{-1}][X_0, X_1]$ be homogeneous of degree r. Suppose*

$$F(t) = \Phi(t, I_0(2t), \partial_t I_0(2t))$$

is nonzero, holomorphic at zero, and satisfies $F(-t) = (-1)^p F(t)$. Define

$$a_n = (2n + p)!\,[t^{2n+p}]F(t).$$

For every $d \ge 1$ and $0 \le b < d$, the sequence (a_{dn+b}) is not eventually zero and has exact eventual recurrence width $\lceil r/2 \rceil$ over $\mathbb{C}(n)$.

Proof. Lemma 26.5 supplies the upper bound, including all further sampling. For the lower bound let $\zeta = e^{2\pi i/d}$ and $\omega_j = e^{\pi i j/d}$ for $0 \le j < d$. Then

$$H(t) = \frac{1}{d}\sum_{j=0}^{d-1} \zeta^{-bj}\omega_j^{-p} F(\omega_j t) = \sum_{n\ge 0} a_{dn+b}\frac{t^{2(dn+b)+p}}{(2(dn+b)+p)!}.$$

The nonzero homogeneous form F generates the entire simple rth symmetric-power Bessel module: it is a nonzero element of that module, and the SL_2 symmetric-power representation is irreducible. After scaling by ω_j, the formal exponential rates are $2\omega_j(r - 2i)$, $0 \le i \le r$. The scales are distinct modulo sign, so the corresponding simple modules are pairwise nonisomorphic. Lemma 26.3 therefore makes the module generated by H their direct sum. In particular H is not a polynomial: a nonzero polynomial spans a trivial rank-one module over $\mathbb{C}(t)$, whereas this direct sum consists of nontrivial simple summands of rank $r + 1 \ge 2$. Thus (a_{dn+b}) is not eventually zero.

The direct sum has $2d\lceil r/2 \rceil$ distinct nonzero formal exponential rates. Apply Lemma 26.4 with one factorial, sampling step $h = 2d$, offset $c = 2b + p$, and $\alpha_1 = 0$. It gives the lower bound $\lceil r/2 \rceil$, completing the proof. □

Example 26.7. The sequences $u_{5,7}(100n+37)$ and $m_5(100n+37)$ both have minimum recurrence width 3. The sequences $y_7(dn + b)$ have minimum width 4 for every d, b, and $y_8(dn + b)$ have minimum width 4 for odd d and width 2 for even d. These assertions concern exact eventual recurrence width, not merely the order of a particular elimination output. They make no claim about the smallest coefficient degree after sampling.

27 A local Fourier proof and sharpening of the Laplace–Jordan principle

The following local Fourier argument controls the origin monodromy of iterated factorial Borel transforms. It uses formal stationary phase and the nearby-cycle/vanishing-cycle description at one point. Only the unit eigenvalue can acquire an additional Jordan step.

Lemma 27.1 (Nearby and vanishing monodromy)**.** *Let V, W be finite-dimensional complex vector spaces, with maps $u : V \to W$ and $v : W \to V$, and suppose*

$$T = I_V + vu, \qquad S = I_W + uv$$

are invertible. For every polynomial $p \in \mathbb{C}[z]$,

$$(T - I_V)p(T) = v\,p(S)u. \tag{27.1}$$

In particular, $p(S) = 0$ implies $(T - I_V)p(T) = 0$. For each $\lambda \neq 1$, the restrictions of T and S to their λ-generalized eigenspaces are isomorphic. Only the 1-eigenspace can acquire a further Jordan step in passing from S to T.

Proof. The relation $uT = Su$ implies $up(T) = p(S)u$. Multiplication by v proves (27.1). On a λ-generalized eigenspace with $\lambda \neq 1$, $T - I_V$ and $S - I_W$ are invertible. Thus u and v give mutually inverse isomorphisms after composing one of them with the appropriate inverse of $T - I_V$ or $S - I_W$. The identity then proves the assertion at $\lambda = 1$ as well. □

We use local monodromy at infinity in the positive local coordinate $1/s$. With this convention, regular local Fourier transform preserves its monodromy conjugacy class. A change of Fourier sign is a homothety of the target and does not alter that class.

Proposition 27.2 (The Fourier monodromy bound)**.** *Let $\mathcal{M}$ be a holonomic module on the complex affine s-line, whose generic connection is regular singular at infinity. Let M be the monodromy of that regular germ. Write T for the monodromy at $x = 0$ of the generic connection underlying its Fourier transform. Then the transformed connection has no singularity at a finite $x \neq 0$, is regular singular or ordinary at 0, and*

$$p(M) = 0 \quad \Longrightarrow \quad (T - I)p(T) = 0.$$

The same annihilating polynomial applies to the monodromy of every differential subquotient of the transformed generic connection. No middle-extension hypothesis on $\mathcal{M}$ is required.

Proof. Formal stationary phase gives

$$\Phi_x(\mathcal{F}\mathcal{M}) \simeq \mathcal{F}^{(\infty,x)}\big((\Psi_\infty \mathcal{M})^{\leq 1,(x)}\big)$$

for every finite x; this is [25, Corollary 2.3(1)]. Because $\Psi_\infty\mathcal{M}$ is regular, its exponential-type summand is zero unless $x = 0$. At $x = 0$ its local Fourier transform is again regular and has monodromy M: [26, Remark 5.2 and Remark 5.3(4)]. These statements concern the extended formal local Fourier functor, which includes trivial regular summands. For example, a trivial rank-one infinity germ contributes the trivial rank-one vanishing cycles of a delta module; it is not discarded.

For a holonomic module on a formal disc, its irregular part is the same in nearby and vanishing cycles. Thus the regularity of the vanishing cycles proves regularity of the nearby germ, and their vanishing at $x \neq 0$ proves smoothness there. On the regular part, nearby and vanishing cycles have the description (V, W, u, v) with monodromies $I + vu$ and $I + uv$, respectively; see

[25, §3.3, formula (3.3)]. The vanishing monodromy S is conjugate to M. Lemma 27.1 therefore proves the asserted identity and bound. Restricting an annihilating polynomial to an invariant subspace or its quotient preserves the identity. The argument also works for the dual local system of solutions, since local Fourier transform is compatible with duality and both source and target monodromies are dualized. □

Theorem 27.3 (Eigenvalue-resolved iterated Borel bound)**.** *Let $A_0(t) = \sum_{n\ge 0} a_n t^n$ be a nonzero D-finite germ with positive radius of convergence. Suppose its minimal equation is regular singular or ordinary at 0, and denote its monodromy by M. For $r \ge 0$, put*

$$A_r(x) = \sum_{n\ge 0} \frac{a_n x^n}{(n!)^r}.$$

Each minimal equation is regular singular or ordinary at 0. If $p(M) = 0$, then its monodromy T_r satisfies

$$(T_r - I)^r p(T_r) = 0. \tag{27.2}$$

In particular, nonunit monodromy eigenvalues acquire no larger Jordan blocks under any number of Borel transforms. The largest Jordan-block size at eigenvalue 1 increases by at most r. If A_0 is algebraic, every nonunit block of T_r is semisimple, and every block at eigenvalue 1 has size at most $r + 1$.

Proof. It suffices to prove the assertion for one transform and iterate. Put $f(s) = s^{-1}A_0(s^{-1})$ and $\mathcal{M} = \mathbb{C}[s]\langle\partial_s\rangle \cdot f$. Its generic connection is the rational gauge and variable transform of the minimal equation of A_0. It is regular at infinity, and its local monodromy in the coordinate $1/s$ is M.

Here is the exact relation with the series transform. On $\mathbb{C}[s] \oplus s^{-1}\mathbb{C}[[s^{-1}]]$, define

$$\mathcal{B}(s^j) = 0 \quad (j \ge 0), \qquad \mathcal{B}(s^{-n-1}) = x^n/n! \quad (n \ge 0).$$

Direct checking on monomials gives $\mathcal{B}(sf) = \partial_x\mathcal{B}(f)$ and $\mathcal{B}(\partial_s f) = -x\mathcal{B}(f)$. Consequently, with Fourier convention $s \mapsto \partial_x$, $\partial_s \mapsto -x$, there is a surjection of holonomic modules

$$\mathcal{F}(\mathcal{M}) \longrightarrow \mathbb{C}[x]\langle\partial_x\rangle \cdot A_1.$$

The solution local system of the minimal equation of A_1 is therefore a subquotient of the Fourier solution local system. Proposition 27.2 gives $(T_1 - I)p(T_1) = 0$ and regularity. Applying the same argument successively with the polynomial $(z - 1)^j p(z)$ proves (27.2).

For algebraic A_0, the local monodromy of its minimal equation has finite order: its solutions are spanned by finitely many algebraic conjugates. Its minimal polynomial is a product of distinct roots-of-unity factors. Multiplication by $(z - 1)^r$ therefore gives exactly the stated eigenvalue-specific bounds. □

For an algebraic source germ, the theorem gives a unit-eigenvalue block bound of two after one transform and three after two transforms. Multiplication by an integer power of the variable is an invertible rational gauge, so it preserves both the regularity hypothesis and these monodromy bounds.

28 Twisted Catalan identities and ordinary algebraicity

Proposition 28.1 (Two-row twists as Catalan tails)**.** *For every $q, n \ge 0$,*

$$u_{1,q}(n) = 1, \qquad u_{2,q}(n) = C_{n+q}, \qquad C_j = \frac{1}{j+1}\binom{2j}{j}.$$

Consequently, with $C(z) = (1 - \sqrt{1-4z})/(2z)$,

$$\sum_{n\ge 0} u_{2,q}(n) z^n = z^{-q} \left(C(z) - \sum_{j=0}^{q-1} C_j z^j \right),$$

where the sum is empty for $q = 0$ *and the singularity at* $z = 0$ *is removable.*

Proof. There is exactly one standard tableau of every one-row shape, which proves the first identity. At $\ell = 2$, the extremal coefficient formula in Theorem 2.6 gives

$$(n+q+1)u_{2,q}(n) = 2(2n+2q-1)u_{2,q}(n-1), \qquad n \ge 1.$$

The initial value is $u_{2,q}(0) = f^{(q,q)} = C_q$ by the two-row hook-length formula. The sequence C_{n+q} has the same recurrence and initial value, proving the claim. Subtracting the first q coefficients of $C(z)$ gives the ordinary generating function. □

Theorem 28.2 (Twisted determinants under arithmetic sampling)**.** *For integers* $\ell, d \ge 1$, $q \ge 0$, *and* $0 \le b < d$, *the ordinary generating function*

$$A_{\ell,q;d,b}(z) = \sum_{n\ge 0} u_{\ell,q}(dn+b) z^n$$

is algebraic over $\mathbb{C}(z)$ *if and only if* $\ell \le 2$.

Proof. For $\ell = 1$ the series is $(1-z)^{-1}$. For $\ell = 2$, Proposition 28.1 gives an algebraic full series. Every arithmetic section of an algebraic series is algebraic: the root-of-unity filter expresses $z^b A_{\ell,q;d,b}(z^d)$ as a finite sum of algebraic functions, and z is algebraic over z^d.

Suppose now $\ell \ge 3$ and $A_{\ell,q;d,b}$ were algebraic. Then

$$F(z) = z^b A_{\ell,q;d,b}(z^d) = \sum_{n\ge 0} u_{\ell,q}(dn+b) z^{dn+b}$$

is algebraic. Write $\mathcal{B}(\sum a_n z^n) = \sum a_n z^n/n!$ and $Q = q\ell$. An exact coefficient calculation gives

$$\mathcal{B}(z^Q \mathcal{B} F) = z^Q H(z), \qquad H(z) = \sum_{n\ge 0} \frac{u_{\ell,q}(dn+b) z^{dn+b}}{(dn+b)!(dn+b+Q)!}.$$

The regular-singular Borel block bound (Theorem 27.3) applies twice. Algebraic functions have semisimple local monodromy, so the largest unipotent Jordan block at zero is initially of size at most one. Each Borel transform increases the permitted size by at most one; multiplication by the integer power z^Q is a rational gauge and changes no local monodromy. Thus the minimal equation of H has no Jordan block at zero larger than three: its nonunit blocks remain semisimple, and its unit-eigenvalue blocks have size at most three. The hypotheses needed at the intermediate step hold: an algebraic module is regular singular everywhere; its Borel transform is regular singular at zero by the cited bound; rational gauges preserve this property.

On the other hand,

$$H(t^2) = \frac{1}{d} \sum_{j=0}^{d-1} \zeta^{-bj} G_{\ell,q}(\zeta^j t^2)$$

is exactly the direct sum of d pairwise nonisomorphic simple symmetric-Bessel modules used in the proof of Theorem 26.1. A Bessel module at zero has one nontrivial unipotent block of size two. Its ℓth symmetric power has one unipotent block of size $\ell+1$: if the local basis is $v, w = v \log t + \text{meromorphic}$, then w^ℓ has nonzero coefficient v^ℓ at $(\log t)^\ell$. The rational gauge defining $G_{\ell,q}$ does not change this block. Consequently $H(t^2)$ has a block of size $\ell + 1 \ge 4$. The

finite pullback $z = t^2$ squares monodromy. For every nonzero eigenvalue, squaring preserves each Jordan-block size, since the derivative of z^2 at that eigenvalue is nonzero. It therefore cannot create a block larger than three, which is the required contradiction. Hence $A_{\ell,q;d,b}$ is transcendental. □

Remark 28.3 (Scope of the two assertions). The recurrence theorem is independent of the Borel block bound: its proof uses only the Bessel modules and the explicit factorial budget. The algebraicity theorem additionally uses the local Fourier–Laplace theorem stated in Theorem 27.3. Neither assertion is an irrationality or transcendence assertion about evaluations of an ordinary generating function at individual algebraic points.

Corollary 28.4 (The remaining tableau algebraicity cases)**.** *The ordinary generating function* $\sum_n y_k(n)z^n$ *is algebraic over* $\mathbb{C}(z)$ *exactly when* $k \le 3$. *The ordinary generating function* $\sum_n m_\ell(n)z^n$ *is algebraic exactly when* $\ell = 1$. *These are the unfiltered classifications developed in the entropy–transport development; their proof is reproduced here to make the present paper independent of that manuscript.*

Proof. The scalar tableau equation has the solution representation $e^{\varepsilon t}\operatorname{Sym}^r V$, with $r = \lfloor k/2 \rfloor$ and V the rank-two Bessel solution space. Thus its origin monodromy has the single block $J_{r+1}(1)$. Algebraicity of the ordinary series would give blocks of size at most two after one Borel transform, by Theorem 27.3. This excludes every $k \ge 4$. For the smaller heights the standard determinant formulas give

$$\sum_n y_1(n)z^n = \frac{1}{1-z}, \qquad \sum_n y_2(n)z^n = \frac{(1+2z)/\sqrt{1-4z^2}-1}{2z},$$

and

$$\sum_n y_3(n)z^n = \frac{1-z-\sqrt{1-2z-3z^2}}{2z^2}.$$

The apparent singularities at zero are removable. For example, the even and odd coefficients of Y_2 are $\binom{2n}{n}$ and $\binom{2n+1}{n}$, and Y_3 is the exponential generating function of the Motzkin numbers, as follows by coefficient extraction in (2.3).

For matchings, algebraicity of $\sum_n m_\ell(n)z^n$ implies algebraicity after $z = t^2$. Its Borel transform is $e^{-t}Y_{2\ell+1}(t)$, whose origin block has size $\ell+1$. The bound two therefore excludes $\ell \ge 2$. For $\ell = 1$, noncrossing matchings are counted by the Catalan numbers, whose generating function is algebraic. □

29 Bessel determinants and the three algebraicity classifications

This section supplies the combinatorial definitions, Bessel reductions, and scalar ordinary-point result used in the differential and arithmetic theorems of Section 30. The Laplace input is Theorem 27.3. In particular, no unpublished tableau operator theorem is needed for the arithmetic specialization.

29.1 A coordinate and indexing dictionary

The combinatorial sequences are those defined in Section 2. In this section the crossing parameter is written k, so

$$f_k(n) = m_{k-1}(n) \quad (k \ge 2), \qquad f_1(0) = 1, \quad f_1(n) = 0 \quad (n \ge 1).$$

Define

$$\widetilde{Y}_k(z)=\sum_{n\ge0}y_k(n)z^n,\quad \widetilde{U}_k(z)=\sum_{n\ge0}u_k(n)z^n,\quad \widetilde{F}_k(z)=\sum_{n\ge0}f_k(n)z^n.$$

Their normalized versions satisfy

$$Y_k(x)=\sum_{n\ge0}y_k(n)x^n/n!,\qquad U_k(s)=\sum_{n\ge0}u_k(n)s^n/(n!)^2=G_k(s),$$

$$F_k(x)=\sum_{n\ge0}f_k(n)x^{2n}/(2n)!=e^{-x}Y_{2k-1}(x).$$

The determinant identities are (2.2), (2.3), and (2.4), with the indicated index shift for matchings. In particular, $U_k(s)=D_k(\sqrt{s})$. These are the Gordon–Gessel and Weyl-chamber enumerations; see also [44, 45].

29.2 Reduction to two Bessel coordinates

Write

$$y(x)=I_0(2x)=\sum_{m\ge0}\frac{x^{2m}}{(m!)^2},\qquad y_s(s)=I_0(2\sqrt{s})=\sum_{m\ge0}\frac{s^m}{(m!)^2}.$$

Their scalar equations and logarithmic companion solutions are

$$y''=4y-x^{-1}y',\qquad \kappa(x):=K_0(2x)=-(\log x)y(x)+h(x),\tag{29.1}$$

$$y_s''=s^{-1}(y_s-y_s'),\qquad \kappa_s(s):=K_0(2\sqrt{s})=-\tfrac12(\log s)y_s(s)+h_s(s),\tag{29.2}$$

where h,h_s are holomorphic at zero. The integer-order Bessel recurrences and these local expansions may be found in [9, §§10.29, 10.31].

Lemma 29.1 (Bessel group and independence)**.** *The two Bessel equations have differential Galois group* SL_2 *over their rational function fields. The pairs* (y,y') *and* (y_s,y_s') *are algebraically independent over* $\mathbb{C}(x)$ *and* $\mathbb{C}(s)$*, respectively.*

Proof. The rank-two case of Theorem 24.1, together with the pullback $s=x^2$, gives both Galois groups. The orbit argument there proves algebraic independence of the two distinguished coordinates. □

Lemma 29.2 (Determinants are homogeneous forms)**.** *Each determinant in* (2.2)*,* (2.3)*, and* (2.4) *is a nonzero homogeneous polynomial in the corresponding Bessel pair, with rational Laurent-polynomial coefficients. The degrees are* $\lfloor k/2\rfloor$ *for* $e^{-\varepsilon x}Y_k(x)$*, where* $k=2\lfloor k/2\rfloor+\varepsilon$*;* k *for* $U_k(s)$*; and* $k-1$ *for* $F_k(x)$*.*

Proof. The recurrences

$$b_0=y,\qquad b_1=y'/2,\qquad b_{j+1}=b_{j-1}-(j/x)b_j$$

express every b_j as a rational Laurent-polynomial linear form in y,y'. Thus a determinant of size d is homogeneous of degree d.

For the s-variable put $J_j(s)=s^{-j/2}I_j(2\sqrt{s})$ for $j\ge0$. Then $J_0=y_s$, $J_1=y_s'$ and $sJ_{j+1}=J_{j-1}-jJ_j$. Multiplying row i of (2.2) by $s^{-i/2}$ and column j by $s^{j/2}$ does not change its determinant, and changes its entry to $s^{\max(0,j-i)}J_{|i-j|}(s)$. All coefficients are therefore in $\mathbb{Q}[s,s^{-1}]$. Each resulting form is nonzero because the normalized counting series has constant coefficient 1. □

Lemma 29.3 (The logarithmic degree is exact)**.** *Let v be either analytic Bessel solution above, in variable z, and let $\Phi \in \mathbb{C}(z)[X_0, X_1]$ be nonzero homogeneous of degree $d \geq 1$. The function $\Phi(v, v')$ has minimal differential order $d+1$, and its minimal equation at zero has one unipotent Jordan block of size $d+1$. Multiplication by $e^{\varepsilon z}$ does not change these two assertions.*

Proof. This is the order and local-monodromy part of Theorem 30.1 below. Its proof uses Lemma 29.1, independently of the present lemma. □

29.3 Ordinary generating functions

Theorem 29.4 (Algebraicity of the three combinatorial families)**.** *For positive integer parameters:*

1. *$\widetilde{Y}_k$ is algebraic if and only if $k \leq 3$;*
2. *the ordinary generating function for permutations avoiding an increasing subsequence of length ℓ is algebraic if and only if $\ell \leq 3$;*
3. *$\widetilde{F}_k$ is algebraic if and only if $k \leq 2$.*

Proof. For tableaux and matchings use Corollary 28.4 and $f_k = m_{k-1}$; the case f_1 has generating function 1. For permutations use Theorem 28.2 at twist zero and sampling interval one, with $k = \ell - 1$. Avoidance of length one permits only the empty permutation. □

29.4 A published Jacobi system and a scalar tableau equation

The arithmetic specialization below requires every nonzero point to be ordinary for the minimal tableau scalar equation. We retain a general-parameter Jacobi elimination and identify its specialization with the scalar equation already proved.

For $N \geq 1$, $\alpha > -1$, $b > 0$, $\beta > 0$, and $0 \leq p \leq N$, let

$$H_p(z) = \binom{N}{p}^{-1} \int_{[0,1]^N} e^{-z\sum u_i} \prod_i u_i^{\alpha}(1-u_i)^{b-1} \prod_{i<j} |u_i - u_j|^{\beta} \cdot e_p(1-u_1, \ldots, 1-u_N)\, du,$$

where e_p is the elementary symmetric polynomial and $e_0 = 1$. The Forrester–Kumar system [10, Corollary 2.5, equations (2.19)–(2.20)] states that

$$(N-p)zH_{p+1} = ((N-p)z + B_p)H_p + zH_p' - D_pH_{p-1}, \tag{29.3}$$

where

$$B_p = p\left(\alpha + b + 1 + \frac{\beta}{2}(2N - p - 1)\right), \qquad D_p = p\left(b + \frac{\beta}{2}(N-p)\right).$$

Terms outside the range $0 \leq p \leq N$ are omitted. This is the only published differential-system input to the construction; the elimination and its specialization are given explicitly next.

Lemma 29.5 (Scalar elimination)**.** *There is a scalar differential operator $\mathcal{A} = \sum_{m=0}^{N+1} a_m(z)\partial_z^m$ annihilating H_0 such that*

$$a_m \in \mathbb{R}[z], \qquad a_{N+1} = z^N, \qquad \deg a_m \leq N, \qquad z^{m-1} \mid a_m \quad (m \geq 1).$$

The coefficients are rational when the parameters are rational.

Proof. Write $D = \partial_z$ and construct rational operators

$$P_0 = 1, \qquad P_1 = 1 + D/N,$$

and, for $1 \le p < N$,

$$(N-p)P_{p+1} = \left(D + N - p + \frac{B_p}{z}\right) P_p - \frac{D_p}{z} P_{p-1}. \tag{29.4}$$

Equation (29.3) gives $H_p = P_p H_0$. Write $P_p = \sum_{m=0}^{p} q_{p,m} D^m$ for $p \ge 1$. Induction using $D \circ (qD^m) = qD^{m+1} + q'D^m$ gives

$$q_{p,p} = \frac{(N-p)!}{N!}, \qquad q_{p,m} \in \operatorname{span}\{z^{-(p-1)}, \dots, 1\}, \qquad \operatorname{ord}_0 q_{p,m} \ge m - p.$$

For the induction, differentiation or division by z lowers the least Laurent exponent by at most one, and no positive exponent is introduced. The highest derivative comes only from $D \circ P_p$. The coefficient of D^m in that product is $q'_{p,m} + q_{p,m-1}$, whose order is at least $m - p - 1$; the other terms in (29.4) have the same bound. The case $p = 1$ involving P_0 is immediate.

The $p = N$ equation in (29.3) now shows that

$$\mathcal{A} = N! z^{N-1} (zD \circ P_N + B_N P_N - D_N P_{N-1}) \tag{29.5}$$

annihilates H_0. The bracket has Laurent exponents between $-(N-1)$ and 1, so multiplication by z^{N-1} makes its coefficients polynomials of degree at most N. Its top coefficient is $z/N!$, giving $a_{N+1} = z^N$. The coefficient of D^m in the bracket has order at least $m - N$, proving the stated divisibility after multiplication. The formulas also prove rationality. □

Proposition 29.6 (No nonzero tableau scalar singularities)**.** *Let $k = 2r + \varepsilon \ge 2$, with $\varepsilon \in \{0, 1\}$. There is an order-$r+1$ operator over $\mathbb{Q}[x]$ annihilating Y_k whose leading coefficient is x^r. The minimal monic operator has order $r + 1$ and is ordinary at every $x \ne 0$.*

Proof. The ordinary-point and order assertions are Theorem 8.2, proved in Appendix B. The relation with the general Jacobi elimination above is explicit: at $N = r$, $\alpha = \frac{1}{2}$, $b = \varepsilon + \frac{1}{2}$, $\beta = 2$,

$$Y_k(x) = \frac{4^{r(r+\varepsilon)}}{r!(2\pi)^r} e^{(\varepsilon - 2r)x} H_0(-4x). \tag{29.6}$$

Substituting $z = -4x$ and $\partial_z = -\frac{1}{4}(\partial_x - (\varepsilon - 2r))$ in (29.5), and multiplying by -4, gives leading coefficient x^r. This retains the normalization while avoiding a second proof of the same tableau integral. □

Only the ordinary-point conclusion of Proposition 29.6 is needed in Theorem 30.4. Possible cancellation of a common power of x in a primitive polynomial normalization is irrelevant to that conclusion. Both the general elimination and its specialized minimality proof are included here.

30 Exact Bessel modules, algebraic base changes, and tableau values

The algebraicity classifications of Theorem 29.4 admit stronger differential and arithmetic conclusions. The general homogeneous-form theorem below determines the normalized equations, their complete local partition, and all algebraic relations between their initial jets. The ordinary-point result needed for arithmetic specialization was proved explicitly in Proposition 29.6.

30.1 A homogeneous-form theorem

Let $K = \mathbb{C}(z)$ and let y denote either

$$y(z) = I_0(2z), \qquad y'' = 4y - z^{-1}y', \quad \text{or} \quad y(z) = I_0(2\sqrt{z}), \qquad y'' = z^{-1}(y - y').$$

Both rank-two connections have differential Galois group SL_2. Their local monodromy at zero is a nontrivial unipotent block of size two. These Bessel inputs, including their primary references, are established in Lemma 29.1. For $d \geq 1$, let $\Phi \in K[X_0, X_1]$ be a nonzero homogeneous form of degree d, and put

$$F(z) = e^{\varepsilon z}\Phi(y(z), y'(z)), \qquad \varepsilon \in \{0, 1\}.$$

Theorem 30.1 (Exact module and stability)**.** *The minimal differential operator of F over K is irreducible of order $d + 1$. Its differential Galois group, in its solution representation, is*

$$\begin{cases} \mathrm{Sym}^d(\mathrm{SL}_2), & \varepsilon = 0, \\ \{c\,\mathrm{Sym}^d(g) : c \in \mathbb{C}^*,\ g \in \mathrm{SL}_2\}, & \varepsilon = 1. \end{cases}$$

In the first case this is isomorphic to SL_2 for odd d and to PSL_2 for even d. At zero the complete local monodromy partition consists of one unipotent block $J_{d+1}(1)$.

All these orders and irreducibility assertions remain valid over every finite algebraic extension of K. After a ramified algebraic change of variable, the local block at a point above zero still has size $d + 1$.

Proof. Let V be the two-dimensional Bessel solution space in a Picard–Vessiot extension. Polarizing $u \mapsto \Phi(u, u')$ gives a SL_2-equivariant map from $\mathrm{Sym}^d V$ to the differential field. It is nonzero because y, y' are algebraically independent over K, by Lemma 29.1. The representation $\mathrm{Sym}^d V$ is irreducible, so this map is injective. The Galois orbit of any nonzero image spans the entire image. The standard Galois-orbit characterization of a minimal scalar operator therefore gives the claimed order, irreducibility, and solution representation.

For $\varepsilon = 1$, the compositum of the Bessel Picard–Vessiot extension and that of $u' = u$ has group $\mathrm{SL}_2 \times \mathbb{G}_m$. Indeed, a proper subdirect product would give a nontrivial common quotient of SL_2 and $\mathbb{G}_m$, which is impossible. The preceding argument applies to the irreducible representation $\mathrm{Sym}^d V$ tensored with the standard character of $\mathbb{G}_m$.

The symmetric power of a nontrivial $J_2(1)$ is a single $J_{d+1}(1)$: in the usual degree-d polynomial representation, its logarithm acts by a nonzero multiple of $X_0\partial_{X_1}$, with one chain of length $d + 1$. The exponential twist is single-valued and nonvanishing at zero and does not alter this partition.

Both displayed Galois groups are connected. Finite algebraic extension of the base leaves the identity component unchanged, and hence leaves the full connected group and its irreducible representation unchanged. A ramification of index e replaces the local monodromy by its eth power; a unipotent Jordan block in characteristic zero retains its size under this operation. □

Corollary 30.2 (The three determinant families)**.** *With the notation of Section 29, the following orders are exact:*

function	d	*minimal order*	*monodromy at zero*
$Y_k(x),\ k \geq 2$	$\lfloor k/2 \rfloor$	$\lfloor k/2 \rfloor + 1$	$J_{\lfloor k/2 \rfloor + 1}(1)$
$U_k(s),\ k \geq 1$	k	$k+1$	$J_{k+1}(1)$
$F_k(x),\ k \geq 2$	$k-1$	k	$J_k(1)$.

The permutation and matching groups are respectively $\mathrm{Sym}^k(\mathrm{SL}_2)$ *and* $\mathrm{Sym}^{k-1}(\mathrm{SL}_2)$*. For tableaux, write* $k = 2d + \varepsilon$*; the group is the corresponding group in Theorem 30.1. Every listed minimal order persists after finite algebraic extension of the coefficient field.*

Proof. The determinant reductions in Lemma 29.2 give nonzero homogeneous forms of exactly the displayed degrees. Their nonvanishing also follows from the constant term 1 of each generating function. Apply Theorem 30.1. □

30.2 Algebraic relations among initial jets

Write $\mathbf{J}_F = (F, F', \dots, F^{(d)})^{\mathsf{T}}$ and

$$\mathbf{m}_d = (y^d, y^{d-1}y', \dots, (y')^d)^{\mathsf{T}}.$$

Differentiating and reducing y'' by the Bessel equation produces a rational matrix $P(z)$ with

$$\mathbf{J}_F = e^{\varepsilon z} P(z) \mathbf{m}_d. \tag{30.1}$$

Theorem 30.3 (Jet cone)**.** *The matrix* P *belongs to* $\mathrm{GL}_{d+1}(K)$*. After the invertible rational change of jet coordinates* $\mathbf{Z} = P^{-1}\mathbf{J}_F$*, the ideal of all polynomial relations over* K *is generated by*

$$Z_i Z_{j+1} - Z_{i+1} Z_j, \qquad 0 \leq i < j \leq d-1. \tag{30.2}$$

For $d = 1$ *the ideal is zero. Consequently*

$$\mathrm{trdeg}_K\, K(F, F', \dots, F^{(d)}) = 2.$$

Every two distinct entries of this jet vector are algebraically independent over K*, and remain so over every finite algebraic extension of* K*. When* $\varepsilon = 0$*, one has the further exact field identity*

$$K(F, F', \dots, F^{(d)}) = K(y^d, y^{d-1}y', \dots, (y')^d) = K(y, y')^{\mu_d},$$

where μ_d *acts by simultaneous scalar multiplication on* y, y'*. Thus its index in* $K(y, y')$ *is exactly* d*.*

Proof. If the rows of P were dependent, (30.1) would give a nonzero rational differential equation for F of order at most d, contradicting Theorem 30.1. Therefore P is invertible.

The kernel of the substitution $Z_j \mapsto X_0^{d-j} X_1^j$ is the homogeneous ideal (30.2) of the affine cone over the degree-d rational normal curve. For completeness, these minors identify all products of Z_j having the same number of factors and the same sum of indices; their images are the same monomial in X_0, X_1. Grouping monomials by those two integers proves that they generate the entire kernel. Algebraic independence of y, y' shows that this is exactly the relation ideal for $\varepsilon = 0$. For $\varepsilon = 1$, the three functions y, y', e^z are algebraically independent: the group $\mathrm{SL}_2 \times \mathbb{G}_m$ acts transitively on the nonzero Bessel solution vector and independently on the exponential coordinate. Multiplying every degree-d monomial by the same independent scalar preserves the same homogeneous kernel. This proves the ideal assertion and its dimension.

Two different rows of P are linearly independent, so their degree-d forms A, B are not proportional. Two nonproportional homogeneous forms of the same positive degree in two variables are algebraically independent: their ratio is a nonconstant function of X_1/X_0, while simultaneous scaling changes their common magnitude. Equivalently, their Jacobian is nonzero. The same conclusion holds after multiplying both by the independent scalar e^z. This proves pairwise independence; an algebraic extension of the coefficient field cannot decrease transcendence degree. Finally the displayed monomial field is the fixed field of the faithful scalar action of μ_d on the rational field $K(y, y')$, proving the index statement. □

30.3 Arithmetic specialization without exceptional nonzero points

For tableaux Proposition 29.6 supplies an additional essential input: an order-$d+1$ annihilator of Y_k has leading coefficient x^d, where $d = \lfloor k/2 \rfloor$, and $d+1$ is the exact minimal order. Thus every $\alpha \neq 0$ is an ordinary point of the minimal scalar equation. This property is stronger than having a connection which is merely gauge-equivalent to one regular at α; it is what rules out exceptional zeros of the scalar determinant form.

Theorem 30.4 (Values at nonzero algebraic tableau arguments)**.** *Let $k \geq 2$, $d = \lfloor k/2 \rfloor$, and $\alpha \in \overline{\mathbb{Q}}^*$. Then*

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(Y_k(\alpha), Y_k'(\alpha), \dots, Y_k^{(d)}(\alpha)) = 2.$$

Every two distinct numbers in this displayed tuple are algebraically independent over $\overline{\mathbb{Q}}$. All their algebraic relations are obtained from (30.2) by the invertible algebraic change of coordinates $\mathbf{Z} = P(\alpha)^{-1}\mathbf{J}_{Y_k}(\alpha)$. In particular $Y_k(\alpha)$ is transcendental and nonzero. For $k = 1$, $Y_1(\alpha) = e^\alpha$ is transcendental.

Proof. Put $y = I_0(2x)$. The functions y, y', e^x are E-functions over $\mathbb{Q}$; their first-order system is

$$\frac{d}{dx}\begin{pmatrix} y \\ y' \\ e^x \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 4 & -1/x & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} y \\ y' \\ e^x \end{pmatrix}.$$

They are algebraically independent over $\mathbb{C}(x)$ by the group argument in Theorem 30.3. The Siegel–Shidlovskii theorem in [24, Theorem 1.1], with common denominator x, therefore says that $y(\alpha), y'(\alpha), e^\alpha$ are algebraically independent over $\overline{\mathbb{Q}}$ for every nonzero algebraic α.

The determinant reductions use coefficients in $\mathbb{Q}[x, x^{-1}]$, so P in (30.1) has no pole at α. Moreover $P(\alpha)$ is invertible. Indeed the symmetric-power Bessel system is ordinary there, and so is the minimal scalar equation of Y_k by the preceding input. Evaluation of its first $d+1$ jets is an isomorphism from its solution space to $\mathbb{C}^{d+1}$. The injective polarized solution map of Theorem 30.1 is an isomorphism between solution spaces of that same dimension. Their evaluation matrices are related by $P(\alpha)$, forcing its determinant to be nonzero.

We may therefore specialize the monomial-coordinate argument of Theorem 30.3. The independence of the three basic values gives exactly the Veronese-cone relation ideal and transcendence degree two. Different rows of the invertible $P(\alpha)$ are nonproportional homogeneous forms, giving pairwise algebraic independence by the same Jacobian argument. The case $k = 1$ is the classical transcendence of e^α. □

Remark 30.5 (Scope of the arithmetic statement)*.* No corresponding absence of exceptional nonzero scalar points is assumed here for the permutation and matching determinants. Their functional order, Galois, and jet-cone conclusions hold for all parameters as stated. The arithmetic theorem above uses Proposition 29.6 explicitly; functional Galois information alone would not justify its all-points specialization.

31 Arithmetic subsequences and quasipolynomial coefficient weights

The three algebraicity thresholds persist under every arithmetic subsequence and every nonzero polynomial weighting. In fact, one can allow an arbitrary quasipolynomial weight. The point requiring proof is that the roots-of-unity filter cannot remove the long logarithmic block. We prove this using the simple Bessel modules before applying the Borel bound; no semisimplicity assertion about a Borel transform is needed.

A *nonzero quasipolynomial* is a function $w : \mathbb{Z}_{\geq 0} \to \mathbb{C}$ for which some $q \geq 1$ and polynomials $P_0, \dots, P_{q-1} \in \mathbb{C}[X]$, not all zero, satisfy $w(n) = P_b(n)$ when $n \equiv b \pmod q$. We use the counting sequences and normalizations of Section 29.

Theorem 31.1 (Algebraicity survives no nonzero periodic polynomial filter)**.** *For every nonzero quasipolynomial w, the following equivalences hold:*

$$\sum_{n\geq 0} w(n)y_k(n)t^n \text{ is algebraic over } \mathbb{C}(t) \iff k \leq 3,$$
$$\sum_{n\geq 0} w(n)u_k(n)t^n \text{ is algebraic over } \mathbb{C}(t) \iff k \leq 2,$$
$$\sum_{n\geq 0} w(n)f_k(n)t^n \text{ is algebraic over } \mathbb{C}(t) \iff k \leq 2.$$

Here all parameters k are positive integers. Thus the threshold for permutations avoiding $12\cdots\ell$ remains $\ell \leq 3$. The same statements hold after changing finitely many coefficients.

Corollary 31.2 (Every arithmetic progression and every polynomial weight)**.** *Let $q \geq 1$, $b \geq 0$, and $0 \neq P \in \mathbb{C}[X]$. The generating functions*

$$\sum_{n\geq 0} P(n)y_k(qn+b)t^n, \qquad \sum_{n\geq 0} P(n)u_k(qn+b)t^n, \qquad \sum_{n\geq 0} P(n)f_k(qn+b)t^n$$

are algebraic exactly for $k \leq 3$, $k \leq 2$, and $k \leq 2$, respectively. No exceptional residue classes occur.

31.1 A noncancellation lemma for simple differential modules

Put $K = \mathbb{C}(z)$ and let $\mathcal{E}$ be a differential field extension with field of constants $\mathbb{C}$. A finite-dimensional K-subspace of $\mathcal{E}$ stable under differentiation is viewed as a differential module.

Lemma 31.3 (Multiplicity-free sums retain their local blocks)**.** *Let $M_1, \dots, M_h \subset \mathcal{E}$ be simple differential modules which are pairwise nonisomorphic. Then their sum is direct. If $g_i \in M_i$ and $g = \sum_i g_i$, the differential module generated by g is exactly $\bigoplus_{g_i \neq 0} M_i$. Consequently, the local monodromy of the minimal equation of g is the direct sum of those of the nonzero components.*

Proof. The abstract direct sum of the M_i is semisimple. Every simple submodule is isomorphic to one of the M_i, and projection to every nonisomorphic summand is zero. Because each isomorphism type occurs once, its simple submodules are precisely the individual summands. The kernel of the summation map to $\mathcal{E}$ therefore is a sum of some summands. The map is injective on each M_i, so its kernel is zero. The differential module generated by g is a submodule of this direct sum. Its projection onto every index with $g_i \neq 0$ is nonzero and hence surjective. The same multiplicity-free argument shows that it contains exactly those summands.

The cyclic differential module of a scalar germ determines the solution local system of its minimal equation, with the conventional duality if necessary. Direct sums and their Jordan block sizes are preserved by this correspondence, giving the last assertion. □

Lemma 31.4 (Filtered Bessel blocks)**.** *Let w be a nonzero quasipolynomial. In the ranges specified below, the minimal equations of the following entire functions are regular singular at zero and have at least one block of the indicated size:*

function	*parameter range*	*block size*
$\sum_n w(n)y_k(n)z^n/n!$	$k \geq 2$	$\lfloor k/2 \rfloor + 1$
$\sum_n w(n)u_k(n)z^n/(n!)^2$	$k \geq 1$	$k+1$
$\sum_n w(n)f_k(n)z^{2n}/(2n)!$	$k \geq 2$	k.

More precisely, each complete local partition consists of one or more copies of the displayed unipotent block, with no blocks of other sizes.

Proof. Write $\vartheta = z\,d/dz$. Fourier expansion of the residue-class indicators expresses each weighted series as a sum of terms $Q_\zeta(\vartheta)H(\zeta z)$, where ζ ranges over roots of unity, Q_ζ are polynomials, and H is the appropriate unweighted normalized series. For matchings, use $2q$th roots and $\vartheta/2$ because their normalized series is supported on even powers. For example, for tableaux the exact expression is

$$\sum_{b=0}^{q-1} P_b(\vartheta) \left\{ \frac{1}{q} \sum_{\zeta^q=1} \zeta^{-b} Y_k(\zeta z) \right\}.$$

For the z-Bessel equation, the differential module spanned by the degree-d monomials in $I_0(2az)$ and its derivative is simple, has rank $d+1$, and has local block $J_{d+1}(1)$ at zero, by Theorem 30.1. Its formal exponential factors at infinity are

$$\exp\bigl(2a(d-2j)z\bigr), \qquad 0 \leq j \leq d. \tag{31.1}$$

These follow directly from the two formal Bessel solutions $e^{\pm 2az}z^{-1/2}(1+O(z^{-1}))$ by taking symmetric powers. The multiset is unchanged when a is replaced by $-a$, and determines a up to that sign: its two extreme points are $\pm 2da$. Modules with distinct multisets of formal exponential factors are nonisomorphic, since a rational gauge cannot change these factors. Moreover, the two modules for a and $-a$ are the same embedded module, because I_0 is even and its derivative is odd. We therefore combine those two contributions before applying Lemma 31.3.

For even tableaux and for matchings, this is the entire argument, with $d = \lfloor k/2 \rfloor$ and $d = k-1$, respectively. For odd tableaux, write $k = 2d+1$. The exponential twist in (2.3) changes the factors to

$$\exp\bigl(a(1+2d-4j)z\bigr), \qquad 0 \leq j \leq d. \tag{31.2}$$

These multisets distinguish every nonzero a: their sums of exponents are $(d+1)a$. The modules are again simple by Theorem 30.1, and every homothety has local block $J_{d+1}(1)$.

For permutations in the variable z, the degree-k symmetric power associated with $I_0(2\sqrt{az})$ has, after the common ramification $z = v^2$, the formal factors

$$\exp\bigl(2\sqrt{a}(k-2j)v\bigr), \qquad 0 \leq j \leq k.$$

Their extreme points determine $\sqrt{a}$ up to sign and hence determine a. Thus the modules for distinct roots of unity a are pairwise nonisomorphic. Each has rank $k+1$ and block $J_{k+1}(1)$ before ramification as well, by Theorem 30.1.

Application of a polynomial in ϑ keeps a term in its simple differential module. After grouping the equal modules in the even cases, Lemma 31.3 applies to all remaining terms, including any that become zero. At least one term remains nonzero. Indeed $y_k(n), u_k(n) > 0$ for every n, and $f_k(n) > 0$ for every n when $k \geq 2$, while a nonzero quasipolynomial is nonzero at infinitely many integers. Thus the weighted power series is nonzero. Every surviving summand has the same indicated local block, proving both the lower bound and the complete partition assertion. □

Proof of Theorem 31.1. Algebraic series are closed under homotheties, finite sums, differentiation, and multiplication by rational functions. The roots-of-unity formula and the Euler operators above show that quasipolynomial coefficient weighting preserves algebraicity. The algebraic directions therefore follow from Theorem 29.4.

For $k \geq 4$ the weighted tableau exponential generating function has a block of size $\lfloor k/2 \rfloor + 1 \geq 3$ by Lemma 31.4. Algebraicity of its ordinary generating function would bound that size by two after one Borel transform, by Theorem 27.3. This is a contradiction. For $k \geq 3$, the weighted permutation function has a block of size $k+1 \geq 4$, whereas two Borel transforms of an algebraic series permit size at most three. This is again impossible. For matchings with $k \geq 3$, algebraicity of the ordinary series would imply algebraicity after $t \mapsto t^2$. One Borel transform of that substituted series is the weighted matching function in Lemma 31.4, whose block has size $k \geq 3$, contradicting the upper bound two. Finally, changing finitely many ordinary coefficients adds a polynomial, and therefore cannot change algebraicity. □

Proof of Corollary 31.2. First suppose $0 \leq b < q$. Set $w(m) = 0$ outside the residue class $b \pmod q$ and $w(m) = P((m-b)/q)$ on that class. Then, for each of the three sequences (a_n), the weighted ordinary generating function is

$$\sum_{m \geq 0} w(m) a_m t^m = t^b \sum_{n \geq 0} P(n) a_{qn+b} (t^q)^n.$$

The two series on the right and left are algebraic simultaneously, because substitution by t^q and its algebraic inverse preserve algebraicity. Apply Theorem 31.1. For general $b = b_0 + qh$, remove the first h terms from the corresponding b_0-subsequence and replace $P(n)$ by $P(n-h)$. Removing a finite initial segment and multiplying by a power of t do not change algebraicity. □

Corollary 31.5 (Exact differential orders after a single residue filter)**.** *Fix $q \geq 1$, $0 \leq b < q$, and $0 \neq P \in \mathbb{C}[X]$. Use the inherited factorial normalization, namely*

$$\mathcal{Y}(z) = \sum_{n \geq 0} P(n) y_k(qn+b) \frac{z^{qn+b}}{(qn+b)!}, \qquad \mathcal{U}(z) = \sum_{n \geq 0} P(n) u_k(qn+b) \frac{z^{qn+b}}{((qn+b)!)^2},$$

$$\mathcal{F}(z) = \sum_{n \geq 0} P(n) f_k(qn+b) \frac{z^{2(qn+b)}}{(2(qn+b))!}.$$

The following minimal differential orders over $\mathbb{C}(z)$ are exact:

function and parameters	*order*	*partition at zero*
$\mathcal{Y},\ k = 2d \geq 2$	$q(d+1)/\gcd(q,2)$	$J_{d+1}(1)^{\oplus q/\gcd(q,2)}$
$\mathcal{Y},\ k = 2d+1 \geq 3$	$q(d+1)$	$J_{d+1}(1)^{\oplus q}$
$\mathcal{U},\ k \geq 1$	$q(k+1)$	$J_{k+1}(1)^{\oplus q}$
$\mathcal{F},\ k \geq 2$	qk	$J_k(1)^{\oplus q}$.

These are orders of the displayed normalized differential equations, not assertions about the shift width of a recurrence for the sampled coefficient sequence.

Proof. For a single residue filter, every root-of-unity coefficient is nonzero. In the even-tableau case there are $q/\gcd(q,2)$ classes modulo sign. If q is even, each class contributes a nonzero multiple of $Y_k(az) + (-1)^b Y_k(-az)$. This is nonzero because its surviving parity has strictly positive counting coefficients. If q is odd, each class has a single nonzero contribution. The odd-tableau and permutation cases have q nonisomorphic simple summands. For matchings use the $2q$th-root filter and the residue $2b$; combining opposite roots gives q nonzero summands because F_k is even.

It remains to justify that applying the weight polynomial kills none of these summands. Before weighting, each nonzero component is holomorphic at zero and generates a simple module of rank at least two. It cannot be a polynomial: a nonzero polynomial generates the rank-one differential module $\mathbb{C}(z)$. For a holomorphic germ $g(z) = \sum c_n z^n$, the identity $Q(\vartheta)g = 0$ with a nonzero polynomial Q forces $c_n = 0$ unless $Q(n) = 0$, and would therefore make g a polynomial. Thus every component survives the relevant operator $P((\vartheta - b)/q)$, or $P((\vartheta/2 - b)/q)$ for matchings. Apply Lemma 31.3 and count the surviving simple summands, each with its already known local block. □

Remark 31.6 (Exact scope). These statements concern algebraicity of generating functions over $\mathbb{C}(t)$, not arithmetic properties of their values at particular arguments. They extend the unfiltered classification and use the classical enumeration identities cited above. They do not require a joint differential-Galois group theorem for all Bessel scales.

32 An algebraicity dichotomy for homogeneous Bessel observables

Write $y(t) = I_0(2t)$. A *homogeneous Laurent Bessel observable* of degree r is a function

$$F(t) = \Phi(t, y(t), y'(t)), \qquad \Phi \in \mathbb{C}[t, t^{-1}][X_0, X_1],$$

where Φ is homogeneous of degree r in X_0, X_1. The Laurent condition allows powers t^j with integer j, but no other denominators. The following theorem uses the holomorphic cancellation of all poles at $t = 0$.

For clarity, a quasipolynomial weight is a function w for which there are an integer $h \geq 1$ and polynomials $P_0, \ldots, P_{h-1} \in \mathbb{C}[X]$ such that

$$w(n) = P_s(n) \quad \text{when } n \equiv s \pmod{h}.$$

It is nonzero if at least one P_s is nonzero. No positivity of these polynomials is assumed.

Theorem 32.1 (Homogeneous-observable dichotomy)**.** *Let $r \geq 1$ and $p \in \{0, 1\}$. Suppose that a homogeneous Laurent Bessel observable F of degree r is nonzero, holomorphic at zero, and satisfies $F(-t) = (-1)^p F(t)$. Put*

$$a_n = (2n + p)!\,[t^{2n+p}]F(t).$$

For every $d \geq 1$, $0 \leq b < d$, and every nonzero quasipolynomial weight w, the series

$$A_{d,b,w}(z) = \sum_{n \geq 0} w(n) a_{dn+b} z^n$$

is algebraic over $\mathbb{C}(z)$ if and only if $r = 1$. None of these weighted sampled sequences is eventually zero. Moreover, when $r = 1$, the unweighted unsampled series satisfies the sharper conclusion

$$\sum_{n \geq 0} a_n z^n \in \mathbb{C}(z, \sqrt{1 - 4z}).$$

The exact-width result of Corollary 26.6 and the algebraicity dichotomy measure different phenomena. For example, degree-two observables have coefficient recurrence width one, but their ordinary generating functions are transcendental even after arbitrary nonzero quasipolynomial weighting and arithmetic sampling.

32.1 The holomorphic lattice in degree one

For $m \geq 0$, define

$$J_m(z) = \sum_{n \geq 0} \frac{z^n}{n!(n+m)!}.$$

Then $J_0(t^2) = y(t)$ and $y'(t) = 2tJ_1(t^2)$. Direct coefficient comparison gives

$$zJ_{m+2}(z) = J_m(z) - (m+1)J_{m+1}(z), \qquad m \geq 0. \tag{32.1}$$

Lemma 32.2 (Holomorphic division in the Bessel lattice)**.** *Let*

$$\mathcal{S} = \left\{ \sum_{m=0}^{M} P_m(z) J_m(z) : M < \infty,\ P_m \in \mathbb{C}[z] \right\}.$$

If $g \in \mathcal{S}$ and $g(0) = 0$, then $g(z)/z \in \mathcal{S}$. Consequently every function $a(z)J_0(z) + b(z)J_1(z)$ with $a, b \in \mathbb{C}[z, z^{-1}]$ that is holomorphic at zero belongs to $\mathcal{S}$.

Proof. Write $P_m(z) = P_m(0) + zQ_m(z)$. It remains to divide a constant linear combination $\sum_m c_m J_m$ whose value at zero vanishes. Put $K_m = m!J_m$ and $d_m = c_m/m!$, so $\sum_m d_m = 0$. If $D_j = \sum_{m=0}^{j} d_m$, telescoping gives

$$\sum_{m=0}^{M} d_m K_m = \sum_{j=0}^{M-1} D_j (K_j - K_{j+1}) = z \sum_{j=0}^{M-1} D_j j! J_{j+2},$$

where the last identity is (32.1). This proves the closure under division by z.

For the last assertion, choose N so that $z^N(aJ_0 + bJ_1) \in \mathcal{S}$. Its first N Taylor coefficients vanish, because the function before multiplication is holomorphic. Apply the division result successively N times. □

Lemma 32.3 (Algebraic factorial transforms of the lattice)**.** *For every $m \geq 0$,*

$$H_m(z) = \sum_{n \geq 0} \frac{(2n)!}{n!(n+m)!} z^n \quad \textit{belongs to} \quad \mathbb{C}(z, \sqrt{1-4z}).$$

If $f \in \mathcal{S}$ and $p \in \{0, 1\}$, the same is true of $\sum_{n \geq 0} (2n+p)![z^n]f(z)\, z^n$.

Proof. One has $H_0 = (1-4z)^{-1/2}$. For $m \geq 1$, the beta integral, evaluated term by term near $z = 0$, gives

$$H_m(z) = \frac{1}{(m-1)!} \int_0^1 \frac{(1-u)^{m-1}}{\sqrt{1-4zu}}\, du.$$

Set $v = \sqrt{1-4zu}$, taking the branch equal to 1 at $z = 0$. The resulting identity is

$$H_m(z) = \frac{1}{2z(m-1)!(4z)^{m-1}} \int_{\sqrt{1-4z}}^{1} (v^2 + 4z - 1)^{m-1}\, dv.$$

The integrand is a polynomial in v, so evaluating its polynomial antiderivative at the two endpoints proves the claimed quadratic-field membership. The displayed apparent pole at zero is removable by the original power series.

For a generator $z^k J_m(z)$ of $\mathcal{S}$, with $k \geq 0$, its factorial transform of parity p is

$$z^k Q_{k,p}(\theta) H_m(z), \qquad Q_{k,p}(X) = \prod_{j=1}^{2k+p} (2X + j), \quad \theta = z\partial_z.$$

This follows by putting $n = \nu + k$ in its coefficients. The quadratic function field is closed under differentiation, and finite linear combinations prove the second assertion. □

32.2 Why polynomial weights cannot remove the obstruction

Lemma 32.4 (Nonzero Euler polynomials preserve the module)**.** *Let V be a simple differential module over $\mathbb{C}(t)$ realized by meromorphic functions. Suppose V has a nonzero formal exponential rate at infinity. For every $0 \neq f \in V$ and every nonzero polynomial $Q \in \mathbb{C}[X]$, $Q(t\partial_t)f$ is nonzero and generates V.*

Proof. If $Q(t\partial_t)f = 0$, the minimal scalar equation of f is a right divisor of this Euler operator. Its local solutions at infinity would therefore be finite linear combinations of powers of t times powers of $\log t$, with no nonzero exponential rate. This contradicts the hypothesis, since a nonzero element of the simple module is cyclic. Thus the resulting element is nonzero, and simplicity makes it cyclic as well. □

Proof of Theorem 32.1. First take $r = 1$, and write $F = A(t)y + B(t)y'$ with Laurent polynomials A, B. The functions y, y' are linearly independent over $\mathbb{C}(t)$, as follows from the Bessel SL_2 group. The parity assumption therefore implies

$$A(t) = t^p a(t^2), \qquad B(t) = t^{p-1} b(t^2), \qquad a, b \in \mathbb{C}[z, z^{-1}].$$

Consequently $F(t) = t^p f(t^2)$ with $f = aJ_0 + 2bJ_1$. Holomorphicity of F implies holomorphicity of f. Lemmas 32.2 and 32.3 now prove the stronger quadratic-field assertion for the full coefficient series.

Arithmetic sections of algebraic series are algebraic by root-of-unity filtering and algebraic substitution. Polynomial coefficient weighting is application of a polynomial in $z\partial_z$, which preserves algebraicity. Finally a quasipolynomial can be expressed as a finite sum $\sum_{\omega^h=1} \omega^n Q_\omega(n)$ with polynomial Q_ω, by Fourier inversion on its residue polynomials. Its weighted generating function is a finite sum of $Q_\omega(z\partial_z)A(\omega z)$, and is algebraic. This proves all the upper assertions for $r = 1$.

For the converse, first consider a polynomial weight $P(n)$ on an arithmetic section, where $P \neq 0$. Replacing F by

$$F_P(t) = P\left(\frac{(t\partial_t - p)/2 - b}{d}\right) F(t)$$

multiplies its coefficient at t^{2m+p} by $P((m-b)/d)$. It remains a holomorphic homogeneous Laurent Bessel observable of degree r and parity p. It is nonzero by Lemma 32.4, since the underlying simple symmetric-power Bessel module has nonzero rates $2r$ and $-2r$. Thus it suffices to rule out algebraicity of unweighted arithmetic sections for any observable with these properties.

Suppose $A(z) = \sum_{n\geq 0} a_{dn+b} z^n$ is algebraic. The padded series $z^{2b+p}A(z^{2d})$ is algebraic, and its onefold factorial Borel transform is

$$H(t) = \sum_{n\geq 0} a_{dn+b} \frac{t^{2(dn+b)+p}}{(2(dn+b)+p)!}.$$

Theorem 27.3 implies that all its local Jordan blocks at zero have size at most two. On the other hand, the root-of-unity decomposition in the proof of Corollary 26.6 identifies its module with a direct sum of d pairwise nonisomorphic simple rth symmetric-power Bessel modules. Each has a unipotent block of size $r+1$ at zero. This contradicts the Borel bound when $r \geq 2$.

Now let w be a nonzero quasipolynomial of period h and choose a residue s with $P_s \neq 0$. If $A_{d,b,w}$ were algebraic, so would its section at $n = hm + s$, whose coefficients are

$$P_s(hm+s)\, a_{dhm+ds+b}.$$

This is a nonzero polynomial weighting of an arithmetic section, already excluded for $r \geq 2$. For every $r \geq 1$, the same choice of residue also proves that the weighted sequence is not eventually

zero: Corollary 26.6 gives that property for $a_{dhm+ds+b}$, and the nonzero polynomial $P_s(hm+s)$ has only finitely many integer roots. □

Remark 32.5 (The homogeneity assumption cannot be omitted). The function

$$F(t) = \frac{I_0(2t) - 1}{t}$$

is holomorphic and odd, but is not homogeneous of degree one in the Bessel coordinates. Its factorial coefficient series is

$$\sum_{n\geq 0} \frac{(2n+1)!}{((n+1)!)^2} z^n = \frac{1}{z} \log\left(\frac{2}{1+\sqrt{1-4z}}\right).$$

The identity follows by differentiating after multiplication by z and matching the zero constant term. This function is not algebraic: at infinity, on a sector with fixed branches, it has a nonzero $(\log z)/z$ term, whereas algebraic functions have Puiseux expansions without logarithms. Thus arbitrary integration or subtraction of constants cannot replace the holomorphic-lattice argument above.

33 Quasipolynomial robustness of the twisted classification

Theorem 33.1 (Weighted sections of twisted determinants). *Let $\ell, d \geq 1$, $q \geq 0$, and $0 \leq b < d$ be integers, and let w be a nonzero quasipolynomial. Then*

$$\sum_{n\geq 0} w(n) u_{\ell,q}(dn+b) z^n$$

is algebraic over $\mathbb{C}(z)$ if and only if $\ell \leq 2$. For $\ell = 1$ it is rational.

Proof. The identities $u_{1,q}(n) = 1$ and $u_{2,q}(n) = C_{n+q}$ from Proposition 28.1, together with the algebraic closure operations just proved, handle $\ell \leq 2$. The generating function of a quasipolynomial is rational, which gives the stronger statement for $\ell = 1$.

For $\ell \geq 3$, extraction of a residue on which w is a nonzero polynomial reduces the claim to a polynomial weight $P(n)$ on a section $u_{\ell,q}(dn+b)$. Suppose its generating function $A(z)$ were algebraic. Put $Q = q\ell$ and $F_0(z) = z^b A(z^d)$, which is algebraic. Two Borel transforms with an intervening monomial gauge give

$$\mathcal{B}(z^Q \mathcal{B} F_0(z)) = z^Q H(z), \quad H(z) = \sum_{n\geq 0} \frac{P(n) u_{\ell,q}(dn+b) z^{dn+b}}{(dn+b)!(dn+b+Q)!}.$$

Theorem 27.3 and the rational gauge z^Q imply that every Jordan block of H at zero has size at most three; nonunit blocks are semisimple.

After $z = t^2$, the unweighted filtered function is the direct sum of d simple ℓth symmetric-power Bessel modules at scales distinct modulo sign. The polynomial weight is application of $P(((t\partial_t)/2 - b)/d)$ to this filter. Lemma 32.4 shows that each component remains nonzero, and Lemma 26.3 preserves their direct sum. Each component has a unipotent block of size $\ell + 1 \geq 4$ at zero. Squaring monodromy under $z = t^2$ cannot increase any Jordan-block size, since every monodromy eigenvalue is nonzero. This contradicts the upper bound three and proves the theorem. □

Remark 33.2. These classifications concern algebraicity over a rational function field. They do not assert irrationality or transcendence of individual numerical values of the generating functions. The new scope here is uniformity in the homogeneous observable, together with robustness under every arithmetic sampling and every nonzero quasipolynomial weighting.

34 Multipoint arithmetic of determinant jets

The all-twist Laurent gauge can be combined with the classical transcendence theory of scaled Bessel functions. The resulting statement describes *all* relations among normalized Toeplitz jets at any finite set of distinct nonzero algebraic arguments. The differential-Galois independence step is proved explicitly below; pairwise distinct exponential rates alone would not establish a product Galois group. Multipoint independence of the scalar Bessel values themselves is classical: Siegel's 1929 theorem for J_0, recalled in [49, §1], already covers distinct nonzero squared algebraic arguments. The purpose here is the exact all-twist jet ideal and field structure, including the derivatives.

34.1 The product group for scaled Bessel connections

For $a \in \mathbb{C}^*$ let M_a be the rank-two connection

$$\frac{d}{dt}\begin{pmatrix} f \\ f' \end{pmatrix} = A_a(t)\begin{pmatrix} f \\ f' \end{pmatrix}, \qquad A_a(t) = \begin{pmatrix} 0 & 1 \\ 4a^2 & -1/t \end{pmatrix}. \tag{34.1}$$

It has differential Galois group SL_2, by the Bessel result already used above and the nonzero linear change of variable $t \mapsto at$. Its entire distinguished solution is $b_a(t) = I_0(2at)$.

Lemma 34.1 (Subdirect products of the rank-two groups)**.** *Let H be an algebraic subgroup of SL_2^m whose projection onto each factor is surjective. Then H° also projects surjectively onto every factor. If $H \neq \mathrm{SL}_2^m$, there are two distinct factors whose three-dimensional adjoint representations restrict to isomorphic representations of H°.*

Proof. The image of H° in a factor is a connected normal subgroup with finite quotient in SL_2. Since SL_2 is connected, this image is the whole factor. Consequently $\mathfrak{h} = \mathrm{Lie}(H^\circ)$ is a subdirect Lie subalgebra of $\mathfrak{sl}_2^m$. Its solvable radical projects into a solvable ideal of each $\mathfrak{sl}_2$, hence projects to zero everywhere and is zero. Thus $\mathfrak{h}$ is a direct sum of simple ideals.

The image of a simple ideal under any coordinate projection is an ideal of $\mathfrak{sl}_2$. If nonzero, the projection is an isomorphism onto $\mathfrak{sl}_2$: its kernel is an ideal of the simple source. Exactly one simple ideal maps onto a given factor, since distinct simple ideals commute and $\mathfrak{sl}_2$ does not commute with itself. Every simple ideal maps nontrivially to at least one factor. If none maps to two factors, there are m simple ideals and $\mathfrak{h} = \mathfrak{sl}_2^m$; this forces $H^\circ = H = \mathrm{SL}_2^m$. Otherwise two factors share a simple ideal. The isomorphism between their two copies of $\mathfrak{sl}_2$ intertwines the adjoint actions of $\mathfrak{h}$, and hence those of the connected group H°. This proves the assertion. □

Lemma 34.2 (Adjoint scales cannot merge algebraically)**.** *If M_a and M_b have isomorphic adjoint differential modules after a finite algebraic extension of $\mathbb{C}(t)$, then $a = b$ or $a = -b$.*

Proof. The constant term of $A_a(t)$ at infinity has the distinct eigenvalues $2a, -2a$. The induced adjoint connection on traceless endomorphisms therefore has the three distinct leading eigenvalues

$$0, \quad 4a, \quad -4a.$$

A formal change of basis over $\mathbb{C}((t^{-1}))$ diagonalizes this connection with diagonal entries $\lambda + O(t^{-1})$, one for each displayed λ. Indeed, after diagonalizing the constant term, successive coefficients of the off-diagonal gauge are solved uniquely by dividing by the nonzero eigenvalue differences. The same applies to the adjoint connection at scale b.

Choose a place above infinity in the finite algebraic extension. Its completion embeds in a Puiseux field $\mathbb{C}((t^{-1/e}))$ for some $e \geq 1$. After the two formal diagonalizations, any entry h of an intertwining matrix obeys

$$h' = (\mu - \lambda + O(t^{-1}))h, \qquad \lambda \in \{0, 4a, -4a\}, \quad \mu \in \{0, 4b, -4b\}.$$

For a nonzero Puiseux Laurent series, $h'/h = O(t^{-1})$. Thus such an entry can be nonzero only when $\lambda = \mu$. An invertible intertwiner requires equality of the two multisets of leading eigenvalues. Since $a, b \neq 0$, this gives $a = \pm b$. □

Theorem 34.3 (Full product group and functional independence)**.** *Let $a_1, \dots, a_m \in \mathbb{C}^*$ satisfy $a_i \neq \pm a_j$ for $i \neq j$. The differential Galois group of $M_{a_1} \oplus \cdots \oplus M_{a_m}$ over $\mathbb{C}(t)$ is*

$$\mathrm{SL}_2^m.$$

The $2m$ functions $b_{a_1}, b'_{a_1}, \dots, b_{a_m}, b'_{a_m}$ are algebraically independent over $\mathbb{C}(t)$.

Proof. Let E be a Picard–Vessiot field for the direct sum and H its Galois group. Restriction to each summand makes H a subdirect algebraic subgroup of SL_2^m. If it were proper, Lemma 34.1 would supply two adjoint representations isomorphic on H°. Let $F = E^{H^\circ}$; the differential Galois correspondence makes $F/\mathbb{C}(t)$ finite algebraic. The two adjoint differential modules would then be isomorphic over F. Explicitly, if U_i, U_j are their fundamental matrices and C is a constant intertwiner of the H°-representations, the matrix $U_j C U_i^{-1}$ is fixed by H° and is the required invertible differential gauge over F. Lemma 34.2 contradicts the condition on the scales. Hence $H = \mathrm{SL}_2^m$.

For each i, complete b_{a_i} to a solution basis b_{a_i}, v_i of M_{a_i}. The product group moves these first solution columns independently to

$$c_i \begin{pmatrix} b_{a_i} \\ b'_{a_i} \end{pmatrix} + d_i \begin{pmatrix} v_i \\ v_i' \end{pmatrix}, \qquad (c_i, d_i) \neq (0, 0).$$

A polynomial relation among the $2m$ functions would therefore vanish after these substitutions for all constants c_i, d_i. Their product parameter set is Zariski dense in $\mathbb{C}^{2m}$, and each two-by-two substitution matrix has nonzero Wronskian. The relation must be the zero polynomial. This proves functional independence. □

Remark 34.4 (Why central or projective identifications do not escape). The proof passes to H° and compares adjoint modules over its finite algebraic fixed field. A finite central ambiguity or a rank-one projective twist disappears in the adjoint comparison; it cannot change the leading-rate multiset used in Lemma 34.2. Once the Lie algebra is the full product, H° is already the full connected ambient group. No separate assumption that the original subdirect group is connected is used.

34.2 Simultaneous arithmetic specialization

Recall $y(x) = I_0(2\sqrt{x})$ and the Laurent matrices $P_{\ell,q}$ of (24.4). For the remainder of this section put $k = \overline{\mathbb{Q}}$.

Theorem 34.5 (Independent Bessel coordinates at distinct arguments)**.** *If $\alpha_1, \dots, \alpha_m \in k^*$ are pairwise distinct, then the $2m$ numbers*

$$y(\alpha_i),\ y'(\alpha_i), \qquad 1 \leq i \leq m,$$

are algebraically independent over k.

Proof. Choose $a_i \in k^*$ with $a_i^2 = \alpha_i$ and set $b_i(t) = I_0(2a_i t) = y(\alpha_i t^2)$. Distinctness of the α_i is exactly the condition $a_i \neq \pm a_j$. Each b_i, b_i' is an E-function: its coefficients are algebraic scalar multiples of the central binomial coefficients, with heights and common-denominator growth bounded exponentially. The block-diagonal first-order system (34.1) has common denominator t and no other finite singularity. By Theorem 34.3 its $2m$ distinguished coordinate functions are

algebraically independent. The Siegel–Shidlovskii theorem, in the formulation [24, Theorem 1.1], applied at $t = 1$, proves independence of $b_i(1), b_i'(1)$ for all i simultaneously. Finally

$$b_i(1) = y(\alpha_i), \qquad b_i'(1) = 2\alpha_i y'(\alpha_i).$$

The nonzero algebraic factors preserve independence. □

Theorem 34.6 (The multipoint relation ideal)**.** *Let the $\alpha_i \in k^*$ be pairwise distinct. Independently choose integers $\ell_i \geq 1$ and $q_i \geq 0$, and put*

$$\mathbf{J}_i = \left(G_{\ell_i,q_i}(\alpha_i), G'_{\ell_i,q_i}(\alpha_i), \ldots, G^{(\ell_i)}_{\ell_i,q_i}(\alpha_i)\right)^{\mathsf{T}}.$$

Then

$$\operatorname{trdeg}_k k(\mathbf{J}_1, \ldots, \mathbf{J}_m) = 2m.$$

For each i, use the invertible algebraic coordinate change $\mathbf{Z}_i = P_{\ell_i,q_i}(\alpha_i)^{-1}\mathbf{J}_i$. The ideal of all polynomial relations among the combined coordinates is the sum of the m ideals generated by

$$Z_{i,r}Z_{i,s+1} - Z_{i,r+1}Z_{i,s}, \quad 0 \leq r < s \leq \ell_i - 1, \quad 1 \leq i \leq m. \tag{34.2}$$

Thus every relation is generated by the individual-argument Veronese relations; no additional generator connecting distinct arguments is needed. Choosing any two distinct initial jet entries at each argument gives $2m$ algebraically independent numbers over k.

Proof. By Theorem 24.2, every specialized gauge is invertible. Set $u_i = y(\alpha_i)$ and $v_i = y'(\alpha_i)$. Then

$$Z_{i,r} = u_i^{\ell_i - r} v_i^r.$$

The $2m$ base numbers are independent by Theorem 34.5. The polynomial substitution above uses disjoint pairs of independent variables. Within each pair, the kernel is the Veronese ideal proved in Theorem 24.4. Tensoring these monomial subalgebras introduces no new relation: distinct products of their monomial bases have distinct exponent vectors in the independent variables $u_1, v_1, \ldots, u_m, v_m$. Hence the combined kernel is exactly (34.2), and its dimension is $2m$.

Two distinct jet entries at argument α_i are nonproportional homogeneous forms of the same positive degree in u_i, v_i, because the specialized gauge has independent rows. Their two-variable Jacobian is nonzero, as in the proof of Theorem 24.4. The Jacobian for all the chosen pairs is block diagonal with these nonzero determinants. In characteristic zero this proves their joint algebraic independence. □

Corollary 34.7 (Arbitrarily many independent determinant values)**.** *With the hypotheses above, the m values $G_{\ell_i,q_i}(\alpha_i)$ are algebraically independent over k. The same holds if one chooses, independently at each argument, any one of the derivatives of orders $0, \ldots, \ell_i$. In particular any family of such choices indexed by distinct nonzero algebraic arguments is algebraically independent in the sense that every finite subfamily is independent.*

Proof. Complete each choice to a pair of distinct initial jet entries and apply Theorem 34.6. Algebraic independence of an arbitrary family is, by definition, a finite-subfamily property. □

34.3 The exact field laws across arguments and dimensions

Fix distinct $\alpha_1, \ldots, \alpha_m \in k^*$ and write

$$\mathcal{E} = k(u_1, v_1, \ldots, u_m, v_m), \qquad (u_i, v_i) = (y(\alpha_i), y'(\alpha_i)).$$

For a vector $\mathbf{d} = (d_1, \ldots, d_m)$ of positive integers, let $\mathcal{F}_{\mathbf{d}}$ be the field generated by complete initial jets of dimension d_i at α_i, choosing arbitrary nonnegative integer twists at the different arguments.

Theorem 34.8 (Product cyclic fields and componentwise gcd–lcm)**.** *The field $\mathcal{F}_{\mathbf{d}}$ is independent of all chosen twists, and*

$$\mathcal{F}_{\mathbf{d}} = k(u_i^{d_i}, v_i/u_i : 1 \le i \le m) = \mathcal{E}^{\mu_{d_1} \times \cdots \times \mu_{d_m}}.$$

The ith cyclic group scales u_i, v_i simultaneously and fixes the other pairs. The extension $\mathcal{E}/\mathcal{F}_{\mathbf{d}}$ is Galois of degree $\prod_i d_i$, with precisely that product group. For any two positive dimension vectors $\mathbf{d}, \mathbf{e}$,

$$\mathcal{F}_{\mathbf{d}}\mathcal{F}_{\mathbf{e}} = \mathcal{F}_{\gcd(\mathbf{d},\mathbf{e})}, \qquad \mathcal{F}_{\mathbf{d}} \cap \mathcal{F}_{\mathbf{e}} = \mathcal{F}_{\mathrm{lcm}(\mathbf{d},\mathbf{e})},$$

where gcd and lcm are taken componentwise. In particular dimension choices coprime at every argument recover all $2m$ basic Bessel values exactly.

Proof. The exact single-argument field identity (24.7), specialized by the invertible gauge, gives the displayed monomial generators. Theorem 34.5 identifies $\mathcal{E}$ with a rational function field in $2m$ variables over k. Therefore the independent scalar actions have the stated fixed field and degree. Put $L_i = \mathrm{lcm}(d_i, e_i)$ and work in the finite Galois extension $\mathcal{E}/\mathcal{F}_{\mathbf{L}}$. The intersection of the two fixing subgroups is $\prod_i \mu_{\gcd(d_i,e_i)}$, and the subgroup they generate is $\prod_i \mu_{L_i}$. The Galois correspondence gives the two field identities. □

Remark 34.9 (Scope and attribution). The use of scaled Bessel equations and Siegel–Shidlovskii specialization belongs to classical transcendence theory. The contribution of this section is the explicit combination with the all-twist, all-nonzero-argument Laurent gauge: it supplies complete multipoint determinant-jet ideals and exact field laws. These statements do not assert a new transcendence criterion for unrelated constants, nor do they certify external priority for the classical Bessel independence input.

Part III

Entropy transport and certified finite-dimensional recovery

35 The constant-one Chebyshev–Bernstein theorem

We prove the constant-one coefficient bound by an exact rational certificate, printed below, and then derive an energy refinement. Section 36 determines the optimal exponential constant when a fixed universal prefactor is allowed. These are different normalizations of the coefficient inequality.

Write $B_j^t(x) = \binom{t}{j}x^j(1-x)^{t-j}$ and define $C(t,m,j)$ by

$$T_m(2x-1) = \sum_{j=0}^{t} C(t,m,j)B_j^t(x), \qquad 0 \le m \le t.$$

Theorem 35.1 (The coefficient conjecture)**.** *For all integers $t \ge 1$, $0 \le m \le t$, and $0 \le j \le t$,*

$$|C(t,m,j)| \le \exp(m^2/t). \tag{35.1}$$

The inequality is strict for $m > 0$. The proof below is a finite-certificate proof: its only computational part uses exact rational arithmetic for $t < 512$. The complete verifier is included in the text.

This is the precise constant-one assertion posed in [30, Supplement, Remark 4.1, equation (23)], rather than a bound with an unspecified multiplicative constant. The distinction is discussed in [31, Appendix C.2]. The universal exponent with an unspecified fixed prefactor is determined in Theorem 36.1; the prefactor-one optimum remains separate.

Lemma 35.2 (Three complementary coefficient bounds)**.** *For $1 \le m \le t$,*

$$|C(t,m,j)| \le 2^m, \tag{35.2}$$

$$|C(t,m,j)| \le (t+1)\exp\{tH(m/t)\} \le (t+1)\exp\{(\log 2)m^2/t\}, \tag{35.3}$$

where $H(a) = \frac{1}{2}[(1+a)\log(1+a) + (1-a)\log(1-a)]$ and $0\log 0 = 0$. For $m < t$ there is also the bound

$$|C(t,m,j)| \le D(t,m) := \prod_{l=1}^{m-1} \frac{t+l}{t-l}. \tag{35.4}$$

In particular, $D(t,m) \le e^{m^2/t}$ whenever

$$(m+6)(m-1)^2 \le 6t^2. \tag{35.5}$$

Proof. At degree m the coefficients are $(-1)^{m-i}\binom{2m}{2i}/\binom{m}{i}$. The successive absolute-value ratio is $(2m-2i-1)/(2i+1)$, so their maximum occurs at a middle index. Vandermonde's identity bounds the numerator at that index by $2^m \max_i \binom{m}{i}$, proving (35.2) at degree m. Degree elevation replaces coefficients by convex combinations and proves the bound at every $t \ge m$.

For (35.3), put

$$F(z) = (1+z)^t T_m\left(\frac{z-1}{z+1}\right) = \sum_{j=0}^{t}\binom{t}{j}C(t,m,j)z^j.$$

Endpoint coefficients have absolute value one. For $0 < j < t$, set $r = j/(t-j)$ and, on $z = re^{i\theta}$, set $u = 2\sqrt{r}\,|\sin(\theta/2)|/(1+r)$. The larger inverse Joukowski modulus of $(z-1)/(z+1)$ is $\sqrt{(1+u)/(1-u)}$. Consequently

$$\frac{|F(re^{i\theta})|}{(1+r)^t} \le (1-u^2)^{t/2}\left(\frac{1+u}{1-u}\right)^{m/2} \le \exp\{tH(m/t)\}.$$

The last maximum is attained at $u = m/t$; limiting values handle $u = 1$. Cauchy's formula and $\binom{t}{j} r^j/(1+r)^t \ge 1/(t+1)$ give the first inequality in (35.3). This probability is a mode of a Binomial$(t, j/t)$ law, which proves the stated lower bound. Finally

$$H(a) = \sum_{k\ge 1} \frac{a^{2k}}{2k(2k-1)} \le a^2 \sum_{k\ge 1} \frac{1}{2k(2k-1)} = a^2 \log 2.$$

To apply the older bound (35.4), combine the hypergeometric formula with the identity

$$x^r = \sum_j (j)_{\underline{r}} B_j^t(x)/(t)_{\underline{r}}.$$

This gives

$$C(t,m,j) = (-1)^m\, {}_3F_2\left(\begin{matrix}-m, m, -j \\ 1/2, -t\end{matrix}; 1\right) = (-1)^m Q_m(j; -1/2, -1/2, t).$$

Dette's Hahn bound [33, Corollary 3.6(b)] is exactly (35.4). Its stipulated range $m < t$ is respected here. Thus this inequality itself is prior art. For the final assertion, using the positive series for arctanh gives

$$\begin{aligned}\log D(t,m) &= 2\sum_{l=1}^{m-1} \operatorname{arctanh}(l/t) \\ &\le \frac{m(m-1)}{t} + \frac{m^2(m-1)^2}{6t^3\{1-(m-1)^2/t^2\}}.\end{aligned}$$

The second term is at most m/t precisely under (35.5). □

Proof of Theorem 35.1. The case $m = 0$ is immediate. Suppose first $t \ge 512$. If $m^2/t \ge \frac{10}{3}\log(t+1)$, the contour bound proves the assertion because $\log 2 < 7/10$. Otherwise,

$$m^2 < \frac{10}{3} t\log(t+1) \le \frac{25}{9} t^{4/3}, \qquad m < \frac{5}{3} t^{2/3}.$$

Here $\log(t+1) \le (5/6)t^{1/3}$ for $t \ge 512$: at 512 it follows from $513 < 729$, $\log 3 < 11/10$, and $33/5 < 20/3$; the difference $(5/6)t^{1/3} - \log(t+1)$ has positive derivative thereafter. In particular $m < t$, and

$$(m+6)(m-1)^2 \le m^3 + 6m^2 \le \left(\frac{125}{27} + \frac{25}{96}\right) t^2 < 6t^2.$$

The last estimate uses $t^{2/3} \ge 64$. The Hahn bound therefore applies. The elementary logarithmic inequalities just used can, for example, be verified by positive Taylor lower bounds for $e^{7/10}$ and $e^{11/10}$.

For $1 \le t < 512$, the following verifier covers every pair (t, m). Its first branch uses (35.5); its second uses $2^m \le e^{7m/10} \le e^{m^2/t}$; its third uses the contour bound with $\log 2 < 7/10$. Every other pair is checked with the exact identity

$$C(t,m,j) = \frac{1}{\binom{t}{j}} \sum_{i=\max(0,j-t+m)}^{\min(m,j)} (-1)^{m-i} \binom{2m}{2i}\binom{t-m}{j-i}. \tag{35.6}$$

The function `exp_lower` returns a rational number no larger than e^x for $x \ge 0$, so each successful assertion is a rigorous inequality in the required direction. There are no floating-point comparisons. The complete output is

$$(34784, 39518, 56329, 185), \qquad 12734 \text{ direct coefficient comparisons}, \quad \max t_{\mathrm{direct}} = 145.$$

The four counts sum to $511 \cdot 512/2$, as required. □

```
from fractions import Fraction
from math import comb

def exp_lower(x, degree=32):
    term = total = Fraction(1)
    for k in range(1, degree + 1):
        term *= x / k
        total += term
    return total

counts = [0, 0, 0, 0]
comparisons = largest_direct = 0
for t in range(1, 512):
    for m in range(1, t + 1):
        if m < t and (m+6)*(m-1)**2 <= 6*t*t:
            counts[0] += 1
            continue
        if 10*m >= 7*t:
            counts[1] += 1
            continue
        if exp_lower(Fraction(3*m*m, 10*t)) >= t+1:
            counts[2] += 1
            continue
        lower = exp_lower(Fraction(m*m, t))
        for j in range(t + 1):
            numerator = sum(
                (-1)**(m-i)*comb(2*m, 2*i)*comb(t-m, j-i)
                for i in range(max(0,j-t+m), min(m,j)+1)
            )
            assert Fraction(abs(numerator), comb(t,j)) <= lower
            comparisons += 1
        counts[3] += 1
        largest_direct = max(largest_direct, t)
print(counts, comparisons, largest_direct)
assert sum(counts) == 511*512//2
```

35.1 Krawtchouk identities and the energy refinement

The following identities also justify the weighted norm used in the next section. Define

$$K_k(x; N) = \sum_{h} (-1)^h \binom{x}{h}\binom{N-x}{k-h}, \qquad \sum_{k=0}^{N} K_k(x; N) u^k = (1-u)^x (1+u)^{N-x}.$$

Lemma 35.3. *For $0 \le m, j \le t$,*

$$\binom{t}{j} C(t, m, j) = (-1)^{m+j} K_{2j}(t+m; 2t).$$

Moreover

$$\sum_{x=0}^{N} \binom{N}{x} K_k(x; N) K_\ell(x; N) = 2^N \binom{N}{k} \mathbf{1}_{k=\ell}.$$

For fixed t, j, write $y = 2j/t - 1$ and $C_m = C(t, m, j)$. Then

$$C_0 = 1, \quad C_1 = y, \qquad (t-m)C_{m+1} = 2tyC_m - (t+m)C_{m-1} \quad (1 \le m \le t-1).$$

Proof. In the coefficient generating polynomial $F(z)$ above, put $z = w^2$. The even part of $(1+iw)^{t+m}(1-iw)^{t-m}$ is

$$\frac{1}{2}\{(1+iw)^{t+m}(1-iw)^{t-m} + (1-iw)^{t+m}(1+iw)^{t-m}\}.$$

Multiplication by $(-1)^m$ gives $F(z)$. Comparing its even powers with the Krawtchouk generating function, and using $K_k(N-x;N) = (-1)^k K_k(x;N)$, gives the first identity. Multiplying two generating functions and summing with weight $\binom{N}{x}$ gives $(2+2uv)^N$; comparison of coefficients gives orthogonality. Finally, the same generating function verifies

$$(N-x)K_k(x+1;N) + xK_k(x-1;N) = (N-2k)K_k(x;N).$$

Take $N = 2t$, $k = 2j$, $x = t+m$ and insert the signs in the first identity to obtain the recurrence. □

A one-term consequence of this orthogonality identity is the squared bound

$$C(t,m,j)^2 \le \frac{4^t\binom{2t}{2j}}{\binom{2t}{t+m}\binom{t}{j}^2}.$$

It is not needed by the rational-certificate proof above.

Proposition 35.4 (An energy bound)**.** *For fixed $t \ge 2, j$, set $y = 2j/t - 1$, $C_m = C(t,m,j)$, and*

$$\alpha_0 = 1, \qquad \alpha_{m+1} = \alpha_m \frac{(t-m)^2}{(t+m)(t-m+1)}.$$

For $1 \le m \le t$ define

$$Q_m = \alpha_m C_m^2 + \alpha_{m-1}C_{m-1}^2 - \frac{2t\alpha_m}{t-m+1} yC_mC_{m-1}, \qquad \kappa_m = \frac{t^2}{(t+m-1)(t-m+2)}.$$

Then, for $1 \le m < t$,

$$Q_{m+1} - Q_m = -\frac{2(m-1)\alpha_{m-1}}{(t+m-1)(t-m+2)} C_{m-1}^2 \le 0, \qquad Q_1 = 1 - \frac{t}{t+1}y^2.$$

If $\kappa_m y^2 < 1$, this yields

$$C(t,m,j)^2 \le D(t,m)\frac{t+1}{t-m+1}\frac{1 - ty^2/(t+1)}{1-\kappa_m y^2},$$

where $D(t,m) = \prod_{i=1}^{m-1}(t+i)/(t-i)$ also at $m = t$. In particular, for each fixed $0 < \delta < 1$, uniformly for $|y| \le 1-\delta$ and $m \le \delta t/2$,

$$|C(t,m,j)| \le K_\delta \exp\{m(m-1)/(2t) + O(m^4/t^3)\}.$$

Proof. Substitute the three-term recurrence into Q_{m+1} and cancel its $y^2C_m^2$ terms. The result is

$$Q_{m+1} = \alpha_m C_m^2 - \frac{2t\alpha_m}{t-m+1} yC_mC_{m-1} + \alpha_m \frac{t+m}{t-m+1} C_{m-1}^2.$$

Subtracting Q_m and using the ratio of consecutive α's gives the displayed difference. Completing the square gives

$$Q_m = \alpha_{m-1}\left(C_{m-1} - \frac{t\alpha_m y}{(t-m+1)\alpha_{m-1}} C_m\right)^2 + \alpha_m C_m^2(1-\kappa_m y^2).$$

Telescoping the defining product gives $\alpha_m^{-1} = D(t,m)(t+1)/(t-m+1)$. These identities and $Q_m \le Q_1$ prove the bound. In the stated bulk range, $(t+m-1)(t-m+2) \ge t^2 - m^2$, so $\kappa_m y^2 \le (1-\delta)^2/(1-\delta^2/4) < 1$ uniformly. The remaining rational factors are bounded, and the positive series for artanh gives $\log D(t,m) = m(m-1)/t + O(m^4/t^3)$ for $m \le t/2$. □

36 The optimal exponent with a universal prefactor

Let $M_{t,m} = \max_{0\le j\le t} |C(t,m,j)|$ and put

$$H(a) = \tfrac{1}{2}\{(1+a)\log(1+a) + (1-a)\log(1-a)\}, \qquad 0 \le a \le 1,$$

where $0\log 0 = 0$. The following improves the exponential constant in Theorem 35.1 when a universal multiplicative prefactor is allowed. The distinction between that prefactor and prefactor one is essential.

Theorem 36.1 (A sharp exponential constant)**.** *There is an absolute constant K such that, for all $t \ge 1$, $0 \le m \le t$ and $0 \le j \le t$,*

$$|C(t,m,j)| \le K(1+m^2/t)^{1/4}\exp\{tH(m/t)\}, \tag{36.1}$$

$$|C(t,m,j)| \le K\,2^{m^2/t}. \tag{36.2}$$

The exponent $\log 2$ in the second inequality is optimal among inequalities $|C(t,m,j)| \le K_c\exp(cm^2/t)$ with a finite constant K_c independent of t,m,j. This does not prove the same inequality with prefactor exactly one.

The improvement comes from integrating the coefficient contour, rather than replacing its integrand by its supremum. The resulting prefactor depends on m^2/t, instead of on t alone.

Lemma 36.2 (A Gaussian integral against the arcsine law)**.** *For every $A, B \ge 0$,*

$$(1+B)\int_0^{\pi/2} \exp\{-\tfrac{1}{2}(B\sin v - A)^2\}\,dv \le K\sqrt{1+A}.$$

Proof. The case $B \le 1$ is immediate. Suppose $B > 1$. The substitution $y = B\sin v$ gives the measure $dy/\sqrt{B^2-y^2}$ on $[0,B]$. If $B \le 2(A+1)$, use $\sqrt{B^2-y^2} \ge \sqrt{B(B-y)}$ to bound the integral by

$$B^{-1/2}\int_0^B e^{-(y-A)^2/2}(B-y)^{-1/2}\,dy \le B^{-1/2}(2+\sqrt{2\pi}).$$

For the last inequality put $u = B-y$, split at $u=1$, and use $u^{-1/2} \le 1$ on the remaining interval. Multiplication by $1+B \le 2B$ proves the claim in this case.

If $B > 2(A+1)$, the part $0 \le y \le 3B/4$ is at most K/B, because the denominator is at least $\sqrt{7}B/4$ and the Gaussian integral is bounded. On $3B/4 \le y \le B$ we have $y - A > B/4$; this part is at most $(\pi/2)e^{-B^2/32}$. Multiplying by $1+B$ gives a uniform constant, which is at most $K\sqrt{1+A}$. □

Proof of (36.1). Endpoint coefficients have absolute value one. For $0 < j < t$, put $p = j/t$, $r = p/(1-p)$, $v = p(1-p)$, $a = m/t$, and

$$F(z) = (1+z)^t T_m\left(\frac{z-1}{z+1}\right) = \sum_{j=0}^{t}\binom{t}{j}C(t,m,j)z^j.$$

For $z = re^{i\theta}$ set $u = 2\sqrt{v}\,|\sin(\theta/2)|$. The inverse Joukowski formula and $|T_m(w)| \le |\zeta|^m$ when $w = (\zeta+\zeta^{-1})/2$, $|\zeta| \ge 1$, give

$$\frac{|F(re^{i\theta})|}{(1+r)^t} \le \exp\{tf_a(u)\}, \qquad f_a(u) = \tfrac{1}{2}\log(1-u^2) + a\,\operatorname{artanh} u.$$

Explicitly, one may take $\zeta = (\sqrt{z}+i)/(\sqrt{z}-i)$ or its reciprocal; choosing the larger modulus gives $|\zeta| = \sqrt{(1+u)/(1-u)}$. Limits cover the removable endpoint configurations. Since

$f_a'(u) = (a-u)/(1-u^2)$ and $f_a(a) = H(a)$, integration of this derivative on the interval between u and a shows

$$f_a(u) \le H(a) - \tfrac{1}{2}(u-a)^2. \tag{*}$$

This uses $1/(1-w^2) \ge 1$ and also holds at $a = 1$ by a limit.

The binomial mass $\pi_j = \binom{t}{j} p^j (1-p)^{t-j}$ is a mode of $\mathrm{Bin}(t,p)$. Chebyshev's probability inequality on the interval of radius $2\sqrt{tv}$ around its mean j implies

$$\pi_j \ge \frac{3/4}{4\sqrt{tv}+1} \ge \frac{c}{1+\sqrt{tv}}.$$

Cauchy's coefficient formula, (*) and Lemma 36.2, with $A = m/\sqrt{t}$ and $B = 2\sqrt{tv}$, therefore yield

$$|C(t,m,j)| \le K e^{tH(a)} \sqrt{1+m/\sqrt{t}} \le K(1+m^2/t)^{1/4} e^{tH(a)}.$$

The final inequality follows from $(1+u)^2 \le 2(1+u^2)$. □

Proof of (36.2) and optimality. Write $x = m^2/t$ and $\Delta(a) = (\log 2)a^2 - H(a)$. The positive power series

$$H(a) = \sum_{h \ge 1} \frac{a^{2h}}{2h(2h-1)}$$

shows that $H(a)/a^2$ increases strictly on $(0,1)$ and tends to $\log 2$ at one. Hence for $0 \le a \le 99/100$ there is an absolute $c_0 > 0$ such that $t\Delta(a) \ge c_0 x$. The factor $(1+x)^{1/4} e^{-c_0 x}$ is bounded, proving the assertion in this range.

Suppose $99/100 \le a < 1$ and write $\delta = 1-a$, $k = t-m$. Then k is an integer with $1 \le k \le t/100$. Since $H'(a) = \operatorname{artanh} a$,

$$\begin{aligned} \Delta(1-\delta) &\ge -2(\log 2)\delta + \tfrac{1}{2}\int_0^\delta \log(1/u)\,du \\ &= \delta\{\tfrac{1}{2}\log(1/\delta) + \tfrac{1}{2} - 2\log 2\} \ge \tfrac{1}{4}\delta \log(1/\delta). \end{aligned}$$

The last inequality follows from $\delta \le 1/100$; for example $\log 100 > 4$ and $\log 2 < 7/10$ suffice. The function $k \mapsto k\log(t/k)$ increases for $1 \le k \le t/100$, so

$$t\Delta(a) \ge \tfrac{1}{4} k \log(t/k) \ge \tfrac{1}{4}\log t.$$

Since $x \le t$, this absorbs the prefactor in (36.1). At $m = t$, the elementary degree bound $|C(t,t,j)| \le 2^t$ proves the result directly.

Finally, for even t the exact central coefficient is

$$|C(t,t,t/2)| = \frac{\binom{2t}{t}}{\binom{t}{t/2}} = 2^{t-1/2}(1+o(1)).$$

Thus no fixed prefactor permits an exponent $c < \log 2$. □

36.1 Exact weighted energy and matching exponential profiles

The next identity is the classical Hahn orthogonality in this normalization. We prove it directly from the Krawtchouk identities; its sharp-profile consequences are the additional conclusions needed here. Put

$$w_{t,j} = 4^{-t}\binom{2j}{j}\binom{2t-2j}{t-j}, \qquad A_{t,m} = \frac{(t+m)!(t-m)!}{(t!)^2}.$$

Proposition 36.3 (Exact weighted inverse norm)**.** *The weights $w_{t,j}$ sum to one. If $P(x) = \sum_{m=0}^{t} a_m T_m(2x-1) = \sum_{j=0}^{t} b_j B_j^t(x)$, with real coefficients, then*

$$\sum_{j=0}^{t} w_{t,j} b_j^2 = a_0^2 + \frac{1}{2}\sum_{m=1}^{t} A_{t,m} a_m^2.$$

In particular, for $m \geq 1$,

$$M_{t,m} \geq \sqrt{A_{t,m}/2}. \tag{36.3}$$

Proof. The generating series of $4^{-j}\binom{2j}{j}$ is $(1-z)^{-1/2}$, so convolution proves the assertion about the weights. Write $K_k(x;N)$ for the Krawtchouk polynomials defined in the preceding section. Their combinatorial formula gives the duality

$$\frac{K_k(x;N)}{\binom{N}{k}} = \frac{K_x(k;N)}{\binom{N}{x}}.$$

Indeed, multiplying either side by $\binom{N}{x}\binom{N}{k}$ counts ordered subsets of sizes x, k, with the sign of their intersection. Together with the orthogonality identity from that section this gives

$$\sum_{k=0}^{N} \frac{K_k(x;N)K_k(y;N)}{\binom{N}{k}} = \frac{2^N}{\binom{N}{x}}\mathbf{1}_{x=y}.$$

Using $K_k(N-y;N) = (-1)^k K_k(y;N)$, its even-$k$ part is half the sum of the right sides for y and $N-y$.

Take $N = 2t$, $x = t+m$, $y = t+\ell$, and use $\binom{t}{j}C(t,m,j) = (-1)^{m+j}K_{2j}(t+m;2t)$ and

$$\frac{w_{t,j}}{\binom{t}{j}^2} = \frac{(2t)!}{4^t(t!)^2}\frac{1}{\binom{2t}{2j}}.$$

For $0 \leq m, \ell \leq t$, the resulting weighted inner product is zero unless $m = \ell$. Its value is one for $m = \ell = 0$ and $A_{t,m}/2$ otherwise. Expanding $b_j = \sum_m a_m C(t,m,j)$ proves the identity. Averaging $C(t,m,j)^2 \leq M_{t,m}^2$ proves the lower bound. □

Corollary 36.4 (Sharp proportional-degree and moderate-degree scales)**.** *If $t \to \infty$ and $m/t \to a \in [0,1]$, then*

$$\frac{1}{t}\log M_{t,m} \longrightarrow H(a).$$

If instead $m = o(t)$ and $m^2/t \to \infty$, then

$$\log M_{t,m} = (\tfrac{1}{2} + o(1))m^2/t.$$

The latter maximum is over every coefficient, without a restriction keeping j/t away from zero and one.

Proof. The elementary factorial estimate $\log(n!) = n\log n - n + O(\log(n+1))$, valid also at $n = 0$ with $0\log 0 = 0$, gives $\log A_{t,m} = 2tH(m/t) + O(\log(t+1))$ uniformly. Combine (36.3) with (36.1); the case $m = 0$ is immediate. For $m = o(t)$, the product

$$A_{t,m} = \prod_{i=1}^{m} \frac{t+i}{t-i+1}$$

and $\log(1+u) = u + O(u^2)$ give $\log A_{t,m} = m^2/t + O(m^3/t^2)$. Also $tH(m/t) = m^2/(2t) + O(m^4/t^3)$. The logarithm of the prefactor in (36.1) is $O(1+\log(1+m^2/t)) = o(m^2/t)$, proving the second assertion. □

Remark 36.5 (What remains open in this argument). The conjecturally stronger statement $M_{t,m} \le 2^{m^2/t}$ with prefactor one is not proved here. Conversely, a global exponent $1/2$ with either a fixed or polynomial prefactor is impossible: the degree-t central coefficient grows as $2^{t-1/2}$. The exponent $1/2$ is the sharp moderate-degree limit, whereas $\log 2$ is the sharp universal exponent when a fixed prefactor is allowed.

37 An entropy–transport inequality for mixture likelihood

Let $\mathcal{X}$ be a countable metric space, let K_θ be component probability kernels, and write $p_H = \int K_\theta \, dH(\theta)$. For an empirical law q, assume the exact mixing NPMLE H exists and has positive probability on the observed support. Zero empirical cells are omitted in score sums.

Theorem 37.1 (Component concentration transfers through likelihood contact)**.** *Suppose that $\psi : [0,\infty) \to [0,\infty)$ is nondecreasing and concave and that*

$$W_1(R, K_\theta) \le \psi(D(R\|K_\theta))$$

for every component and every probability law with finite entropy. Then

$$W_1(q, p_H) \le \psi(D(q\|p_H)) \le \psi(D(q\|p_G))$$

for every admissible comparator mixing law G for which the final entropy is finite.

Proof. The directional derivative toward a point mass gives $S(\theta) = \sum_x q(x)K_\theta(x)/p_H(x) \le 1$. Its H-integral equals one, so $S = 1$ almost surely under H. Thus $R_\theta(x) = q(x)K_\theta(x)/p_H(x)$ is a probability kernel H-almost surely, and $\int R_\theta \, dH = q$. The entropy chain rule gives

$$\int D(R_\theta\|K_\theta) \, dH(\theta) = D(q\|p_H).$$

Mixing couplings, applying the component bound, and then Jensen's inequality prove the first conclusion. Likelihood optimality gives $D(q\|p_H) \le D(q\|p_G)$, proving the second. All applications below have finite right sides; truncation of transport costs gives the same conclusion before passing to an unbounded cost. □

For normalized binomial counts, the component bound has $\psi(u) = \sqrt{u/(2t)}$ for normalized count cost. For Poisson mixtures, a parameter-free occupancy channel will import that same inequality while preserving the original likelihood fit.

The argument does not normalize every stratum of an arbitrary heterogeneous joint likelihood: only the aggregate score is normalized. Consequently an arbitrary-profile theorem does not follow verbatim. The results proved below concern common known trial counts or common known exposures.

38 Closure of the Poisson critical window

Consider the experiment

$$p_G(j) = \int_0^B e^{-\lambda} \frac{\lambda^j}{j!}\, dG(\lambda), \qquad X_1, \dots, X_N \overset{\text{iid}}{\sim} p_G,$$

where the support radius $B > 0$ is known. Let $\widehat{G}$ be any measurable maximizer of the original Poisson mixture likelihood over all probability laws on $[0, B]$. There is no restriction on the number of mixing atoms. Set $L = \log(eN)$ and, for $N \geq 1$, define

$$r(N, B) = \begin{cases} B \min\{1, L^{-1} \log(e + L/B)\}, & 0 < B \leq L, \\ \max\{\sqrt{B/L}, B/\sqrt{N}\}, & B > L. \end{cases} \tag{38.1}$$

The two expressions at the breakpoint agree up to absolute constants.

Theorem 38.1 (Global Poisson population recovery)**.** *There exist absolute constants $0 < c < C < \infty$ such that, for every $N \geq 1$ and every $B > 0$,*

$$c\, r(N, B) \leq \inf_{\widetilde{G}} \sup_{G \in \mathcal{P}([0,B])} \mathbb{E}_G W_1(\widetilde{G}, G) \leq \sup_{G \in \mathcal{P}([0,B])} \mathbb{E}_G W_1(\widehat{G}, G) \leq C\, r(N, B).$$

In particular, the ordinary NPMLE is minimax up to absolute factors throughout the entire support/sample-size plane. For observations $X_i \mid \Theta_i \sim \mathrm{Poi}(t\Theta_i)$ with $G \in \mathcal{P}([0, M])$, the corresponding rate is $r(N, tM)/t$.

The growing-support crossover in Theorem 38.1 is $B \asymp N/\log(eN)$: below that scale the deconvolution term $\sqrt{B/L}$ dominates, and above it the latent-sampling term $B/\sqrt{N}$ dominates. The statement uses the ordinary likelihood estimator throughout, with no logarithmic loss in this transition.

Miao et al. [43, Theorem 6.1(a)] established the small-support rate for the NPMLE, and their Theorem 6.1(b) gives $\sqrt{B/\log N}$ under $B \leq C_0 N^{1/3-\varepsilon_0}$. The rate includes diameter clipping for arbitrarily small B: an unclipped formula cannot be extrapolated beyond the maximum possible loss B. All ingredients used below are proved here, including the two-scale binomial inverse inequality in Proposition 38.6. The estimator is never replaced by a transformed-data MLE.

For $N \in \{1, 2\}$ the rate in (38.1) is comparable to B. The diameter bound and the two-point latent-sampling lower bound given below therefore suffice. In the proof of the nontrivial regimes we may assume $N \geq 3$.

38.1 A parameter-free occupancy channel

For an integer $m \geq 1$, define a Markov kernel A_m as follows. Given a count x, independently assign x balls uniformly to m boxes, and output the number of occupied boxes. Poisson splitting proves

$$A_m \,\mathrm{Poi}(\lambda) = \mathrm{Bin}(m, \phi_m(\lambda)), \qquad \phi_m(\lambda) = 1 - e^{-\lambda/m}. \tag{38.2}$$

Indeed the box counts are independent Poisson random variables of mean λ/m, so their nonempty indicators are independent Bernoulli variables. In particular, if $F = \phi_m \# G$, then $A_m p_G$ is exactly the binomial mixture with mixing law F. On $[0, \phi_m(B)]$ the inverse map has derivative at most $me^{B/m}$, and consequently

$$W_1(G, H) \leq me^{B/m} W_1(\phi_m \# G, \phi_m \# H). \tag{38.3}$$

This channel is a proof device. We do not transform the data before fitting, or assert that a transformed MLE remains an MLE.

We use $W_{1,m}$ for transportation on $\{0, \dots, m\}$ with cost $|i - j|/m$, and $h(p, q)^2 = \sum_j (\sqrt{p_j} - \sqrt{q_j})^2$.

Lemma 38.2 (Empirical entropy on the infinite alphabet)**.** *Let q be the empirical law of N observations from p_G. Uniformly over $G \in \mathcal{P}([0,B])$,*

$$\mathbb{E}D(q\|p_G) \le C(B+L)/N. \tag{38.4}$$

Proof. For every coordinate with probability $p_j > 0$, convexity and $\log u \le u-1$ give

$$0 \le \mathbb{E}\{q_j \log(q_j/p_j) - q_j + p_j\} \le \min\{(1-p_j)/N,\ p_j \log(1/p_j)\}.$$

The second bound also uses $q_j \log q_j \le 0$. The expression inside braces is nonnegative, so its infinite sum can be averaged by Tonelli's theorem. Coordinates with $p_j = 0$ contribute zero almost surely. For $j \ge B$, the Poisson mass is increasing in its mean throughout $[0,B]$, and hence

$$p_G(j) \le e^{-B}B^j/j! \le (eB/j)^j \le e^{-j} \quad (j \ge e^2 B).$$

Take $J = \lceil e^2 B + 3L \rceil$. The first $J+1$ coordinates contribute at most $(J+1)/N$. Since $u\log(1/u)$ is increasing on $(0, e^{-1}]$, the remaining contribution is bounded by $\sum_{j>J} j e^{-j} \le C(J+1)e^{-J} \le C/N$. This proves (38.4). Notice that the expected empirical chi-square divergence on the full infinite alphabet would not give this conclusion. □

Lemma 38.3 (Root-N prediction after occupancy)**.** *For $m = \lceil \max(B, L) \rceil$ and every Poisson NPMLE $\widehat{G}$,*

$$\mathbb{E}W_{1,m}(A_m p_{\widehat{G}}, A_m p_G) \le C/\sqrt{N}. \tag{38.5}$$

Proof. Existence follows from compactness of the mixing parameter space and upper semicontinuity of the likelihood; the admissible fit δ_B is strictly positive at every observed count. Write $H = \widehat{G}$. The likelihood directional derivative gives

$$S(\lambda) = \sum_{j: q_j > 0} \frac{q_j K_\lambda(j)}{p_H(j)} \le 1 \quad (0 \le \lambda \le B).$$

Since $\int S\, dH = 1$, equality holds H-almost everywhere. Thus $R_\lambda(j) = q_j K_\lambda(j)/p_H(j)$ is a probability kernel H-almost everywhere and $\int R_\lambda\, dH = q$. Its entropy chain rule and the likelihood comparison yield

$$\int D(R_\lambda \| K_\lambda)\, dH(\lambda) = D(q\|p_H) \le D(q\|p_G). \tag{38.6}$$

The binomial transport-entropy inequality is

$$W_{1,m}(R, \mathrm{Bin}(m,p))^2 \le D(R\|\,\mathrm{Bin}(m,p))/(2m).$$

For completeness, a normalized-count 1-Lipschitz function changes by at most $1/m$ on changing one of the m Bernoulli bits. The conditional Hoeffding bound gives centered log-mgf at most $u^2/(8m)$; entropy duality and Wasserstein duality give the displayed inequality, including degenerate endpoints whenever the entropy is finite. Apply it to $A_m R_\lambda$, use entropy data processing, and integrate (38.6). Convexity of transportation under mixtures and Cauchy–Schwarz give

$$W_{1,m}(A_m q, A_m p_H) \le \sqrt{D(q\|p_G)/(2m)}.$$

For each count threshold, the CDF of $A_m q$ is an average of independent $[0,1]$-valued functions of the X_i. Its expected absolute error is at most $1/(2\sqrt{N})$. The CDF formula for $W_{1,m}$ therefore gives $\mathbb{E}W_{1,m}(A_m q, A_m p_G) \le 1/(2\sqrt{N})$. Now use Lemma 38.2 and $(B+L)/m \le 2$. □

38.2 A growing-support Hellinger bound

Lemma 38.4 (Poisson mixture entropy and likelihood control)**.** *For every fixed absolute $b_0 > 0$, uniformly over $b_0 L \le B \le N$,*

$$\mathbb{E}h(p_{\widehat{G}}, p_G) \le C_{b_0} B^{1/4} L^{3/4} / \sqrt{N}. \tag{38.7}$$

Proof. For complex z the vector $K_z = (e^{-z} z^j / j!)_{j \ge 0}$ belongs to ℓ_1 and satisfies $\|K_z\|_1 = e^{|z| - \operatorname{Re} z}$. On the ellipse $z = B(1 + \cos(\theta + iu))/2$, direct calculation gives

$$|z| - \operatorname{Re} z = \tfrac{B}{2}(\cosh u - 1)(1 - \cos\theta) \le B(\cosh u - 1).$$

The kth vector Chebyshev coefficient v_k of $x \mapsto K_{Bx}$ therefore satisfies, by contour shifting,

$$\|v_k\|_1 \le 2\exp\{B(\cosh u - 1) - ku\} \le 2\exp\{-c\min(k^2/B, k)\}.$$

For the second inequality take $u = k/(2B)$ if $k \le 2B$, using $\cosh u - 1 \le u^2$ for $0 \le u \le 1$, and take $u = 1$ otherwise. It follows by summing the Gaussian and exponential tails that truncating the expansion at

$$d = \left\lceil C\left\{\sqrt{B\log(C(B+1)/\beta)} + \log(C(B+1)/\beta)\right\}\right\rceil$$

has uniform ℓ_1 error at most $\beta/8$. Every Chebyshev moment of a probability law on $[0, B]$ belongs to $[-1, 1]$, and $\|v_k\|_1 \le 2$ also follows directly from its real integral formula. Grid the first d moments with mesh $\beta/(16(d+1))$. Choose one actual mixture in each occupied grid box. This gives an ℓ_1 cover of radius β, with centers in the mixture class and cardinality M satisfying

$$\log M \le Cd\log(C(d+1)/\beta). \tag{38.8}$$

Take $J = \lceil e^2 B + 6L \rceil$ and $\beta = N^{-3}/(J+1)$. In the indicated range, $d \le C_{b_0}\sqrt{BL}$ and $\log M \le C_{b_0}\sqrt{B}L^{3/2}$. For every cover center p_a, define the pointwise envelope

$$u_a(j) = \begin{cases} p_a(j) + \beta, & j \le J, \\ e^{-j}, & j > J. \end{cases}$$

Every mixture is bounded coordinatewise by its associated envelope, whose total mass U_a lies in $[1, 1+\eta]$ for $\eta = CN^{-3}$, whenever that envelope is used for at least one mixture. Unused envelopes can be discarded. Normalize $r_a = u_a/U_a$. An associated mixture p satisfies $h(p, r_a) \le \sqrt{2\eta}$.

Here is the elementary likelihood argument, to specify the treatment of the infinite alphabet. If $\widehat{p}$ is a likelihood maximizer and $u_a \ge \widehat{p}$, then

$$\prod_{i=1}^{N} \frac{r_a(X_i)}{p_G(X_i)} \ge e^{-N\eta}.$$

Under p_G, the expectation of the square root of this product is the Nth power of the affinity, at most $\exp\{-Nh(p_G, r_a)^2/2\}$. Markov's inequality followed by a union bound shows, with probability at least $1 - e^{-x}$,

$$h(p_G, \widehat{p}) \le \sqrt{2\eta} + \sqrt{\eta + 2(\log M + x)/N}.$$

Integration gives $\mathbb{E}h(p_G, \widehat{p})^2 \le C\eta + C(\log M + 1)/N$, which proves (38.7). □

38.3 The two-scale argument closes every growing-support regime

The binomial inverse inequality proved below states that if F, H are mixing laws on $[0,1]$, b_F, b_H their m-trial binomial mixtures, and $1 \le S \le K \le m/2$, then

$$W_1(F,H) \le C/K + Ce^{4S^2/m} W_{1,m}(b_F, b_H) + Ce^{2K^2/m} h(b_F, b_H)/S. \tag{38.9}$$

Its small high-frequency amplitude $1/S$ is indispensable. The proof approximates a Lipschitz test function at two polynomial degrees, preserves a bounded derivative at the lower degree, and uses the degree-elevated Bernstein coefficient bounds. This is Proposition 38.6.

Suppose $b_0 L \le B \le N$, with $b_0 > 0$ an absolute constant to be fixed below. Set $m = \lceil \max(B, L) \rceil$, so $m \asymp_{b_0} B$. For sufficiently large absolute N, take

$$S = \lfloor \sqrt{m}/16 \rfloor, \qquad K = \left\lfloor \min\{m/2, \sqrt{amL}\} \right\rfloor, \qquad a = 1/64.$$

These are admissible, $S \asymp \sqrt{m}$, $e^{4S^2/m} \le C$, and $m/K \le C\sqrt{m/L}$ because $m \ge L$. Apply (38.9) to the pushforwards under ϕ_m, then use (38.3), Lemmas 38.3 and 38.4, and Hellinger data processing. This gives

$$\mathbb{E} W_1(\widehat{G}, G) \le C_{b_0} \left\{ \sqrt{B/L} + \frac{B}{\sqrt{N}} + \frac{e^{2aL} B^{3/4} L^{3/4}}{\sqrt{N}} \right\}.$$

The ratio of the last term to $\sqrt{B/L}$ is at most $Ce^{2aL} N^{-1/4} L^{5/4}$, which is uniformly bounded since $B \le N$ and $2a < 1/4$. We have proved

$$\mathbb{E} W_1(\widehat{G}, G) \le C_{b_0} \{ \sqrt{B/L} + B/\sqrt{N} \}. \tag{38.10}$$

Bounded N is absorbed in the absolute constant by the diameter bound; this is uniform in B because $B \ge b_0 L$ in this step.

If $B > N$, no Hellinger estimate or approximation is necessary. For any binomial mixing law, coupling its parameter p with its normalized count gives expected cost at most $1/(2\sqrt{m})$. Thus

$$W_1(F,H) \le m^{-1/2} + W_{1,m}(b_F, b_H).$$

Together with (38.3) and Lemma 38.3, this gives $\mathbb{E} W_1(\widehat{G}, G) \le C(\sqrt{B} + B/\sqrt{N}) \le CB/\sqrt{N}$. Every $B \ge L$ is now covered.

38.4 The small-support regime, including arbitrarily small radii

We include a direct proof to ensure uniformity as $B \downarrow 0$. Let f be 1-Lipschitz on $[0, B]$ and normalize $f(0) = 0$. Jackson approximation gives a degree-k polynomial $P(\lambda) = \sum_{l=0}^{k} a_l \lambda^l$ such that

$$\|f - P\|_{[0,B]} \le CB/k, \qquad \|P\|_{[0,B]} \le CB.$$

Its monomial coefficients satisfy

$$|a_l| \le B C_1^k B^{-l}. \tag{38.11}$$

Indeed expand $P(Bx)$ in shifted Chebyshev polynomials: each Chebyshev coefficient has magnitude at most $2\|P\|$, and the sum of absolute monomial coefficients of $T_j(2x-1)$ is at most 7^j by its three-term recurrence. Summing the finite geometric series proves (38.11) after enlarging C_1.

Take $k \ge 2eB$ and put $Q_k(\lambda) = \sum_{r=0}^{k} \lambda^r / r!$. The exponential remainder implies

$$\|e^{-\lambda} P(\lambda) Q_k(\lambda) - f(\lambda)\|_{[0,B]} \le CB/k.$$

For example, $0 \le 1 - e^{-\lambda} Q_k(\lambda) \le C(eB/(k+1))^{k+1} \le C2^{-k}$, which suffices. Write this weighted polynomial as $\sum_{j=0}^{2k} b_j K_\lambda(j)$. Since

$$b_j = \sum_{\substack{0 \le l \le k \\ 0 \le j-l \le k}} a_l \frac{j!}{(j-l)!},$$

equation (38.11) gives

$$\max_{0 \le j \le 2k} |b_j| \le B(C_2 k/B)^k. \tag{38.12}$$

The factors $k+1$ and C_1^k have been absorbed into C_2^k; $2k/B \ge 1$ by hypothesis. Wasserstein duality consequently gives the deterministic inequality

$$W_1(F, H) \le CB/k + B(C_2 k/B)^k \|p_F - p_H\|_1. \tag{38.13}$$

For the MLE, likelihood comparison, $h^2 \le D$, the Hellinger triangle inequality and $\|p-q\|_1 \le 2h(p,q)$ imply

$$\mathbb{E}\|p_{\widehat{G}} - p_G\|_1 \le 4\sqrt{\mathbb{E} D(q\|p_G)} \le C\sqrt{L/N} \qquad (B \le L).$$

Put $d_0 = \log(e + L/B)$ and $k = \lfloor \gamma L/d_0 \rfloor$, where $\gamma > 0$ is a sufficiently small absolute constant. If $\gamma L/d_0 < 2$, the diameter bound $W_1 \le B$ is already at most a constant times $B\min(1, d_0/L)$. Otherwise $k \ge \gamma L/(2d_0)$. Choose a fixed $b_0 > 0$ sufficiently small that $B \le b_0 L$ implies $\gamma(L/B)/(2\log(e + L/B)) \ge 2e$; then $k \ge 2eB$. Furthermore

$$k \log(C_2 k/B) \le C_3 \gamma L \le L/8$$

after decreasing γ once, since $\log(C_2\gamma(L/B)/d_0) \le C_3 d_0$. The stochastic term in (38.13) is therefore at most $CBe^{L/8}\sqrt{L/N} \le CB/k$: here $k \le \gamma L$ and $e^{-3L/8}L^{3/2}$ is bounded. This proves the required rate for $B \le b_0 L$. On $b_0 L \le B \le L$, equation (38.10) is bounded by an absolute constant, whereas $B\min(1, d_0/L)$ is bounded below by a positive absolute constant. This proves the upper bound in Theorem 38.1 for every B.

38.5 Matching lower bounds without support-range restrictions

We first record a convenient moment-matching pair. On an interval $[a-M, a+M]$, push forward the probability measures

$$(1 \pm \tfrac{1}{2} T_k(x)) \frac{dx}{\pi\sqrt{1-x^2}}, \qquad -1 < x < 1,$$

under $x \mapsto a + Mx$. Denote them by F_k^+, F_k^-. They match all moments of degree less than k. Their CDF difference is, up to orientation, $\sin(k\theta)/(\pi k)$ for $x = \cos\theta$. Consequently

$$W_1(F_k^+, F_k^-) = \frac{M}{\pi k} \int_0^\pi |\sin(k\theta)| \sin\theta \, d\theta \ge \frac{M}{\pi k}. \tag{38.14}$$

The last inequality follows from $|\sin u| \ge \sin^2 u$ and $\int_0^\pi \sin^2(k\theta)\sin\theta\, d\theta = 1 + (4k^2-1)^{-1}$.

For $B \le L$, use this pair on $[0, B]$. Expanding $e^{-\lambda}$ and using the matched moments gives the elementary bound

$$\|p_{F_k^+} - p_{F_k^-}\|_1 \le 2 \sum_{r \ge k} \frac{(2B)^r}{r!}. \tag{38.15}$$

Indeed the term with observed count j and exponential-expansion index l vanishes if $j + l < k$; summing the absolute values of all remaining terms gives (38.15). Set $d_0 = \log(e + L/B)$ and

$k = \lceil 100L/d_0 \rceil$. Writing $t = L/B \geq 1$, we have $t = e^{d_0} - e \geq e^{d_0}/(e+1)$ and $e^{d_0/2}/d_0 \geq e/2$. These inequalities show

$$\log \frac{k}{2eB} \geq \frac{d_0}{2}, \qquad k \log \frac{k}{2eB} \geq 50L.$$

They also ensure $k \geq 4B$. Thus the tail in (38.15) is at most $4(2eB/k)^k \leq 4e^{-50L}$. The two N-sample laws have TV at most $2Ne^{-50L}$, whereas (38.14) is at least $cB \min\{1, d_0/L\}$. Le Cam's two-point inequality proves the small-support lower bound, including arbitrarily tiny B.

For $B \geq L$, set $a = B/2$, $M = \sqrt{BL}/8$, and $k = \lceil 20L \rceil$. The moment-matching interval lies inside $[0, B]$. The exact Poisson identity

$$\sum_{x=0}^{\infty} \frac{K_\lambda(x) K_\mu(x)}{K_a(x)} = \exp\{(\lambda - a)(\mu - a)/a\}$$

shows, upon integrating against the signed difference $F_k^+ - F_k^-$, that

$$\sum_x \frac{(p_{F_k^+}(x) - p_{F_k^-}(x))^2}{K_a(x)} \leq 4 \sum_{r \geq k} \frac{(M^2/a)^r}{r!} = 4 \sum_{r \geq k} \frac{(L/32)^r}{r!}.$$

All exchanges are justified by boundedness of λ, μ on the integration interval and the absolutely convergent exponential series. The last tail is at most Ce^{-40L}, for instance by $r! \geq (r/e)^r$ and a geometric-tail bound. Cauchy–Schwarz bounds the predictive TV by half the square root of this quantity. Le Cam's inequality and (38.14) therefore give the lower rate $c\sqrt{B/L}$.

Finally consider mixing laws on $\{0, B\}$ with respective weights $1/2 \pm c_0/\sqrt{N}$ at B, for a sufficiently small absolute $c_0 > 0$. Their latent N-sample laws have TV bounded away from one, by the Bernoulli KL formula and Pinsker's inequality. The Poisson sample is a Markov image of that latent sample, so its TV is no larger. The mixing-law distance is $2c_0 B/\sqrt{N}$. Le Cam's inequality yields the remaining lower rate $cB/\sqrt{N}$. Taking the maximum completes the proof of Theorem 38.1.

The proof uses a common known exposure and a known compact support. It makes no assertion for heterogeneous exposures, unknown support radii, or an MLE constrained to a prescribed number of components. The gain over a direct one-scale inverse estimate is structural: the occupancy channel imports a binomial concentration inequality, likelihood contact supplies posterior entropy, and the two-scale inverse places predictive transportation and high-frequency Hellinger error in different terms.

38.6 Self-contained binomial inversion tools

We give the approximation tools used in the proof. Write $B_j^t(x) = \binom{t}{j} x^j (1-x)^{t-j}$ and $p_F(j) = \int B_j^t \, dF$. The integer t is a binomial trial count, $W_{1,t}$ uses cost $|i-j|/t$, and $h(p,q)^2 = \sum_j (\sqrt{p_j} - \sqrt{q_j})^2$.

Lemma 38.5 (Stable approximation and bounded-polynomial inversion)**.** *There is an absolute C with the following properties. For every 1-Lipschitz $f : [0,1] \to \mathbb{R}$ and integer $k \geq 1$, there is a polynomial P_k of degree at most k such that*

$$\|P_k - f\|_\infty \leq C/k, \qquad \|P_k'\|_\infty \leq 2.$$

If $P = \sum_{j=0}^t b_j B_j^t$ has degree $k \leq t/2$, then

$$\max_j |b_j| \leq Ce^{2k^2/t} \|P\|_\infty. \tag{38.16}$$

For the derivative-stable P_k, its coefficient vector also satisfies

$$\max_{j<t} t|b_{j+1} - b_j| \leq Ce^{4k^2/t}. \tag{38.17}$$

Proof. Subtract $f(0)$, reflect f on $[0,1]$ to a period-two 1-Lipschitz function, and convolve with the normalized positive Jackson kernel $c_s[\sin(s\pi u/2)/\sin(\pi u/2)]^4$. Its first absolute moment is $O(1/s)$, so the resulting trigonometric polynomial Q_s has error $O(1/s)$ and derivative norm at most one. Its frequencies have magnitude at most $2s$, and its Fourier coefficients have modulus at most one. Replacing each exponential by its Taylor polynomial at $1/2$ of degree $100s$ changes the function and derivative by at most e^{-100s}. Indeed $|\pi l(x-1/2)| \le \pi s$ and the factorial inequality $M! \ge (M/e)^M$ bounds both remainders, including the $4s+1$ terms and the extra derivative factor. Take $s = \lfloor k/100 \rfloor$ for $k \ge 100$; use a constant polynomial and enlarge C for smaller k.

For (38.16), put $M = \|P\|_\infty$ and $F(z) = (1+z)^t P(z/(1+z))$. The maximum principle in the inverse Joukowski variable gives $|P((2+w+w^{-1})/4)| \le M|w|^k$ for $|w| \ge 1$. For $0 < j < t$, set $r = j/(t-j)$, $v = r/(1+r)^2$, and $u = 2\sqrt{v}\,|\sin(\phi/2)|$. The larger inverse Joukowski modulus for $re^{i\phi}/(1+re^{i\phi})$ is $(1+u)/\sqrt{1-u^2}$. Hence

$$\frac{|F(re^{i\phi})|}{M(1+r)^t} \le (1-u^2)^{(t-k)/2}(1+u)^k \le e^{2k^2/t}e^{-tu^2/8}.$$

The last inequality follows from $k \le t/2$ and completing the square. The average of the final Gaussian factor is at most $C/\sqrt{1+tv}$. Cauchy's coefficient formula, together with $\binom{t}{j}(j/t)^j(1-j/t)^{t-j} \ge c/\sqrt{1+tv}$, proves (38.16). The last lower bound follows directly from Stirling's factorial bounds; endpoint coefficients are $P(0)$ and $P(1)$. Finally differentiate the Bernstein expansion: $P_k' = t\sum_{j<t}(b_{j+1}-b_j)B_j^{t-1}$. Apply (38.16) at degree $k-1 \le (t-1)/2$ and use $2(k-1)^2/(t-1) \le 4k^2/t$. □

Proposition 38.6 (Two-scale deterministic inversion)**.** *For integers $1 \le s \le k \le t/2$ and any two mixing laws G, H,*

$$W_1(G,H) \le \frac{C}{k} + Ce^{4s^2/t}W_{1,t}(p_G,p_H) + \frac{Ce^{2k^2/t}}{s}h(p_G,p_H). \tag{38.18}$$

Proof. For any 1-Lipschitz test f, take the polynomials P_k, P_s from Lemma 38.5. The approximation error in integrating P_k instead of f against $G-H$ is at most C/k. The Bernstein coefficients of P_s have normalized-grid Lipschitz constant at most $Ce^{4s^2/t}$. On the other hand,

$$\|P_k - P_s\|_\infty \le C/k + C/s \le C/s.$$

Thus the Bernstein coefficients of $P_k - P_s$ have supremum norm at most $Ce^{2k^2/t}/s$. Test P_s by count transportation and $P_k - P_s$ by $\|p_G - p_H\|_1 \le 2h(p_G,p_H)$. All constants are uniform over f, so taking the Wasserstein dual supremum proves the result. □

39 Binomial mixture estimation and testing across the replication range

The critical replication scale is $t \asymp N/\log N$. The estimation proof uses two statistical norms on different parts of a test polynomial. The low-degree part is controlled by likelihood transportation. The remaining polynomial has small amplitude, and is controlled by a predictive Hellinger bound. This yields the full replication curve, including its transition and its parametric plateau.

For a probability measure G on $[0,1]$, put

$$K_x(j) = B_j^t(x) = \binom{t}{j} x^j (1-x)^{t-j}, \qquad p_G(j) = \int B_j^t(x)\, dG(x), \quad 0 \le j \le t.$$

We observe N independent counts from p_G. Their empirical law is q. The unrestricted mixing-measure MLE is any maximizer $\widehat{G}$ of $\ell(H) = \sum_j q_j \log p_H(j)$. Write $\widehat{p} = p_{\widehat{G}}$, $L_N = \log(eN)$, and

$$\mathcal{R}_{N,t} = \inf_{\widetilde{G}} \sup_{G} \mathbb{E}_G W_1(\widetilde{G}, G), \qquad r_{N,t} = \min\left\{1, \frac{1}{t} + \frac{1}{\sqrt{tL_N}} + \frac{1}{\sqrt{N}}\right\}.$$

The infimum allows randomized estimators. Wasserstein distance on the count space uses the normalized cost $|j-l|/t$ and is denoted $W_{1,t}$. Our Hellinger convention is $h(p,v)^2 = \sum_j (\sqrt{p_j} - \sqrt{v_j})^2$, also for nonnormalized nonnegative vectors v.

Theorem 39.1 (All-replication optimality of the unrestricted MLE)**.** *There are absolute constants $0 < c < C < \infty$ such that, for every pair of integers $N, t \ge 1$,*

$$c r_{N,t} \le \mathcal{R}_{N,t} \le \sup_G \mathbb{E}_G W_1(\widehat{G}, G) \le C r_{N,t}. \tag{39.1}$$

The upper bound holds for every measurable choice of an unrestricted MLE. In particular, at $t \asymp N/\log(eN)$ its risk is $O(N^{-1/2})$, and the same order holds for every larger t.

The statistical question extends the replication-range conjectures of Vinayak, Kong, Valiant and Kakade [30] and the discussion in Section 6.2 of Musco, Musco, Rosenblatt and Singh [31]. The preceding near-linear result already answered the fixed-polynomial-range question in that discussion. The estimation theorem is reproduced from the preceding compilation; its consequence for the full testing curve is derived below. It is not counted as a new estimation result here.

39.1 An elementary finite-envelope bound for predictive likelihood

The following entropy argument is stated with its details because it must apply even when the true distribution has zero cells. Neither an interior mixing density nor a regular parametric likelihood is assumed.

Lemma 39.2 (Small covers of the binomial moment body)**.** *For $0 < \beta < 1$ the set $\{p_G : G \in \mathcal{P}([0,1])\}$ has a finite ℓ_1 cover of radius β, with centers in that same set, whose cardinality M satisfies*

$$\log M \le m \log \frac{C(m+1)}{\beta}, \qquad m = \min\left\{t, \left\lceil \sqrt{2t \log(16t/\beta)} \right\rceil\right\}. \tag{39.2}$$

Proof. View $K_x = (B_0^t(x), \dots, B_t^t(x))$ as a vector-valued polynomial. Its shifted Chebyshev expansion is

$$K_x = \sum_{k=0}^{t} v_k T_k(2x-1).$$

We claim $\|v_k\|_1 \le 2\exp\{-k^2/(2t)\}$ for $k \ge 1$. Indeed, the even periodic function $F(z) = K_{(1+\cos z)/2}$ has Fourier coefficient $v_k/2$. For $z = \theta + iu$, the ellipse identity gives

$$\left|\frac{1+\cos z}{2}\right| + \left|\frac{1-\cos z}{2}\right| = \cosh u.$$

The binomial formula therefore yields $\|F(\theta+iu)\|_1 = (\cosh u)^t \le e^{tu^2/2}$. Shift the Fourier contour in the appropriate direction and take $u = k/t$. This proves the claim. In particular, truncating after degree $m < t$ has vector ℓ_1 error at most $2te^{-(m+1)^2/(2t)} \le \beta/8$, uniformly in x. For $m = t$ the error is zero.

Each mixing law contributes its first m shifted Chebyshev moments in $[-1,1]^m$. Partition that cube into cells of coordinate diameter at most $\beta/[8(m+1)]$, and choose one mixing-law representative from every nonempty cell. If G, H lie in one cell, their truncated transforms differ in ℓ_1 norm by at most $2m\beta/[8(m+1)] \le \beta/4$. The two tails contribute at most $\beta/4$. Thus their predictive laws have distance at most β. The number of cells is at most $[C(m+1)/\beta]^m$, as asserted. □

Lemma 39.3 (Likelihood envelopes, including zero cells)**.** *Let $\varepsilon \ge 0$ and $\eta > 0$. Let p be the true law on a finite alphabet with d cells, and suppose a model admits an ℓ_1 cover of radius η/d, with M probability centers in the model. Let $\widetilde{p}$ be any fitted model law satisfying*

$$\sum_{i=1}^{N} \log \widetilde{p}(X_i) \ge \sum_{i=1}^{N} \log p(X_i) - N\varepsilon.$$

For every $x \ge 0$, simultaneously for all such fitted laws, except on an event of probability at most e^{-x},

$$h(p, \widetilde{p}) \le \sqrt{\eta} + \sqrt{\varepsilon + \eta + 2(\log M + x)/N}. \tag{39.3}$$

Consequently

$$\mathbb{E}h(p, \widetilde{p})^2 \le C\left\{\varepsilon + \eta + \frac{\log M + 1}{N}\right\}. \tag{39.4}$$

Proof. For each center p_a, form the nonnegative vector $u_a(j) = p_a(j) + \eta/d$. Every model law is bounded coordinatewise by at least one u_a, and $\sum_j u_a(j) = 1 + \eta$. If $\widetilde{p} \le u_a$, then

$$h(u_a, \widetilde{p})^2 \le \sum_j (u_a(j) - \widetilde{p}(j)) = \eta.$$

Thus $h(p, \widetilde{p}) > \sqrt{\eta} + r$ implies $h(p, u_a) > r$. Moreover the likelihood comparison implies $\prod_i u_a(X_i)/p(X_i) \ge e^{-N\varepsilon}$. The ratios need only be defined on the support of p, where all sampled observations lie almost surely. Markov's inequality with exponent $1/2$ gives

$$\begin{aligned}
\Pr\left\{\prod_i \frac{u_a(X_i)}{p(X_i)} \ge e^{-N\varepsilon}\right\} &\le e^{N\varepsilon/2}\left(\sum_j \sqrt{p(j)u_a(j)}\right)^N \\
&= e^{N\varepsilon/2}\left(1 + \frac{\eta}{2} - \frac{h(p,u_a)^2}{2}\right)^N \\
&\le \exp\{N(\varepsilon + \eta - r^2)/2\}
\end{aligned}$$

when $h(p, u_a) > r$. Taking the union over the M envelopes proves (39.3). Integrating that tail, and using $(a+b)^2 \le 2a^2 + 2b^2$, proves (39.4). □

Corollary 39.4 (A sufficient predictive Hellinger rate)**.** *For $1 \le t \le N$, every exact MLE satisfies*

$$\mathbb{E}_G h(p_G, \widehat{p}) \le \frac{C t^{1/4} L_N^{3/4}}{\sqrt{N}}. \tag{39.5}$$

More generally, if $\ell(\widetilde{G}) \ge \ell^* - \varepsilon$, the right side can be increased by $C\sqrt{\varepsilon}$.

Proof. Set $\eta = (eN)^{-2}$ and $\beta = \eta/(t+1)$ in the preceding lemmas. For $t \le N$, (39.2) gives $m \le C\sqrt{tL_N}$ and $\log M \le C\sqrt{t}\, L_N^{3/2}$. The MLE likelihood is at least that of the true law. The approximate likelihood condition loses at most $N\varepsilon$. Apply (39.4) and Jensen's inequality. □

The logarithmic factors in (39.5) need not be sharp. The argument below only needs the strict improvement from effective dimension t to a quantity of order $\sqrt{t}$ up to logarithms.

Lemma 39.5 (Likelihood transport, recalled)**.** *For every binomial-mixture MLE,*

$$W_{1,t}(q, \widehat{p})^2 \le \frac{D(q\|\widehat{p})}{2t} \le \frac{D(q\|p_G)}{2t}. \tag{39.6}$$

Consequently

$$\mathbb{E}_G W_{1,t}(\widehat{p}, p_G) \le C/\sqrt{N}. \tag{39.7}$$

Proof. In score sums, terms with $q_j = 0$ are set to zero; every observed cell has positive fitted probability. The likelihood score condition gives $S(x) = \sum_j q_j K_x(j)/\widehat{p}(j) \le 1$, and its integral against $\widehat{G}$ is one. Thus $S = 1$ almost surely under $\widehat{G}$. The posterior vectors $R_x(j) = q_j K_x(j)/\widehat{p}(j)$ are therefore probability laws, with mixture q, and

$$\int D(R_x\|K_x)\, d\widehat{G}(x) = D(q\|\widehat{p}).$$

Every normalized-count binomial kernel satisfies $W_{1,t}(R, K_x)^2 \le D(R\|K_x)/(2t)$: a 1-Lipschitz test of the normalized sum of t Bernoulli variables has centered logarithmic moment generating function at most $\lambda^2/(8t)$ by bounded differences; entropy duality and optimization over λ give the inequality. Mix the couplings, use Cauchy–Schwarz, and then use likelihood optimality to prove (39.6). Also $\mathbb{E}D(q\|p_G) \le t/N$, by $D(q\|p_G) \le \sum_{j:p_G(j)>0}(q_j - p_G(j))^2/p_G(j)$. Finally $\mathbb{E}W_{1,t}(q, p_G) \le 1/(2\sqrt{N})$ by integrating the standard deviation of the empirical distribution function. The triangle inequality proves (39.7). □

Upper bound in Theorem 39.1. It suffices first to treat $t \le N$. Bounded N or bounded t are handled by the loss bound one. Fix a sufficiently small absolute $a > 0$, for example $a = 1/64$, and take

$$k = \left\lfloor \min\{t/2, \sqrt{atL_N}\} \right\rfloor, \qquad s = \lfloor \sqrt{t}/4 \rfloor.$$

For sufficiently large N and t, these satisfy $1 \le s \le k \le t/2$, $s \asymp \sqrt{t}$, and

$$1/k \le C\{1/t + (tL_N)^{-1/2}\}, \qquad e^{4s^2/t} \le C.$$

Use Proposition 38.6, Lemma 39.5, and Corollary 39.4. This gives

$$\mathbb{E}_G W_1(\widehat{G}, G) \le \frac{C}{k} + \frac{C}{\sqrt{N}} + \frac{C e^{2k^2/t} L_N^{3/4}}{\sqrt{N}\, t^{1/4}}. \tag{39.8}$$

The final term, multiplied by $\sqrt{tL_N}$, is at most

$$C(eN)^{2a} N^{-1/2} t^{1/4} L_N^{5/4} \le C(eN)^{2a} N^{-1/4} L_N^{5/4} \le C.$$

The last bound is uniform because $2a < 1/4$. This proves the desired upper bound when $t \le N$.

For $t \ge N$, couple every mixing parameter x to its normalized binomial count. The expected absolute difference is at most $\sqrt{x(1-x)/t} \le 1/(2\sqrt{t})$. Hence, after identifying count laws with measures on $[0,1]$,

$$W_1(\widehat{G}, G) \le 1/\sqrt{t} + W_{1,t}(\widehat{p}, p_G).$$

Take expectations and apply (39.7). This is $O(N^{-1/2})$ for $t \ge N$, and completes the proof of the upper bound. □

39.2 Matching lower bounds

Lower bound in Theorem 39.1. The $N^{-1/2}$ term follows from mixing laws supported on $\{0,1\}$ with weights $1/2 \pm c/\sqrt{N}$. Each count then reveals exactly one endpoint label, regardless of t. Their N-sample total variation stays bounded away from one for sufficiently small absolute c.

For the other terms let $\pi(dx) = dx/[\pi\sqrt{x(1-x)}]$ and $h_k(x) = T_k(2x-1)$. Define $G_k^{\pm} = (1 \pm h_k/2)\pi$. Integration of their signed cumulative function gives

$$W_1(G_k^+, G_k^-) = \frac{1}{2\pi k}\int_0^\pi |\sin(k\theta)| \sin\theta \, d\theta \asymp 1/k.$$

If $k = t+1$, the count laws are identical, proving the $1/t$ lower bound. For $k \le t$, draw $X \sim \pi$, $J \mid X \sim \mathrm{Binomial}(t, X)$ and put $Qh(j) = \mathbb{E}[h(X) \mid J = j]$. The posterior is $\mathrm{Beta}(j+1/2, t-j+1/2)$. The self-adjoint operator Q^*Q preserves polynomial degrees and has leading degree-k coefficient

$$\lambda_{t,k} = \prod_{i=0}^{k-1} \frac{t-i}{t+1+i}.$$

Orthogonality of the Chebyshev polynomials therefore gives $Q^*Qh_k = \lambda_{t,k}h_k$ and $\mathbb{E}[Qh_k(J)^2] = \lambda_{t,k}/2$. This is the classical beta–binomial diagonalization [32, Section 4.1]. The count laws are $p_\pi(1 \pm Qh_k/2)$, so

$$\chi^2(p_{G_k^+} \| p_{G_k^-}) \le 2\mathbb{E}[Qh_k(J)^2] = \lambda_{t,k} \le e^{-k^2/(2t)}.$$

Take $k = \lceil\sqrt{8t\log(8N)}\rceil$ if this is at most t. The chi-squared divergence of the N-fold products is $(1+\chi^2)^N - 1 \le e^{N(8N)^{-4}} - 1$, so their total variation is bounded by an absolute constant smaller than one. The two-point testing bound yields risk at least $c/k \ge c'/\sqrt{tL_N}$. If this choice has $k > t$, then $t \le CL_N$ apart from bounded t, and $1/\sqrt{tL_N} \le C'/t$; the exact moment-matching lower bound already suffices. The maximum of the three lower bounds is comparable to their sum clipped at one, proving (39.1). □

39.3 The full worst-null testing curve

Fix $\alpha, \beta \in (0,1)$ with $\alpha + \beta < 1$. A null-dependent test ψ_{G_0} has level α if $\mathbb{E}_{G_0}\psi_{G_0} \le \alpha$. Define $\varepsilon^*_{N,t}(\alpha,\beta)$ as the infimum of the separations ε for which such tests exist for every G_0, with $\mathbb{E}_G(1-\psi_{G_0}) \le \beta$ whenever $W_1(G, G_0) \ge \varepsilon$. Thus the rate is uniform in the null law; it need not describe the difficulty at each particular fixed null.

Theorem 39.6 (Testing across the replication range)**.** *For all integers $N, t \ge 1$,*

$$\varepsilon^*_{N,t}(\alpha,\beta) \asymp_{\alpha,\beta} \min\left\{1, \frac{1}{t} + \frac{1}{\sqrt{t\log(eN)}} + \frac{1}{\sqrt{N}}\right\}.$$

An upper bound is attained by the plug-in test based on any unrestricted NPMLE. In particular the parametric plateau begins at $t \asymp N/\log(eN)$, with no logarithmic loss in the transition.

Proof. Let $r = r_{N,t}$ and let C be the upper-bound constant in Theorem 39.1. Reject when $W_1(\widehat{G}, G_0) > Cr/\alpha$. Markov's inequality gives level α. If $W_1(G, G_0) > Cr(\alpha^{-1} + \beta^{-1})$, the triangle inequality bounds acceptance probability by $\Pr_G\{W_1(\widehat{G}, G) > Cr/\beta\} \le \beta$. Using a slightly larger constant handles non-strict separation.

For the lower bound use the same three pairs constructed in the lower proof of Theorem 39.1, taking one member as the null and the other as the alternative. The degree-$t{+}1$ pair gives identical sample laws and separation at least c/t. For the middle term choose $k = \lceil \sqrt{C_{\alpha,\beta} t \log(eN)} \rceil$ when $k \le t$; the displayed chi-square estimate makes the product-law total variation smaller than $1 - \alpha - \beta$ by taking $C_{\alpha,\beta}$ large. If $k > t$, the $1/t$ pair already gives the required order. Finally choose the endpoint-weight perturbation of size $c_{\alpha,\beta}/\sqrt{N}$ with $c_{\alpha,\beta}$ small, giving the $N^{-1/2}$ obstruction. For any two sample laws P_0, P_1, the sum of type I and type II errors is at least $1 - \|P_0 - P_1\|_{\mathrm{TV}}$. These pairs therefore prove the maximum of the three lower bounds. Their maximum is comparable to their sum clipped at one, as required. □

Kania, Wasserman and Balakrishnan [29, Section 3, Theorem 3] give the corresponding worst-null lower curve and discuss the gap in the intermediate replication range. Theorem 39.6 combines the estimation bound of Theorem 39.1 with an elementary Markov argument to obtain the stated testing curve.

40 Certified approximate likelihood fits

This section extends the rates in Theorems 38.1 and 39.1 to approximate likelihood maximizers. The entropy chain rule, posterior reweighting, and likelihood monotonicity are classical EM ingredients [46]; the contribution here is their quantitative combination with the two-scale inverse inequalities and the resulting uniform computational tolerance. The result concerns the known compact parameter interval. It does not assert analogous stability on an unbounded parameter space.

40.1 An exact posterior-defect identity

Let q be a finitely supported probability law, let H be a mixing law with $p_H > 0$ on the support of q, and put

$$\ell(H) = \sum_x q(x) \log p_H(x), \qquad S_H(\theta) = \sum_x \frac{q(x)K_\theta(x)}{p_H(x)}, \qquad dH^+(\theta) = S_H(\theta)\, dH(\theta).$$

The supremum ℓ^* below is over the same admissible mixing class, which is assumed to contain H^+ and all point masses. Assume all component first moments are finite and $D_K := \sup_{\theta,\vartheta} W_1(K_\theta, K_\vartheta) < \infty$.

Theorem 40.1 (Transport with an explicit likelihood defect)**.** *Suppose the component transport bound of Theorem 37.1 holds with concave nondecreasing function ψ. Put $D_0 = D(q\|p_H)$ and $d_0 = D(H^+\|H)$. Then*

$$d_0 \le \ell(H^+) - \ell(H) \le \ell^* - \ell(H), \tag{40.1}$$

$$W_1(q, p_H) \le \psi(D_0 - d_0) + D_K\sqrt{d_0/2}. \tag{40.2}$$

In particular, if $\ell^* - \ell(H) \le \delta$, then for every comparator G in the model,

$$W_1(q, p_H) \le \psi(D(q\|p_G) + \delta) + D_K\sqrt{\delta/2}. \tag{40.3}$$

The conclusions also hold after a fixed Markov channel A, with q, K_θ, p_H replaced in the transport distances by Aq, AK_θ, Ap_H, provided the transport hypothesis and component diameter refer to AK_θ. The entropy on the right remains that of the original data.

Proof. Since $\int S_H\, dH = 1$, H^+ is a probability law. On $S_H > 0$, set

$$R_\theta(x) = \frac{q(x)K_\theta(x)}{p_H(x)S_H(\theta)}.$$

These are probability laws, their H^+ mixture is q, and the entropy chain rule gives the exact identity

$$D(q\|p_H) = D(H^+\|H) + \int D(R_\theta\|K_\theta)\, dH^+(\theta). \tag{40.4}$$

For every observed x, the posterior $\Pi_x(d\theta) = K_\theta(x)\, dH(\theta)/p_H(x)$ satisfies $p_{H^+}(x)/p_H(x) = \int S_H\, d\Pi_x$. Jensen's inequality gives

$$\ell(H^+) - \ell(H) \ge \sum_x q(x) \int \log S_H\, d\Pi_x = \int S_H \log S_H\, dH = d_0.$$

Zero-score points carry no mass in the posterior joint law; the formulas follow by the convention $0 \log 0 = 0$ or by truncation. This is the usual EM ascent argument, specialized to mixing measures. Mix the component couplings and use concavity of ψ in (40.4) to obtain $W_1(q, p_{H^+}) \le \psi(D_0 - d_0)$.

Couple the common mass of H, H^+ identically. The remaining component mass costs at most $D_K \operatorname{TV}(H, H^+)$, which is at most $D_K\sqrt{d_0/2}$ by Pinsker's inequality. This proves (40.2). Finally $D_0 \le D(q\|p_G) + \delta$ by likelihood comparison. For a channel, apply entropy data processing to each R_θ before mixing; its mixture remains Aq. The posterior reweighting and its likelihood bound are still computed from the original observations. □

Proposition 40.2 (A global stopping certificate)**.** *For every such fit,*

$$\ell^* - \ell(H) \le \log \sup_\theta S_H(\theta). \tag{40.5}$$

Thus $\sup_\theta S_H(\theta) \le \exp(A/N)$ *certifies a total log-likelihood deficit of at most* A *for* N *observations.*

Proof. For every G, Jensen's inequality yields

$$\ell(G) - \ell(H) \le \log \sum_x q(x) \frac{p_G(x)}{p_H(x)} = \log \int S_H \, dG \le \log \sup_\theta S_H(\theta).$$

Take the supremum over G. The integral of S_H under H is one, so its supremum is at least one. □

For binomial data S_H is a polynomial of degree at most t on $[0,1]$; its maximum is among the endpoints and zeros of its derivative. For Poisson data it is $e^{-\lambda}P(\lambda)$, where P is a polynomial of degree at most the largest observed count. Its maximum on $[0, B]$ is among the endpoints and real zeros of $P' - P$. These are finite candidate sets for the mathematical supremum certificate. A numerical implementation must bound the candidate values with controlled error. No polynomial-time complexity or certification from a small change between consecutive iterations is asserted.

40.2 The same estimation and testing curves at finite precision

Theorem 40.3 (A bounded total likelihood error costs only a constant)**.** *Let* $A \ge 0$. *In the binomial model, let* $\widetilde{G}$ *be any measurable fit on* $[0,1]$ *satisfying, for every data set,*

$$\sum_{i=1}^N \log p_{\widetilde{G}}(X_i) \ge \sup_H \sum_{i=1}^N \log p_H(X_i) - A.$$

Then, for every $N, t \ge 1$,

$$\sup_G \mathbb{E}_G W_1(\widetilde{G}, G) \le C(1 + \sqrt{A})\, r_{N,t}. \tag{40.6}$$

In the Poisson model with known $[0, B]$*, the same total likelihood condition over that model implies, for every* $N \ge 1$ *and* $B > 0$,

$$\sup_G \mathbb{E}_G W_1(\widetilde{G}, G) \le C(1 + \sqrt{A})\, r(N, B). \tag{40.7}$$

Here $r_{N,t}$ *and* $r(N,B)$ *are exactly the complete curves defined before Theorems 39.1 and 38.1; C is absolute. In particular, a fixed total log-likelihood tolerance preserves minimax optimality throughout both parameter planes.*

Proof. Write $\delta = A/N$. For normalized binomial counts, $\psi(u) = \sqrt{u/(2t)}$ and $D_K = 1$. Equation (40.3), the bound $\mathbb{E}D(q\|p_G) \le t/N$, and the empirical CDF bound give

$$\mathbb{E}W_{1,t}(p_{\widetilde{G}}, p_G) \le \sqrt{\frac{t/N + \delta}{2t}} + \sqrt{\delta/2} + \frac{1}{2\sqrt{N}} \le \frac{C(1+\sqrt{A})}{\sqrt{N}}. \tag{40.8}$$

For $t \le N$, Corollary 39.4, which explicitly allows approximate fits, gives

$$\mathbb{E}h(p_{\widetilde{G}}, p_G) \le C\{t^{1/4}L_N^{3/4}/\sqrt{N} + \sqrt{\delta}\} \le C(1+\sqrt{A})t^{1/4}L_N^{3/4}/\sqrt{N}.$$

Use the same s, k as in the proof of Theorem 39.1. Every term in (39.8) increases by at most $C(1+\sqrt{A})$, while the deterministic approximation term is unchanged. This proves (40.6) for $t \le N$. For $t \ge N$, use $W_1(\widetilde{G}, G) \le t^{-1/2} + W_{1,t}(p_{\widetilde{G}}, p_G)$ and (40.8).

For Poisson data, take $m = \lceil \max(B, L_N) \rceil$ and use the occupancy channel of (38.2). The transformed components are binomial, with $\psi(u) = \sqrt{u/(2m)}$ and diameter at most one. The channel version of Theorem 40.1 gives

$$\mathbb{E}W_{1,m}(A_m p_{\widetilde{G}}, A_m p_G) \le \sqrt{\frac{C(B+L_N)/N + \delta}{2m}} + \sqrt{\delta/2} + \frac{1}{2\sqrt{N}} \le C(1+\sqrt{A})/\sqrt{N}.$$

The finite-envelope proof of Lemma 38.4 also applies: the likelihood ratio threshold becomes $e^{-N\eta - A}$, so the Hellinger upper bound acquires the additional term $C\sqrt{\delta}$. For $b_0 L_N \le B \le N$ this is absorbed into $C(1+\sqrt{A})B^{1/4}L_N^{3/4}/\sqrt{N}$. Substitution into the same two-scale bound proves (38.10) with that factor. For $B > N$, the coupling and occupancy argument following (38.10) uses only the just-proved predictive bound.

Finally, when $B \le b_0 L_N$, likelihood comparison gives $D(q\|p_{\widetilde{G}}) \le D(q\|p_G) + \delta$. The Hellinger triangle inequality therefore implies

$$\mathbb{E}\|p_{\widetilde{G}} - p_G\|_1 \le C\sqrt{L_N/N} + 2\sqrt{\delta} \le C(1+\sqrt{A})\sqrt{L_N/N}.$$

The small-support inverse inequality (38.13) with the same k as before proves (40.7), including the diameter clipping for arbitrarily small B. All intermediate parameter regions and bounded sample sizes were already covered by those same choices and the diameter bound. □

For fixed testing levels α, β, the plug-in binomial test based on $\widetilde{G}$ therefore has the full worst-null separation curve $r_{N,t}$ for every fixed A. Indeed use the threshold $C(1+\sqrt{A})r_{N,t}/\alpha$ and the same Markov/triangle proof as in the exact-fit testing theorem. The existing minimax testing lower bound is unchanged. This closes the finite-computation version as well as the exact-optimizer version, without claiming EM convergence at a particular iteration count.

41 Arbitrary known replication profiles: exact reduction and lower certificates

Let independent units have $P_i \sim G$ and $X_i \mid P_i \sim \mathrm{Binomial}(t_i, P_i)$, where the positive integer depths t_i are known. Write $N_m = \#\{i : t_i \ge m\}$. The following result concerns an explicitly constructed estimator, not the unmodified heterogeneous joint likelihood.

Theorem 41.1 (Thinning upper bound and spectral lower bound)**.** *Let $\mathcal{R}(t_1, \dots, t_N)$ be the minimax expected W_1 risk over all mixing laws on $[0,1]$. Then*

$$\mathcal{R}(t_1, \dots, t_N) \le C \min_{m:\, N_m > 0} r_{N_m, m}. \tag{41.1}$$

Set

$$\lambda_{t,k} = \begin{cases} \displaystyle\prod_{j=0}^{k-1} \frac{t-j}{t+1+j}, & 1 \le k \le t, \\ 0, & k > t, \end{cases} \qquad \Lambda_k = \sum_{i=1}^{N} \lambda_{t_i,k}.$$

For an absolute $c > 0$,

$$\mathcal{R}(t_1, \dots, t_N) \geq c \max\left\{ N^{-1/2}, \sup_{k \geq 1} \frac{1}{k\sqrt{1+\Lambda_k}} \right\}. \tag{41.2}$$

These are design-specific upper and lower certificates. Equality up to an absolute factor for every arbitrary profile is not asserted.

Proof. Fix m and discard units with $t_i < m$. For each retained unit, independently draw

$$Y_i \mid X_i \sim \text{Hypergeometric}(t_i, X_i, m).$$

Viewing X_i as the total of t_i independent Bernoulli trials shows that $Y_i \mid P_i \sim \text{Binomial}(m, P_i)$. Thus the retained transformed observations are exactly N_m independent observations from the common-depth mixture. Theorem 39.1 proves (41.1). The minimization is deterministic; it can be restricted to the distinct observed depths, since N_m is constant between successive depths and $r_{N_m,m}$ decreases in m.

For the lower bound, use the arcsine law π and $h_k(x) = T_k(2x-1)$. As in the proof of Theorem 39.1, the beta–binomial posterior operator has squared norm $\mathbb{E}[\mathbb{E}(h_k(P_i) \mid X_i)^2] = \lambda_{t_i,k}/2$; for $k > t_i$ it is zero by polynomial orthogonality. Take $G_\pm = (1 \pm a h_k)\pi$ with $a = 1/(4\sqrt{1+\Lambda_k})$. These are probability laws. The individual predictive chi-squared divergences satisfy

$$\chi^2(p_{G_+,t_i} \| p_{G_-,t_i}) \leq 4a^2 \lambda_{t_i,k}.$$

Consequently the product chi-squared divergence is at most $\exp(4a^2\Lambda_k) - 1 \leq e^{1/4} - 1$, and product total variation is bounded away from one. Direct integration gives

$$W_1(G_+, G_-) = \frac{a}{\pi k} \int_0^\pi |\sin(k\theta)| \sin\theta \, d\theta \geq \frac{a}{\pi k}.$$

For the last inequality, use $|\sin(k\theta)| \geq \sin^2(k\theta)$ and $\int_0^\pi \sin^2(k\theta)\sin\theta\, d\theta = 1 + (4k^2-1)^{-1} \geq 1$. Le Cam's two-point inequality proves the second lower certificate. Endpoint mixing laws with weights $1/2 \pm c/\sqrt{N}$ give the $N^{-1/2}$ certificate, since every positive depth reveals the endpoint label exactly. □

Corollary 41.2 (Any number of bounded-depth observations)**.** *Fix $d \geq 1$. Suppose $n \geq 1$ units have a common depth $T > d$ and an arbitrary additional number of units have arbitrary depths in $\{1, \dots, d\}$. There exist $c_d > 0$ and absolute C such that*

$$c_d r_{n,T} \leq \mathcal{R}(t_1, \dots, t_N) \leq C r_{n,T}.$$

Thus an arbitrarily large bounded-depth sample does not change the worst-case Wasserstein rate supplied by the deep sample, up to constants depending on the fixed depth bound d. An estimator ignoring the shallow sample attains this rate. No claim about the heterogeneous joint NPMLE is needed.

Proof. The upper bound discards the shallow sample. In the lower bound, every $k > d$ makes all shallow contributions to Λ_k vanish. Choose $k = T+1$ to obtain c/T, and $k = d+1$ to obtain $c/((d+1)\sqrt{n})$ since $\Lambda_k \leq n$. For the remaining term, let $k = \max\{d+1, \lceil\sqrt{8T\log(8n)}\rceil\}$. If $k \leq T$, the estimate $\lambda_{T,k} \leq e^{-k^2/(2T)}$ gives $\Lambda_k \leq 1$, so (41.2) is at least $c_d/\sqrt{T\log(en)}$. If $k > T$, then $T \leq C_d \log(en)$, and $1/\sqrt{T\log(en)} \leq C_d/T$. The maximum of these three lower bounds is comparable to $r_{n,T}$. □

Proposition 41.3 (Known heterogeneous Poisson exposures)**.** *Suppose $\Theta_i \sim G$ on the known interval $[0, M]$ and, conditionally, $X_i \sim \mathrm{Poi}(u_i\Theta_i)$, independently, where $u_i > 0$ are known. For $u > 0$ let $N_u = \#\{i : u_i \geq u\}$. There is a randomized estimator with risk at most*

$$C \inf_{u:\, N_u>0} \frac{r(N_u, uM)}{u}.$$

Any fixed choice of u attains its corresponding upper bound.

Proof. Retain the units with $u_i \geq u$ and draw $Y_i \mid X_i \sim \mathrm{Binomial}(X_i, u/u_i)$ independently. Poisson thinning gives $Y_i \mid \Theta_i \sim \mathrm{Poi}(u\Theta_i)$. Fit the common-exposure model on $[0, uM]$ and divide the estimated mixing parameter by u. Theorem 38.1 gives the bound. The design is fixed and known, so choose a threshold whose displayed bound is within a factor of two of the infimum, absorbing that factor into C. □

41.1 Why separate likelihood strata cannot be normalized silently

For stratum weights $w_j > 0$ summing to one, empirical laws q_j, and an exact joint fit H, write $S_j(\theta) = \sum_x q_j(x)K_{j,\theta}(x)/p_{H,j}(x)$ and $dH_j^+ = S_j\, dH$. The joint score gives only

$$\sum_j w_j H_j^+ = H.$$

Applying (40.4) separately yields the exact decomposition

$$\sum_j w_j D(q_j \| p_{H,j}) = \underbrace{\sum_j w_j D(H_j^+ \| H)}_{I(J;\Theta)} + \sum_j w_j \int D(R_{j,\theta} \| K_{j,\theta})\, dH_j^+(\theta).$$

The first term is the mutual information of the stratum label and the reweighted parameter under the joint law $w_j H_j^+$. It is the precise normalization defect. For example, take two equally weighted Bernoulli strata, $q_1 = \delta_0$, $q_2 = \delta_1$, and $H = (\delta_0 + \delta_1)/2$. This is a joint likelihood maximizer, but $H_1^+ = \delta_0$, $H_2^+ = \delta_1$ and the defect is $\log 2$. Each stratum's within-component entropy is zero, whereas $W_1(q_j, p_{H,j}) = 1/2$. Therefore the common-depth transfer theorem cannot be applied separately to the unaltered strata. The thinning results above avoid this defect through an exact experiment reduction.

42 Data-independent finite grids with certified full likelihood accuracy

The preceding approximate-likelihood theorem turns a bounded total log-likelihood error into the complete binomial and known-support Poisson estimation rates. We now give explicit, data-independent finite parameter grids for which ordinary finite-dimensional likelihood maximization has exactly this accuracy. The mechanism is a quadratic loss at an interior score maximum, expressed by the relative entropy between two component laws. The endpoint nodes are essential to the statement.

Finite-grid nonparametric likelihood estimation and quantitative discretization guarantees have an established literature. Dicker and Zhao [47] analyze a convex approximation to the NPMLE in empirical Bayes classification; Soloff, Guntuboyina and Sen [48, Section 2.3] give quantitative fixed-data grid optimality-gap bounds in Gaussian mixture estimation. The purpose here is a self-contained stationary-score derivation of the three explicit grids below and their combination with this manuscript's mixing-distribution risk bounds. The use of a finite grid itself is not claimed as a new method.

42.1 A stationary-score inequality for exponential families

Let K_θ have mass function or density k_θ relative to a fixed dominating measure. The parameter interval is $I = [a, b]$ and the functions $\theta \mapsto k_\theta(x_i)$ are continuous for every observation x_i. In the interior, suppose this is a regular one-parameter exponential family in a smooth coordinate $\eta(\theta)$:

$$k_\theta(x) = h(x) \exp\{\eta(\theta)T(x) - B(\eta(\theta))\}, \qquad \eta'(\theta) \neq 0.$$

The endpoint distributions may be continuous limiting distributions of this family. For a mixing law H on I, define

$$p_H(x) = \int_I k_\theta(x)\, dH(\theta), \qquad \mathcal{L}_N(H) = \sum_{i=1}^{N} \log p_H(x_i), \qquad S_H(\theta) = \frac{1}{N} \sum_{i=1}^{N} \frac{k_\theta(x_i)}{p_H(x_i)}.$$

Throughout, $p_H(x_i) > 0$ and $k_\theta(x_i)$ is finite. Repeated observations are allowed. All three models considered below satisfy these assumptions, and $\int S_H\, dH = 1$.

Lemma 42.1 (Stationary scores lose at most component relative entropy)**.** *If $\theta_* \in (a, b)$ is an interior stationary point of S_H with $S_H(\theta_*) > 0$, then for every $v \in I$ whose component belongs to the same regular exponential family (including a regular endpoint),*

$$\frac{S_H(v)}{S_H(\theta_*)} \geq \exp\{-D(K_{\theta_*} \| K_v)\}. \tag{42.1}$$

Moreover, for any admissible H,

$$\sup_G \mathcal{L}_N(G) - \mathcal{L}_N(H) \leq N \log \sup_{\theta \in I} S_H(\theta). \tag{42.2}$$

Proof. Set $\eta_* = \eta(\theta_*)$ and

$$w_i = \frac{k_{\theta_*}(x_i)}{N p_H(x_i) S_H(\theta_*)}.$$

These weights are nonnegative and sum to one. Since $\eta'(\theta_*) \neq 0$, differentiating the score at its stationary point gives $\sum_i w_i T(x_i) = B'(\eta_*)$. Thus Jensen's inequality yields

$$\begin{aligned} \frac{S_H(v)}{S_H(\theta_*)} &= \sum_i w_i \exp\{(\eta(v) - \eta_*)T(x_i) - B(\eta(v)) + B(\eta_*)\} \\ &\geq \exp\{(\eta(v) - \eta_*)B'(\eta_*) - B(\eta(v)) + B(\eta_*)\} \\ &= \exp\{-D(K_{\theta_*} \| K_v)\}. \end{aligned}$$

Here the last equality is obtained by integrating the exponential-family log-density ratio under K_{θ_*}. For any mixing law G, a second application of Jensen gives

$$\frac{\mathcal{L}_N(G) - \mathcal{L}_N(H)}{N} \leq \log \left\{ \frac{1}{N} \sum_i \frac{p_G(x_i)}{p_H(x_i)} \right\} = \log \int S_H\, dG \leq \log \sup_I S_H.$$

If p_G vanishes at an observation, its likelihood is minus infinity and the same inequality is immediate. Taking the supremum proves (42.2). □

Proposition 42.2 (Finite grids certify the continuous likelihood)**.** *Let $V \subset I$ be finite and contain both endpoints. Suppose every interior θ has a node $v \in V$ belonging to the same regular exponential family and satisfying $D(K_\theta \| K_v) \leq \rho$. Then every admissible fit H satisfies*

$$\sup_G \mathcal{L}_N(G) - \mathcal{L}_N(H) \leq N \log \max_{v \in V} S_H(v) + N\rho. \tag{42.3}$$

It is enough that the stated relative-entropy comparison hold for every interior global score maximizer, instead of every interior parameter.

Suppose some mixture supported on V has positive likelihood. Every exact likelihood maximizer H_V over mixtures on V exists, has positive fitted probability at each observation, and satisfies

$$\sup_G \mathcal{L}_N(G) - \mathcal{L}_N(H_V) \le N\rho. \tag{42.4}$$

More generally, for $A_{\mathrm{grid}} \ge 0$, either of the following conditions suffices for a bound $A_{\mathrm{grid}} + N\rho$ on the continuous-model likelihood deficit:

1. *the fit satisfies the finite score certificate $\max_{v\in V} S_H(v) \le \exp(A_{\mathrm{grid}}/N)$;*
2. *the fit is supported on V and its total log likelihood is within A_{grid} of the finite-grid optimum.*

Proof. Continuity and compactness give a global score maximizer θ_*. Its score is positive since $\int S_H\, dH = 1$. If it is an endpoint, it belongs to V. Otherwise its score derivative is zero, so Lemma 42.1 gives $\sup_I S_H \le e^{\rho} \max_V S_H$. This proves (42.3).

The finite-grid parameter space is a compact probability simplex. The likelihood product is continuous and attains a positive maximum by the assumed positive candidate. Hence its maximizers are positive at all observations. Differentiating the normalized log likelihood towards the point mass at any grid node gives $S_{H_V}(v) - 1 \le 0$. Since the score has H_V-average one, $\max_V S_{H_V} = 1$. Substitute into (42.3) to obtain (42.4). Condition (1) follows directly from that same inequality. Under (2), add the finite-grid optimization deficit to (42.4). The two certificates are sufficient conditions; we do not infer a score bound merely from an objective-value gap. □

42.2 Binomial angular grids

Theorem 42.3 (A deterministic grid for every binomial trial count)**.** *For integers $N, t, m \ge 1$, use the grid*

$$V_m^{\mathrm{bin}} = \left\{ \sin^2\left(\frac{j\pi}{4m}\right) : j = 0, 1, \dots, 2m \right\}.$$

Every exact maximum-likelihood fit H_m on this grid satisfies

$$\sup_{G\in\mathcal{P}([0,1])} \mathcal{L}_N(G) - \mathcal{L}_N(H_m) \le \frac{\pi^2 N t}{8m^2}. \tag{42.5}$$

A finite-grid objective or score certificate as in Proposition 42.2 increases the right side by at most A_{grid}. The grid is fixed before observing the data and contains $0, 1/2, 1$.

Proof. We first prove the needed one-sided Bernoulli divergence bound. For $0 \le a \le b \le \pi/4$, put $p = \sin^2 a$ and $q = \sin^2 b$. Differentiation at $a < v \le b$ gives

$$\frac{d}{dv} D(\mathrm{Ber}(p) \| \mathrm{Ber}(\sin^2 v)) = \frac{4\sin(v+a)\sin(v-a)}{\sin(2v)} \le 4(v-a).$$

Indeed $0 \le v + a \le 2v \le \pi/2$ implies $\sin(v+a) \le \sin(2v)$, while $\sin(v-a) \le v - a$. Integration from a to b proves

$$D(\mathrm{Ber}(p) \| \mathrm{Ber}(q)) \le 2(b-a)^2 \quad (0 \le p \le q \le 1/2). \tag{42.6}$$

The value at $a = 0$ follows by the same integral or continuity. Reflection $p \mapsto 1-p$, $q \mapsto 1-q$ gives the corresponding bound when $1/2 \le q \le p \le 1$.

Let $p_* \in (0,1)$ maximize the score. If $p_* \le 1/2$, round $\arcsin\sqrt{p_*}$ up to the next angular grid node; if $p_* \ge 1/2$, round it down. Thus rounding is always towards $1/2$, and the chosen node lies

in $(0,1)$ with angular displacement at most $\pi/(4m)$. The binomial log-density ratio, integrated using $\mathbb{E}X = tp_*$, gives

$$D(\mathrm{Bin}(t,p_*)\|\,\mathrm{Bin}(t,q)) = tD(\mathrm{Ber}(p_*)\|\,\mathrm{Ber}(q)) \le \frac{\pi^2 t}{8m^2}.$$

Interior stationarity is legitimate because the natural parameter is $\log(p/(1-p))$. Endpoint maxima are already grid nodes. The grid node $1/2$ assigns positive probability to every count $0,\dots,t$, so finite-grid likelihood maximizers exist and are positive on the observed counts. Proposition 42.2 proves the claim. In particular, all-zero or all-t data cause no exceptional case: the relevant endpoint score maximum is captured exactly. □

42.3 Poisson square-root grids and Gaussian uniform grids

Theorem 42.4 (Poisson square-root discretization)**.** *For $M > 0$ and integers $N, m \ge 1$, take*

$$V_m^{\mathrm{poi}} = \{M(j/m)^2 : j = 0,1,\dots,m\}.$$

Every exact finite-grid Poisson mixture MLE on the known interval $[0,M]$ satisfies

$$\sup_{G\in\mathcal{P}([0,M])} \mathcal{L}_N(G) - \mathcal{L}_N(H_m) \le \frac{2NM}{m^2}. \tag{42.7}$$

Either finite-grid certificate in Proposition 42.2 adds at most A_{grid} to this bound.

Proof. For $0 \le a \le b$,

$$D(\mathrm{Poi}(a^2)\|\,\mathrm{Poi}(b^2)) = b^2 - a^2 + a^2\log(a^2/b^2) \le 2(b-a)^2.$$

For $a > 0$, the derivative of the left side with respect to b is $2(b-a)(b+a)/b \le 4(b-a)$, so integration from a proves the inequality. At $a = 0$ it follows directly, with $0\log 0 = 0$. At an interior score maximum $\lambda_* > 0$, choose the next grid node to its right in square-root coordinates. Its displacement is at most $\sqrt{M}/m$, and it is positive. The score is stationary in the natural parameter $\log\lambda$, equivalently its score-weighted observations have mean λ_*. The relative-entropy loss is at most $2M/m^2$. Both endpoints are included. The grid node $M > 0$ assigns positive probability to every nonnegative count, which proves existence and positivity of finite-grid maximizers. This also covers all-zero data: a maximum at $\lambda = 0$ is an included endpoint. Apply Proposition 42.2. □

Theorem 42.5 (Gaussian discretization on a known mean interval)**.** *For known $M, \sigma > 0$, integers $N, m \ge 1$, Gaussian components $K_\mu = \mathcal{N}(\mu,\sigma^2)$ on $\mu \in [-M,M]$, and the grid*

$$V_m^{\mathrm{gau}} = \{-M + 2Mj/m : j = 0,1,\dots,m\},$$

every exact finite-grid MLE satisfies

$$\sup_{G\in\mathcal{P}([-M,M])} \mathcal{L}_N(G) - \mathcal{L}_N(H_m) \le \frac{NM^2}{2\sigma^2 m^2}. \tag{42.8}$$

Either finite-grid certificate adds at most A_{grid}.

Proof. Every Gaussian kernel is positive at every real observation, so the finite-grid likelihood maximizer exists and has positive fitted values. At an interior maximum the score is stationary in the natural parameter μ/σ^2. The nearest grid node v has $|v - \mu_*| \le M/m$, and

$$D(\mathcal{N}(\mu_*,\sigma^2)\|\mathcal{N}(v,\sigma^2)) = \frac{(v-\mu_*)^2}{2\sigma^2} \le \frac{M^2}{2\sigma^2 m^2}.$$

Here a nearest node may be an endpoint, but both endpoint Gaussian components are regular members of the same exponential family; the proof of Lemma 42.1 applies unchanged to them. Endpoint maxima are captured exactly. Proposition 42.2 therefore gives the result. No restriction on the values of the observations is required. □

42.4 Finite-dimensional fits and population recovery

Corollary 42.6 (Minimax recovery on explicit grids)**.** *For binomial mixtures, take* $m = \lceil \sqrt{Nt} \rceil$ *in Theorem 42.3. Any measurable exact grid MLE, or a grid fit with fixed total objective deficit* A_{grid}*, satisfies*

$$\sup_G \mathbb{E}_G W_1(H_m, G) \le C(1 + \sqrt{A_{\mathrm{grid}}}) \min\left\{1, \frac{1}{t} + \frac{1}{\sqrt{t \log(eN)}} + \frac{1}{\sqrt{N}}\right\}.$$

This uses $2\lceil \sqrt{Nt} \rceil + 1$ *deterministic grid nodes and attains the full minimax curve, including the transition and plateau.*

For Poisson mixtures on known $[0, M]$*, take* $m = \lceil \sqrt{NM} \rceil$*. Any such finite-grid fit satisfies*

$$\sup_G \mathbb{E}_G W_1(H_m, G) \le C(1 + \sqrt{A_{\mathrm{grid}}})\, r(N, M),$$

where, with $L = \log(eN)$*,*

$$r(N, M) = \begin{cases} M \min\{1, L^{-1} \log(e + L/M)\}, & 0 < M \le L, \\ \max\{\sqrt{M/L}, M/\sqrt{N}\}, & M > L. \end{cases}$$

This uses $\lceil \sqrt{NM} \rceil + 1$ *nodes. Both conclusions also hold when the corresponding finite score certificate replaces the finite-grid objective-gap certificate.*

Proof. The binomial total continuous-model likelihood deficit is at most $A_{\mathrm{grid}} + \pi^2/8$; the Poisson deficit is at most $A_{\mathrm{grid}} + 2$. Apply Theorem 40.3, absorbing the fixed constants into $C(1 + \sqrt{A_{\mathrm{grid}}})$. The matching minimax lower bounds already hold for all estimators, so they apply to these grid estimators as well. This argument does not assume that the true mixing distribution is supported on the grid. □

The Gaussian grid similarly has a bounded total likelihood deficit when $m \ge \lceil M\sqrt{N}/\sigma \rceil$. Only likelihood accuracy is asserted for that Gaussian conclusion here. The grid sizes above are sufficient, not claimed minimal, and no iteration-complexity bound is claimed for a particular numerical optimizer. These are consequences proved within this manuscript; no claim of external historical priority is made for the discretization principle.

Part IV
Further arithmetic classifications and scope

43 Global Franel recurrences and arithmetic Apéry limits

Definite critical values give one criterion for recurrence complexity. The Franel classification below instead uses multiplicities of formal exponential factors, which also cover complex weights and colliding critical values. We classify arithmetic subsequences and determine the space of Apéry limits over the algebraic numbers.

43.1 The critical-value criterion and its scope

The width of an eventual nonzero recurrence $\sum_{j\in J} p_j(n)u_{n+j} = 0$, with $p_j \in \mathbb{C}(n)$, is $\max J - \min J$ after zero coefficients are removed. Here J is a finite set of integers; gaps are allowed. The zero sequence has minimum width zero. A sequence with positive minimum width is understood modulo finitely many initial terms. Width is not the number of its nonzero terms and is not the order of an ordinary differential equation.

Proposition 43.1 (Visibility of same-side definite critical values)**.** *Let Φ be real analytic on a compact real analytic manifold of positive dimension, equipped with a positive analytic density $d\mu$. Set $u_n = \int \Phi^n \, d\mu$. Suppose that a nonzero critical value v has finitely many critical points above it, all nondegenerate local maxima, or all nondegenerate local minima. Regular points in the same fiber are allowed. Then some analytic continuation of $\sum_{n\geq 0} u_n z^n$ is singular at $z = 1/v$. Consequently, s distinct such nonzero critical values force minimum recurrence width at least s.*

Proof. The analytic Morse lemma and radial integration give, near v, a pushforward density with singular part $\mathbf{1}_{\{\epsilon u>0\}}(\epsilon u)^{d/2-1}h(\epsilon u)$, where $u = x - v$, d is the manifold dimension, ϵ is the common side and h is analytic with $h(0) > 0$. The regular contribution is smooth. Positivity and the common side prevent cancellation of the first nonsmooth term. This remains true when $d/2-1$ is an integer: the one-sided polynomial then has a derivative jump. The Cauchy transform of this measure cannot be holomorphic through v, because its boundary imaginary part is a nonzero constant multiple of this nonsmooth density. Inverting the Cauchy variable gives the asserted singularity of the moment generating function.

For the recurrence assertion, translate shifts to $0, \dots, L$ and clear denominators. With $\theta = z\partial_z$, the coefficient recurrence gives an inhomogeneous differential equation whose operator has the form $\sum_{j=0}^{L} z^{L-j}p_j(\theta - j)$. Its highest derivative coefficient is $z^D H(z)$ with $\deg H \leq L$. An operator polynomial in θ removes the polynomial inhomogeneous term without creating new nonzero finite zeros in that leading coefficient. Every branch is regular away from those zeros and zero. Thus at most L distinct nonzero finite singularities are possible. □

This proposition requires the stated real positivity and Hessian conditions. It does not assert that every complex critical value is visible. None of those conditions is needed in the next theorem.

43.2 Every weight and every arithmetic subsequence

For $r \geq 2$ and $\lambda \in \mathbb{C}^*$ write

$$a_{r,\lambda}(N) = \sum_{k=0}^{N} \binom{N}{k}^r \lambda^k, \qquad u_{r,\lambda;q,b}(n) = a_{r,\lambda}(qn + b), \quad q \geq 1, \quad 0 \leq b < q.$$

Theorem 43.2 (Weighted Franel subsequences)**.** *For every integer $r \geq 2$, every $\lambda \in \mathbb{C}^*$, and all q, b as above, the minimum eventual recurrence width of $u_{r,\lambda;q,b}$ over $\mathbb{C}(n)$ is*

parameter stratum	*minimum width*
$\lambda \notin \{1, -1\}$	r
$\lambda = 1$	$\lceil r/2 \rceil$
$\lambda = -1,\ q$ *odd*	$2\lfloor r/2 \rfloor$
$\lambda = -1,\ q$ *even,* b *even*	$\lfloor r/2 \rfloor$
$\lambda = -1,\ q$ *even,* b *odd*	0.

In the last case the sequence is identically zero. The same minimum holds over every field $K(n)$ with $\mathbb{Q}(\lambda) \subseteq K \subseteq \mathbb{C}$.

The $q = 2, b = 0$ case contains the all-weight centered classification. For arbitrary q, b, the conclusion also rules out a loss of width under sampling, except for the stated alternating parity symmetry. Stoll's and McIntosh's constructions provide the symmetry-weight upper bounds [36, 37]. Weighted low-degree recurrences and the higher-Bessel product interpretation have established precedents [38, 39]. The theorem above is proved here by a uniform lower bound, including multiplicity at exceptional weights; it is not presented as a previously named conjecture about subsequences.

43.3 The differential Galois input

Define the entire function

$$\mathcal{F}_r(x) = \sum_{n \geq 0} \frac{x^n}{(n!)^r}.$$

It solves $((x\partial_x)^r - x)y = 0$. Work over $K = \mathbb{C}(t)$, and write M_a for the rank-r differential module whose scalar solutions include

$$f_a(t) = \mathcal{F}_r(at^r), \qquad a \in \mathbb{C}^*.$$

Equivalently its scalar equation is

$$\left\{ \left(\frac{t\partial_t}{r} \right)^r - at^r \right\} y = 0.$$

We use the following precise classical input. The differential Galois group of M_a in its standard representation is

$$G_r = \begin{cases} \mathrm{SL}_r(\mathbb{C}), & r \text{ odd}, \\ \mathrm{Sp}_r(\mathbb{C}), & r \text{ even}. \end{cases} \tag{43.1}$$

This is Katz's Kloosterman-group calculation [34, Corollary 4.4.8]; the connection and its r-fold pullback are identified explicitly in [35, Proposition 2.4 and the proof of Proposition 2.7]. Scaling t by a nonzero constant gives the assertion for every a. In particular these groups are connected and their standard representations are irreducible.

The formal decomposition at $t = \infty$ consists of r rank-one factors with exponential parts

$$\exp(r\alpha\zeta t), \qquad \zeta^r = 1, \quad \alpha^r = a. \tag{43.2}$$

Their remaining parts are regular singular. Direct substitution gives formal solutions

$$e^{r\alpha\zeta t} t^{-(r-1)/2} (1 + O(t^{-1}));$$

the distinct leading exponential roots recursively determine the formal series. Only the exponential parts, with their multiplicities, will be used below. The duality relation is

$$M_a^* \simeq M_{(-1)^r a}. \tag{43.3}$$

For example, this follows by reversing the cyclic basis for the companion system and alternating its signs, or by formal adjunction with respect to dx/x before the pullback.

Lemma 43.3 (Independence of distinct scale classes)**.** *Choose finitely many nonzero scales $a_1, \dots, a_s$. If r is even, assume they are pairwise distinct. If r is odd, assume $a_i \neq \pm a_j$ for $i \neq j$. The connected differential Galois group of $\bigoplus_i M_{a_i}$ is the full product G_r^s.*

Proof. The connected joint group projects onto each individual group: its image has finite index in the connected group G_r. Its Lie algebra is therefore a subdirect product of simple Lie algebras. By the Lie-algebra form of Goursat's lemma, a proper subdirect product identifies two of these simple factors by an automorphism. Automorphisms of $\mathfrak{sp}_r$ are inner; automorphisms of $\mathfrak{sl}_r$ are inner or the standard duality automorphism. Consequently the two standard representations on the connected joint group would be isomorphic, or in the odd case dual.

Passing to the finite algebraic extension corresponding to the identity component makes this an isomorphism of the associated differential modules. But an isomorphism preserves the formal exponential parts. After any common finite ramification at infinity, (43.2) shows that M_a and M_b can be isomorphic only if $a = b$, and that M_a and M_b^* can be isomorphic only if $a = (-1)^r b$. Indeed equality of the multisets of coefficients of the exponential terms, followed by taking their rth powers, gives these equalities. Both possibilities contradict the scale hypotheses. Thus no pair of simple factors is identified and the Lie algebra is the full product. A connected closed subgroup with this Lie algebra is the full product group. □

43.4 A factorial-normalization budget with an arbitrary sampling interval

For a nonzero differential operator $P = \sum_i b_i(t)\partial_t^i$ with formal Laurent coefficients at infinity, let v be the largest leading power of t among its coefficients and put

$$\sigma_\infty(P)(\xi) = \sum_i \left(\lim_{t\to\infty} t^{-v} b_i(t) \right) \xi^i.$$

The limit means the constant coefficient after Laurent normalization. Differentiating a Laurent coefficient lowers its degree, so

$$\sigma_\infty(PQ) = \sigma_\infty(P)\sigma_\infty(Q) \tag{43.4}$$

(up to a harmless nonzero constant if another normalization is used).

Lemma 43.4 (Nonzero exponential multiplicities bound width)**.** *Let*

$$H(t) = \sum_{n\geq 0} \frac{u_n}{((qn+b)!)^r} t^{r(qn+b)}.$$

Suppose the differential module generated by this scalar function has at infinity a decomposition into rank-one factors with exponential parts $e^{\beta t}$ and regular-singular remaining parts. If ν of these factors have $\beta \neq 0$, counted with multiplicity, then every eventual recurrence of u_n has width at least $\nu/(qr)$.

Proof. Write a proposed width-L recurrence as $\sum_{j=0}^{L} p_j(n)u_{n-j} = 0$, with $p_j \in \mathbb{C}[n]$. Let $D = \max \deg p_j$ and let c_j be the leading coefficient when $\deg p_j = D$. Multiplication by $((qn+b)_{\underline{qL}})^r/((qn+b)!)^r$ gives $PH = Q(t)$, with Q polynomial and

$$P = \sum_{j=0}^{L} t^{qrj} p_j \left(\frac{t\partial_t}{qr} + j - \frac{b}{q}\right)^{q(L-j)-1} \prod_{\ell=0} \left(\frac{t\partial_t}{r} - \ell\right)^r.$$

Indeed the coefficient contributed by the jth term at exponent $r(qn+b)$ is $p_j(n)u_{n-j}/((qn+b-qL)!)^r$ for sufficiently large n. The finitely many other coefficients account for Q. The largest coefficient degree in t is $D + qrL$, and

$$\sigma_\infty(P)(\xi) = \sum_{\deg p_j = D} c_j \left(\frac{\xi}{qr}\right)^D \left(\frac{\xi}{r}\right)^{qr(L-j)}. \tag{43.5}$$

Distinct j give distinct powers, so this is nonzero and has at most qrL nonzero roots with multiplicity. Left multiplication by a polynomial in $t\partial_t$ killing Q makes the equation homogeneous and adds only zero roots to the principal symbol.

Let R be the monic minimal scalar annihilator of H. Its formal rank-one decomposition gives a first-order factorization with factors $\partial_t - \beta_i + O(t^{-1})$: take a formal rank-one solution, right-divide by its logarithmic-derivative factor, and repeat on the quotient. Laurent gauge changes add only $O(t^{-1})$. Thus $\sigma_\infty(R) = \prod_i(\xi - \beta_i)$. The homogeneous annihilator is a left multiple of R. Equation (43.4) then forces its principal symbol to contain all ν nonzero factors of R, with multiplicity. Consequently $\nu \le qrL$. □

43.5 Cyclicity after root-of-unity filtering

The Cauchy-product identity gives

$$\mathcal{F}_r(x)\mathcal{F}_r(\lambda x) = \sum_{N\ge 0} \frac{a_{r,\lambda}(N)}{(N!)^r} x^N.$$

Writing $\mu_q = \{\epsilon : \epsilon^q = 1\}$, its root-of-unity filter is

$$H_{q,b}(t) = \frac{1}{q} \sum_{\epsilon\in\mu_q} \epsilon^{-b} f_\epsilon(t) f_{\lambda\epsilon}(t) = \sum_{n\ge 0} \frac{a_{r,\lambda}(qn+b)}{((qn+b)!)^r} t^{r(qn+b)}. \tag{43.6}$$

All products lie in one Picard–Vessiot field. Multiplication of scalar solutions is an equivariant linear map from their tensor product. On an irreducible representation a nonzero such map is injective. Distinct nonisomorphic irreducible images have direct sum, and a vector with a nonzero component in each generates the whole direct sum. These facts establish cyclicity for the specific filtered function, not just for an ambient system that annihilates it.

Lower bounds in Theorem 43.2. First suppose $\lambda \ne \pm 1$. Each product in (43.6) belongs to an irreducible rank-r^2 tensor representation. For even r the two scales are distinct independent symplectic factors. For odd r they are distinct modulo sign, and hence independent special-linear factors. Use Lemma 43.3 for the finite set of all scale classes occurring in the filter.

The q product representations are pairwise nonisomorphic even when their scales overlap. For even r, their two-factor supports agree only if the unordered pairs $\{\epsilon, \lambda\epsilon\}$ agree. For odd r, a representation also records whether it is standard or dual at each signed scale class: these are inequivalent for odd $r \ge 3$. Hence isomorphism again requires equality of the unordered pairs of actual scales. For two distinct indices ϵ, η, such equality would require $\eta = \lambda\epsilon$ and $\lambda\eta = \epsilon$, forcing $\lambda^2 = 1$, which is excluded. Since every coefficient ϵ^{-b}/q is nonzero, the filter is cyclic

in all q irreducible summands. A formal exponential coefficient in $M_a \otimes M_c$ can vanish only if $c = (-1)^r a$. This too is excluded. Therefore $\nu = qr^2$ and the budget gives $L \geq r$.

For $\lambda = 1$, the summands are $\operatorname{Sym}^2 M_\epsilon$. Their representations are irreducible and pairwise nonisomorphic. For even r this follows from distinct symplectic scale factors. For odd r, opposite scales if present give $\operatorname{Sym}^2 V$ and $\operatorname{Sym}^2 V^*$, which are nonisomorphic for $r \geq 3$. In each summand the formal exponents are the unordered sums of two rth-root exponents. There are $r/2$ zero sums when r is even, and none when r is odd. The nonzero rank per summand is thus $r^2/2$ for even r, and $r(r+1)/2$ for odd r. The q nonzero filter components give respectively $L \geq r/2$ and $L \geq (r+1)/2$.

Now let $\lambda = -1$. The products indexed by ϵ and $-\epsilon$ coincide. If q is even, their filter coefficients sum to zero for odd b, and to $2\epsilon^{-b}/q$ for even b. This proves the asserted vanishing in the first case. In the second case there are $q/2$ surviving products. If q is odd, none of these pairs repeats and there are q surviving products.

For even r, each surviving product is an irreducible tensor of two independent symplectic modules. Distinct surviving products have disjoint scale pairs, so they are nonisomorphic. Each has r^2 nonzero formal factors, since zero would require the scales to be equal rather than opposite.

For odd r, each product lies in $V \otimes V^* = \mathbf{1} \oplus \mathrm{Ad}$ for the special-linear factor indexed by the class $\{\epsilon, -\epsilon\}$. The multiplication map kills the trivial summand. To check this without assuming that invariant scalar functions are constant, put $\delta = (t/r)\partial_t$. For solutions y, z of $\delta^r y = \epsilon t^r y$, $\delta^r z = -\epsilon t^r z$, the nondegenerate concomitant

$$\mathcal{B}(y,z) = \sum_{j=0}^{r-1} (-1)^j (\delta^j y)(\delta^{r-1-j} z)$$

is constant. Indeed its derivative telescopes to zero. Choose dual solution bases y_i, z_i for this pairing and jet matrices Y, Z whose rows are δ^j, $0 \leq j < r$. Then $Y^{\mathsf{T}} J Z = I$, with J invertible and anti-diagonal, whence

$$\sum_i y_i z_i = (Y Z^{\mathsf{T}})_{00} = (J^{-\mathsf{T}})_{00} = 0 \qquad (r \geq 2).$$

The selected product is nonzero, so the multiplication map is injective on the irreducible adjoint summand. Different surviving products use different independent scale classes. Each therefore contributes the $r(r-1)$ nonzero ordered differences of distinct rth-root exponents. There are q such products for odd q, or $q/2$ for even q and even b. Dividing their nonzero ranks by qr gives respectively $2\lfloor r/2 \rfloor$ and $\lfloor r/2 \rfloor$, in both parities of r. □

Upper bounds and coefficient fields. For $\lambda \neq \pm 1$ set $\mathcal{L}_N[p] = \sum_{k=0}^{N} p(k) \binom{N}{k}^r \lambda^k$. An index shift gives

$$\mathcal{L}_N[k^{r+h} - \lambda (N-k)^r (k+1)^h] = 0 \qquad (h \geq 0).$$

The polynomial displayed here has leading coefficient $1 - (-1)^r \lambda \neq 0$. Successive elimination reduces every moment to $\mathcal{L}_N[1], \ldots, \mathcal{L}_N[k^{r-1}]$. The identity $\binom{N-1}{k}^r = N^{-r}(N-k)^r \binom{N}{k}^r$ therefore gives a backward r-dimensional state relation over $\mathbb{Q}(\lambda)(N)$. Apply it q times between observations: $r+1$ consecutive observations on the progression $qn + b$ are rows on the same r-dimensional state, so they satisfy a recurrence of width at most r. This argument needs no matrix inverse.

For $\lambda = 1$, Stoll's ordinary Franel recurrence has width at most $m = \lceil r/2 \rceil$ over $\mathbb{Q}(n)$ [36]. A rational companion system has dimension m; its q-step system gives the same upper bound on every progression. For $\lambda = -1$, symmetry gives $a_{r,-1}(N) = 0$ for odd N. McIntosh's recurrence [37, Theorem 1] gives width at most $h = \lfloor r/2 \rfloor$ for $A_r(n) = (-1)^n a_{r,-1}(2n)$. Geometric rescaling preserves width, so even progressions have the claimed upper bound h. Interlacing this recurrence

with the zero odd sequence gives a width-$2h$ recurrence for $a_{r,-1}(N)$: a recurrence on its even entries with shifts j becomes one with shifts $2j$, valid also on the zero odd entries. Taking a q-step companion system gives the upper bound $2h$ for odd q. All upper constructions have coefficients in $\mathbb{Q}(\lambda)(n)$. Since the lower bounds hold over $\mathbb{C}(n)$, extending the coefficient field anywhere between these fields cannot change the minimum. □

Corollary 43.5 (Franel and centered alternating sums)**.** *For every $r \geq 2$,*

$$\operatorname{minwidth}_{\mathbb{C}(n)} \sum_{k=0}^{n} \binom{n}{k}^r = \lceil r/2 \rceil, \qquad \operatorname{minwidth}_{\mathbb{C}(n)} \sum_{k=-n}^{n} (-1)^k \binom{2n}{n+k}^r = \lfloor r/2 \rfloor.$$

More generally, the centered weighted sequence $\lambda^{-n} a_{r,\lambda}(2n)$ has width $\lceil r/2 \rceil$ at $\lambda = 1$, $\lfloor r/2 \rfloor$ at $\lambda = -1$, and r elsewhere.

Proof. Use Theorem 43.2 with $(q, b) = (1, 0)$ or $(2, 0)$. The centered alternating sum equals $(-1)^n a_{r,-1}(2n)$, and geometric rescaling preserves width. □

The alternating conclusion does not require a critical-value visibility argument. The unweighted conclusion removes the restriction to telescoping recurrences in [40, Theorem 1.1]. Critical-value collisions do not create exceptions in Theorem 43.2: the budget counts formal exponential factors with multiplicity, not distinct critical values.

43.6 Arithmetic spaces of Apéry limits

Put $A_r(N) = a_{r,1}(N)$ and $m = \lceil r/2 \rceil$. In this subsection the notation A_r denotes the ordinary, not the alternating, Franel sequence. Define rational sequences $A_{r,j}(N)$ by

$$\sum_{j \geq 0} A_{r,j}(N) z^{2j} = \sum_{k=0}^{N} \binom{N}{k}^r \left\{ \prod_{\ell=1}^{k} \left(1 - \frac{z}{\ell}\right) \prod_{\ell=1}^{N-k} \left(1 + \frac{z}{\ell}\right) \right\}^{-r}. \tag{43.7}$$

The right side is even by replacing k with $N - k$, and $A_{r,0} = A_r$. Let

$$\phi_{r,j} = [z^{2j}] \left(\frac{z}{\sin z}\right)^r \in \mathbb{Q}_{>0}.$$

The input from Straub–Zudilin [40, Theorem 1.3] is that $A_{r,j}$, $0 \leq j < m$, satisfy every telescoping recurrence for $\binom{N}{k}^r$ on a sufficiently late tail, and have limits

$$\lim_{N \to \infty} \frac{A_{r,j}(N)}{A_r(N)} = \phi_{r,j} \pi^{2j}. \tag{43.8}$$

Van Hoeij's invariant-submodule calculation [42, Remark 6] gives a telescoper of order m for this summand. These are external inputs; the transfer to the unrestricted minimal recurrence is supplied by the preceding theorem.

For $q \geq 1$ and $0 \leq b < q$, let $L_{r;q,b}$ denote the monic minimal eventual recurrence of $A_r(qn+b)$, with shifts normalized to $0, \ldots, m$. If K is a field between $\mathbb{Q}$ and $\overline{\mathbb{Q}}$, write $\mathcal{S}_{r;q,b}(K)$ for the K-valued solutions on a sufficiently late regular tail, identifying sequences that agree eventually.

Theorem 43.6 (The Apéry-limit map on algebraic-valued solutions)**.** *For every $r \geq 2$, $q \geq 1$, $0 \leq b < q$ and $\mathbb{Q} \subseteq K \subseteq \overline{\mathbb{Q}}$, the following statements hold.*

1. *$L_{r;q,b}$ has order m and coefficients in $\mathbb{Q}(n)$. For $(q, b) = (1, 0)$ it is also the monic minimal telescoper for the summand $\binom{n}{k}^r$.*

2. *The sequences $A_{r,j}(qn+b)$, $0 \le j < m$, are a basis of $\mathcal{S}_{r;q,b}(K)$. Every complex tail solution also has a unique expansion in this same basis, with complex constants.*

3. *Every $X \in \mathcal{S}_{r;q,b}(K)$ has an Apéry limit, and*

$$X \longmapsto \lim_{n\to\infty} \frac{X(n)}{A_r(qn+b)} \quad \text{is a } K\text{-linear isomorphism onto} \quad \bigoplus_{j=0}^{m-1} K\pi^{2j}. \tag{43.9}$$

In particular two algebraic-number-valued tail solutions with the same limit are equal eventually.

Proof. For $q = 1, b = 0$, the minimal recurrence has order m by Corollary 43.5. Stoll provides an annihilator of this order over $\mathbb{Q}(n)$, so its monic normalization is the unique minimal operator even over $\mathbb{C}(n)$. The minimal telescoper supplied by van Hoeij has the same order and annihilates the sequence. Right division by the monic minimal recurrence leaves a lower-order annihilator as remainder, necessarily zero. Equality of orders thus makes the two monic operators equal. Equation (43.8) therefore applies to solutions of this minimal recurrence.

Pass to a tail beyond all poles and integer zeros of its leading and trailing coefficients, and beyond the finitely many exceptional indices for the $A_{r,j}$. A companion system has dimension m. Its q-step system yields one common rational-coefficient recurrence of order at most m for every sampled solution. By Theorem 43.2, the sampled sequence $A_r(qn+b)$ has minimum order exactly m. Hence this common operator has order m and, after normalization, is $L_{r;q,b}$.

The m rational sequences $A_{r,j}(qn+b)$ are linearly independent over $\mathbb{C}$. Indeed a complex dependence among rational columns of an infinite matrix implies a rational dependence: the row space is finite-dimensional and can be spanned by finitely many rational rows, whose nullspace is defined over $\mathbb{Q}$. A rational dependence on any tail would, after division by $A_r(qn+b)$ and passage to the limit, contradict the transcendence of π in (43.8). The solution space of a regular order-m tail recurrence has dimension m, so these columns are a basis over $\mathbb{C}$.

Evaluation at m consecutive sufficiently late indices is an isomorphism from this solution space to $\mathbb{C}^m$. The resulting basis matrix has rational entries and nonzero determinant, hence rational inverse. A solution taking values in K consequently has all its basis coefficients in K, proving the basis assertion over K. Its limit is $\sum_{j=0}^{m-1} c_j \phi_{r,j} \pi^{2j}$. The numbers $1, \pi^2, \dots, \pi^{2m-2}$ are linearly independent over $\overline{\mathbb{Q}}$, and every $\phi_{r,j}$ is nonzero rational. This proves injectivity and the stated image. □

An algebraic-number-valued solution need not be assumed in advance to lie in one number field: its m initial tail values generate a finite extension of $\mathbb{Q}$, and the rational recurrence then keeps its entire tail in that field. Thus the theorem covers all such solutions, including those initially described only as $\overline{\mathbb{Q}}$-valued.

Corollary 43.7 (Normalized Apéry limits and uniqueness)**.** *For $1 \le j < m$ put*

$$D_{r,j}(N) = \frac{A_{r,j}(N)}{2\binom{r+2j-1}{2j}}.$$

Then $D_{r,j}(0) = 0$, $D_{r,j}(1) = 1$, and it is a solution of the minimal Franel recurrence on a tail, with

$$\lim_{N\to\infty} \frac{D_{r,j}(N)}{A_r(N)} = \frac{\phi_{r,j}\pi^{2j}}{2\binom{r+2j-1}{2j}}.$$

It is the unique algebraic-number-valued tail solution with that limit. The same uniqueness holds for its sampled sequence $D_{r,j}(qn+b)$ relative to $L_{r;q,b}$ and $A_r(qn+b)$. In particular, for $r \ge 3$ and $r \ge 5$, respectively,

$$\lim \frac{D_{r,1}(N)}{A_r(N)} = \frac{\zeta(2)}{r+1}, \qquad \lim \frac{D_{r,2}(N)}{A_r(N)} = \frac{3(5r+2)\zeta(4)}{(r+1)(r+2)(r+3)}.$$

Proof. At $N = 0$ the deformation is 1, while at $N = 1$ it is $(1-z)^{-r}+(1+z)^{-r}$. The stated initial values follow by coefficient extraction. The limits and uniqueness follow from Theorem 43.6. Finally $\phi_{r,1} = r/6$ and $\phi_{r,2} = r(5r+2)/360$ give the two special formulas, already computed in [40, Examples 1.5 and 1.7]. □

43.7 Exact scope of the conjectural statements

Chamberland–Straub [41, Conjectures 9 and 11] state the normalized $\zeta(2)$ and $\zeta(4)$ limits, respectively for $r \geq 3$ and $r \geq 5$, and include uniqueness. Their wording does not specify a coefficient field for the secondary solution. The numerical values of the limits and the deformed rational sequences are already in Straub–Zudilin. What the argument here supplies is their membership in the unrestricted minimal recurrence, complete uniqueness over $\overline{\mathbb{Q}}$, and the same conclusions for every arithmetic subsequence. It does not establish unrestricted complex uniqueness.

In fact, on the complex tail solution space the limit map has a kernel of dimension $m-1$. For $r \geq 7$ this obstruction persists even with the two displayed initial normalizations in the eventual sense. To make it explicit, set $c_j = A_{r,j}(1)$ and $\ell_j = \phi_{r,j}\pi^{2j}$, $j = 1, 2, 3$, and put

$$Z = (c_2\ell_3 - c_3\ell_2)A_{r,1} + (c_3\ell_1 - c_1\ell_3)A_{r,2} + (c_1\ell_2 - c_2\ell_1)A_{r,3}.$$

Then $Z(0) = Z(1) = 0$, $Z/A_r \to 0$, and Z is not the zero tail solution: its coefficient of $A_{r,3}$ is nonzero by the transcendence of π, and the basis is independent. Thus $D_{r,1} + cZ$ has the same two initial values and limit for every $c \in \mathbb{C}$, and satisfies the recurrence eventually. An assertion requiring the recurrence to hold at every small nonnegative index needs a separate regularity argument; no such assertion is inferred here from eventual annihilation. The coefficient-degree prediction in [40, Remark 4.1] is likewise not settled by a width bound.

44 An arithmetic classification for split beta orthogonality

The critical-value argument uses positivity to prevent cancellation of local contributions. A different use of positivity gives an arithmetic classification for multiple orthogonal polynomials: once a logarithmic moment identity forces orthogonality on both sides of a cut, the required sign changes can exceed the degree. The classification below is uniform in both orthogonality indices, in the algebraic cut, and in every positive algebraic polynomial modification of the weight.

44.1 Parameter and index classification

Let $a, b > 0$ be rational with $a + b \in \mathbb{Z}$, let $\xi \in \overline{\mathbb{Q}} \cap (0, 1)$, and let $w \in (\overline{\mathbb{Q}} \cap \mathbb{R})[X]$ satisfy $w(x) > 0$ for $0 < x < 1$. Define the positive integrable weight

$$\rho(x) = w(x)x^{a-1}(1 - x)^{b-1}.$$

For $m, n \geq 0$, the monic type-II Angelesco polynomial $P_{m,n}$ has degree $m + n$ and satisfies

$$\int_0^\xi P_{m,n}(x)x^j\rho(x)\,dx = 0 \qquad (0 \leq j < m), \tag{44.1}$$

$$\int_\xi^1 P_{m,n}(x)x^j\rho(x)\,dx = 0 \qquad (0 \leq j < n). \tag{44.2}$$

Empty lists of conditions are allowed, and $P_{0,0} = 1$. Existence and uniqueness will be proved below.

Theorem 44.1 (Exact algebraic-coefficient classification)**.** *For the preceding family, the following are equivalent:*

1. *Every coefficient of $P_{m,n}$ is algebraic.*
2. *Every zero of $P_{m,n}$ is algebraic.*
3. *Either $a, b \in \mathbb{Z}$, or $m = n$.*

In the algebraic cases the coefficients belong to

$$K = \mathbb{Q}(\xi, w_0, \ldots, w_d), \qquad w(X) = \sum_{j=0}^{d} w_j X^j.$$

Consequently, when a, b are nonintegers and $m \neq n$, $P_{m,n}$ has at least one transcendental coefficient and at least one transcendental zero. All its zeros are nevertheless real and simple, with precisely m in $(0, \xi)$ and n in $(\xi, 1)$.

The rationality of a, b and the integer-sum condition are essential to the stated arithmetic reduction. The theorem includes the arcsine pair $a = b = 1/2$; no exclusion of that pair is needed. We assert that at least one zero is transcendental off the diagonal, not that every zero is.

44.2 Positivity and normality

Lemma 44.2 (Sign changes and existence)**.** *For a positive integrable weight on $(0, 1)$, each system (44.1)–(44.2) has a unique monic solution of degree $m + n$. That solution has exactly the zero distribution stated in Theorem 44.1.*

Proof. A nonzero real polynomial Q orthogonal to $1, x, \dots, x^{k-1}$ on an interval with a positive weight has at least k sign changes in its interior. Indeed, if the sign-change points were $t_1, \dots, t_s$ with $s < k$, the polynomial $q(x) = \prod_{i=1}^{s}(x - t_i)$ would make Qq have one constant nonzero sign except at finitely many points. Its weighted integral could not vanish, contradicting orthogonality to q.

A nonzero polynomial satisfying both lists therefore has at least $m + n$ distinct real zeros in the two disjoint interiors. In particular, there is no such polynomial of degree at most $m + n - 1$. The linear map from that $(m + n)$-dimensional coefficient space to its $m + n$ moments is injective, hence bijective. It follows that the lower coefficients of a monic degree-$m + n$ solution exist uniquely. The same sign-change count exhausts its degree, proving simplicity and the precise locations of all zeros. The case $m + n = 0$ is immediate. □

44.3 A logarithmic moment forces orthogonality on both sides

The transcendence input is the classical consequence of Baker's theorem: a nonzero linear form, with algebraic coefficients, in fixed logarithms of nonzero algebraic numbers is transcendental [27]. We reproduce the needed integral reduction, so the argument does not require any unproved period-independence assertion.

For $r \in \mathbb{Q} \cap (0, 1)$ write

$$b_r(x) = x^{r-1}(1 - x)^{-r}, \qquad U_r(y) = \int_0^y b_r(x)\, dx.$$

Lemma 44.3 (Two transcendental base integrals)**.** *If $\xi \in \overline{\mathbb{Q}} \cap (0, 1)$, then both $U_r(\xi)$ and $\int_\xi^1 b_r(x)\, dx$ are transcendental. This includes $r = 1/2$.*

Proof. Write $r = p/q$ with $1 \le p < q$ and put $s = (\xi/(1 - \xi))^{1/q} > 0$. The substitution $x = z^q/(1 + z^q)$ and partial fractions give

$$U_r(\xi) = q \int_0^s \frac{z^{p-1}}{1 + z^q}\, dz = -\sum_{\zeta^q = -1} \zeta^p \log(1 - s/\zeta).$$

For each root ζ, take the continuous logarithm along positive real s, starting at zero when $s = 0$. Its argument never vanishes because no root of $\zeta^q = -1$ is positive real. The displayed identity follows by differentiation and the common value at $s = 0$. All logarithm arguments at the specified s are nonzero algebraic numbers, and all coefficients are algebraic. The integral is strictly positive, so Baker's theorem proves its transcendence. Finally,

$$\int_\xi^1 b_r(x)\, dx = U_{1-r}(1 - \xi),$$

and the same argument applies. □

Lemma 44.4 (Arithmetic synchronization of moments)**.** *Let $r \in \mathbb{Q} \cap (0, 1)$ and $S \in (\overline{\mathbb{Q}} \cap \mathbb{R})[X]$. Then*

$$\int_0^\xi S(x) b_r(x)\, dx = 0 \quad \Longleftrightarrow \quad \int_\xi^1 S(x) b_r(x)\, dx = 0.$$

More precisely, define the unique constant $c(S)$ and polynomial R_S by

$$S(X) = c(S) + X(1 - X)R_S'(X) + (r - X)R_S(X). \tag{44.3}$$

Either vanishing condition is equivalent to the two algebraic linear conditions

$$c(S) = 0, \qquad R_S(\xi) = 0. \tag{44.4}$$

If S has coefficients in a field $K \supset \mathbb{Q}$ containing ξ, then $c(S)$ and all coefficients of R_S belong to K.

Proof. The operator in (44.3) sends X^j to $(j+r)X^j-(j+1)X^{j+1}$. Descending elimination therefore gives a unique R_S and constant $c(S)$, over the coefficient field. Put $H_S(x)=x^r(1-x)^{1-r}R_S(x)$. Direct differentiation and $H_S(0)=H_S(1)=0$ give

$$\int_0^\xi Sb_r = c(S)U_r(\xi)+H_S(\xi),$$
$$\int_\xi^1 Sb_r = c(S)\{U_r(1)-U_r(\xi)\}-H_S(\xi).$$

Here $H_S(\xi)$ is algebraic. Either integral can vanish only if $c(S)=0$, by Lemma 44.3. It then forces $H_S(\xi)=0$, equivalently $R_S(\xi)=0$ since $0<\xi<1$. Conversely, these two conditions make both integrals zero. All operations in descending elimination take place over K. □

44.4 Proof of the classification

Proof of Theorem 44.1. Suppose first that a,b are integers. Since they are positive, ρ is a polynomial with coefficients in K. Each moment on either interval belongs to K, by termwise integration and $\xi\in K$. The invertible linear system in Lemma 44.2 consequently has its monic solution in $K[X]$.

Otherwise neither shape is an integer, since their sum is an integer. Write

$$a=A+r,\qquad b=B+1-r,\qquad A,B\in\mathbb{Z}_{\ge 0},\quad r\in\mathbb{Q}\cap(0,1).$$

Then $\rho=gb_r$, where $g(X)=w(X)X^A(1-X)^B\in K[X]$ is positive on $(0,1)$.

Assume that $P_{m,n}$ has algebraic coefficients and set $h=\max(m,n)$. For every $0\le j<h$, at least one of (44.1) and (44.2) holds. Apply Lemma 44.4 to the algebraic polynomial $S=X^jP_{m,n}g$. Both integrals then vanish. Thus $P_{m,n}$ satisfies h orthogonality conditions on each interval. Positivity forces at least $2h$ distinct zeros, so

$$m+n=\deg P_{m,n}\ge 2\max(m,n).$$

This is possible only when $m=n$.

Conversely let $m=n=k$. The $2k$ equations

$$c(X^jPg)=0,\qquad R_{X^jPg}(\xi)=0,\qquad 0\le j<k, \tag{44.5}$$

are homogeneous linear equations over K in the $2k+1$ coefficients of P of degree at most $2k$. They have a nonzero solution in $K[X]$. Lemma 44.4 shows that this solution satisfies both orthogonality lists. By the sign-change argument it has degree exactly $2k$. Divide by its nonzero leading coefficient. Uniqueness identifies the resulting monic polynomial with $P_{k,k}$, proving $P_{k,k}\in K[X]$.

Finally, a monic polynomial has all roots algebraic if and only if all its coefficients are algebraic: one direction uses elementary symmetric functions, and the other uses the algebraic closedness of $\overline{\mathbb{Q}}$. The zero distribution follows from Lemma 44.2. □

44.5 Quadrature, an explicit example, and the literature boundary

Corollary 44.5 (Algebraic cuts destroy all-algebraic Gaussian nodes)**.** *If $a,b>0$ are rational nonintegers with integer sum, then for every algebraic $\xi\in(0,1)$ and every $N\ge 1$, the N-node Gaussian quadrature rule for ρ restricted to either $[0,\xi]$ or $[\xi,1]$ has at least one transcendental node. In contrast, all zeros of every balanced $P_{N,N}$ are algebraic. These assertions remain true under every positive algebraic polynomial factor w.*

Proof. The Gaussian nodes are the zeros of the degree-N monic orthogonal polynomial, namely $P_{N,0}$ or $P_{0,N}$. Apply the theorem. The balanced assertion is the case $(m, n) = (N, N)$. □

For $w = 1$ and $(a, b) = (r, 1 - r)$, the classical diagonal Rodrigues formula specializes to

$$P_{k,k}(x) = \frac{(-1)^k}{(2k)_k\, b_r(x)} \frac{d^k}{dx^k} \left[x^{k+r-1}(1-x)^{k-r}(x-\xi)^k \right], \qquad k \geq 1. \tag{44.6}$$

Leibniz' rule makes the right side a polynomial of degree at most $2k$. Integration by parts k times on each interval proves the orthogonality: derivatives of order at most $k - 1$ of the expression in brackets vanish at all relevant endpoints, and $\deg(x^j) < k$. Its leading coefficient before normalization is $(-1)^k(2k)_k$, as can be read from the highest-degree terms. This establishes (44.6) and its monic normalization. For example,

$$P_{1,1}(x) = x^2 - \frac{1 + r + \xi}{2} x + \frac{r\xi}{2}.$$

At $r = 1/3$ and $\xi = 1/2$ its two zeros are

$$\frac{11 - \sqrt{73}}{24}, \qquad \frac{11 + \sqrt{73}}{24},$$

one on each side of the cut. Every off-diagonal polynomial for the same measures has a transcendental coefficient and a transcendental zero, including the degree-one polynomial for either single interval.

The diagonal formula is classical. After an affine change of variable, this is the Jacobi–Angelesco system with zero exponent at the common endpoint; see Van Assche and Coussement [28, §3.5, Eq. (3.45)]. Their equations (3.48)–(3.50) also express near-diagonal polynomials through incomplete moment ratios. We do not count the diagonal Rodrigues formula, Angelesco normality, polynomial beta reduction, or Baker's theorem as new. The addition proved here is the exact arithmetic exclusion of every unbalanced index throughout the specified parameter family, including positive polynomial modifications. We make no historical-priority claim. Nor does this result assert algebraic independence of different zeros or settle an unrestricted problem for arbitrary rational beta shapes with noninteger sum.

45 Recurrence scope and remaining questions

Remark 45.1 (Random walks). For the ℓ-step planar walk with unit steps, the even moments $W_\ell(2n) = \sum_{a_1+\cdots+a_\ell=n} \binom{n}{a_1,\dots,a_\ell}^2$ have the generating function $\sum_n W_\ell(2n)t^{2n}/(n!)^2 = I_0(2t)^\ell$. Its minimal operator is the ℓ-th symmetric power of (5.1), by the Galois argument of Proposition 5.3. The solution space is spanned by $i_0^{\ell-b}k_0^b$, with rates $2(\ell-2b)$, exponents $\ell/2$ at infinity and exponent 0 (with logarithms) at the origin.

The identity $\binom{\ell+1}{2} - 0 - (\ell+1)\frac{\ell}{2} = 0$ again excludes apparent singularities. In the variable x one finds $P_0 = \theta^{\ell+1}$ and $\deg P_k \le \ell + 1 - 2k$. This recovers the recurrence of order $\lceil \ell/2 \rceil$ and degree $\ell - 1$, with leading coefficient $n^{\ell-1}$, found by Verrill [22]; see [5, Thms. 1 and 2] and also [6]. Lemma 3.6 excludes recurrences of smaller order. The exponent-divisibility argument alone does not determine the least degree, because all exponents at the origin vanish. The logarithmic monodromy argument in Theorem 45.3 below proves that the least degree is $\ell - 1$, even among recurrences of arbitrary order valid only for sufficiently large indices.

The unitary and the random-walk operators thus have the same generic structure: both have solution spaces isomorphic to Sym^ℓ of the Bessel module. The difference is entirely in the lattice at the origin. For the walk the local monodromy at 0 is a single unipotent Jordan block and all exponents are 0. For $U(\ell)$ the same Jordan type is realized with exponents $-b(\ell-b)$, which provide the direct divisibility argument for degree $\ell - 1$. For planar walks, the proof below instead uses Jordan-block length.

45.1 The polynomial degree in the planar-walk recurrence

The ordinary generating function retains information that is lost by using only the numerical exponents of the double exponential generating function. The relevant invariant is the size of its unipotent monodromy block.

Lemma 45.2 (Recurrence degree and local monodromy)**.** *Let $A(z) = \sum_{n\ge 0} a_n z^n$ be a convergent germ whose minimal differential equation is regular singular or ordinary at zero. Suppose (a_n) satisfies an eventual nonzero polynomial-coefficient recurrence of degree at most d. Then every Jordan block of the local monodromy of the minimal equation of A has size at most* $\max\{d, 1\}$. *If $d = 0$, A is rational.*

Proof. Write the recurrence as $\sum_{s=0}^{M} c_s(n)a_{n+s} = 0$ for all sufficiently large n, where $\deg c_s \le d$, and put $\theta = z\partial_z$. The nonzero operator

$$R = \sum_{s=0}^{M} z^{M-s} c_s(\theta - s)$$

has order $e \le d$, and $RA = P$ for a polynomial P. If $P = 0$, the minimal equation of A is a right factor of R, so its solution space has dimension at most e. If $P \neq 0$, set $Q = \partial_z - P'/P$. On a sufficiently small punctured disc about zero, the solution space of QR is

$$\mathrm{Sol}(QR) = \mathrm{Sol}(R) \oplus \mathbb{C}A.$$

Indeed, $QRf = 0$ gives $Rf = cP$, and hence $f - cA \in \mathrm{Sol}(R)$; the sum is direct since $RA = P \neq 0$. The second summand is fixed by local monodromy because A is holomorphic at zero. The first has dimension e. Thus every Jordan block on this direct sum has size at most $\max\{e, 1\}$. The minimal equation of A right divides QR, so its local solution space is an invariant subspace and obeys the same bound. When $d = 0$, R is multiplication by a nonzero polynomial, and $A = P/R$ is rational. □

Theorem 45.3 (Minimal degree of the planar-walk recurrence)**.** *For $\ell \geq 1$, let*

$$a_n = W_\ell(2n) = \sum_{a_1+\cdots+a_\ell=n} \binom{n}{a_1,\ldots,a_\ell}^2.$$

Every nonzero polynomial-coefficient recurrence for (a_n) valid for all sufficiently large n has degree at least $\ell-1$. Together with the Verrill recurrence and the order bound in Remark 45.1, the smallest possible recurrence order and the smallest possible coefficient degree are respectively $\lceil \ell/2 \rceil$ and $\ell-1$; the same recurrence attains both.

Proof. Put $A(z) = \sum_{n\geq 0} a_n z^n$. The estimate $a_n \leq \ell^{2n}$ gives convergence near zero. The recurrence in Remark 45.1, originally due to Verrill [22, 5], has the form

$$\sum_{k=0}^{K} R_k(n) a_{n-k} = 0 \quad (n \geq 1), \qquad R_0(n) = n^{\ell-1}, \qquad \deg R_k \leq \ell-1, \quad K = \lceil \ell/2 \rceil,$$

with negative-index terms omitted. For $\ell \geq 2$ it gives the homogeneous equation

$$\left(\theta^{\ell-1} + \sum_{k=1}^{K} z^k R_k(\theta+k)\right) A = 0.$$

The $n=0$ coefficient vanishes as well. This operator has order $\ell-1$ and is Fuchsian at zero, since its lowest θ-coefficient is $\theta^{\ell-1}$ and all the other θ-coefficients have degree at most $\ell-1$. Its minimal right factor, the minimal equation of A, is consequently regular singular or ordinary there.

The second factorial Borel transform is

$$A_2(x) = \sum_{n\geq 0} \frac{a_n x^n}{(n!)^2} = I_0(2\sqrt{x})^\ell.$$

Its minimal differential module is the ℓth symmetric power of the rank-two Bessel module, as in Remark 45.1. The base module has a nontrivial unipotent monodromy block of size two at zero. Its ℓth symmetric power therefore has a single block $J_{\ell+1}(1)$.

If a recurrence of degree $d \geq 1$ exists, Lemma 45.2 bounds the unit-eigenvalue Jordan blocks of A by d. Theorem 27.3, applied twice, bounds those of A_2 by $d+2$. Hence $\ell+1 \leq d+2$, or $d \geq \ell-1$. For $d=0$, A would be rational and the same Borel theorem would bound every block of A_2 by three; this excludes $\ell \geq 3$. For $\ell = 2$, the identity $a_n = \binom{2n}{n}$ gives $A(z) = (1-4z)^{-1/2}$, which is not rational and thus also excludes degree zero. Finally, for $\ell = 1$ the sequence is constantly one and has a degree-zero recurrence. These cases prove the lower bound for every ℓ, and the stated Verrill recurrence supplies the upper bound. □

Remark 45.4. This degree bound uses the logarithmic monodromy at the origin, whereas the exponent-divisibility argument alone sees only the repeated exponent zero. The sharper dictionary in Theorem 11.7(c) additionally proves that $N^{\ell-1}$ divides the leading coefficient of every backward recurrence, at arbitrary order.

Remark 45.5 (Scope of the recurrence classifications).

1. Theorem 11.1 proves minimality for all positive β and nonnegative integer twists. The Bessel-module and jet-field classification extends to every positive even β by Theorems 20.1–20.3; no such classification is asserted for all non-even β.

2. The minimum degree of every twisted Toeplitz sequence is $\ell-1$, including the exceptional twists, by Theorem 21.2. For non-even β, the generic degree theorem and the explicit degree-drop family leave the residual exceptional cases specified in Section 23.

3. Remark A.9 of [11] concerns auxiliary functions of the matrix system and is not addressed here.

4. The exponent at every nonzero radial-unitary threshold is determined in Theorem 25.1. Its value is $\frac{1}{2}((\ell-j)^2+j^2-3)$. This indicial calculation does not determine which logarithmic terms occur in a particular physical density branch.

5. The ordinary tableau algebraicity classification is established by Corollary 28.4. The proof uses local monodromy and the fully stated Borel theorem. The twisted permutation classification is stronger and includes every arithmetic section.

6. Theorem 26.1 includes every tableau height, sampling interval and residue. The sampled coefficient degree is not classified. Likewise, a polynomial factor of a global distributional annihilator cannot be cancelled at its zeros without controlling possible point-supported residues.

46 Statistical and arithmetic scope

The statistical results specify estimation rates across the stated parameter ranges and a worst-null testing consequence. The local Fourier argument controls Jordan blocks under explicit regularity hypotheses. The Franel classification includes arithmetic subsequences. The Angelesco and tableau-value theorems have their own parameter restrictions, recorded below.

The following distinctions remain part of the mathematical conclusions.

1. The inequality $|C(t,m,j)| \le 2^{m^2/t}$ with prefactor exactly one is not proved. The exponent $\log 2$ is optimal when an absolute prefactor is allowed. An exponent $1/2$ is valid in the moderate-degree asymptotic regime but impossible globally with a polynomial prefactor.

2. The mixture theorems use a known support interval and either an ordinary unrestricted maximum-likelihood maximizer or a certified approximate fit. Approximate fits require the certificates in Section 40. They do not establish an unknown-support adaptation theorem. The testing rate is worst-null, not a claim that every fixed null is equally difficult. Arbitrary heterogeneous observation kernels still require a separate normalization or comparison argument; the homogeneous posterior identity cannot simply be applied within each stratum.

3. Recurrence width and differential order are different invariants. The Franel results classify width over rational-function coefficient fields. They do not determine the least possible coefficient degrees of every minimum-width recurrence.

4. The Apéry results concern regular eventual tails. Initial singular indices of an arbitrary chosen polynomial recurrence are not silently filled in. Complex linear combinations can share a limit; injectivity is asserted over subfields of the algebraic numbers.

5. The iterated Fourier statement assumes the specified regularity at the source point. It is not a bound for arbitrary irregular source connections. For Bessel determinant values, absence of exceptional nonzero points is proved for tableaux and every integer-twisted Toeplitz family. The multipoint theorem retains distinct nonzero algebraic arguments. These statements do not remove exceptional points for arbitrary rational homogeneous observables.

6. For Angelesco polynomials, a noninteger parameter sum, an arbitrary nonpolynomial tilt, or more than two intervals lies outside the theorem. In the nonalgebraic cases the proof gives at least one transcendental coefficient and at least one transcendental zero, not transcendence of every zero.

The finite verification in Section 35 is a proof certificate, not numerical evidence: every comparison uses rational arithmetic and a one-sided Taylor bound. It checks all 130816 pairs with $1 \leq m \leq t < 512$ through four exhaustive branches, including 12734 direct coefficient comparisons. Symbolic checks of representative Angelesco formulas supplement, but do not replace, the all-parameter proofs. The analytic, arithmetic and statistical arguments underwent separate internal checks. These checks improve the manuscript's auditability; they do not constitute external peer review or establish historical priority.

A Computational verification

The proofs do not depend on the numerical checks listed here. The following paragraphs preserve the computational record reported in the supplied original manuscript. The original supplementary scripts were not supplied with the two attachments, so these historical runs are not presented as newly executed checks. The new exact verification for this revision is reported separately below. The original record uses the prime $p = 2^{31} - 1$, Bessel determinant series, and linear algebra over $\mathbb{F}_p$ unless otherwise specified.

1. *The operator L_ℓ^t, $1 \le \ell \le 10$.* We found no operator of order at most ℓ with coefficients of degree at most $3\ell + 2$. For order $\ell + 1$ there is a one-dimensional space of operators with coefficients of degree $\le \ell$, and none with degree $\le \ell - 1$. The operator has leading coefficient t^ℓ, its θ-form contains only even powers of t, and the indicial roots at 0 are exactly $\{-2\lfloor j/2\rfloor(\ell - \lfloor j/2\rfloor)\}_{j=0}^{\ell}$.

2. *The operator L_ℓ in x, $1 \le \ell \le 13$.* The coefficient degrees are exactly $\lfloor(\ell - 1 + m)/2\rfloor$. The leading coefficients of the θ-coefficients P_k agree with Theorem 2.2(b), and the subleading coefficients with (5.2). For even ℓ, the root of P_K is $-\ell^2/8$.

3. *Orders of the vertex integrals at 0, $2 \le \ell \le 6$.* The coefficients of $\det(\alpha T_I + \beta T_K)$ were evaluated with 400 significant digits at $t = 10^{-5}, 10^{-7}, 10^{-9}$. The logarithmic slopes agree with $-2\lfloor b/2\rfloor(\ell - \lfloor b/2\rfloor)$ up to the expected logarithmic corrections.

4. *Recurrences for u_ℓ, $2 \le \ell \le 12$.* The recurrence of Theorem 2.4(a), built from the computed P_k, holds for all $n < 260$. The formulas for R_0, R_K and the characteristic polynomial hold. The space of recurrences of order K and degree $\ell - 1$ is one-dimensional and is spanned by it.

5. *Exact recurrences, $2 \le \ell \le 7$.* The polynomials P_k were reconstructed over $\mathbb{Q}$ from two primes. The resulting recurrences (Table 3) were checked in exact arithmetic against $u_\ell(n) = \sum_\lambda (f^\lambda)^2$, computed by the hook length formula independently of (2.2), for $n < 26$ ($\ell \le 5$) and $n < 20$ ($\ell = 6, 7$).

6. *Tableaux, $2 \le k \le 9$, exactly.* Minimal recurrences were guessed modulo two primes and reconstructed over $\mathbb{Q}$. They were checked exactly against hook length sums for $n < 56$ ($k \le 6$) and $n < 46$ ($k = 7, 8, 9$). Theorem 2.8 predicts the order, the degree, c_0 and c_R, and all predictions are confirmed. On the same data there is no recurrence of order $R - 1$ and degree $\le r + 1$, and none of order R and degree $r - 1$. Modulo p the predicted orders, degrees and extreme coefficients were also confirmed for $k \le 14$.

7. *Matchings, $1 \le \ell \le 6$, exactly.* The values $m_\ell(n)$ were computed by counting oscillating tableaux of height at most ℓ, independently of (2.4). The minimal recurrences have the order, degree and extreme coefficients predicted by Theorem 2.9. Modulo p, from (2.4), the same was confirmed for $\ell \le 10$.

8. *Twisted determinants, variable x.* For $1 \le \ell \le 8$ and $0 \le q \le 5$, for $q = \ell \in \{6, 7, 8\}$ and for $(\ell, q) \in \{(3, 9), (4, 7), (5, 10)\}$, the series $G_{\ell,q}$ was computed from the determinant. Its first nine coefficients agree, in exact arithmetic, with the expansion of Lemma 6.1 evaluated by the hook length formula. In every case there is no operator of order ℓ with coefficients of degree $\le 3\ell + 2$, and there is a unique operator of order $\ell + 1$ and degree ℓ, none of degree $\ell - 1$. Its coefficients satisfy $q_{\ell+1} = x^\ell$, $x^{m-1} \mid q_m$ and $\deg q_m = \lfloor(\ell - 1 + m)/2\rfloor$, and P_0, the $\mathrm{lc}(P_k)$ and P_K are as in Theorem 2.6(a). The Taylor coefficients satisfy no recurrence of order $K - 1$ and degree $\le 3\ell + 3$.

9. *Twisted determinants, variable t.* For 21 pairs (ℓ, q) with $\ell \le 8$, among them the exceptional pairs $(3,4)$, $(4,3)$, $(5,12)$, $(6,8)$ and $(8,6)$, the minimal operator of $D_{\ell,q}$ has order $\ell+1$, leading coefficient t^{ℓ} or $t^{\ell+1}$ as predicted by Theorem 2.6(b), and the predicted exponents at 0.

10. *Twisted recurrences.* For $2 \le \ell \le 7$ with $1 \le q \le 4$, and for $q = \ell \in \{5,6,7\}$, the sequence $u_{\ell,q}$ satisfies a unique recurrence of order K and degree $\ell - 1$, with leading coefficient $\prod_{i=1}^{\ell-1}(n + i(\ell + q - i))$; there is none of order $K-1$ and degree $\le 3\ell+3$, and none of order K and degree $\ell - 2$.

11. *Displacement with drift.* For $\ell \le 6$ and $q \le 8$, the maximum of $E_q(\pi, S)$ over $\pi \in S_\ell$ and $|S| = b$ was computed by enumeration. It equals $f_{(b)}$ for every b.

A second set of scripts, written independently of the first, computes all sequences by dynamic programming over Young diagrams and oscillating tableaux instead of the Bessel determinants. It confirmed the statements of Sections 5–9 for small parameters, including the vertex constants C_b, the orders of the $\Psi_b^{(q)}$ at 0, and Lemma 6.3 for $\ell \le 9$, $q \le 12$. This sentence records the original manuscript's reported supplementary checks; those scripts are not part of the present source-and-PDF deliverable.

Exact checks executed for this revision

The finite continuant and path-matching formulas were compared symbolically for $1 \le \ell \le 8$. Independently computed hook-length sums verified 300 twisted recurrence instances with $1 \le \ell \le 5$, $0 \le q \le 3$ and $1 \le n \le 15$. The exponent-zero condition and the Pythagorean square criterion agreed in all $12,800$ cases with $1 \le \ell \le 80$, $1 \le q \le 160$. A separate exact determinant calculation checked the Catalan-tail identity in 99 pairs (q, n). These finite checks detect normalization errors; the all-parameter claims are established by the preceding proofs.

Additional independent checks use rational arithmetic and symbolic polynomial identities in SymPy. They make no numerical claim of algebraic independence. The checks were executed for this revision, with the following results.

- The Laurent jet determinant, including the closed constant in Proposition 24.3, was checked for $1 \le \ell \le 3$, $0 \le q \le 2$, and $(\ell, q) = (4, 0)$: ten cases. For the nine cases with $\ell \le 3$, the inverse matrices were checked exactly, and all their denominators were confirmed to be monomials in x. The coordinate conversions between twists $q = 0, 1, 2$ were also verified exactly.

- The full indicial polynomials in Theorem 25.1 were checked at all nonzero thresholds for $2 \le \ell \le 8$: nineteen thresholds. The operators were built from the integral Ore continuant and the exact recurrence-to-density dictionary.

- The tableau indicial polynomials in Corollary 25.3 were checked for $2 \le k \le 7$: eighteen thresholds. The symplectic ones were checked for $1 \le \ell \le 3$: nine thresholds. In these checks the scalar operators were constructed independently from the homogeneous Bessel determinant forms, and their Fourier transforms were computed by the Weyl-algebra product rule.

All these finite checks passed. They confirm normalization and indexing in the formulas, while the preceding proofs establish the stated results for all parameters.

B Proof of Theorem 8.2

For completeness we give the proof of Theorem 8.2, following [15, Part I]. Let $k = 2r + \varepsilon$ with $r \geq 1$, $a = \varepsilon - \frac{1}{2}$, $w(u) = (2-u)^a(2+u)^{1/2}$ on $(-2,2)$ and $\tilde{w}(u) = (2-u)^a|2+u|^{1/2}$ on $(-\infty,-2)$.

B.1. One-variable functions

For $t > 0$ put

$$\varphi_I(t) = \int_{-2}^{2} e^{tu} w(u)\,du, \qquad \varphi_K(t) = \int_{-\infty}^{-2} e^{tu}\tilde{w}(u)\,du.$$

Let $g(u) = (4-u^2)w(u)$. Then $g' = w\cdot\big((2-2\varepsilon) - (2+\varepsilon)u\big)$ and $g(\pm 2) = 0$. Integrating $(e^{tu}g)'$ over $[-2,2]$, and similarly over $(-\infty,-2]$, shows that both functions satisfy

$$B_\varepsilon f := tf'' + (2+\varepsilon)f' - (4t + 2 - 2\varepsilon)f = 0.$$

They are independent: $\varphi_I(0)$ is finite, while $\varphi_K(t) \to \infty$ as $t \to 0^+$. A direct computation gives the operator identities

$$B_0 \circ (1 + \tfrac{1}{2}\partial) = (\tfrac{1}{2}\partial + 1)\circ T, \qquad B_1 \circ \big((2t)^{-1}\partial\big) = \big(\partial \circ (2t)^{-1}\big)\circ T,$$

with T as in (5.1). Hence B_ε is gauge equivalent to T over $\mathbb{C}(t)$, and its differential Galois group acts on $\mathrm{Sol}(B_\varepsilon)$ as $\mathrm{SL}_2(\mathbb{C})$ by Kolchin's theorem.

B.2. The Hankel form

For $g_1,\dots,g_r \in \mathrm{Sol}(B_\varepsilon)$ put

$$F(g_1,\dots,g_r) = \sum_{\sigma,\tau\in S_r} \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)\prod_{p=1}^{r} g_p^{(\sigma(p)+\tau(p)-2)}.$$

It is symmetric and multilinear. Suppose $g_p = \int_{\gamma_p} e^{tu}\omega_p(u)\,du$, where each pair (γ_p,ω_p) is either $([-2,2],w)$ or $((-\infty,-2],\tilde{w})$. Expanding $\Delta(u)^2 = \det[u_p^{i-1}]^2$ gives

$$F(g_1,\dots,g_r) = \int_{\gamma_1\times\cdots\times\gamma_r} e^{t\sum u_p}\Delta(u)^2\prod_p \omega_p(u_p)\,du.$$

In particular $F(\varphi_I,\dots,\varphi_I) = r!\,(2\pi)^r e^{-\varepsilon t}Y_k$ by Lemma 8.1. Put $\Psi_b = F(\varphi_K^{(b\text{ times})},\varphi_I^{(r-b\text{ times})})$ for $0 \leq b \leq r$. As in Proposition 5.3, Lemma 3.7 and polarization show that the solution space of the minimal operator of $e^{-\varepsilon t}Y_k$ is $V = \mathrm{span}\{\Psi_0,\dots,\Psi_r\}$. The minimal operator of Y_k then has solution space $e^{\varepsilon t}V$.

B.3. Vertices

On $(-\infty,-2]^b\times[-2,2]^{r-b}$ the function $\sum_p u_p$ attains its maximum $\mu_b = 2r-4b$ only at the vertex with first b coordinates -2 and the others 2. In the local coordinates $y_p = -2-u_p \geq 0$ $(p\leq b)$ and $x_p = 2-u_p \geq 0$ $(p>b)$, the integrand is

$$e^{\mu_b t}e^{-t(\sum x+\sum y)}\,\Delta(x)^2\Delta(y)^2\prod x_p^a\prod y_p^{1/2}\,H(x,y),$$

with H real-analytic near 0 and $H(0) = 4^{2b(r-b)+(r-b)/2+ab} > 0$. The weight in front of H is positively homogeneous of degree $\gamma_b^{(k)} - r$, and Watson's lemma on the orthant gives

$$\Psi_b^{(m)}(t) = C_b e^{\mu_b t}t^{-\gamma_b^{(k)}}\big(\mu_b^m + O(t^{-1})\big), \qquad C_b = H(0)\,S_{r-b}(a)\,S_b(\tfrac{1}{2}) > 0,$$

with $S_n(c) = \int_{(0,\infty)^n} e^{-\sum x}\Delta(x)^2\prod x^c\,dx$. The expansion is in integral powers of t^{-1}, and the expansions of the derivatives are the termwise derivatives. Since the μ_b are distinct, $\dim V = r+1$, and $y_b = e^{\varepsilon t}\Psi_b$ has rate $\lambda_b^{(k)} = \varepsilon + \mu_b$.

B.4. The origin

Elements of V are polynomials in φ_I, φ_K and their derivatives with coefficients in $\mathbb{C}(t)$. The exponents of B_ε at 0 are 0 and $-1-\varepsilon$, so $V \subset \mathcal{N}$.

On the rays substitute $u_p = -s_p/t$. Collecting powers of t, using $b+\varepsilon b+b(b-1)+2b(r-b) = -e_b$, gives

$$t^{-e_b}\Psi_b(t) = \iint e^{t\sum u'} e^{-\sum s}\Delta(s)^2 \prod_{p,i}(s_p+tu_i')^2 \prod_p (s_p+2t)^a (s_p-2t)^{1/2}\,\Delta(u')^2 \prod w(u_i')\,ds\,du',$$

over $s_p \geq 2t$ and $u' \in [-2,2]^{r-b}$. For $0 < t \leq 1$ the integrand is dominated by an integrable function. By dominated convergence,

$$\lim_{t\to 0^+} t^{-e_b}\Psi_b(t) = S_b(2r-2b+\varepsilon)\int_{[-2,2]^{r-b}} \Delta(u)^2 \prod w(u_i)\,du > 0.$$

Hence y_b has leading part $C_b' t^{e_b}$ with $C_b' \neq 0$. The e_b are distinct, $e_0 = 0 > e_1 > \cdots > e_r$, so by Lemma 3.2 they are exactly the exponents at 0.

B.5. Conclusion

As in the proof of Theorem 5.8, Abel's identity and B.3 give

$$\sum_{\tau\neq 0}\kappa_\tau = \binom{r+1}{2} - \sum_j \gamma_j^{(k)} - \sum_b e_b.$$

Using $a+\frac{1}{2} = \varepsilon$,

$$\sum_j \gamma_j^{(k)} = \frac{r(r+1)(2r+1)}{3} + \varepsilon\frac{r(r+1)}{2}, \qquad \sum_b e_b = \frac{r(r+1)(2r+1)}{6} - r^2(r+1) - \varepsilon\frac{r(r+1)}{2},$$

whose sum is $\frac{r(r+1)}{2} = \binom{r+1}{2}$. Hence there are no apparent singularities. The leading coefficient is ct^A, and Lemma 3.5 gives $\deg p_m \leq A$ and the formula for χ_k. Since all exponents at 0 are ≤ 0 and 0 is an exponent, the argument in the proof of Theorem 5.8 gives $A = r$ and $t^{m-1} \mid p_m$. $\square$

C Verification of the ensemble and moment formulas

The proofs of Sections 11–23 are analytic and algebraic. The supplementary calculations associated with the ensemble formulas provide finite checks, not general minimality proofs. The following checks were reproduced for this integrated version.

1. The symbolic circular operators for $n \le 3$ and Jacobi operators for $n \le 2$ agree exactly with the supplied symbolic output.

2. Nine circular and nine Jacobi parameter cases with $n \le 3$ satisfy the scalar coefficient recursion through degree 17 modulo $2^{31} - 1$. These include fractional positive β, integer and fractional twists, and Jacobi endpoint exponents $-1/2$.

3. For $n = 1, 2$, the Bessel moment recurrence agrees exactly with the gamma-integral ratios $c_1(k+2)/c_1(k) = (k+1)^2$ and $c_2(k+2)/c_2(k) = (k+1)^3/[4(k+2)]$.

4. At $(n, \beta, q) = (2, 1/2, -1/2)$ the scalar operator is $x^2\partial^3 + (7/4)x\partial^2 - 4x\partial$, verifying the stated failure of the degree formula at that boundary.

5. Polynomiality of the sharpened dictionary coefficients was checked for $1 \le M \le 4$, $0 \le n_0 \le 3$, $0 \le r \le 3$.

The supplied archive additionally reports larger finite sweeps, quadrature tests and recurrence searches. These were not all rerun. An operator search with bounded coefficient degree and a finite Taylor prefix establishes only a statement about that tested ansatz; it does not exclude operators of arbitrary degree or recurrences beginning at an arbitrary later index. In particular the archived circular search tests coefficient degree at most $2n + 2$, and the planar-walk search uses a 161-term prefix. The general lower bounds in this volume come from visibility, Laurent transfer and monodromy.

References

[1] J. Baik and E. M. Rains, *Algebraic aspects of increasing subsequences*, Duke Math. J. **109** (2001), 1–65.

[2] A. Berele, *A Schensted-type correspondence for the symplectic group*, J. Combin. Theory Ser. A **43** (1986), 320–328.

[3] F. Bergeron, L. Favreau and D. Krob, *Conjectures on the enumeration of tableaux of bounded height*, Discrete Math. **139** (1995), 463–468.

[4] F. Bergeron and F. Gascon, *Counting Young tableaux of bounded height*, J. Integer Seq. **3** (2000), Article 00.1.7.

[5] J. M. Borwein, D. Nuyens, A. Straub and J. Wan, *Some arithmetic properties of short random walk integrals*, Ramanujan J. **26** (2011), 109–132.

[6] J. M. Borwein, A. Straub, J. Wan and W. Zudilin, *Densities of short uniform random walks*, Canad. J. Math. **64** (2012), 961–990.

[7] W. Y. C. Chen, E. Y. P. Deng, R. R. X. Du, R. P. Stanley and C. H. Yan, *Crossings and nestings of matchings and partitions*, Trans. Amer. Math. Soc. **359** (2007), 1555–1575.

[8] W. Y. C. Chen, J. Qin, C. M. Reidys and D. Zeilberger, *Efficient counting and asymptotics of k-noncrossing tangled-diagrams*, preprint (2008), arXiv:0802.3491.

[9] *NIST Digital Library of Mathematical Functions*, `https://dlmf.nist.gov`, Chapter 10.

[10] P. J. Forrester and S. Kumar, *Differential recurrences for the distribution of the trace of the β-Jacobi ensemble*, Phys. D **434** (2022), 133220; arXiv:2011.00787.

[11] P. J. Forrester and F. Wei, *Higher-order linear differential equations for unitary matrix integrals: applications and generalisations*, SIGMA **22** (2026), 015; arXiv:2508.20797.

[12] I. M. Gessel, *Symmetric functions and P-recursiveness*, J. Combin. Theory Ser. A **53** (1990), 257–285.

[13] E. L. Ince, *Ordinary Differential Equations*, Dover, New York, 1956.

[14] K. Johansson, *On random matrices from the compact classical groups*, Ann. of Math. (2) **145** (1997), 519–545.

[15] Keio, *Minimal differential and recurrence orders: Bessel periods, bounded-height tableaux, and banana integrals*, manuscript, October 2026.

[16] J. P. Keating and F. Wei, *Joint moments of higher order derivatives of CUE characteristic polynomials II: Structures, recursive relations, and applications*, Nonlinearity **37** (2024), 085009; arXiv:2307.02831.

[17] E. R. Kolchin, *Algebraic groups and algebraic dependence*, Amer. J. Math. **90** (1968), 1151–1164.

[18] M. Mishna, *On standard Young tableaux of bounded height*, in: Recent Trends in Algebraic Combinatorics, Springer, 2019; arXiv:1805.08130.

[19] E. M. Rains, *Increasing subsequences and the classical groups*, Electron. J. Combin. **5** (1998), R12.

[20] A. Regev, *Asymptotic values for degrees associated with strips of Young diagrams*, Adv. Math. **41** (1981), 115–136.

[21] S. Sundaram, *Tableaux in the representation theory of the classical Lie groups*, in: Invariant Theory and Tableaux, IMA Vol. Math. Appl. **19**, Springer, 1990, 191–225.

[22] H. A. Verrill, *Sums of squares of binomial coefficients, with applications to Picard–Fuchs equations*, preprint (2004), arXiv:math/0407327.

[23] M. van der Put and M. F. Singer, *Galois Theory of Linear Differential Equations*, Grundlehren **328**, Springer, 2003.

[24] F. Beukers, *A refined version of the Siegel–Shidlovskii theorem*, Ann. of Math. (2) **163** (2006), 369–379, https://doi.org/10.4007/annals.2006.163.369.

[25] D. Arinkin, *Fourier transform and middle convolution for irregular D-modules*, 2008, especially Corollary 2.3 and §3.3. https://arxiv.org/abs/0808.0699.

[26] C. Sabbah, *An explicit stationary phase formula for the local formal Fourier–Laplace transform*, Contemp. Math. **474** (2008), 309–330, especially Remarks 5.2 and 5.3(4). https://arxiv.org/abs/0706.3570.

[27] A. Baker, Linear forms in the logarithms of algebraic numbers. III, *Mathematika* **14** (1967), 220–228. doi:10.1112/S0025579300003843.

[28] W. Van Assche and E. Coussement, Some classical multiple orthogonal polynomials, *Journal of Computational and Applied Mathematics* **127** (2001), 317–347. doi:10.1016/S0377-0427(00)00503-3. Author manuscript: https://wis.kuleuven.be/analyse/members/walter/clasmult.pdf.

[29] L. Kania, L. Wasserman and S. Balakrishnan, *Testing Random Effects for Binomial Data*, arXiv:2504.13977v3, 2025, Section 3 and Theorem 3. https://arxiv.org/pdf/2504.13977.

[30] R. Korlakai Vinayak, W. Kong, G. Valiant and S. Kakade, *Maximum likelihood estimation for learning populations of parameters*, Proceedings of ICML 2019, PMLR **97**, 6448–6457. https://proceedings.mlr.press/v97/vinayak19a.html.

[31] C. Musco, C. Musco, L. Rosenblatt and A. V. Singh, *Sharper Bounds for Chebyshev Moment Matching, with Applications*, Proceedings of COLT 2025, PMLR **291**, 4309–4358. https://proceedings.mlr.press/v291/musco25a.html.

[32] P. Diaconis, K. Khare and L. Saloff-Coste, *Gibbs sampling, exponential families and orthogonal polynomials*, Statistical Science **23** (2008), 151–178.

[33] H. Dette, *New bounds for Hahn and Krawtchouk polynomials*, SIAM Journal on Mathematical Analysis **26** (1995), 1647–1659. https://arxiv.org/abs/math/9406223; Corollary 3.6(b).

[34] N. M. Katz, On the calculation of some differential Galois groups, *Invent. Math.* **87** (1987), 13–61, Corollary 4.4.8. https://web.math.princeton.edu/~nmk/old/calcdiffgal.pdf.

[35] J. Fresán, C. Sabbah and J.-D. Yu, Hodge theory of Kloosterman connections, *Duke Math. J.* **171** (2022), 1649–1747. https://perso.pages.math.cnrs.fr/users/claude.sabbah/articles/kloosterman-revised10.pdf.

[36] M. Stoll, Bounds for the length of recurrence relations for convolutions of P-recursive sequences, *European J. Combin.* **18** (1997), 707–712. https://doi.org/10.1006/eujc.1996.0123.

[37] R. J. McIntosh, Recurrences for alternating sums of powers of binomial coefficients, *J. Combin. Theory Ser. A* **63** (1993), 223–233. https://doi.org/10.1016/0097-3165(93)90058-G.

[38] T. Matala-Aho, On recurrences for some hypergeometric type polynomials, *Astérisque* **198–199–200** (1991), 237–244. https://www.numdam.org/item/AST_1991__198-199-200__237_0/.

[39] I. Gaiur, V. Rubtsov and D. van Straten, Product formulas for the higher Bessel functions, *Selecta Math. (N.S.)* **32** (2026), article 12. https://doi.org/10.1007/s00029-025-01114-w. Preprint: https://arxiv.org/abs/2405.03015.

[40] A. Straub and W. Zudilin, Sums of powers of binomials, their Apéry limits, and Franel's suspicions, *Int. Math. Res. Not.* **2023** (2023), no. 11, 9861–9879. https://arminstraub.com/downloads/pub/franel-aperylimits.pdf.

[41] M. Chamberland and A. Straub, Apéry limits: experiments and proofs, *Amer. Math. Monthly* **128** (2021), no. 9, 811–824. https://arminstraub.com/downloads/pub/aperylimits.pdf.

[42] M. van Hoeij, Submodule approach to creative telescoping, preprint, 2024, Remark 6. https://arxiv.org/abs/2401.08455.

[43] Z. Miao, W. Kong, R. Korlakai Vinayak, W. Sun and F. Han, *Fisher–Pitman permutation tests based on nonparametric Poisson mixtures with application to single cell genomics*, author version dated 8 June 2021, arXiv:2106.03022v1, Theorems 6.1–6.2 and Remark 6.1. https://ramyakv.github.io/testing-poisson-npmle.pdf.

[44] D. J. Grabiner and P. Magyar, *Random walks in Weyl chambers and the decomposition of tensor powers*, J. Algebraic Combin. **2** (1993), 239–260.

[45] E. Y. Jin, C. M. Reidys, and R. R. Wang, *Asymptotic analysis of k-noncrossing matchings*, arXiv:0803.0848 (2008), equation (1.4). https://arxiv.org/abs/0803.0848.

[46] A. P. Dempster, N. M. Laird and D. B. Rubin, Maximum likelihood from incomplete data via the EM algorithm, *Journal of the Royal Statistical Society, Series B* **39** (1977), 1–22. doi:10.1111/j.2517-6161.1977.tb01600.x.

[47] L. H. Dicker and S. D. Zhao, High-dimensional classification via nonparametric empirical Bayes and maximum likelihood inference, *Biometrika* **103** (2016), 21–34. doi:10.1093/biomet/asv067.

[48] J. A. Soloff, A. Guntuboyina and B. Sen, Multivariate, heteroscedastic empirical Bayes via nonparametric maximum likelihood, *Journal of the Royal Statistical Society, Series B* **87** (2025), 1–32. doi:10.1093/jrsssb/qkae040.

[49] E. Delaygue, *A Lindemann–Weierstrass theorem for E-functions*, arXiv:2210.12046v2, 7 March 2025, https://arxiv.org/abs/2210.12046.

[50] D. H. Bailey, J. M. Borwein and R. E. Crandall, *Integrals of the Ising class*, J. Phys. A **39** (2006), 12271–12302.

[51] D. H. Bailey, D. Borwein, J. M. Borwein and R. E. Crandall, *Hypergeometric forms for Ising-class integrals*, Experiment. Math. **16** (2007), 257–276.

[52] D. H. Bailey, J. M. Borwein, D. Broadhurst and M. L. Glasser, *Elliptic integral evaluations of Bessel moments and applications*, J. Phys. A **41** (2008), 205203.

[53] W. Balser, *Formal Power Series and Linear Systems of Meromorphic Ordinary Differential Equations*, Universitext, Springer, New York, 2000.

[54] J. M. Borwein and B. Salvy, *A proof of a recursion for Bessel moments*, Experiment. Math. **17** (2008), 223–230.

[55] A. Erdélyi, *Asymptotic Expansions*, Dover, New York, 1956.

[56] D. J. Grabiner and P. Magyar, *Random walks in Weyl chambers and the decomposition of tensor powers*, J. Algebraic Combin. **2** (1993), 239–260.

[57] L. Hörmander, *The Analysis of Linear Partial Differential Operators I*, 2nd ed., Springer, Berlin, 1990.

[58] M. Kac, *Random walk and the theory of Brownian motion*, Amer. Math. Monthly **54** (1947), 369–391.

[59] J. Kaneko, *Selberg integrals and hypergeometric functions associated with Jack polynomials*, SIAM J. Math. Anal. **24** (1993), 1086–1110.

[60] OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org.

[61] J. J. Sylvester, *Théorème sur les déterminants de M. Sylvester*, Nouv. Ann. Math. **13** (1854), 305.

[62] O. Taussky and J. Todd, *Another look at a matrix of Mark Kac*, Linear Algebra Appl. **150** (1991), 341–360.

[63] E. C. Titchmarsh, *The Theory of Functions*, 2nd ed., Oxford University Press, 1939.

[64] W. Wasow, *Asymptotic Expansions for Ordinary Differential Equations*, Interscience, New York, 1965.

[65] R. Wong, *Asymptotic Approximations of Integrals*, Academic Press, Boston, 1989.